TEAMCENTER

# Engineering BOM Management — Usage

Software Version 2606

SIEMENS

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# 1. About engineering BOM management

Managing the part and design data of a product within a single structure results in a complex BOM, which is difficult to maintain, resulting in rework, delays, and losses. It is more efficient to enable independent evolution of part and design lifecycles. Teamcenter Engineering BOM address this challenge by maintaining the parts and designs as separate structures. Aligning the two structures allows you to perform BOM-driven digital mockups.

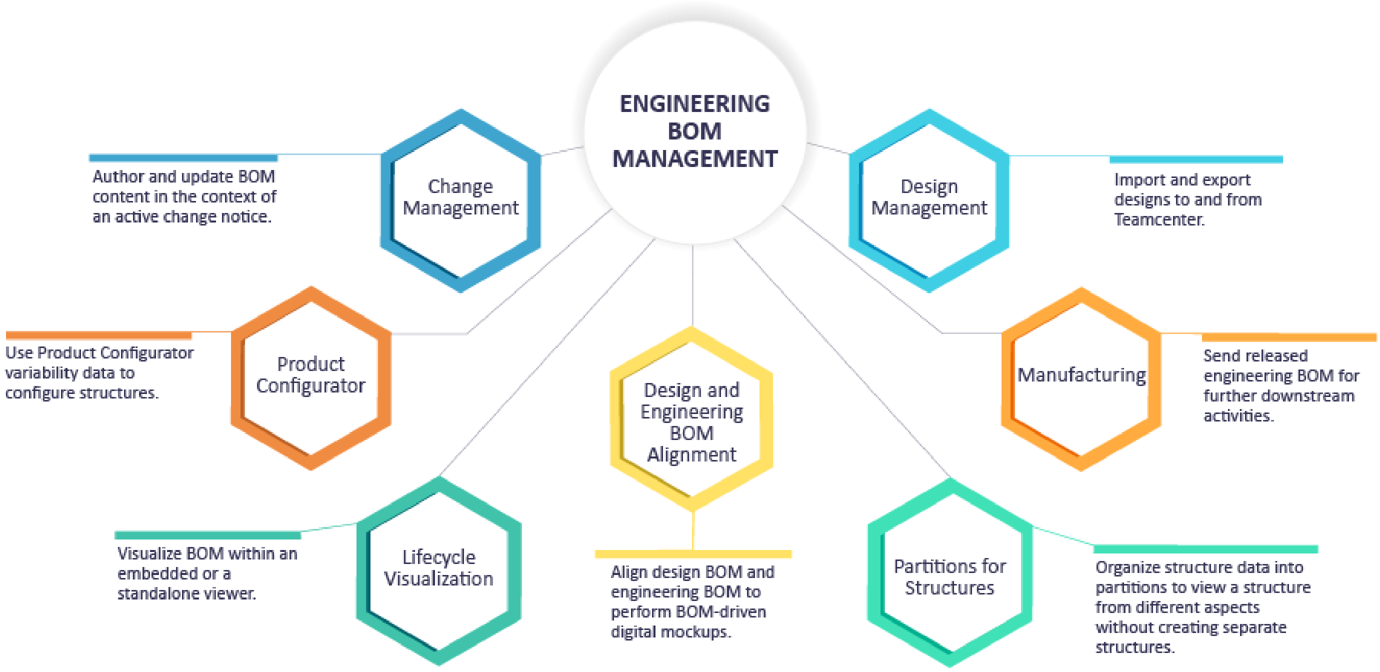
> **Note**
>
> Videos are not included in PDF. To access videos, use HTML.

The engineering BOM management is a multi-domain engineering bill of materials that help you manage mechanical, electrical, electronics, software parts, and documents required for a product. The engineering BOM management further helps you align the part structure with the design, and then visualize the product structure in real-time. In Teamcenter, you can manage your engineering BOM either as an Assembly BOM or as a Usage BOM.

If you want the parts not to have their independent lifecycle, instead the parts must be used based on their occurrence properties, you must use Assembly BOM. In Assembly BOM mode of engineering BOM management, the assembly parent owns the lifecycle of the child occurrences.

If you want the parts to have their independent lifecycle and change control, referred to as *part usages*, you must use Usage BOM. In Usage BOM mode of engineering BOM management, a part usage can be revised and released independent of its parent assembly. The engineering BOM management works on the principle that every product uses parts that are drawn from a much larger master part list.

You can use several Teamcenter applications that provide additional features to manage the engineering BOM data.




## Where do I go from here?

| | |
|---|---|
| Administrator | See Engineering BOM Management — Deployment and Administration. |
| Business User | |
| New user | Familiarize yourself with the engineering BOM **terms** used in Teamcenter. You can also look at the **engineering BOM management business process**. |
| Set an active change notice | It is recommended that you **set an active change** before you start creating or updating the engineering BOM data. |
| Manage the engineering BOM data of a product | Considering your business requirement, you can manage the engineering BOM data by creating and maintaining either a **Usage BOM** or an **Assembly BOM**. |
| Verify if the engineering BOM data is created correctly | Review the engineering BOM data to validate if the data is created correctly. You can **compare** the data, **validate the alignment**, and **visualize** structures. |
| Release engineering BOM data | The engineering BOM **release process** includes validation steps to ensure that the BOM adheres to your business rules before it is released. |

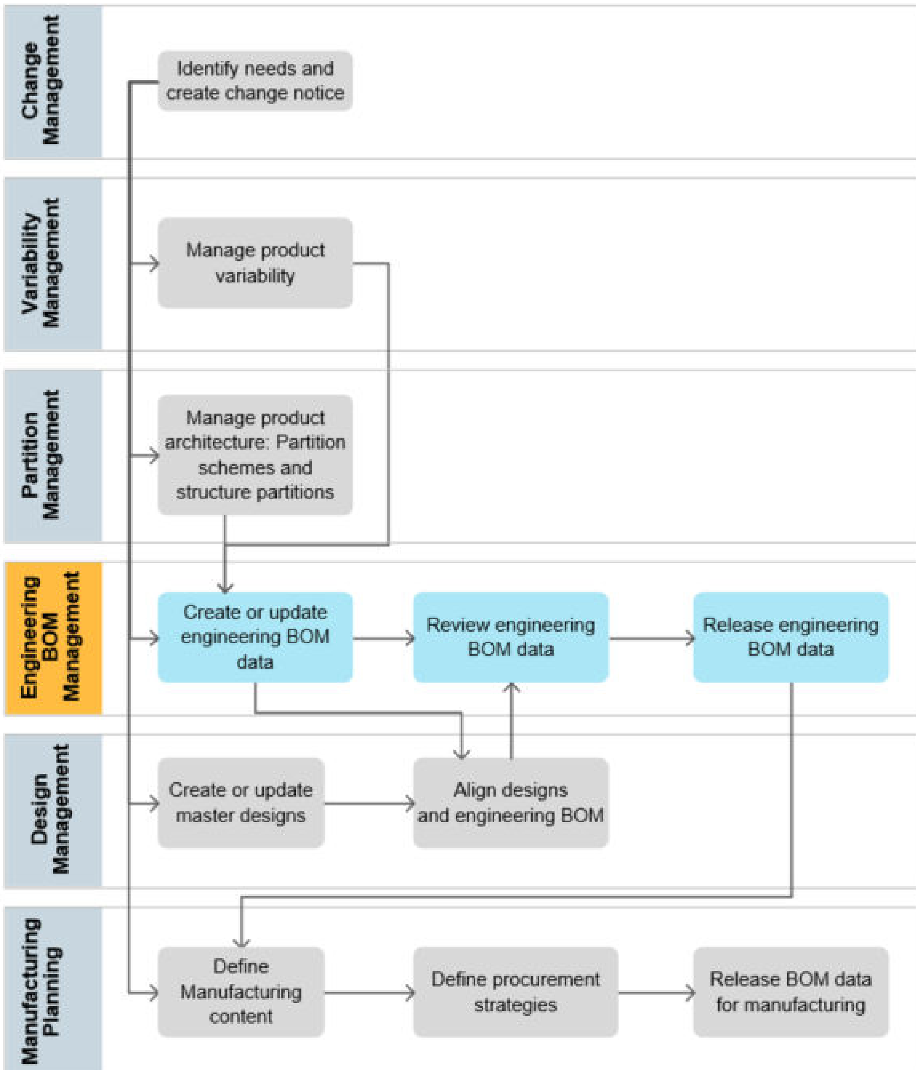
# 2. Engineering BOM management terms

The terms explained in this topic help you manage your bill of materials (BOM) data.

<table>
<tr><th>The term</th><th>Refers to</th></tr>
<tr><td>Design</td><td>The virtual representation of a part. It represents what a part will look like in its floor position. For example,  represents a wheel. Designs are attached to design revisions in Teamcenter.</td></tr>
<tr><td>Part</td><td>This is the primary component of a product. For example, the product CrossKart consists of different parts such as Wheel, Engine, and Chassis. Each part is saved as a part revision in Teamcenter.<br><br>A part can be of the following types:<br>
<table>
<tr><th>The type</th><th>Refers to</th></tr>
<tr><td>Generic part</td><td>A placeholder part that cannot be ordered or procured, or assembled in a product. It has variability associated with it. It is used in a configurable assembly and is replaced with a specific part when a 100% product variant is derived. You can add child parts to a generic part but they neither participate in alignment process nor corresponding manufacturing BOM creation.</td></tr>
<tr><td>Configurable assembly</td><td>An assembly part that has variability associated with its child parts.<br><br>A configurable assembly can contain other configurable assemblies and generic parts.</td></tr>
<tr><td>Supplier Part</td><td>A supplier part is an assembly that is defined in the design structure. It can contain parts and subassemblies. It is purchased as a complete or whole assembly. The terms piece part, bought out part, purchased assembly, and foreign part are synonyms for supplier part.<br><br>You can add child parts to a supplier part. The supplier part participates in the alignment process but not in the corresponding manufacturing BOM creation. You can add a spare part as a child to a supplier part.</td></tr>
</table></td></tr>
<tr><td>Alternate part</td><td>A part that can be used in place of another part in all products.</td></tr>
<tr><td>Substitute part</td><td>A part that can be used in place of an occurrence of another part within a single assembly.</td></tr>
</table>

| The term | Refers to |
|---|---|
| Flexible part | A part that can be used in a product in various states. Examples include a coil spring, hose pipe, and wire harness. |
| Bought out part | A part that is purchased and not manufactured in-house. |
| Engineering BOM | The term engineering BOM refers to the structure that is a Usage BOM and an Assembly BOM structure. |
| Usage BOM | A 150% BOM that represents the complete content of a product and its variants, if available. This 150% BOM uses item revisions and occurrences, called collaborative part and part usages. The part usages have independent lifecyle and change control. The part usages in a Usage BOM can be revised and released independently of the parent assembly. |
| Part Usage | The usage of a part in a Usage BOM. |
| Part structure (Assembly BOM) | An Assembly BOM is a BOM structure that is based on occurrence in which an assembly owns the lifecycle of the occurrence of the child parts. |
| Part occurrence | An instance of a part in a Assembly BOM. For example, the wheel assembly contains the part occurrences of *Tire*, *Rim*, and *Valve*. |
| Partition | A container to logically organize the engineering BOM data. |
| Design | A virtual representation of a part. |
| Design structure | A design structure of a product to represent it virtually from a geometrical standpoint. It consists of designs, which contain the design information such as CAD and product manufacturing information (PMI). |
| Design occurrence | An occurrence of a design in a design structure. A design occurrence contains the position information of a part. For example, the design occurrence of a wheel contains the information related to its placement in *Crosskart*. |
| Design-part alignment | The alignment between a design and a part. It shows what a part will look like in its floor position. |
| Design occurrence-part occurrence alignment | The alignment between a design occurrence and a part occurrence. It shows what a part will look like when placed in a product. |
| Design-Usage BOM link | This is the link between a design structure and its corresponding Usage BOM. |
| Solution variant | A 100% buildable BOM with unique part numbers. |

# 3. Engineering BOM management business process



# 4. Access the engineering BOM workspace

As a release engineer you can access your engineering BOM data inside a dedicated workspace. To access the **Engineering BOM** workspace, follow these steps.

**Procedure**

1. Click your profile icon on the session controls.
2. From the **Group** list, select **Engineering**.
3. From the **Role** list, select a role. For example, select **BOM Engineer**.
4. From the **Workspace** list, select **Engineering BOM**.

# 5. Find, filter, and update structures using Teamcenter Copilot for Structures

Use Teamcenter Copilot for Structures to find, filter, and change structures and review the updated structure or visual representation in real time.

#### Prerequisites

Your administrator must install and configure Teamcenter Artificial Intelligence (AI) services.

#### Procedure

1. Click Copilot ✦.
2. Select **Teamcenter Structures**.
3. Use natural language to type your prompt.

   **Example**

   **`Locate and open the most recent hard drive BOM.`**

4. Type additional requests to filter the structure, find specific parts, complete weight and balance rollups, or replace parts.
5. Click **Submit** or press **Enter** to perform the action.
   Copilot completes the action and updates the structure and visual representation based on your request.

   Copilot also suggests a list of follow-up prompts for the next steps. You can either use the suggested follow-up prompts or type a different prompt.

6. (Optional) You can **Copy** the Copilot response for future use.
7. (Optional) To clear the current conversation and start a new chat, click **New Chat**.

#### Examples

- You have learned that one of your BOM components needs a clearance of 5 cm. You need to quickly identify if there are any other pieces within 5 cm of this component. When you select the applicable component and ask Copilot to show you any content within 5 cm of the selected item, the BOM structure and visual representation are filtered according to these parameters.
- You want to work on a chassis and any parts within 30 mm of this chassis. When you ask Copilot to show you any parts within 30mm of the chassis, the structure and visual representation are filtered according to your request. You can then ask Copilot to create and name a workset. Copilot provides a link to the workset after it is created.

- You have learned that one of your structures exceeds the recommended weight limit. You need to find a way to reduce the weight. Therefore, you can ask Copilot to identify the heaviest item in the structure by creating a rollup report, suggest replacements, and even replace the item in the structure.
- You know that the *Aerospace Fastener Torque Standards - Rev C* document, stored in Teamcenter, contains the definitive torque specifications for various materials and fastener types. Specifically, this document mandates that for *Titanium Alloy (Ti-6Al-4V) fasteners* used in critical shear applications, the torque value must be within the range of 150 Nm to 160 Nm.

  Instead of manually searching through hundreds of BOM lines, you can use Copilot to efficiently identify non-compliant parts by asking it to find all the parts in your current wing assembly BOM that are made of Titanium Alloy (Ti-6Al-4V), have a specified torque value, and whose torque value falls outside the range of 150 Nm to 160 Nm, as specified in the *Aerospace Fastener Torque Standards - Rev C* document.
- You can ask Copilot to save a series of your favorite actions, and then run them by prompting a keyword. For example, you could ask Copilot to remember the command *Open in context*, which would then perform the following sequence of actions: open the BOM structure, filter the electrical components, and finally, highlight all items that are assigned to you. The next time you prompt *Open in context*, Copilot automatically performs these tasks for you.
- You have identified that certain parts require updates, which must be processed through an Engineering Change Notice (ECN). You can instruct Copilot to manage these structural updates within an engineering change. This includes asking Copilot to creating an engineering change, automatically associating affected items to a change, replacing identified parts with recommendations, creating an impact analysis workset for affected items, and finally, visualizing change markups across multiple structures.
- You have learned that a part of an assembly is impacted as a part of the requirement R123. You can prompt Copilot and understand all the impacted items due to this requirement. Copilot analyzes the cross-structure relationships, and displays all affected items, which can include linked designs, associated parts, supplier components, and dependent assemblies.
- You can ask Copilot to configure a structure by applying the *Any Status; Working* revision rule and the *Physical* partition scheme.

# 6. Work in the context of a change

## Set a change context to track BOM updates

You can track updates made to an engineering BOM by setting an engineering change notice as an active change context. The updates can be viewed by accessing the change notice.

### Procedure

1. Create a new change notice or a simple change, and select it.For more information on creating a change notice, refer to the *Change Management* documentation.
2. Click **More Commands** > **Manage** > **Submit to Workflow** .
3. In the **Submit to Workflow** panel,
   - For a change notice, select the **EBOM Release Process** template, enter a **Name**, and click **Submit**.
   - For a simple change, select the **EBOM Simple Change Workflow** template, enter a **Name**, and click **Submit**.
4. To set a change context, click **No Active Change** in the global header and select the required change notice.
   The date and unit (end item) effectivity of the change are automatically set on the current revision rule.

## Review active or closed changes for a structure

You can track changes (added, modified, replaced, revised, or deleted) to the assemblies using a change request or a change notice. When the ECN or ECR is an Active change, redlines are shown in the structure. To view the closed changes, use the **Show Redlines** command.

### Prerequisites

Your administrator can configure the various methods to effectively track and display relevant changes to an assembly.

### Procedure

1. Search for and open the structure. If the structure has changes, they are indicated by redlining. Expand the assembly to view the details of the change.

   You can also open the structure in the alignment view in which the design structure and its aligned engineering BOM are displayed side-by-side. In this case, enable the **Show Redlines** command.

| Element | ID | Revision |
|---|---|---|
| Engine | 035611 | *B* ~~A~~ |
| Piston | 035612 | A |
| *Enhanced Valve* ~~Valve~~ | *035617* ~~035613~~ | A |
| ~~Connecting Rod~~ | ~~035614~~ | ~~A~~ |
| *Connecting Rod Assembly* | *035621* | *A* |

- Deleted parts are indicated by a red strikethrough.
- The added parts display in green and are italicized.
- For replaced or revised parts, the old and the new values are shown side by side.

After the structure is revised, the redlines are shown for all changes including the edits that you make for the part attributes.

2. (Optional) To disable highlighting the changes in the structure, go to **More Commands** > **View** icon and turn off **Show Redlines** from the **Page level** toolbar.

   If you are in the alignment view, disable the **Show Redlines** command to stop the changes from being highlighted.

3. Only active changes are shown by default. To view the closed changes, open the design structure or engineering BOM in a separate window and click **More Commands** > **View** > **Show Redlines** from the **Page level** toolbar.

   You can return a redline change back to its original impacted version by reverting the BOM modifications.

# Update multiple structures simultaneously in a change notice

Update multiple impacted structures simultaneously for the problem items that are a part of a change notice by adding siblings, child items, and occurrence substitutes simultaneously as a part of the solution items.

When updating the structure, ensure that the selected item is not in the **Released** state.

When updating multiple structures simultaneously, you can perform the following tasks:

- Add siblings, child items, and occurrence substitutes.
- Revise the structures.
- Replace and remove the items from the structures.

All updates that are made as a part of the change notice are indicated as redline text.

### Restrictions and limitations

You can update multiple structures simultaneously if your Teamcenter setup includes BOM Structure Managementlicense.

### Procedure

1. In an active change notice, perform the impact analysis for the problem items and then select the impacted items.
   The impacted items are the structures or immediate parent assemblies for the problem items. If a problem item is a part of multiple structures, you can select all these structures as impacted items.

2. Click the **Change Content** tab to view all the problem items and their immediate parent assemblies, which are identified as impacted items.

   > **Tip**
   >
   > To navigate to or locate specific items on the **Change Content** tab, use filters and sorting in columns.

   After you apply a filter to a column, all items that satisfy the filter criteria are displayed on the **Change Content** tab. For example, suppose that you are searching for an item with ID as 12. In such a case, in the **ID** column, select the **Contains** operator and type **12**. The **Change Content** tab displays all items that have ID as 12. After a filter is applied, the **Change Content** tab considers the redlining and shows the modified content. If an item is removed on the **Change Content** tab as a part of the change, and its content satisfy the filter criteria, then that item is still shown on the **Change Content** tab. For example, suppose that the 8cm nut is removed and you entered the filter specifying the 8cm size. In such a case, the **Change Content** tab shows all 8cm nuts including the one that you removed.

   You cannot filter the visual indicator columns like V. After applying column filters to locate the required content, if you click another tab in the change and then return to the **Change Content** tab, the column filtering is removed. You must again apply the column filters.

3. Perform the following actions as appropriate to select and show the required items on the **Change Content** tab.

| If you want to... | Perform this actions... |
|---|---|
| **Select all items including the parent and child items.** | Click **Select** ✓ > **All** ✓. |
| **Select all top parents assemblies.** | Click **Select** ✓ > **All top parents**. |
| **Select all occurrences of the selected item.** | Click **Select** ✓ > **All matching elements**. |
| **View all child items for the selected parent node apart from the ones that are a part of the change notice and also view the rest of the siblings and sub-assemblies. After viewing the** | Click **Show** > **All Children**. |

| If you want to... | Perform this actions... |
|---|---|
| **non-impacted child items, you can update their properties and include them in the change.**<br>**You might see all child items by default if your administrator has set it for you. Furthermore, if all child items are shown, the parent are always collapsed. You must manually expand them.** | |
| **View only the child items that are a part of the change notice and do not want to view the siblings and sub-assemblies.** | Click **Show** > **Children in Change**. |

4. To revise one or more items, select them and click **Revise**.

   If you select only one item for revision, the **Revise** panel appears. Review the information on the **Revise** panel and click **Revise**.

   If you select multiple items, a confirmation message appears. Click **Revise** on the confirmation message.

   The selected items are revised and are shown with the redlines.

5. If you want to edit the description, quantity, and any other properties of the selected one or more child items, click **Edit**.
   After you are finished editing the properties, click **Save Edits**.

6. If you want to add a child item to one or more items:

   a. Select the items under which you want to add the child item and click **Add** > **Child**.

   b. In the **Add Child** panel, search an existing child item or create a new one, and then click **Add**.
      The child is added to the selected items.

7. If you want to add a sibling item to one or more items:

   a. Select the items for which you want to add the sibling item and click **Add** > **Sibling**.

      You cannot add sibling to the parent item in the structure.

   b. In the **Add Child** panel, search for an existing child item or create a new one, and then click **Add**.
      The child item is added to the selected items.

8. If you want to add occurrence substitutes:

   a. Select one or more items and click **Add** > **Substitute**.

   b. In the **Add Substitute** panel, search for an item or create a new item and click **Add**.
      The occurrence of the selected item is added as a substitute. The **Substitutes** and **Primary Component** columns show the information of substitutes and primary items. The **Substitutes** section in the **Overview** tab for the primary item also show the list of the substitutes items. You can further set substitutes as primary components.

      After the substitutes are added, if you want to hide them from being shown in the structure, right-click a structure and select **Hide Substitute Components**.

9. If you want to remove some of the items from the structure, select the items and click **Edit Structure** > **Remove** .

   The selected items are removed from the structures and redlines appear for them.

10. If you want to replace the selected items by another item:

    a. Select one or more items and click **Edit Structure** > **Replace** .

    b. In the **Replace** panel, search for an existing item or create a new one, and then click **Replace**.

       All selected items are replaced with the chosen item.

11. If required, select the required revision rules and apply date and unit effectivity on the **Change Content** tab.

#### What to do next

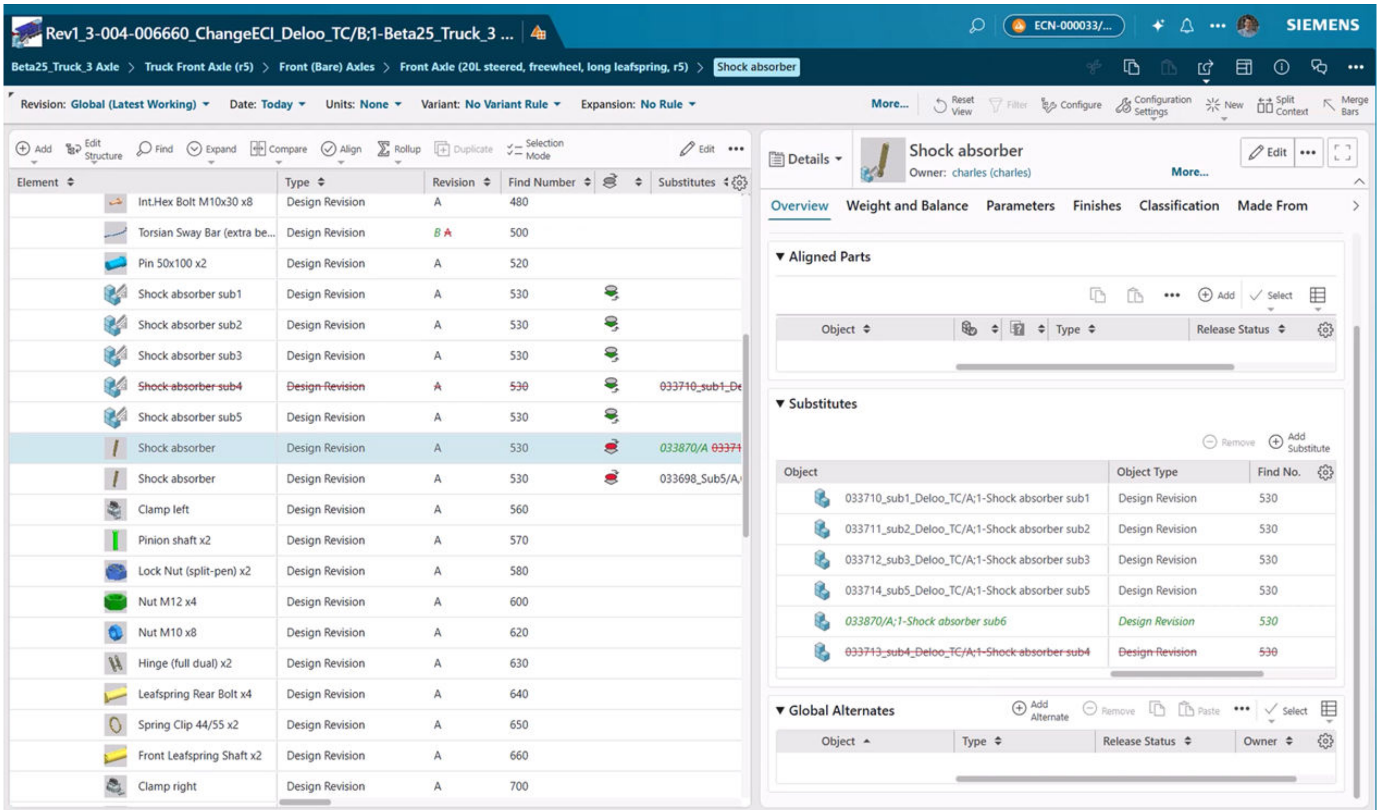
After the impacted structures are updated and the solution items are provided, you can view the change summary and then send the change for downstream process.

## Track BOM modifications in the Change Summary

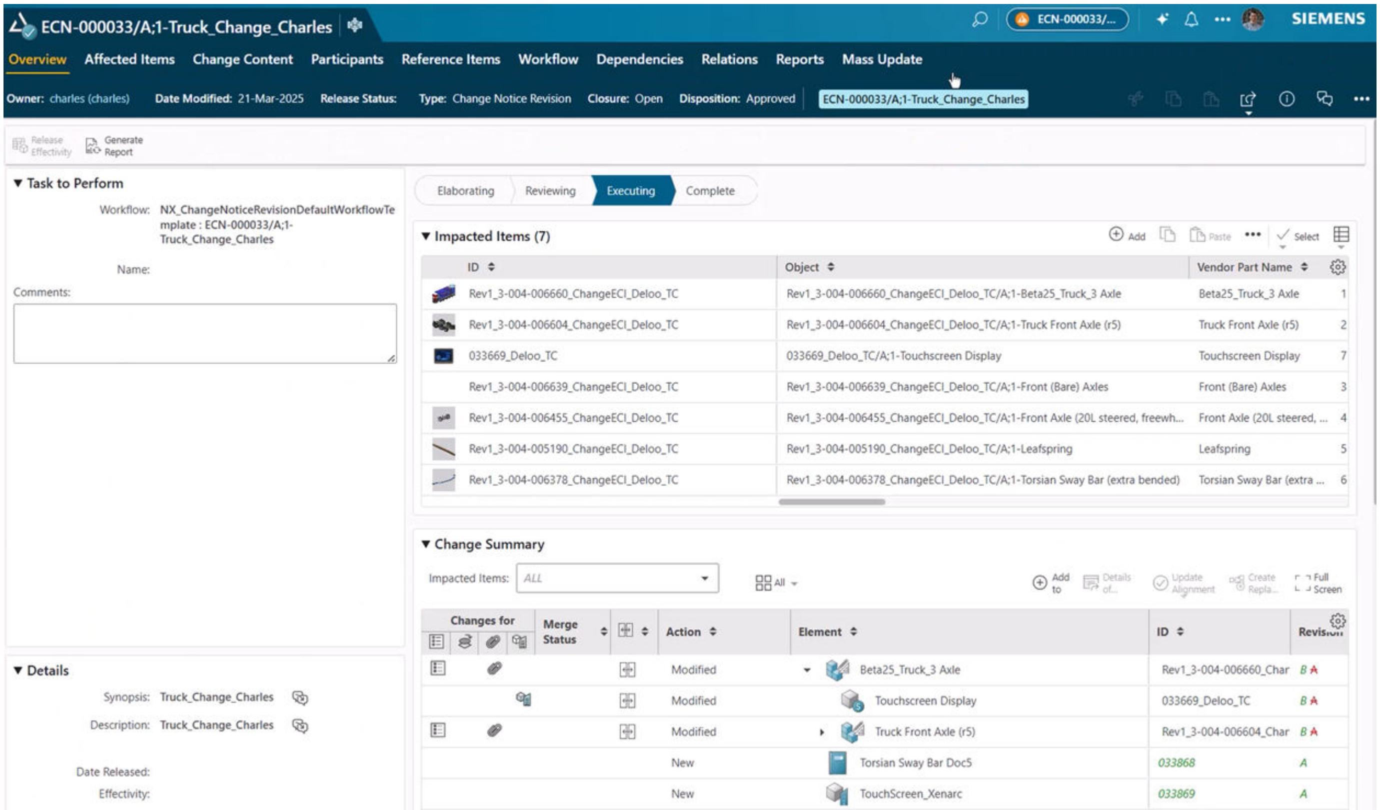
Applying an active change context for BOM revisions automatically tracks the modifications in the Change Summary.

See **set a change context to track structure updates** for more information.

With the change notice set as an active change, BOM modifications and properties are identified by redlines. For more information on the **Show Redlines** feature, refer to **Review active or closed changes for a structure**.



These modifications, not the full assemblies, are tracked in the Change Summary of the change notice.



## View the authoring change notice and the changes made to the part, and the change dashboard for the product

View the authoring change notice for a selected structure element, or access an interactive change dashboard to see how change notices impact a top-level product and its immediate sub-assemblies. You can also review the modification history of a structure element.

### Procedure

1. Open the required structure and select the part.
2. Go to the **Changes** tab. In the **Changes** section, you can see the authoring change notice for the selected part.

   If the selection is top level product, then you can see the interactive change dashboard listing all the change notices under which the top product is impacted along with change notices in which its first level part usages are added or updated. Here, click on the PI chart to filter between the change notices as required. This dashboard is only visible for top product but your administrator can enable it for assemblies below the top product as well.
3. Go to the **History** tab. All the modifications or revisions made to the selected part with their respective change notices are listed under **Change History**.

# Cancel a change notice

Based on a specific product requirement, an engineering change notice is created. The engineering BOM content is authored or updated in the context of the change notice. After authoring or updating the BOM content, the product requirements may get canceled. In such a case, you must also cancel the corresponding change notice before it is released. On canceling the change notice, the updates made to the BOM content also get canceled.

### Procedure

1. From **Launcher**, click **Changes** ⚠.

   If this option is unavailable, search for it. You can pin it from the search results for future access.

2. Select the change under which the structure is created or modified.

3. Click **Cancel Change** in the **Overview** tab.
   All the **Solution Items** of the change notice get canceled and their property **Cancelled** is set as *<true>*.

   Canceling the engineering change notice will cancel all its associated solution objects. A canceled revision can never configure. However, you can create a new revision from a canceled revision. You cannot reopen a canceled change and release its content.

   If there are sequential changes in ECN2 that have been created based on ECN1, and ECN1 is canceled, ECN2 is not canceled automatically. You can choose to cancel it manually.

   When you cancel a change notice which contains a workset or a session having a structure, this cancel change behavior is not supported.

   Every cancellation of ECN may not inherit this behavior. For example, cancellation of ECN having workset and subset, which contains the structure.

# Release engineering BOM data by validating and releasing the active change

If you created or updated the engineering BOM data, by setting an active change notice, the data can be validated and released together by validating and releasing the active change notice.

### Procedure

1. If the change is not yet submitted, **submit the change**.

   You can either use the change notice and the **EBOM Release Process** template or use a simple change and the **EBOM Simple Change Workflow** template.

2. Validate the change notice to perform validation checks on the engineering BOM before its release. This validation is based upon a set of predefined validation rules in the system. Your administrator can also create custom validation rules.

a. From **Launcher**, click **Changes**.

   If this option is unavailable, search for it. You can pin it from the search results for future access.

b. Select the change under which the structure is created or modified.

c. Click **Validate Change** in the **Overview** tab.

If the validation fails, the issues are listed in the **Validation** tab under the **Validation Summary** section. Resolve the issues, as needed.

Once the issues are resolved and the validation is successful, you can release the change.

3. Click **Release Change** to release the change notice.

   While releasing the change, Teamcenter validates the change again to resolve any issues found during the initial validation.

   If no issues found, structure is released. Note that, the workflow (**EBOM Release Process** workflow or the **EBOM Simple Change Workflow**) also releases the BOM view revisions of the *ItemRevisions* that are tracked by the change.

# 7. Access engineering BOM data within partitions

A BOM architect organizes a large product into smaller, accessible sections called *partitions* and arranges the engineering BOM data logically and hierarchically within the partitions. Partitions can belong to different partition schemes such as functional, physical, or spatial.

When the data is organized within partitions, you can view the data and perform various engineering BOM management tasks from within the partitions.

For information on partition schemes and partitions, see *Structure Partitions — Usage*.

# 8. Find and navigate structures

## Finding structures and structure elements

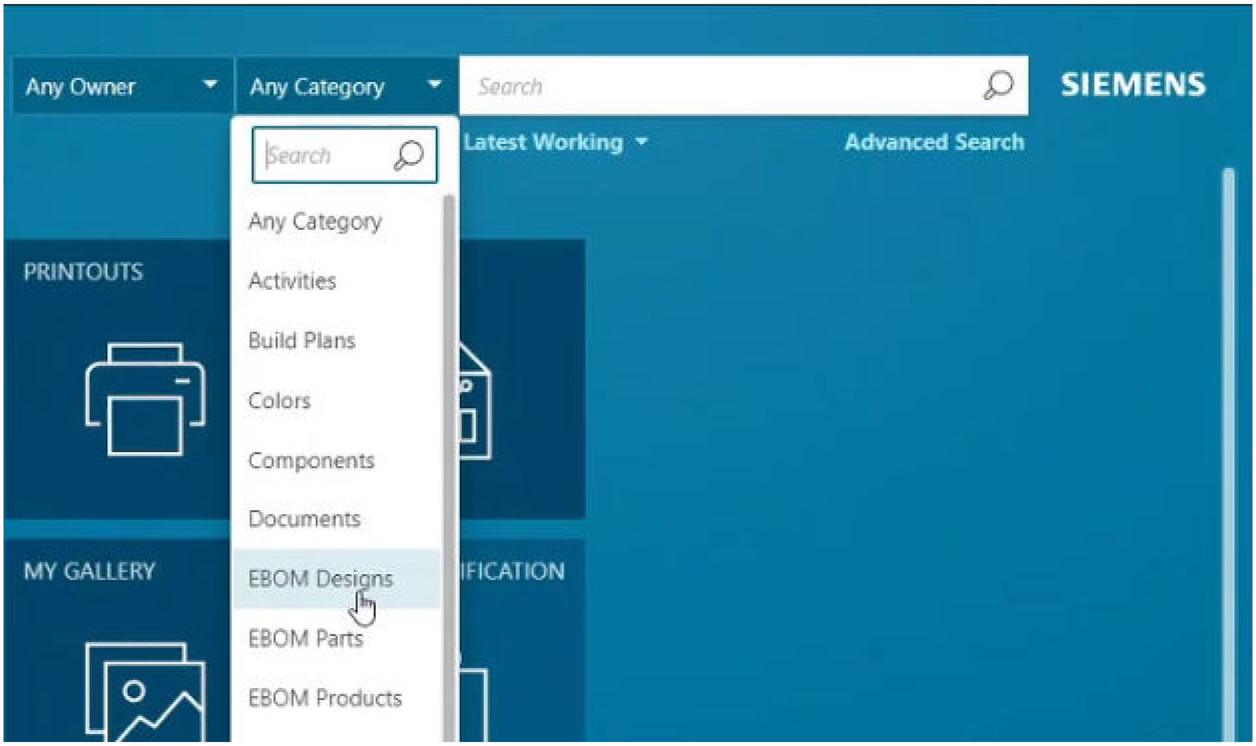
Discover various methods for locating structures and structure elements within Teamcenter, including global, advanced, and in-context searches, to efficiently find the data you need.

You can search for structures or structure components using the following methods:

| | |
|---|---|
| **Global Search** | Searches all indexed data in Teamcenter. |
| **Advanced Search** | Searches for data by using specific criteria based on predefined queries. |
| **In-context search** | Searches for content within an open structure, a working context, or a session that is composed of structures.<br><br>The structure can be indexed by using either Active Content Structure indexing or Smart Discovery Indexing.<br><br>It may be non-indexed as well. The non-indexed search requires a Context Management User license. This is used for Quick Find searches. |

You can perform a simple search to find an occurrence of an object in an open structure. You can enter any attribute or text associated with the occurrence (for example, a name) and then click **Search**. Search results span the entire context.

To define context, you can choose **EBOM Parts**, **EBOM Designs**, or **EBOM Products** from the categories list before doing the search.




You can use the same syntax as for full-text searches, including operators such as OR and AND. You can use the **Filter** tab to narrow the results by selecting specific filters and categories.

# Find elements within a structure

Perform keyword searches within an open structure or a selected subassembly to quickly locate specific elements, including those based on content, and narrow down your search results.

To narrow your search results to a specific branch in the structure, you can perform a keyword search within the selected subassembly from the **Find** panel.

### Procedure

1. Open a structure from the search results.
2. Click **Find** .
3. In the **Find** panel, enter the search parameters, and click **Search** .

   If file content indexing is deployed in your Teamcenter configuration, you can specify dataset filenames as the search parameter. You can also specify the text within a file as the search parameter. For example, consider that the dataset *EngineSpecification.pdf* contains the text **RPM**. If you enter **RPM** as the search parameter, the search result displays elements that have the dataset *EngineSpecification.pdf* as an attachment. The search result also displays elements that have datasets with the text **RPM** in their content.

Additionally, you can specify the name of a *.prt* file as the search parameter.

When a structure is indexed using Smart Discovery and the administrator has enabled indexing for an element's occurrence properties, you can search for elements using those properties. For example, if the **Assembly Instructions** occurrence property is marked for Smart Discovery Indexing and the structure is already indexed using Smart Discovery, you can search for elements based on their **Assembly Instructions** occurrence property.

4. (Optional) To narrow the results:

   a. Select the subassembly to which you want to limit your search.

   b. Select the **Find within** check box, enter the search parameters, and click **Search**.

5. Select an element from the search results to view it in the structure.

   If you have **Partitions for Structure** deployed in your Teamcenter configuration *and* if you select an element from the search results on the **Find** panel, the element is highlighted in the work area within the partition that contains it.

   **Note**

   Clicking **Select All**✓ does not select the corresponding elements in the structure.

## Searching structured content if you do not have access to objects

If you find an element in a structure that you do not have read access to, Teamcenter does not show that occurrence and excludes it from the find-in-context results. If the occurrence exists in an assembly to which you *do* have access, Teamcenter shows an **Access Denied** indicator in its place.

**Note**

Access to occurrences is set by your system administrator by using Access Manager. Access controls protect intellectual property and prevent general access to data. To view restricted content that you do not have access to, you can request the owner of the assembly to use project-level security to allow collaboration.

If you do not have access to a revision configured by the revision rule, Teamcenter searches for the lastest revision you have access to, and configures the access accordingly.

Automatic searching for the lastest accessible revision is not supported in indexed structures. However, it is supported if Smart Discovery Indexing is used.

# Navigate a structure

Learn how to open and navigate through a structure in Teamcenter, including expanding to view child parts and using the breadcrumb trail to move to higher-level assemblies.

You can select a structure from the search results and then click **Open** to view the structure.

## Navigate to child parts

The assemblies or subassemblies that have child parts are listed with the **Show Children** button.

Click **Show Children** to navigate to child parts.

## Navigate to a higher level using the breadcrumb trail

Assembly nodes are visible in the breadcrumb trail. Use the breadcrumb trail to navigate to a higher level (such as parent parts) in a structure.


**Note**

The back button does not take you to the parent parts. It takes you to the previous location visited in Active Workspace. It might also not show the latest structure. You may refresh the structure to view the latest one.

If you use the breadcrumb trail to navigate to a different parent, the first leaf in that structure is displayed, the breadcrumb is updated with the path to that leaf node, and the display is switched back to the hierarchical view.

# 9. Import and export structures

## About importing structures from Excel

You can import structure data into Teamcenter from a Microsoft Excel spreadsheet and create new structures with new or existing components. This activity is useful for integrating data from external sources or vendors.

In Teamcenter, you can import a structure from a Microsoft Excel spreadsheet. This feature is useful when you need to import data from an external source, for example, from a design contractor who does not have Teamcenter. You can import relatively simple and small structures, such as structure data provided by vendors. Importing from Excel might not be suitable for very large and complicated structures.

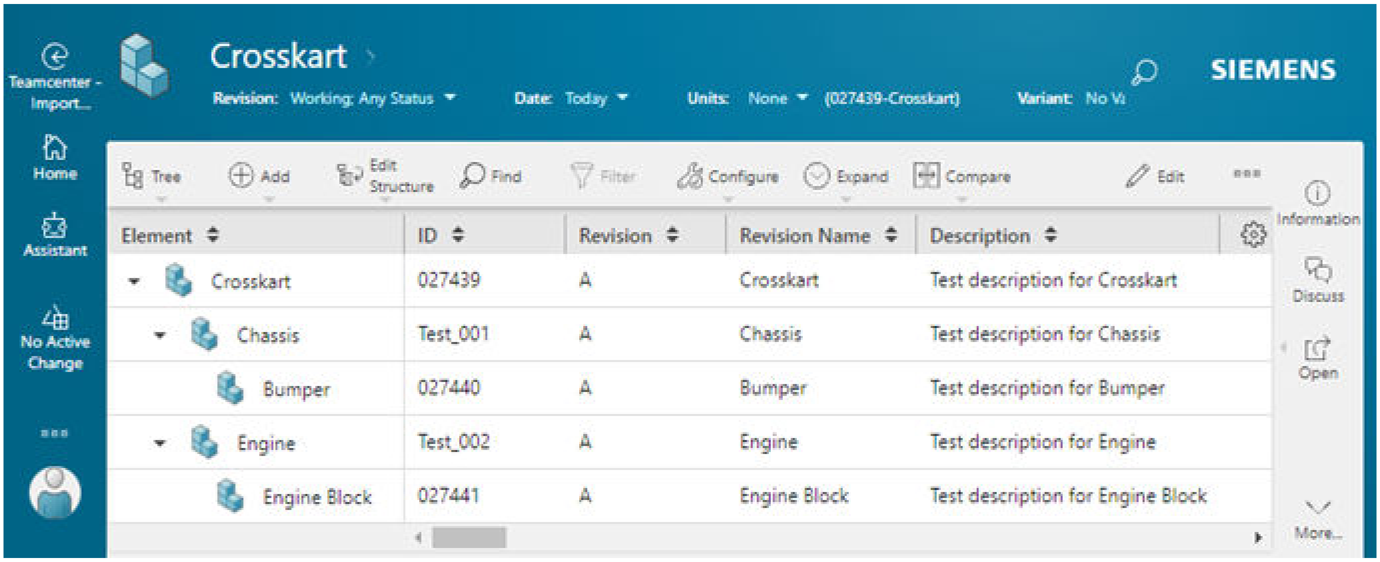
The following scenarios explain how you can **import structures** in Teamcenter.

### Scenario 1

You want to create a new *Crosskart* structure with all new components. To do this, you can create an Excel spreadsheet similar to the following and then import it in Teamcenter:

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | Product Structure | | | | | |
| 2 | Tc_Level | Tc_ObjectType | Name | ID | Revision | Description |
| 3 | 0 | Item | Crosskart | | | Test description for Crosskart |
| 4 | 1 | Item | Chassis | Test_001 | | Test description for Chassis |
| 5 | 2 | Item | Bumper | | | Test description for Bumper |
| 6 | 1 | Item | Engine | Test_002 | | Test description for Engine |
| 7 | 2 | Item | Engine Block | | | Test description for Engine Block |
| 8 | <endtag> | | | | | |

After the import, the following structure is created in Teamcenter:



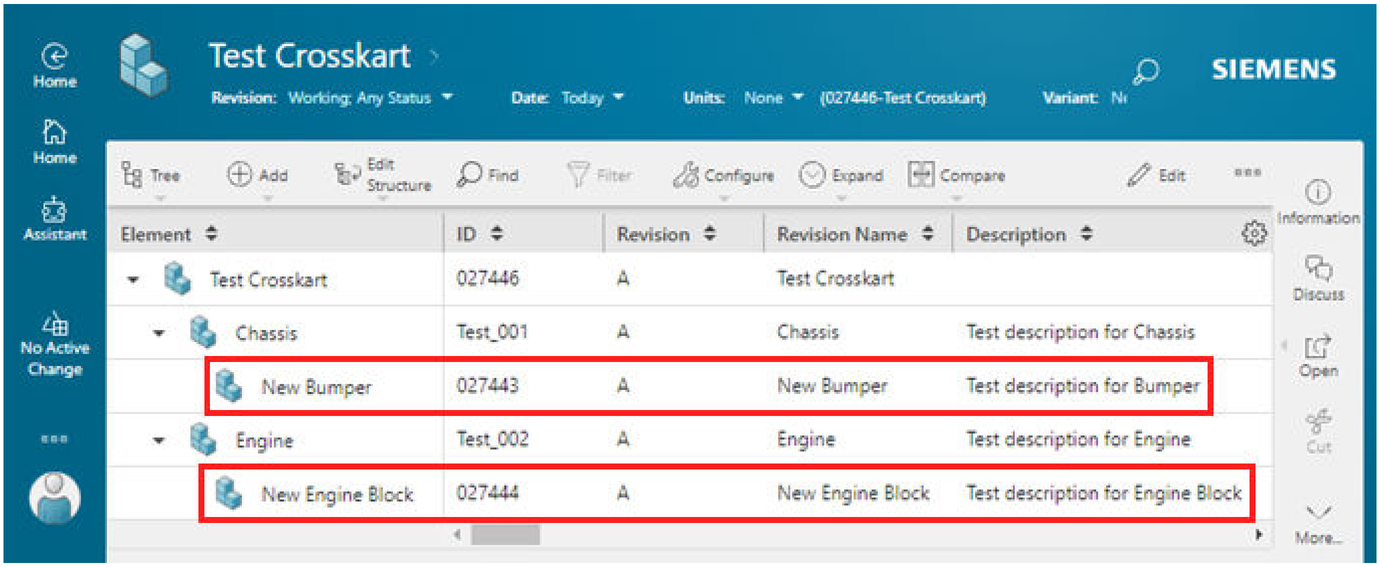

Even if you did not specify an ID for some of the elements in the input spreadsheet, Teamcenter generates the IDs for those elements automatically.

## Scenario 2

You want to create a new structure called *Test Crosskart* by reusing some of the existing elements in Teamcenter. Therefore, in the input spreadsheet, you add the name for the new structure and only the names and IDs of the existing subassemblies. Your input spreadsheet appears as follows.

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | Product Structure | | | | | |
| 2 | Tc_Level | Tc_ObjectType | Name | ID | Revision | Description |
| 3 | 0 | Item | Test Crosskart | | | |
| 4 | 1 | Item | Chassis | Test_001 | | |
| 5 | 1 | Item | Engine | Test_002 | | |
| 6 | <endtag> | | | | | |

After the import, Teamcenter creates the new *Test Crosskart* by reusing the existing elements from the sub-assemblies. You only need to specify the required properties of the subassemblies and not of the existing elements.




## Prepare the Excel file for importing a structure

Prepare an Excel spreadsheet for importing structures into Teamcenter by following specific guidelines for mandatory fields, reference properties, vendor-part data, and handling multiple parts, assemblies, or occurrence-based substitutes.

To import a structure from Excel, you must prepare the Excel file in a specific way. The guidelines for preparing the Excel file are listed here:

- Mandatory points
  - The top row must have a title, for example, **Primary Object**.
  - The cells in the second row must have the headers **Tc_Level** and **Tc_ObjectType**. The header **Tc_Level** must be in column A and **Tc_ObjectType** must be in column B.

    If you are importing a structure for Usage BOM, the cells in the second row must have the headers **Tc_Level**, **Tc_ObjectType** and **Tc_UsageRevisionType**.The header **Tc_Level** must be in column A, **Tc_ObjectType** must be in column B, and **Tc_UsageRevisionType** must be in column C.
  - The properties **Name**, **ID**, and **Revision** are mandatory.

    If you are importing a structure for Usage BOM, the properties **Name**, **ID**, and **Revision** are mandatory. You can also include occurrence properties, for example, **Stock Disposition** and **NX Name**.
  - While you import a structure, the elements that occur multiple times under a single parent in the structure are uniquely identified by certain properties. It is recommended that you use these properties as columns in the input spreadsheet. To get these properties, you may refer to the values in the **AWC_Occ_Unique_Identifier** preference. When you reimport the same structure, make sure that the values for these properties are correctly set. Otherwise, you might not get the expected results.
  - You must include **<endtag>** in column A at the end.

- If you do not want to update the part or occurrence properties, leave the corresponding cells in the input spreadsheet blank. Teamcenter ignores blank cells in the input spreadsheet.

- Import a structure along with reference properties
  - You can import structures with certain reference properties, such as **Unit of Measure**, **Owner**, and **Group**. You can also import any other reference property with an LOV.
  - After you add the display string of a reference property to the Excel sheet, Teamcenter maps this string to the correct reference object.
- Import a structure along with vendor-part data
  - To import vendor-part data, you must specify the secondary objects along with the primary ones. Thus, you must fill in the **Tc_Secondary_ObjectType**, **Tc_Secondary_Relation**, and **Vendor ID** fields.
  - The **Vendor ID** that you specify must exist in the Teamcenter database.
  - **Tc_Secondary_ObjectType** must be the first column in the **Secondary Object** section.
- Import multiple parts
  - To import multiple parts, you can add them, excluding any assemblies, to the Excel file. You must specify the **Tc_Level** of each part as zero (0).
  - As Teamcenter ignores occurrence properties, you need not specify them.
  - After you import multiple parts, Teamcenter opens only the first part.
- Import multiple assemblies
  - To import multiple assemblies, you can add them to the Excel file. You must specify the **Tc_Level** of each top-level part in an assembly as zero (0) and its children as 1 or 2 and so on as per the hierarchy of items.
  - Teamcenter considers each top-level part as a new structure.
  - After you import multiple assemblies, Teamcenter opens only the first assembly.
- Import multiple types of secondary objects
  - You can import multiple types of secondary objects for both an item and an item revision.

    If you are importing a structure for Usage BOM, you can import multiple types of secondary objects for a part.
  - While importing related objects, you can specify the object type and the relation type.
  - If a related object already exists, Teamcenter relates the primary object to the related object. If a related object does not exist, Teamcenter creates a new related object.
  - If you do not specify any value in the **Tc_Primary_To_Secondary_Relation** column, Teamcenter considers the following values, by default: For vendor-manufacturing related parts, the value is **item-to-rev** and for any other parts that are not related to vendor manufacturing, the value is **rev-to-rev**.
- Import occurrence-based substitutes
  - To import the occurrence-based substitutes, set the **Is Substitute** column to **Yes** in Excel for every substitute part.

- The primary part and its substitute must have the same value in the **Tc_Level** column in Excel.
- The primary part must be defined above the substitute in the Excel.
- The primary part is a line with no value in the **Is Substitute** column.
- The substitute part is a line defined below the primary part.

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | **Primary Object** | | | | | |
| 2 | Tc_Level | Tc_ObjectType | Name | ID | Revision | Is Substitute |
| 3 | 0 | Item | Front Door Assembly | FD-100 | A | |
| 4 | 1 | Item | Door shell assembly | DS-200 | A | |
| 5 | 2 | Item | Steel Door Shell | DS-200-ST | A | |
| 6 | 2 | Item | Aluminium Door Shell | DS-200-AL | A | Yes |
| 7 | 1 | Item | Door handle Assembly | DH-300 | A | |
| 8 | 2 | Item | Carbon Fiber Door Shell | DS-200-CF | A | Yes |
| 9 | 2 | Item | Standard Handle Black | DH-300-SH | A | |
| 10 | 2 | Item | Chrome-plated Handle | DH-300-CH | A | Yes |
| 11 | 2 | Item | Sport Handle with Keyless Entry | DH-300-SP | A | Yes |
| 12 | 1 | Item | Window Regulator | WR-400 | A | |
| 13 | 2 | Item | Manual Window Regulator | WR-400-R | A | |
| 14 | 2 | Item | Electric Window Regulator | WR-400-E | A | Yes |
| 15 | 2 | Item | Premium Electric Regulator with Anti-pinch | WR-400-EP | A | Yes |
| 16 | 1 | Item | Side Mirror | SM-500 | A | |
| 17 | 2 | Item | Basic Side Mirror | SM-500-BM | A | |
| 18 | 2 | Item | Heated Mirror | SM-500-H | A | Yes |
| 19 | 2 | Item | Power-folding Mirror with Turn Signal | SM-500-PF | A | Yes |
| 20 | **<endtag>** | | | | | |

The above image shows that the structure of the front door assembly is being imported from Excel. In this structure:

- Aluminum Door Shell and Carbon Fiber Door Shell are substitutes for Steel Door Shell
- Chrome-plated Handle and Sport handle with Keyless Entry are substitutes for Standard Handle Black
- Electric Window Regulator and Premium Electric Regulator with Anti-pinch are substitutes for Manual Window Regulator
- Heated Mirror and Power-folding Mirror with Turn Signal are substitutes for Basic Side Mirror

# Import and export structures and changes using Excel round-trip

## Import structures and structure changes from Excel

Import new structures or apply changes to existing ones in Teamcenter; using data from Excel spreadsheets. This includes adding and removing elements, updating item and element properties, and managing relations, whether from a prepared Excel file for initial import or a round-trip Excel file for modifications.

Use the following guidelines for importing Excel to create structures, elements, and relations in Teamcenter:

- The structure can include all reference properties with an LOV, such as unit of measure, owner, and group.
- You can import structures with the global revision rule and structures using different ID types that use a Multi-Field Key (MFK). You can import vendor-part data, multiple elements and assemblies, and multiple types of secondary objects.
- You can also import the relational properties (like preferred or backup vendor preference status). If an array property contains multiple values in the Excel, you can import them too. For example, if the Excel contains a column for Production Sites and the values are Milford and Troy, both these values are imported in Teamcenter.

When you import changes from a previously exported Excel, you can add or remove structure levels, add or remove elements from the structure. However, do not make changes to the **TCUID** column in Excel. When you import the changes from a previously exported Excel, Teamcenter detects the original structure and imports changes to it.

### Restrictions and limitations

You can perform this task if it is configured for you by your administrator.

### Procedure

1. Prepare the Excel with the structure data, elements, or relations you intend to import.
   - For an external file, ensure that the columns are organized for mapping to Teamcenter properties. These files can represent simple structures or structures with relations.

     To import a structure from Excel, you must **prepare the Excel file in a specific way** for importing a structure.

     To import a structure from Excel, you must prepare the Excel file in a specific way. See *Prepare the Excel file for importing a structure*.
   - For a round-trip file that was previously exported from Teamcenter, you can modify:
     - The item and element properties.
     - Modify structure including adding and removing elements and replacing items.
     - Modify the relation properties, including adding new relations and modifying existing relation properties.
2. In Teamcenter, navigate to the location where you want to import or update the structures.

   The location can be a folder in Teamcenter, the **Content** tab of an engineering change, or an existing open structure in Teamcenter.
3. Perform one of the following actions:
   - Click **Excel Round-trip** > **Import from Excel** .

- Click **More commands** ••• > **Import/Export** > **Import from Excel** .

4. In the Import from Excel panel, click **Choose File** and select the Microsoft Excel file from which you want to import the structure.
   You must review and configure the import options.
   If you are importing an external file, map the Excel columns to Teamcenter columns and properties. Furthermore, the structures, its relations, and elements are imported by default. If you are importing a round-trip file, Teamcenter recognizes the column mapping, and you are not required to map the columns manually.

5. If you are importing an external Excel file, map the properties between Teamcenter and Excel.

   - Select the required mapping from the **Saved Mappings** list. The list shows the mappings created and saved by the administrator or previous users.

     Select a saved mapping that matches your Excel file's column structure. Mapping names typically indicate the structure type or source. For example, BOM_Import_Standard or Vendor_Parts_Mapping. If unsure, create a new mapping or consult your administrator about which mapping to use for your specific import scenario.

   - Alternatively, create a new mapping by entering a name in the **Saved Mappings** field and selecting the **Mapped Attributes** for the **Excel Headers**. Even if an error occurs while importing a structure, the created mapping is saved. You need not select the mapped attributes again for the Excel headers.

   - If a mapped attribute is not available for an Excel header:

     a. In the **Mapped Attributes** list for the corresponding Excel header, click **Add New** to create a new attribute.

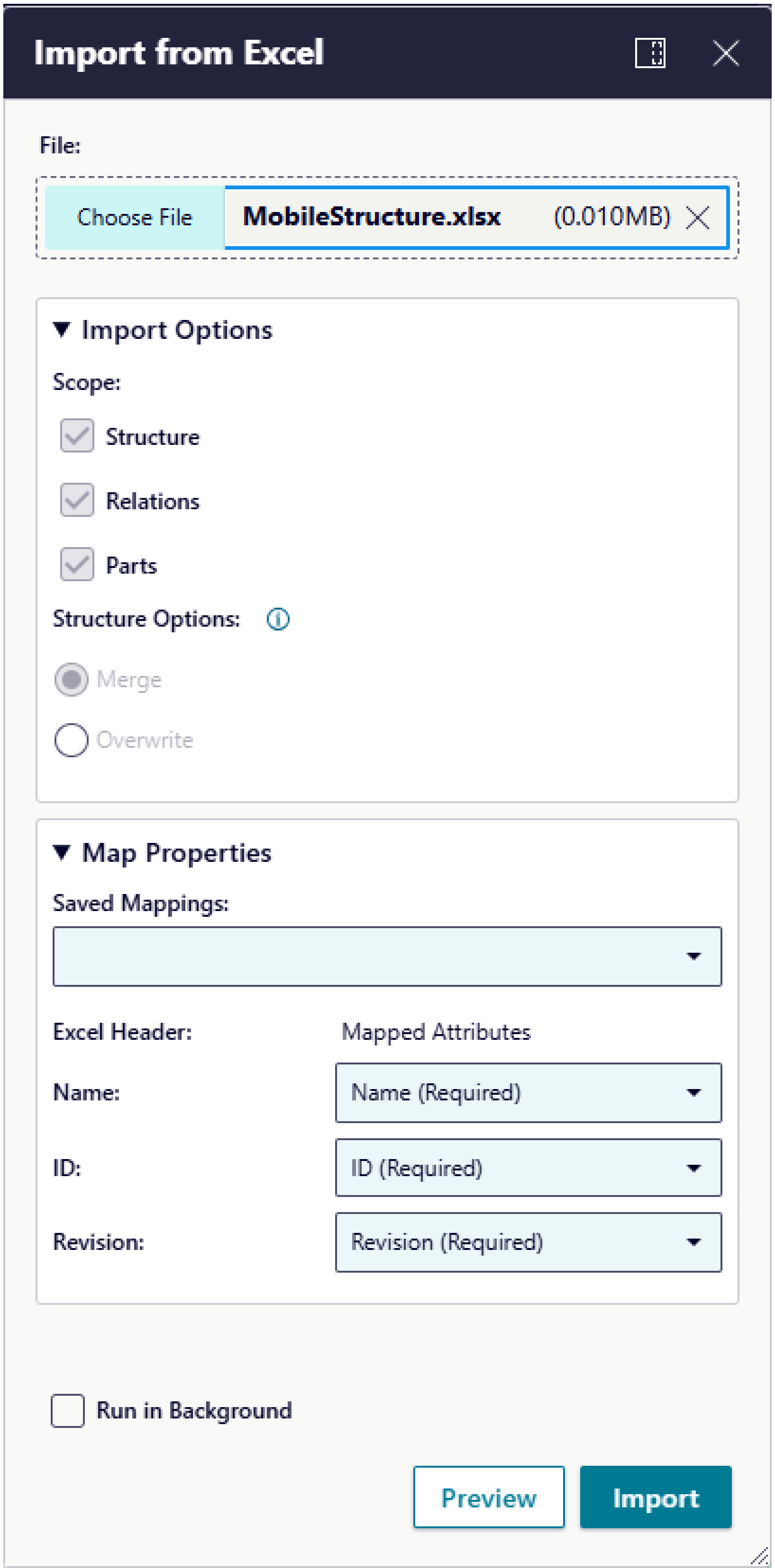
     b. In **Add Properties**, select appropriate **Subtypes**, if you want to change the default one.

        The reference properties **Unit of Measure**, **Owner**, and **Group**, along with other properties, are listed in the **Add Properties** section.

     c. Filter and select the attribute that you want to map.

     d. Click **Add**.

     e. In the **Import Structure** panel, choose the newly added attribute.


Import from Excel
File:
Choose File
MobileStructure.xlsx
(0.010MB)
Import Options
Scope:
Structure
Relations
Parts
Structure Options:
Merge
Overwrite
Map Properties
Saved Mappings:
Excel Header:
Mapped Attributes
Name:
Name (Required)
ID:
ID (Required)
Revision:
Revision (Required)
Run in Background
Preview
Import

**6.** If you are importing a round-trip file that was previously **exported from Teamcenter**, select the following import options:

- Select the **Structure** check box to import the structural changes. The examples of structural change can be a new element is added to the structure, a subassembly is removed from the structure, or the quantity of elements is updated.
- Select the **Relations** check box to import the relational changes between the primary and secondary objects that are made in the Excel.
- Select the **Parts** check box to import the item-specific changes made to the structure. The examples of item-specific changes can be new items or updates to an item's description.
- If you want to merge the changes made to the Excel into the structure and retain all the structure-related data in Teamcenter, click the **Merge** option.

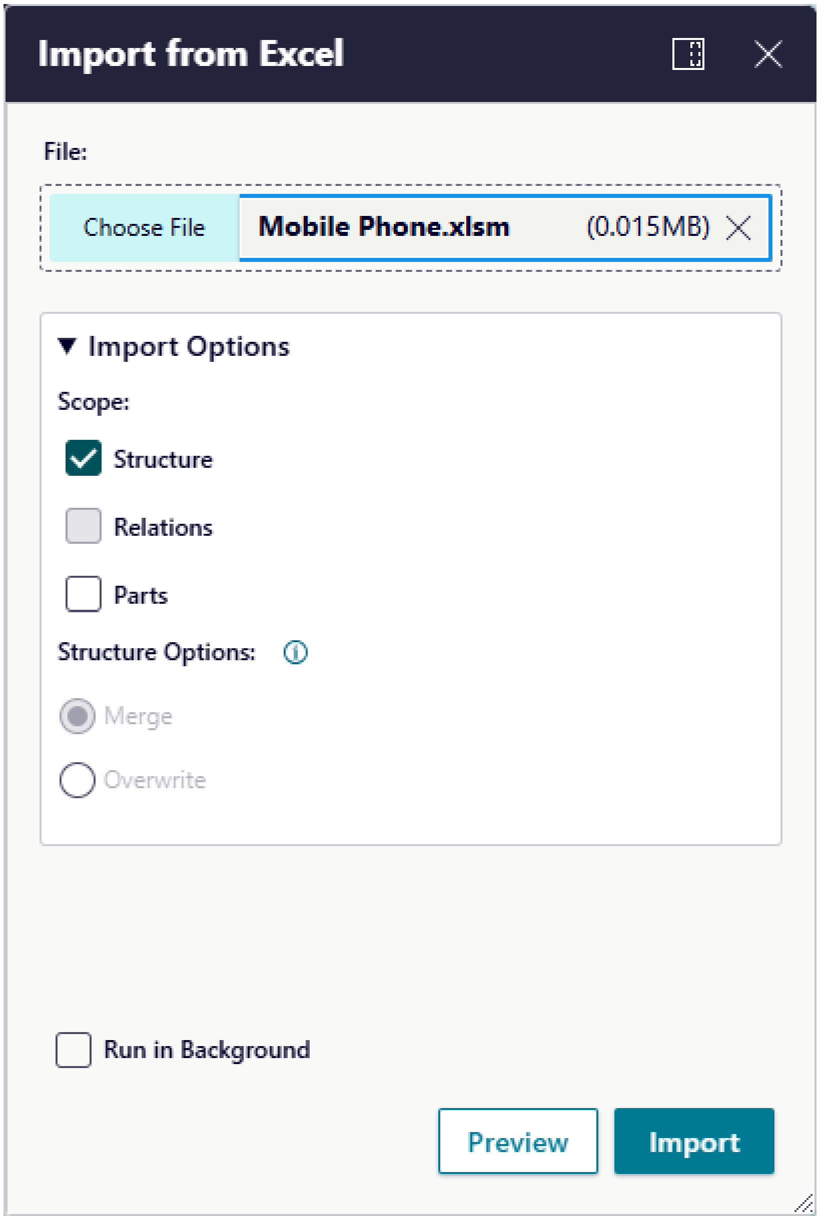
  If you want to replace data in Teamcenter with the data from the Excel file, click the **Overwrite** option.

  **Caution**

  Always ensure that you understand the implications of overwrite operation and have appropriate backups or version control in place before overriding the structure data.

  **Note**

  The **Overwrite** option requires specific settings configured by your administrator. If this option is not available, contact your administrator.




7. (Optional) If you do not want to run the import process in the background, clear the **Run in Background** check box.

   Teamcenter notifies you about the result of the import process in the **Alerts** panel.

8. (Optional) Click **Preview** to check the structure before import.

   Perform the following tasks in the Preview panel:

   a. Review the structure to be imported along with the **Action** that is performed by default and the **Teamcenter Information** column that shows additional information related to an action.

Mobile Phone

| Revision Name | Action | Teamcenter Information |
|---|---|---|
| Mobile Phone | Overwrite | |
| CPU | Revise | |
| Power Supply | New | The object type required to create object at row 4 was not specified. |
| Motherboard | Overwrite | |
| Screen | Reference | |
| Gorilla | New | |
| Camera | Overwrite | |
| 48MP | New | |
| 13MP | New | |
| SIM tray | Overwrite | |

b. (Optional) Change the action for **Revise** and **Overwrite** considering the business requirement. By default, Teamcenter selects actions based on whether items exist and whether they are writable. Change actions when the default action does not match your requirements. To do so, right click an action and select:

- **Revise** when you want to preserve the existing revision and create a new one with updates.
- **Reference** when you want to use existing elements without modifications.
- **Overwrite** when you need to update the existing revisions with new data without creating new revisions.

9. Click **Import** to import the structure in Teamcenter.

### Results

**If there were no errors while importing the structure:**

The structure is imported in Teamcenter. When you import multiple elements or assemblies simultaneously, all the top-level elements and assemblies are added to the folder you specified previously. You receive a notification in the **Alerts** panel when the import is complete. Click **Alerts** to open the **Alerts** panel.

The structure is imported using the global revision rule. The same revision rule is used while opening the structure after import.

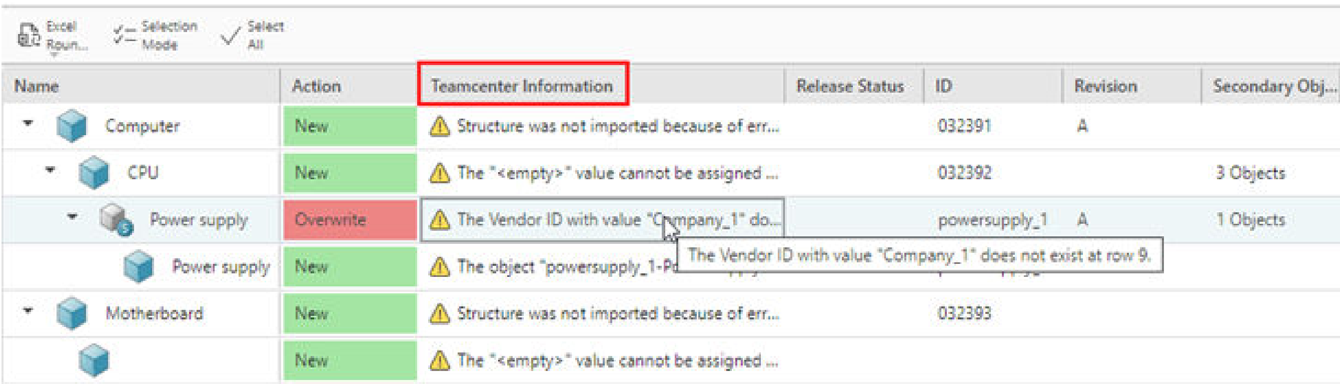
You can also import a structure with different ID types that use a multi-field key, that is, when you have two different elements with the same ID. You can see the elements on the **Import Preview** screen and import them successfully. The import structure functionality searches for and creates elements based on the multi-field key.

**If there were errors while importing the structure:**

You receive an error message stating that the import failed because there were errors while importing the structure. You also receive a notification in the **Alerts** panel, where you can view the error details.

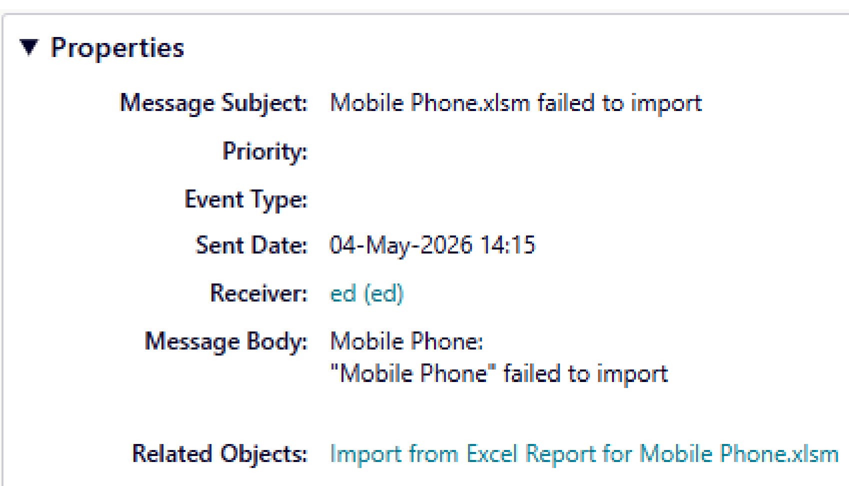
Click **Alerts** 🔔 to open the **Alerts** panel.

In the **Teamcenter Information** section, you can view a list of all potential errors simultaneously. When you hover over an error, you can view detailed information about that error.

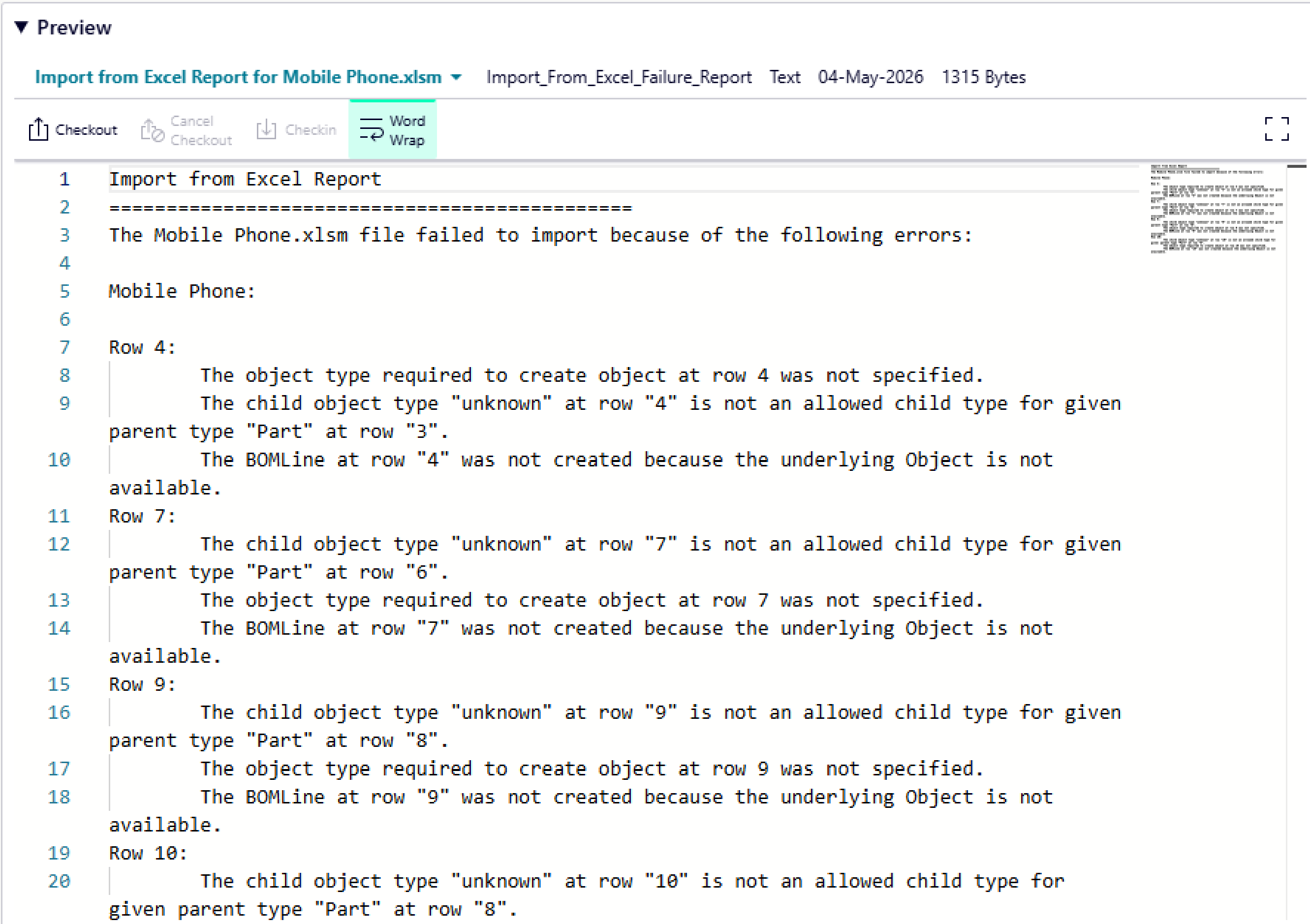



> **Tip**
>
> At this stage, you need not restart the import process from the beginning. You can make the required corrections in the Excel file and then re-upload it by clicking **Choose File** in the **Import Structure** panel. Once this is done, the saved mappings from the previous import are automatically selected and shown in the **Saved Mappings** list. Then, you can click **Import Structure** to proceed with the import.

When you click the link in the **Alerts** panel, you can view the information about the structures that could not be imported and the related objects.



When you click the **Related Objects** link, you receive a detailed report of the errors.




```
Import from Excel Report
=============================================
The Mobile Phone.xlsm file failed to import because of the following errors:

Mobile Phone:

Row 4:
        The object type required to create object at row 4 was not specified.
        The child object type "unknown" at row "4" is not an allowed child type for given parent type "Part" at row "3".
        The BOMLine at row "4" was not created because the underlying Object is not available.
Row 7:
        The child object type "unknown" at row "7" is not an allowed child type for given parent type "Part" at row "6".
        The object type required to create object at row 7 was not specified.
        The BOMLine at row "7" was not created because the underlying Object is not available.
Row 9:
        The child object type "unknown" at row "9" is not an allowed child type for given parent type "Part" at row "8".
        The object type required to create object at row 9 was not specified.
        The BOMLine at row "9" was not created because the underlying Object is not available.
Row 10:
        The child object type "unknown" at row "10" is not an allowed child type for given parent type "Part" at row "8".
```

## Export a structure to Excel for round-trip editing

You can export a structure to Excel and edit its data. You can then import this Excel back to Teamcenter so that these changes are reflected there. This process is called round-trip in Teamcenter.

### Procedure

1. Select a structure element to be exported and click **Excel Round-trip** > **Export to Excel**.
   The **Export to Excel** panel is displayed.
2. (Optional) In the **Settings** section, select one of the following options.
   - Select **All Levels** to export all levels of the structure to Excel.
   - Select **Levels**, and then enter the number of structure levels to be exported to Excel.

- Click the **Include ID as Hyperlink** check box to include a hyperlink in the **Object ID** column in the exported Excel file. You can click the hyperlink to open the object in your active Teamcenter session. You must be logged into Teamcenter for the hyperlink to function.

3. From the **Template** list in the **Properties** section, select a template or select **Custom**.

   - If you select a template from the list, you can export the structure elements into a specified template format. Selecting a template ensures that all required columns are included in Excel.

     Templates define the structure and required columns to be exported. Available templates are configured by your administrator based on your organization's business requirement. If a template is not available, contact your administrator.

   - If you select **Custom** from the list, you can select the columns that are exported to the Excel. When using the **Custom** option, ensure you include the required properties for successful round-trip import.

     Do one of the following:

     - Click to select a saved column arrangement. Saved column arrangements are custom column configurations that you previously saved for reuse.

     - Select one or more columns in the **Available Columns** list on the left. Then click to add the selected column(s) to the **Displayed Columns** list. Select one or more columns in the **Displayed Columns** list on the right. Then, click to remove the selected columns from the list. You can also choose to move a column up or down the list.

   The dynamic compound properties cannot be exported.

4. (Optional) Select the **Run in Background** check box.

   If you select this option, the export runs in the background, allowing you to continue working in Teamcenter. You receive an alert notification when the export process is complete.

5. Click **Export**.

   > **Note**
   >
   > If no objects match the criteria in the Excel template, a blank Excel sheet appears.

### Results

After the export is complete, the Excel file is saved to your browser's default download location.

### What to do next

You can add and remove rows for the elements, modify property values, and update relationships in the exported Excel. Do not add or delete the columns, as this may cause import errors. After you finish the modifications, the Excel can be **imported back** to Teamcenter.

## Guidelines for editing a round-trip Excel file

Use the following guidelines for editing the round-trip Excel file so that it can be imported back to Teamcenter.

### Structural modifications

- You can add new and existing elements to the structure and make structural changes.
- You can remove the existing elements from the structure.
- You can update the element properties.
- You can update the item revision properties.
- You can replace an existing item with another item.
- Suppose that an Excel file does not have any parts in an assembly, and then that Excel is imported back into Teamcenter. In such a case, no parts are removed from that assembly in Teamcenter.

  In other words, do not remove all child elements from the structure or a subassembly. Retain at least one child element.

### Column management

- Do not add or remove columns from a round-trip Excel file.
- The **TCUID** column is hidden. If the **TCUID** column is not hidden for some reasons in the exported Excel, do not edit its values. Teamcenter exports this column as an internal identifier for the exported objects.

### Adding objects

- To add a new object to the structure, add a new row to the exported file and ensure that the **TCUID** column is blank if it is visible.
- If you add a new item, ensure that you add its type in the **Type** column and all other required properties.
- If you want to add an existing item to the structure, enter its key fields (for example, **ID**).

### Macro requirements

Do not turn off the macro while opening the Excel file for editing. This action ensures that any copied lines do not carry forward the **TCUID** value from the original lines.

> **Note**
>
> The paste operation cannot be undone using Ctrl+Z when the macro is present. The newly pasted row can be deleted, or its contents can be deleted.

### Copy and paste operations

If you copy or cut and paste lines in a structure, ensure that the **TCUID** column is blank in the pasted lines. These operations are treated as separate remove and add operations, rather than move operations.

# Export and import structures along with partitions

You can export and import structures along with partitions and partition schemes to and from other Teamcenter sites.

For the import and export, you can use:

- Briefcase files
- Multi-Site Collaboration
- PLM XML

# Export and import structures along with worksets

You can export and import structures along with worksets to and from other Teamcenter sites.

A workset is a context collector for other structures. If your organization has very large structures with a multitude of occurrences, these occurrences can be collated within a workset to do a *what-if* analysis. But when you send the workset to another site by using a Multi-Site environment or a Briefcase, you might not want to export such a large number of occurrences along with the workset. Therefore, while exporting, the **Include entire BOM** option is not enabled by default. Consequently, the TCXML-based Multi-Site functionality works to share workset and related objects but not the product item or item revisions or their content.

Worksets also support site-consolidation activities through the TCXML-based Multi-Site functionality.

To export and import structures along with worksets, you can use:

- Briefcase files
- Multi-Site Collaboration
- PLM XML

# Export a structure to NX

Export a filtered Teamcenter structure to NX, including non-master objects and associated files, to view and work with the assembly within the NX environment.

You can export a filtered structure and open that in NX. That structure does not contain the parts that are filtered out.

### Prerequisites

- **NX for Active Workspace** must be installed.
- Briefcase Browser must be installed.

### Procedure

1. Search for and open the structure that you want to export.
2. Choose **More Commands** ••• > **Import/Export** > **Export NX Assembly**.

   The **Export NX Assembly** command is not available if Briefcase Browser is not installed.
3. In the **Export NX Assembly** panel, choose:
   a. **Export Non-Masters** – To export the object types specified as **Non-Master** in the access control list.
   b. **Export Associated Files** – To export files associated with the selected structure. However, based on the exclusion list defined by your administrator, some file types are not exported.
4. Click **Export**.

   The export operation runs in the background. When the export is complete, a notification is displayed in the **Alerts** panel.
5. Click the **Alerts** icon to see the notification and download the Briefcase **(.BCZ)** file if required.

   **Note**

   For additional information about exporting a structure to NX, see the *Teamcenter Integration for NX* guide.

### What to do next

After the structure is exported, ensure that any revision rules applied to the structure are reflected in the exported structure.
You can also ensure that

- The default variant condition is reflected in the exported structure. Any user-applied variant rules are ignored.
- Any effectivity applied as a part of a revision rule is reflected in the exported structure.
- Any user-applied effectivity other than those applied as a part of the revision rules is ignored.

# Import and export structures and changes individually

## Import a structure from Excel

Import a structure into Teamcenter from a prepared Excel file, including mapping properties, previewing changes, and handling potential errors, to efficiently integrate external structure data.

You can import a structure by using an Excel file. The structure can include all reference properties with an LOV, such as unit of measure, owner, and group. You can import structures with the global revision rule and structures using different ID types that use a Multi-Field Key (MFK). You can import vendor-part data, multiple parts and assemblies, and multiple types of secondary objects. You can also import the relational properties (like preferred or backup vendor preference status). If an array property contains multiple values in the Excel, you can import them too. For example, if the Excel contains a column for Production Sites and the values are Milford and Troy, both these values are imported in Teamcenter.

To import a structure from Excel, you must prepare the Excel file in a specific way. See **Prepare the Excel file for importing a structure**.

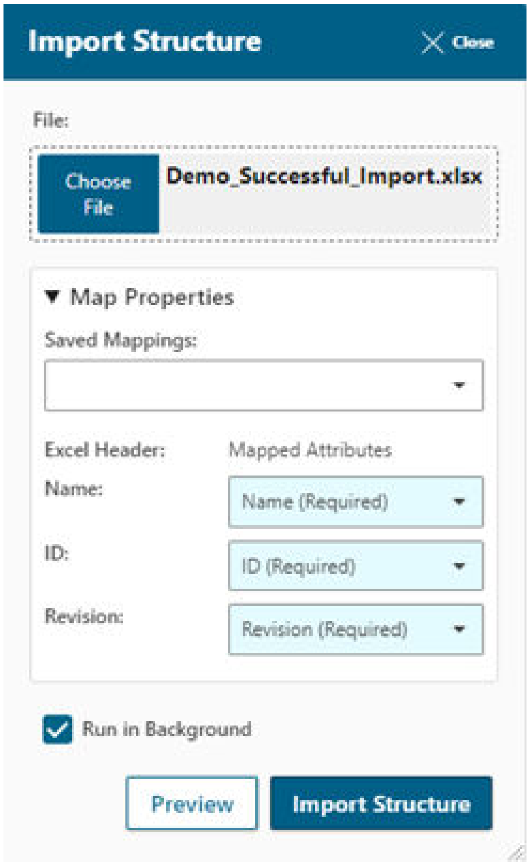
To import a structure from Excel, you must prepare the Excel file in a specific way. See *Prepare the Excel file for importing a structure*.

### Restrictions and limitations

You can perform this task if it is configured for you by your administrator.

### Procedure

1. Navigate to the folder where you want to import the structure, and choose **More Commands** ••• > **Import/Export** > **Import Structure**.
2. In the **Import Structure** panel, select the Microsoft Excel file from which you want to import the structure by clicking **Choose File**.
3. Map the structure properties between Teamcenter and Excel.
   - Select the required mapping from the **Saved Mappings** list. The list shows the mappings created and saved by the administrator or previous users.
   - Alternatively, create a new mapping by entering a name in the **Saved Mappings** field and selecting the **Mapped Attributes** for the **Excel Headers**. Even if an error occurs while importing a structure, the created mapping is saved. You need not select the mapped attributes again for the Excel headers.




- If a mapped attribute is not available for an Excel header:

  **a.** In the **Mapped Attributes** list for the corresponding Excel header, click **Add New** to create a new attribute.

  **b.** In **Add Properties**, select appropriate **Subtypes**, if you want to change the default one.

  The reference properties **Unit of Measure**, **Owner**, and **Group**, along with other properties, are listed in the **Add Properties** section.

  **c.** Filter and select the attribute that you want to map.

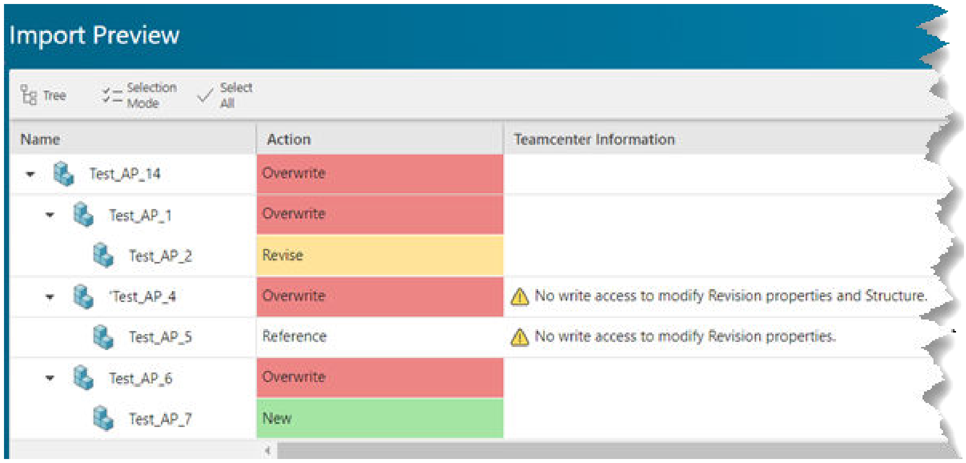
  **d.** Click **Add**.

  **e.** In the **Import Structure** panel, choose the newly added attribute.

**4.** (Optional) If you do not want to run the import process in the background, clear the **Run in Background** check box. The system notifies you about the result of the import process in the **Alerts** panel.

**5.** (Optional) Click **Preview** to check the structure before import.

  **a.** The preview shows the structure to be imported along with the **Action** that is performed by default. **Teamcenter Information** shows additional information related to an action.




| Action | Description |
|---|---|
| New | A new item or occurrence is created. |
| Revise | A new revision of the item or occurrence is created. |
| Overwrite | The existing revision is overwritten with the updated information. |
| Reference | The existing revision is used as is. For example, if the revision is released, and some changes are made to the revision in the Excel file without revising the item, the action is set as **Reference**. This indicates that the changes will not be applied. |

**b.** You can change the action for **Revise** and **Overwrite**. To do so, right click an action and select:

- **Revise**

  To create a new revision of the occurrence.

- **Reference**

  To reuse the existing revision in Teamcenter as is. The existing revision will not be updated with the latest information in the Excel file.

- **Overwrite**

  To update the existing revision in Teamcenter with the latest information from the source Excel file.

**6.** Click **Import Structure** to import the structure in Teamcenter.

**If there were no errors while importing the structure:**

The structure is imported in Teamcenter. When you import multiple elements or assemblies simultaneously, all the top-level elements and assemblies are added to the folder you specified previously. You receive a notification in the **Alerts** panel when the import is complete. Click **Alerts** to open the **Alerts** panel.

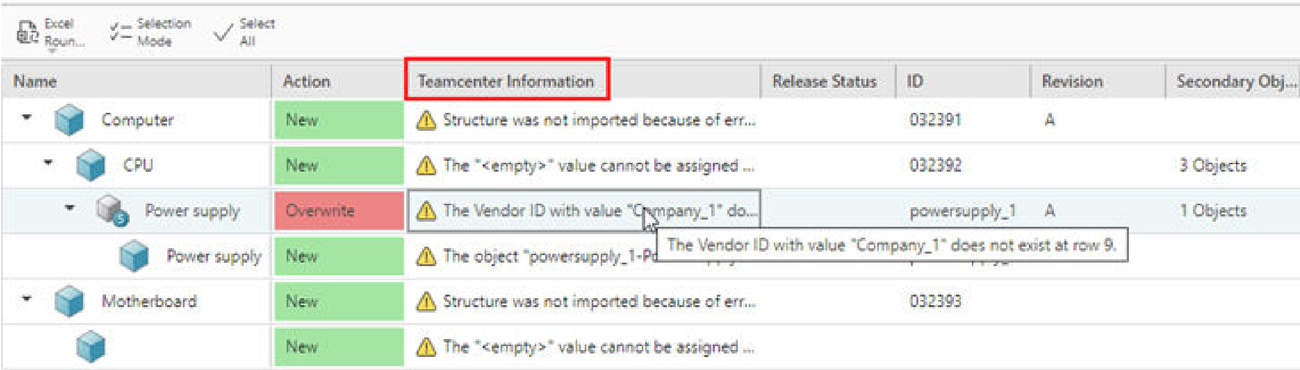
The structure is imported using the global revision rule. The same revision rule is used while opening the structure after import.

You can also import a structure with different ID types that use a multi-field key, that is, when you have two different items with the same ID. You can see the items on the **Import Preview** screen and import them successfully. The import structure functionality searches for and creates items based on the multi-field key.

**If there were errors while importing the structure:**

You receive an error message stating that the import failed because there were errors while importing the structure. You also receive a notification in the **Alerts** panel, where you can view the error details. Click **Alerts** to open the **Alerts** panel.

In the **Teamcenter Information** section, you can view a list of all potential errors simultaneously. When you hover over an error, you can view detailed information about that error.

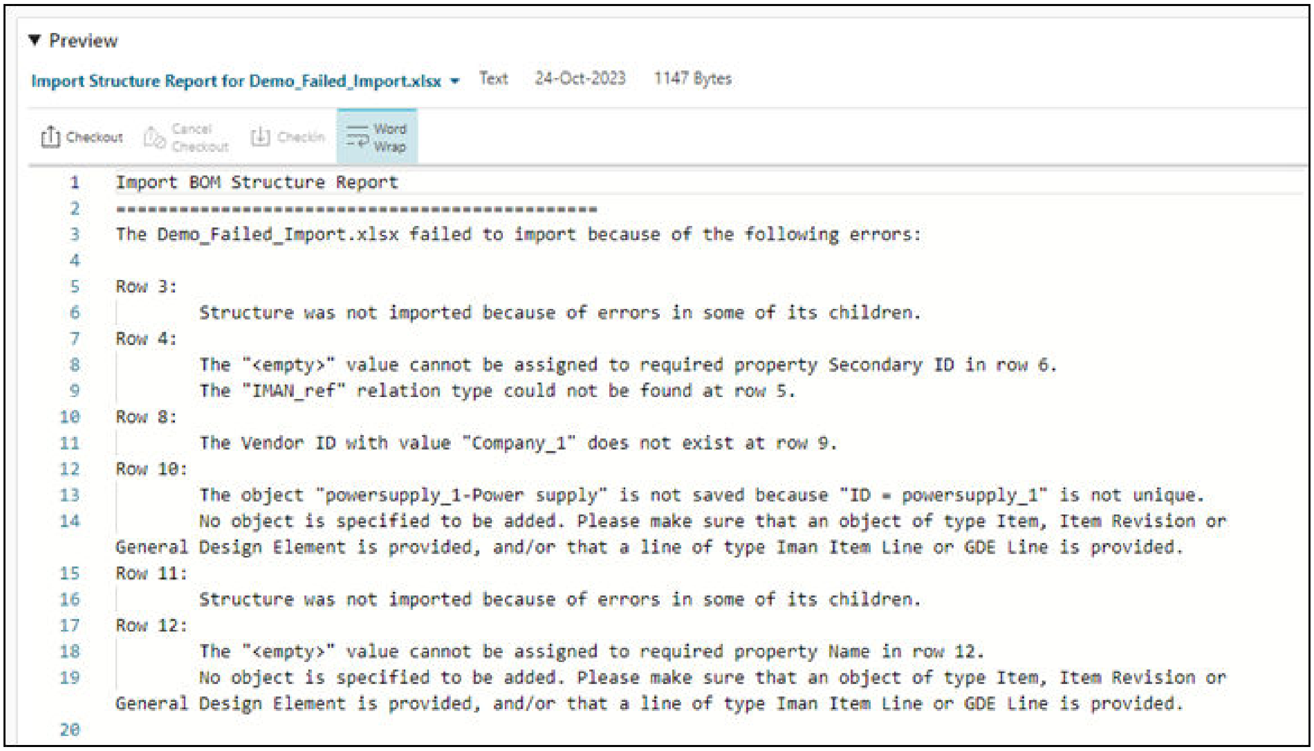



**Tip**

At this stage, you need not restart the import process from the beginning. You can make the required corrections in the Excel file and then reupload it by clicking **Choose File** in the **Import Structure** panel. Once this is done, the saved mappings from the previous import are automatically selected and shown in the **Saved Mappings** list. Then, you can click **Import Structure** to proceed with the import.

When you click the link in the **Alerts** panel, you can view the information about the structures that could not be imported and the related objects.


When you click the **Related Objects** link, you receive a detailed report of the errors.




## Export a structure to Excel

You can export a structure to Excel. Another user who does not have access to Teamcenter might perform some changes in the Excel. You can then import this Excel back to Teamcenter so that these changes are reflected there.

### Restrictions and limitations

You can perform this task if it is configured for you by your administrator.

### Procedure

1. Select the structure elements to be exported and click **Excel Round-trip** > **Export to Excel**. The **Export to Excel** panel is displayed.
2. (Optional) In the **Settings** section, select any of the following check boxes:
   - **Allow Structure Changes in Excel**: Selecting this option allows external users to modify the structure in Excel.
   - **Include ID as Hyperlink**: The **Object ID** column and the link to the object are exported to Excel. You can click the link and open the object in Teamcenter.
   - **Include Outline Numbers**: The numbers denoting the hierarchy are exported to Excel.
3. From the **Template** list in the **Properties** section, select a template or select **Custom**.

- If you select a template from the list, you can export the structure elements into a specified template format.

  If a template is not available, contact your administrator.

- If you select **Custom** from the list, you can select the columns that are exported to the Excel. Do one of the following:

  Select a saved column arrangement. Select one or more columns in the **Available Columns** list on the left. Then click > to add the selected column(s) to the **Displayed Columns** list. Select one or more columns in the **Displayed Columns** list on the right. Then click < to remove the selected column(s) from the list. You can also choose to move a column up or down the list.

**4.** (Optional) Select the **Run in Background** check box. If you select this option, you receive an alert when the export process is complete.

**5.** Click **Export**.
The export process begins, and a progress bar indicates that the process is still running.

**6.** Open the exported Excel and make the required changes to the properties listed, and save the Excel.

> **Note**
>
> If no objects match the criteria in the Excel template, a blank Excel sheet appears.

> **Caution**
>
> Do not alter these columns in the Excel sheet: **TC_ObjectType**, **TC_ObjectID**, and **ID**. One exception to altering **TC_ObjectType** is creating a new object in the structure.

**7.** (Optional) To create a new object in the structure:

**a.** Open the Excel sheet and insert a new row using the Excel insert feature.

**b.** Indicate the object type to be created in the **TC_ObjectType** column, ensuring that the **TC_ObjectID** column is left blank.

**c.** (Optional) To adjust the parent or outline level, use the Excel grouping feature.

**d.** Save the changes to the Excel file.

## Import changes from a previously exported Excel

You can import the changes from a previously exported Excel back to Teamcenter.

### Restrictions and limitations

- You can perform this task if it is configured for you by your administrator.
- Using the Excel Round-trip functionality, you cannot add or remove structure elements. Additionally, you cannot edit some properties, such as element effectivity and variant formula.

### Procedure

1. Click **Excel Round-trip** > **Import Changes** .
   The **Import Changes** panel is displayed.
2. Click **Choose File**, navigate to the Excel file location, and then click **Open**.
3. To ensure that your changes are retained in the event of a conflict, select **Overwrite Conflicts**.
4. Click **Import**.

## View where an element is used across assemblies simultaneously

Export the **Where Used** information for a structure element to an Excel file, allowing you to simultaneously view its usage across all assemblies at all levels or at the product level, for comprehensive analysis.

You can view where a structure element is used across all assemblies up to the top level, starting with the immediate parent and navigating one level up at a time. To do this, you go to the **Used in Structures** section and expand one level at a time. However, if you want to view all this data simultaneously, export it to an Excel file.

Similarly, if you want to see where an element in used at the product level across all assemblies simultaneously, you can export this information to an Excel file as well. However, you must have Smart Discovery Indexing installed to generate this data.

### Procedure

1. Search for and select the structure element.
2. Click the **Where Used** tab.
3. Click **Export to Excel** from the following sections depending on the scenario.

| Scenario | Action |
|---|---|
| View where the element is used across all assemblies at all levels. | In the **Used in Structures** section, click **Export to Excel** .<br><br>**Note**<br>If the number of results in the **Used in Structures** section goes beyond the threshold value, the **Export to Excel** button is unavailable. The threshold value and the number of levels to be exported to Excel are set by your system administrator. |
| View where the element is used across assemblies only at the product level | In the **Top Level** section, click **Export to Excel** . |

| Scenario | Action |
|---|---|
| (works only if you have Smart Discovery Indexing installed). | |

The **Export To Excel** panel is displayed with a list of properties available for export.

4. To change the order of the properties, select a property, and in the **Selected Properties** section, click **Move Up** or **Move Down**.
5. To select the properties to be exported, in the **Selected Properties** section, click **Add Properties**.
6. In the **Add Properties** section, clear the check boxes for the properties that you do not want to export. Select the check boxes for the properties that you want to export.
7. Click **Add**.
8. It is recommended to use the **Run in Background** option while exporting the data from the **Used in Structures** section. However, if you do not want to run the export process in the background, clear the **Run in Background** check box.

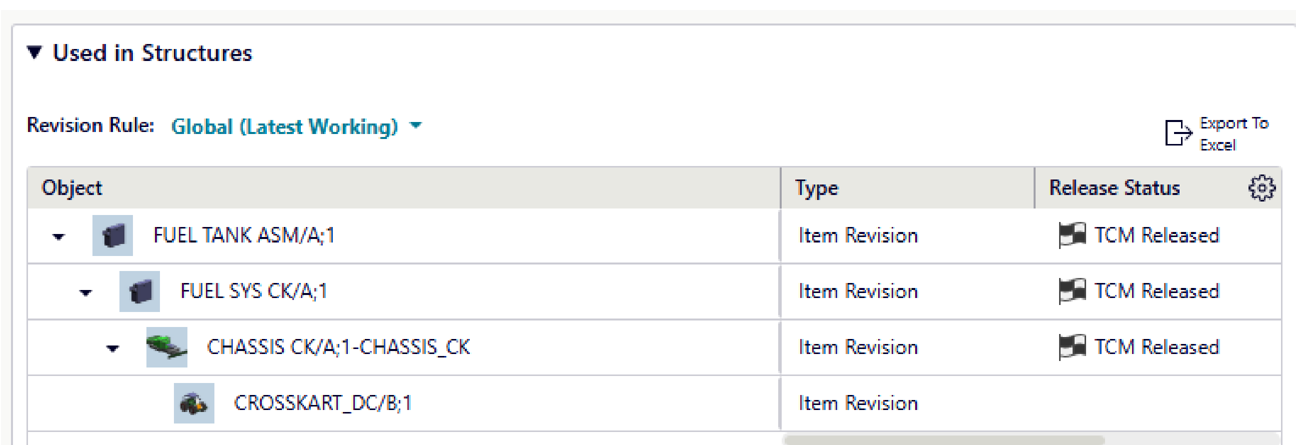
   While exporting the data from the **Top Level** section, you do not get an option to run the export process in the background.
9. Click **Export**.
   If the export process ran in the background, view the notification in the **Alerts** panel and click the link to download the Excel file.

   If the export process did not run in the background, a dialog box is displayed to download the Excel file to your machine.

   **Note**

   You cannot reimport the exported data back into Teamcenter.

The following graphic shows the sample data in the **Used in Structures** section.




The following graphic shows the sample data in the exported Excel file.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Level** | **Object** | **Type** | **Release Status** | **Owner** |
| 2 | 1 | FUEL TANK ASM/A;1 | Item Revision | TCM Released | Doe, John |
| 3 | 2 | FUEL SYS CK/A;1 | Item Revision | TCM Released | Doe, John |
| 4 | 3 | CHASSIS CK/A;1-CHASSIS_CK | Item Revision | TCM Released | Doe, John |
| 5 | 4 | CROSSKART_DC/B;1 | Item Revision | | Doe, John |
| 6 | | | | | |

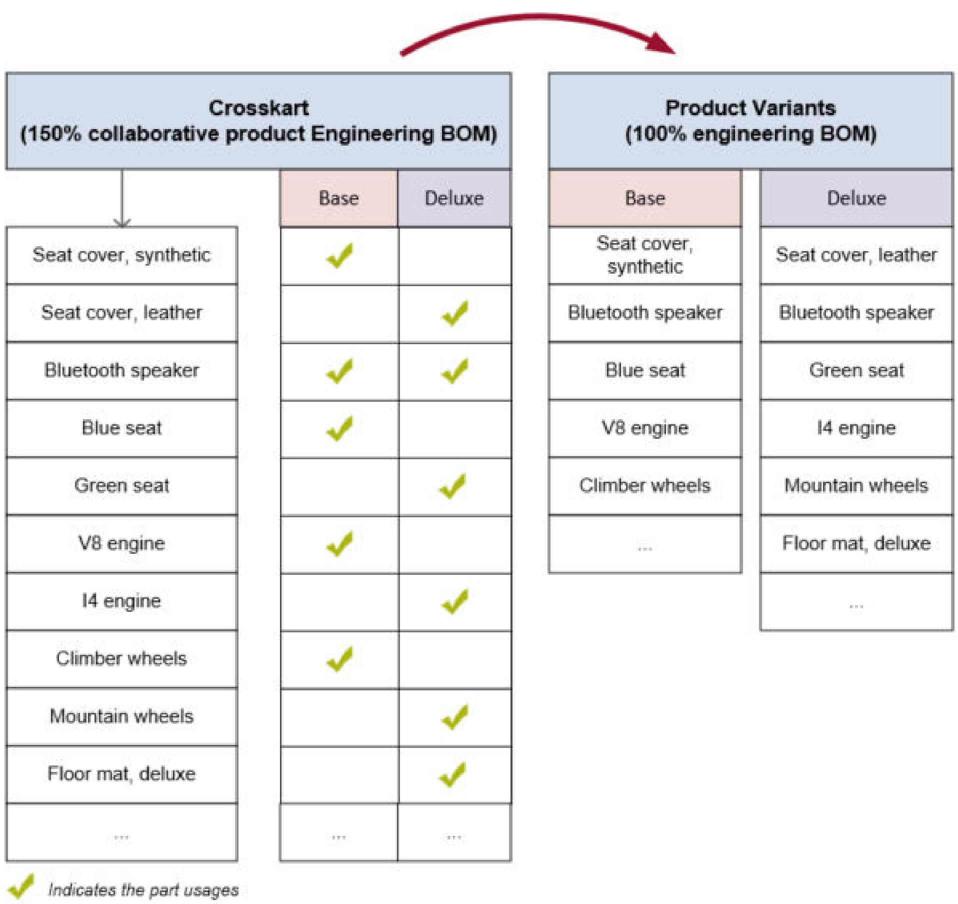
# 10. Create and maintain Usage BOM

## About Usage BOM

A *Usage BOM* is a 150% BOM that represents the complete content of a product and its variants, if available. A Usage BOM consists of item revisions and occurrences, also called collaborative part and part usages. The part usages have independent lifecycle and change control. In other words, the part usages in a Usage BOM can be revised and released independent of its parent assembly.

**Example**

Consider that *Crosskart* is one of the several products that your company manufactures. Crosskart, in turn, has two variants, *Base* and *Deluxe*. The BOM content of these two variants is maintained within a single Usage BOM. The BOM for each variant is derived based on the usage of a part in a specific variant.



# The different ways to create a Usage BOM

In usage-based engineering BOM management, a *Usage BOM* is a 150% BOM that represents the complete content of a product and its variants, if available.

There are several ways to create a Usage BOM:

- Automatically generate an aligned Usage BOM from an existing design BOM.

  The variability scope associated with the design BOM and the alignment data are carried over to the Usage BOM.

- Manually **create** a Usage BOM or **save** an existing BOM to create a new one. You cannot duplicate a Usage BOM to create a new one.

  After creating the BOM, you must generate its aligned design BOM or align it with an existing design BOM so that you can visualize the Usage BOM.

The effectivity is carried over to the Usage BOM from the active change notice.

## Generate a Usage BOM from a design BOM automatically

Learn how to automatically generate a Usage BOM from a design BOM (design structure), align and then update it. You can also manually correct the alignment of parts in case of a mismatch.

You can automatically generate a Usage BOM from a design BOM (design structure) that already exists in Teamcenter. The variability scope associated with the design BOM is carried over to the generated Usage BOM.

If the aligned design structure is updated, you must also update the Usage BOM.

After generating or updating the aligned Usage BOM, you can validate if the alignment is done correctly. In case of an alignment mismatch, you can realign the parts and designs, and the part occurrences and design occurrences manually.

For more information on generating and updating a Usage BOM, validating the alignment, and realigning parts with designs and part occurrences with design occurrences, see *Design and BOM Alignment — Deployment and Administration*.

## Create a Usage BOM manually

You can create a Usage BOM to represent the complete content of a product and its variants, if available.

### Procedure

1. Open the folder where you want to create the Usage BOM and click **Add to** ⊕.
2. In the **Add** panel, filter to select **Collaborative Product EBOM**.
3. Enter the required information and click **Add**.

### What to do next

After creating a Usage BOM, you must add **parts** to it. You can also add a **variability scope** and generate its aligned design BOM, if required.

## Save an existing Usage BOM to create a new one

You can save an existing Usage BOM to create a new one if the new product BOM is very similar to the existing one.

### Procedure

1. Search for **Collaborative Product EBOM** and select it.
2. Click **More Commands** ••• > **New** > **Save As** .
3. In the **Save As** panel, enter the required information and click **Save**.

### Results

- The parts of the existing Usage BOM are reused in the new one. New part usages are created for the reused parts. If a part is an assembly part, a new part usage is only created for the parent part.
- The new Usage BOM is aligned to the same design data as the original one. However, the alignment between the part occurrences and design occurrences are not carried over.
- The effectivity of the new part usages is derived from the active change.

  The effectivity of the active change is not applied to the child parts of an assembly part.

### What to do next

- Add a **variability scope** to the Usage BOM.
- Generate an aligned design BOM, if not already generated. You can also manually align the Usage BOM with an existing design BOM.
- Update the Usage BOM to **add**, **remove**, or **replace** parts. You can also reuse parts from another Usage BOM by **carrying over** the engineering BOM data to the new BOM.

# Associate a variability scope with a Usage BOM or a configurable part

To share the variability defined for a product with other users to author part usages, you associate the product's variability scope with its Usage BOM. You can also associate the variability scope with a part that is set as **Configurable Assembly** or **Generic Part**.

### Procedure

1. Search for **Collaborative Product EBOM** and open it.
2. Click **More Commands** ••• > **Edit** > **Start Edit** .
3. In **Variability Scope**:
   a. Click **Add** .
   b. Search for variability scope, select it, and click **Add**.

   You can associate only one variability scope with a Usage BOM.

   Similarly, select a part set as **Configurable Assembly** or **Generic Part** and add a variability scope. The variability scope that you add to the part must preferably be subsets of the variability scope that you added to the Usage BOM.
4. Click > **More Commands** ••• > **Edit** > **Save Edits** .

# Add parts and part usages to build a Usage BOM

A part usage defines how a part is used in a product, which is represented as a Usage BOM. Add parts and part usages to build a Usage BOM.

### Procedure

1. Set an active change.
2. Search for **Collaborative Product EBOM** and open the structure.
3. If you are adding parts to the Usage BOM for the first time, go to the **Content** tab.
4. Click **Add** > **Child**.
5. In the **Add Part** panel, to add an existing part, go to **Search** or **Palette**.

   OR

   To add a new part, click the **New** tab in the panel, and:
   a. Enter the required information.
   b. Select the **Design Required** check box if you need a corresponding design.

c. Select the **Automatically Create Design** check box if you want the design to be automatically created and aligned to the part.

   OR

   Click **Add Design** to manually create a design. In the **Add Design** panel, enter the required information, and click **Add**.

   After manually adding a design, if you select the **Automatically Create Design** check box, the manually added design is ignored, and a new design will be automatically created by Teamcenter and aligned to the part.

6. In the **Usage Properties** section, enter the required information for part usage.

7. Click **Add** to include the part and its usage in the product BOM.

   If your administrator has enabled effectivity propagation on your structure, then you can see **Occurrence ECN In** and **Occurrence ECN Out** properties for your parts. They show the responsible change notice names due to which the part has derived its element effectivity start and end date respectively.

| Requirement | Action |
|---|---|
| Add parts at the same level | • Select a part and click **Add** > **Sibling**.<br>OR<br>• Right-click the part and click **Add Sibling**:<br>1. In the **Part** column, click the cell and click **Add**.<br>2. After adding the part, click **Save Edits**. |
| Add child parts | • Select a part and click **Add** > **Child**.<br>OR<br>• Right-click a part and click **Add Child**:<br>1. Click the cell in the **Part** column, and click **Add**.<br>2. After adding the part, click **Save Edits**.<br>Adding child parts to a parent part builds a structure. Open the structure in a separate window or a separate tab to maintain it independently or duplicate it. |
| Reuse parts from other Usage BOM | Carry over the parts to this BOM, along with other Usage BOM data. |
| Include software artifacts | Include embedded software artifacts that are made available as engineering parts by the administrator.<br>**Example**<br>The part *Tire Pressure Sensor* in the product *Crosskart* can be programmed such that when the air pressure |

| Requirement | Action |
| --- | --- |
| | drops below the product's recommended level, the sensor captures this information, and the dashboard indicator lights up.<br><br>Software developers create and maintain the software artifacts using the functionality. |
| Include parts that already exist in Teamcenter | To reuse existing parts/assemblies (revision controlled as well as part usage controlled), ensure that these parts are qualified as engineering **Parts** by the administrator.<br><br>**1.** Click **Add** > **Child**.<br><br>**2.** Search for and add the part from **Search** or **Palette**.<br><br>A part usage is created for the part to define its usage in the Usage BOM.<br><br>**Note**<br><br>In case of revision controlled assembly, **effectivity propagation** or **cutback** is not supported. |
| Include ECAD parts | Reuse ECAD parts created and managed in integrated Electronic Design Automation (EDA) applications in the Usage BOM.<br><br>ECAD parts are represented both as designs (in the design BOM) and as parts (in the engineering BOM), hence they are shared between design and part structure.<br><br>**Note**<br><br>It is recommended to edit these parts in the integrated EDA application, and not directly in the BOM. |
| Include an existing solution variant | To reuse an existing solution variant in the Usage BOM:<br><br>**1.** Click **Add** > **Child**.<br><br>**2.** Search for and add the solution variant from **Search** or **Palette**. |
| Remove a part from the Usage BOM | • Select the part and click **Edit Structure** > **Remove**.<br><br>OR<br><br>• Right click the part and click **Remove**.<br><br>The element may split due to **effectivity cutback**.<br><br>The part usage of the selected part may get **discontinued** depending upon effectivity overlap. |

**What to do next**

1. Set the **maturity level** of a part and a part usage.
2. If needed, edit the part to change its assembly indicator. If you choose **Configurable Assembly** or **Generic Part** for the assembly indicator:
   a. **Add a variability scope** to the part to get the variability data.
   b. **Author a variant condition** for the part usage to define when the part must be used in the product structure.

# Discontinue a part usage from a Usage BOM

If a part becomes unavailable because of a logistic issue or raw material shortage, you can temporarily remove its part usage and reintroduce it into the product later.

> **Example**
>
> Imagine that you have a released part usage for a car GPS module which is effective from January 1, 2023 to December 31, 2024. Due to a semiconductor shortage, you want to discontinue the part usage for the car GPS module for 7 months, starting June 1, 2023 until December 31, 2023. After December 31, 2023, you want to reintroduce this module back into the product.

You can do this by removing the released part usage of the GPS module for a specific duration.

To discontinue a part usage:

**Procedure**

1. Create a change notice with a **release effectivity** range that covers the duration of the part usage discontinuation, and **set it as an active change**.
2. Open the released Usage BOM. Right-click the impacted part usage and click **Remove** ⊖.

   The impacted part usage is split due to **effectivity cutback**, creating a new **Usage Revision** which is effective for the required period. A new part usage is created with a value of <*True*> for the **Discontinue** property.

   The discontinued part usage is never configured in the product. You cannot include it by using the **Show excluded by effectivity** toggle, and it is not available in the **History** tab, but you can find it using search.

   > **Note**
   >
   > Discontinuing a part usage does not discontinue its corresponding part, because that part may be reused in other products.

# Discontinue a part

Sometimes a part is unavailable due to logistic issues or raw material shortages. In such cases, you can remove that part for a finite amount of time and reintroduce into the product later.

> **Example**
>
> Imagine that you have a released car GPS module which is effective from January 1, 2023 to December 31, 2024. Due to a semiconductor shortage, you want to discontinue the part for 6 months, starting June 1, 2023 until December 31, 2023. After December 31, 2023, you want to reintroduce the part into to the product.

### Procedure

1. Create a change notice with a **release effectivity** range for the duration of the part discontinuation, and **set it as an active change**.
2. Open the engineering BOM and select the released part.
   a. Click **More Commands** > **New** > **Revise** to first revise the part.
   b. Next, in the **Part Properties** section of the **Overview** tab, select the **Discontinued** check box.

   OR

   a. Click **More Commands** > **Manage** > **Submit to Workflow**.
   b. In the **Submit to Workflow** panel, select **Part Discontinue** in the **Template** section.
   c. Click **Submit**.

   You can continue this part again by revising it and clearing the **Discontinue** check box.

   > **Note**
   >
   > Discontinuing a part does not discontinue its corresponding part usage.

# Carry over BOM data from one Usage BOM to another

You can quickly populate the Usage BOM of a product by carrying over the BOM data such as parts, part usages, and alignment data from another Usage BOM.

### Procedure

1. Search for Usage BOMs.
2. From the search results, select the two required Usage BOMs.
3. Click **More Commands** > **View** > **Compare Structures**.
   The two Usage BOMs open side-by-side.

4. Drag the parts from one Usage BOM to the other in order to carry over the BOM data. You can also copy parts from one BOM and paste the parts on the other BOM.

#### Results

The properties of the carried over part remain the same in the target Usage BOM. New part usages are created for the carried over parts, but the variant condition and quantity of the new part usage remain the same as those of the source part usage. If the target Usage BOM is aligned to a design BOM, the design occurrence of a part is carried over and is aligned with the new part usage (part occurrence). Therefore, you can visualize the new part usage. If the target Usage BOM is not aligned to a design BOM, only the BOM data is carried over. The design data is not carried over. The design data is also not carried over if the **Design Required** flag of the carried over part is **false**.

You can also carry over BOM data by:

- Opening two Usage BOMs in the split view to carry over BOM data from one BOM to another.
- Copying the part usages from one BOM to another.

## Replace a part

You can replace a part with another part. You can also replace an existing part or a sub-assembly in an assembly with its copy using the **Save As And Replace** command. If the **Save As And Replace** option is not visible, you may not have the required privileges.

#### Procedure

1. Search for the element that you want to replace.
2. Select the occurrence and click **Edit Structure** > **Replace** .
3. In the **Replace** pane, select the replacement and click **Replace**.

   Teamcenter replaces the element and refreshes the display.

   In case of a released Usage BOM, these modifications followed by alignment update might result in the element split due to **effectivity cutback**.

### Replace a structure component with its copy

A copy is a new item with identical properties but a new ID. You can copy a part or a subassembly, with or without children, save it with a new name, and replace the existing subassembly in the structure with its copy. If the subassembly is copied with children, the child parts are renamed based on the naming convention specified using the naming rule.

### Procedure

1. Select the structure from which you want to copy the element, and click **Open**.
2. Choose **More Commands** > **New** > **Save As and Replace**.

   The **Save As And Replace** pane is displayed.
3. (Optional) Select the **Clone Children** check box.

   If you choose the **Clone Children** option, the child elements in the assembly are replaced along with the top element.
4. Select **ID Naming Rules** to automatically assign new IDs to the copied elements. The **ID Naming Rules** option is visible only when the **Clone Children** check box is selected.

   - **Prefix**

     The text string provided is added to the start of the existing IDs, and new IDs are assigned.
   - **Suffix**

     The text string provided is added to the end of the existing IDs, and new IDs are assigned.
   - **Replace/With**

     The text string specified in the **Replace** field is replaced with the text string specified in the **With** field in the new IDs.
5. (Optional) Select the **Run in Background** check box.

   > **Note**
   >
   > For the save and replace operation to run in background, your system administrator must configure Dispatcher services.
6. If you are replacing a part, select the **Copy Design Alignment** check box. If you select this check box, another revision of the existing aligned design is automatically created and aligned with the part. If you do not select this check box, a new revision of the existing aligned design is created at the time of alignment.

   If you are replacing a design, select the **Copy Part Alignment** check box. If you select this check box, another revision of the existing aligned part is automatically created and aligned with the design. If you do not select this check box, a new revision of the existing aligned part is created at the time of alignment.
7. Click **Save As And Replace** at the bottom of the **Save As And Replace** pane.

   Teamcenter replaces the element and refreshes the display.

   If you choose to run the replace operation in the background, a notification is generated once it is complete. Click the notification to view the details of the replacement.

   In case of a released Usage BOM, these modifications followed by alignment update might result in the element split due to **effectivity cutback**.

# Author a variant condition for a part usage

A *variant condition* defines how to use a part in a product or a structure. You author a variant condition for a part usage in the context of a Usage BOM or a Assembly BOM within it, which is marked as **Configurable Assembly** or **Generic Part**.

### Procedure

1. Search for **Collaborative Product EBOM** and open it.
2. Associate a variability scope with the Usage BOM or the Assembly BOM, if not done already.
3. Select the part for which you want to author the variant condition.
4. In the **Variant Conditions** tab, click **Edit** .
5. For each **Variability Content**, click a cell to set the variant condition:
   - Click one time to display a green check mark to include the variability content in the variant condition.
   - Click two times to display a circle backslash to exclude the variability content from the variant condition.
   - Click three times to display a blank cell to indicate that the variability content is not used in the variant condition.

   Multiple variant conditions are connected using a logical **AND** operation.
6. Click **Validate All** to validate if the variant condition is correctly authored as per the Product Configurator rules.
7. Click **Save Edits** .

### Results

The variant condition of a part usage is displayed in the **Variant Formula** column. The variant condition of a part usage can be different from the variant condition of its aligned design occurrence.

# Duplicate a structure

You can create a new structure by duplicating an existing structure. You can also duplicate structures that you do not own. When you duplicate a structure that you do not own, you become the owner of the new structure.

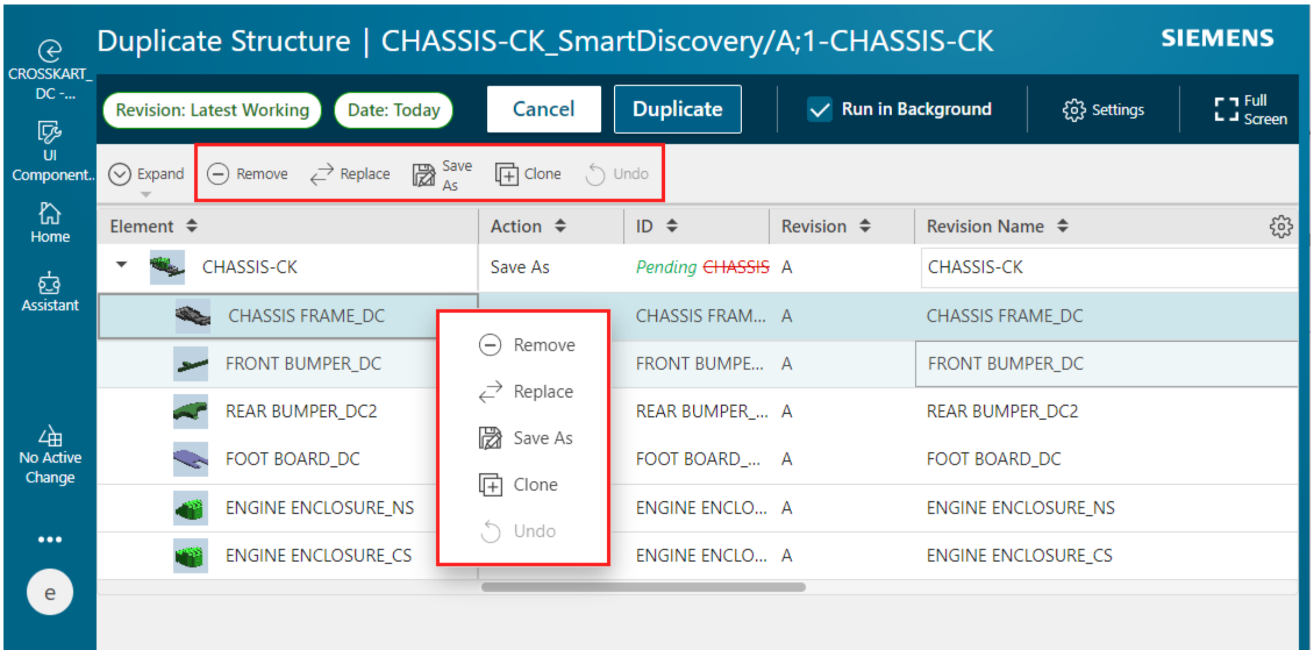
For an Assembly BOM or Usage BOM, duplicate the child items instead of the top-level structure. Design structures can be duplicated in their entirety.

### Procedure

1. For an Assembly BOM or Usage BOM, select the child items. For a design structure, select the top line of an assembly in the structure that you want to duplicate.

2. Click **Duplicate** .

3. Right-click each element and select one of the following actions. You can also select these actions from the work area toolbar.




| Action | Description |
|---|---|
| **Remove** | The element is not included in the new structure. |
| **Replace** | The element is replaced with the replacement element that you specified in the **Replace** panel. |
| **Save As** | A copy of the element is added to the new structure. You own this new element.<br><br>If the element is a structure, all its child elements are only referenced in the new structure. These are still owned by the user who created the source structure.<br><br>You can edit the **Revision Name** and **Description** of the element. You can edit the custom properties if your administrator has configured those custom properties accordingly. |
| **Clone** | A copy of the element is added to the new structure. You own this new element.<br><br>If the element is a structure, copies of its child elements are used in the new structure. |

| Action | Description |
|---|---|
| | You can edit the **Revision Name** and **Description** of each element. You can edit the custom properties if your administrator has configured those custom properties accordingly. |

**Note**

If any child element is a structure, all its children are also marked with the same action. However, if you change the action of one of its child elements later, the action of the parent element changes accordingly.

If you do not select an action, and the **Action** column is blank, the element is only referenced in the new structure. The element continues to be owned by the user who created the source structure.

4. To undo an action set on an element, select the element, and click **Undo**.

5. (Optional) Click **Settings** ⚙, and then perform these additional actions on the duplicated structure:

   a. Choose an option for the ID naming in the Settings panel:

      - Click the **Default IDs** option to retain the original element IDs for the cloned elements and for any elements that are saved as new elements.
      - Alternatively, click the **ID Naming Rule** option and enter the following details.

      | ID naming rule | Description |
      |---|---|
      | **Prefix** | The text is prefixed at the start of the existing IDs. |
      | **Suffix** | The text is appended at the end of the existing IDs. |
      | **Replace/With** | The text specified in **Replace** is replaced with the text specified in **With**.<br>The text specified in **Replace** must be text that is part of the existing ID, and it is case-sensitive. |

   b. To retain the original ownership for cloned elements and the elements that are saved as new elements, click the **Retain Original Ownership** check box.

      The elements on which no action is taken are referenced in the duplicated structure and their original ownership is retained.

   c. Ensure that the **Owning Project** section displays the project that is a part of your active Teamcenter session.

      You can change your project so that the duplicated structure becomes a part of a different project. If no project is selected in the active Teamcenter session, you do not see the **Owning Project** section.

   d. To include the Designcenter NX dependencies, if any, that are associated with the existing elements in the newly duplicated structure, click the **Include all Designcenter Dependencies** check box.

   e. Click **Close**.

6. To run the duplicate process in the background or foreground, select or clear the **Run in Background** check box. If you choose to run the duplicate process in the background, you receive a notification in **Alerts** once the structure is duplicated.

   > **Note**
   >
   > If your administrator has enabled the parallel duplication of large structures, the duplication process always runs in the background. Contact your administrator to adjust this setting.

7. Click **Duplicate**.

### Results

The alignment data is carried over from the original structure to the new structure. The element effectivity for the new structure is derived from the active change.

By default, your active Teamcenter session project is assigned to the duplicated structure and its elements. If the administrator has enabled the project assignments in the Teamcenter environment, the following actions are performed on the duplicate structure:

- If an item in the original structure is associated to a project, that project is replaced with the project selected in the active Teamcenter session for the corresponding duplicated item.
- If no project is associated to an item in the original structure, no project is assigned to the corresponding duplicated item, even though a project is selected in the Teamcenter session.
- For referenced items that are not cloned but are reused in the new structure, if the original item is associated to a project, the project that is selected in the active Teamcenter session is appended to its existing project list.

## Create standalone parts

Each component that makes up a product is identified as a part in Teamcenter. You can create standalone parts and later add these parts to an engineering BOM.

### Procedure

1. Open the folder where you want to create a part and click **Add to** ⊕.
2. In the **Add** panel, filter to select **Part**.
3. Enter the required information and click **Add**.

   By default:

   - **Finish Type** is set to **None**.
   - **Part Maturity** is set to **Concept**.

You can update the part properties later as needed.

While updating the **Assembly Indicator** for the newly added part, if the part is not a *Generic Part*, *Configurable Assembly*, or a *Supplier Part*, leave it blank. If the newly added part is a *bought out* part (that is, a piece part, purchased assembly, or a foreign part), you can edit it and select **Supplier Part** in the **Assembly Indicator** list. If the selected parent part is a supplier part, the child part that you add to it is always treated as a spare part and is not visible in Teamcenter. The spare part does not participate in alignment, structure configuration, and visualization.

# Set the maturity level of a part and part usage

Learn how to set the maturity level for a part and part usage. Understand how part maturity reflects its development cycle (for example, conceptualization and production) and how part usage maturity indicates its readiness for a specific product.

The maturity of a part indicates its progression in its development cycle. For example, the maturity of a part can indicate if it is being conceptualized, if its prototype is created, or if it has entered production.

The maturity of a part usage indicates how ready a part is to be used in a product. The maturity of a part can be different from that of its part usage. For example, a part can already be in production and be used in a product. The same part can be reused in another product. However, the usage of the part in the new product may be in the conceptualization stage.

### Procedure

1. Open a **Collaborative Product EBOM** and select the part for which you want to set the maturity level.
2. Click **Edit** near the **Overview** tab.
3. In the **Overview** tab, set the **Part Maturity** of the part in the **Part Properties** section and the **Usage Maturity** in the **Part Usage Properties** section.
4. Click **Save**.

# Revise a part

At times, you may need to edit an existing part, requiring a revision. You can modify the information in the newer revision. Revisions are also needed to change a part when it has been released.

### Procedure

1. Open a Usage BOM structure.
2. Select the part that you want to revise.
3. Click **More Commands** > **New** > **Revise**.

4. In the **Revise** panel, enter the required information and click **Revise**.

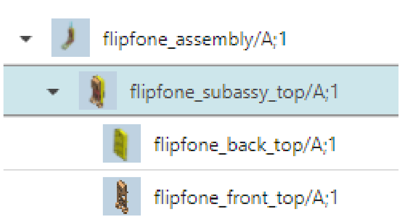
# Insert or remove levels in a structure

You must have write access to the structure to insert or remove a level. The insert action adds a level between the selected object and its parent object. The remove action removes the selected element's parent object one level above and attaches the element and all its siblings to its current grandparent object two levels above.

Search for the structure you want to edit and click **Open**.

| To | Do this | Result |
| --- | --- | --- |
| Insert a level | 1. Click the element above which you want to insert a level and click **Edit Structure** >**Insert Level** .<br>2. Fill in the details and click **Insert**. | A new level is added above, as the new parent of the selected element. |
| Remove a level | 1. (Optional) If any child of the level you want to remove has effectivity or variant condition set on it or is suppressed, enable the following options:<br>• **Show Excluded by Effectivity**<br>• **Show Excluded by Variants**<br>• **Show Suppressed**<br>2. Click the element that you want to remove from the assembly and select **Edit Structure** >**Remove Level** . | The selected element is removed from the structure, and its child elements are attached to what was previously their grandparent elements. |

The following example shows how the remove-action works in the three-level assembly in the graphic:




In both cases, removing or adding levels, the element may split due to **effectivity cutback**.

If you are not able to remove an aligned occurrence, contact your administrator.

# Update a Usage BOM

There are several scenarios where you might want to update the information in a Usage BOM. For example, you might want to do this if the design BOM aligned to it has changed.

There are several ways to update the aligned Usage BOM, such as automated update and guided update.

> **Note**
>
> Consider that a design BOM and a Usage BOM are released and aligned. Later, due to market requirements, the design BOM is revised and one or more child designs is modified (added, removed, or replaced). When you try to align the design BOM with Usage BOM, the child parts are modified accordingly, and the Usage BOM is always revised.

**Procedure**

1. Open a **Collaborative Product EBOM**.
2. Click **Edit**.
3. Update the required information, and click **Save**.

# Update parts and part usages

You can update the properties in the part and part usage, with specific considerations for released revisions and design-part alignments. You can also understand the guidelines for selecting the **Assembly Indicator** value based on various scenarios.

- You can make the following updates to a part and its part usage:
  1. Select a part from a Usage BOM.
  2. Click **Edit** near the **Overview** tab.

     In case of parts or usages having a released current or older revisions and no design-part alignments, you can edit following properties. If alignment exist, you can unalign first and then edit the properties.

     - **Part Required**
     - **Design Required**
     - **Skip alignment**
  3. In the **Overview** tab, update the part information, as needed, in the **Part Properties** section.

     While updating Assembly Indicator, all the options **Configurable Assembly**, **Generic Part**, and **Supplier Part** are available. Use the guidelines for selecting the **Assembly Indicator** value.

4. Update the part usage information, as needed, in the **Part Usage Properties** section.
5. Click **Save**.

   If you updated the properties of a released part usage, the usage is automatically revised.

- Update the **variant condition** of the part usage.

## Guidelines for selecting the Assembly Indicator value

While updating a part, you can use the following guidelines for selecting the **Assembly Indicator**:

| Select the Assembly Indicator value | In the following scenario |
|---|---|
| **Configurable Assembly** | • Its parent part is set as **Configurable Assembly**.<br>• Alternates and substitutes are not created for the part.<br>• Solution variants are not created for the part. |
| **Generic Part** | • Solution variants are not created for the part.<br>• The part does not contain any child parts.<br>• No revision of the part is released yet.<br>• Its parent part is set as **Configurable Assembly**.<br>• Alternates and substitutes are not created for the part.<br>• No revision of the part is released yet. |
| **Supplier Part** | • The part is purchased as a complete or whole assembly.<br>• The part can contain parts and sub-assemblies.<br>• A supplier part is an assembly that is defined in the design structure. |

# Split the occurrences of a part based on quantity

Considering your business requirement, you can split the occurrences of an element based on quantity.

Suppose a *Light Duty conveyor* for small packages needs only four *M8 Hex Bolts* per *Support Leg Assembly*, whereas a *Heavy Duty conveyor* for bulk materials requires 8 *M8 Hex Bolts* for increased structural integrity. In this situation, you can split the *M8 Hex Bolt* occurrence to define different quantities, with each tied to a specific conveyor load capacity.

You can split an occurrence based on quantity if the quantity of the element is at least two. You cannot split the occurrence of an element if the quantity is one.

### Procedure

1. Search for the occurrence and click **Edit Structure** > **Split by Quantity**.
   The Split panel appears.
2. In the **Number of Splits** box, type a number to divide the element by that many splits. Teamcenter suggests the default quantity for each split.
3. Modify the default quantity for the suggested splits in the **Quantity** box as needed.
4. Click **Split**.

### Results

The specified number of occurrences for the element are created with the defined quantity. If you split the occurrences of the element as a part of an active change, the changes, including the creation of new occurrernces and the modification of existing occurrences, are tracked.

> **Note**
>
> After splitting the occurrences of an element, the alignment of designs are retained. However, you must review if the alignments are valid considering your business requirement. Do not use guided or automated updates for adjusting the alignment of these elements.

## Split the occurrences of a part based on variant criteria

Considering your business requirement, you can split the occurrences of an element based on variant criteria.

Suppose the *Front Exterior Assembly* contains a single, generic *Headlight Assembly* occurrence. However, the bus model is sold in different regions, including North America and Europe, with distinct regulatory requirements for headlights. For example, the headlight requirements are amber turn signals in North America, clear in Europe, and different beam patterns. In this situation, you split the *Headlight Assembly* occurrence into multiple variant-specific occurrences. Each new occurrence contains a region-specific headlight part.

### Prerequisites

You can split an occurrence based on variant criteria if the variability scope is defined for the element.

### Procedure

1. Search for the occurrence and click **Edit Structure** > **Split by Variant**.

The Split panel appears and displays that the element is split into two occurrences and it also displays the variant formula.

2. Click **Split**.

### Results

The two occurrences of the element are created with the same variant condition. You can adjust the variant formula. If you split the occurrences of the element as a part of an active change, the changes, including the creation of the new occurrences and the modifications of the existing occurrences, are tracked.

> **Note**
>
> After splitting the occurrences of an element, the alignment of designs are retained. However, you must review if the alignments are valid considering your business requirement. Do not use guided or automated updates for adjusting the alignment of these elements.

### What to do next

You must edit the **variant condition** for the element.

## Classify parts

Classifying parts makes finding and reusing existing parts easier. You classify parts using the **Classification** tab.

> **Example**
>
> You classify the part *Wheel* as follows:
>
> | Property | Value |
> |---|---|
> | Fabric Carcass | Radial |
> | If Tyres With Tube | No Tube Allowed |
> | Season | Studded Winter Tyres |
> | Rim Diameter | 15 Inches |
>
> Later, when you want to use a 15-inch diameter wheel in your product, instead of searching for all the wheels available in the database, you can search for wheels that are 15 inches wide. Based on the other classification properties, you can choose the appropriate wheel for your product.

# Editing occurrence properties in the context of an assembly

## Setting in-context overrides

You can edit the properties of an occurrence in the context of a selected higher level assembly. This in-context override is not defined in the context of the immediate parent but in the context of the next higher level (grandparent level) component.

When a component or sub-assembly is reused in another assembly, some information might have to be overridden to adapt it to fit the context of the new assembly.

**Example**

Properties such as **Sequence** or **Quantity** can be different for a component when the same component is used in the context of two different assemblies.

When component **C1** is used in the context of the assembly **A1**, the quantity could be **4** and Sequence value could be **10**.

However, it is possible that when the same component **C1** is used in another assembly, namely **A2**, the quantity required could change to **6** instead of **4** and the **Sequence** value could change to **20** from **10**.

In case of such requirements, for the same component **C1**, you can use overrides to define the properties **Quantity** and **Sequence** differently in the context of assembly **A1** and **A2**.

For more information about in-context override, refer to *Structure Manager* in the Teamcenter documentation.

## Set in-context overrides

You can set in-context overrides to modify or override certain properties and attributes of elements within a specific structure context without affecting the original item. This helps you add context-specific information, like position and orientation, different mounting positions of an element.

**Procedure**

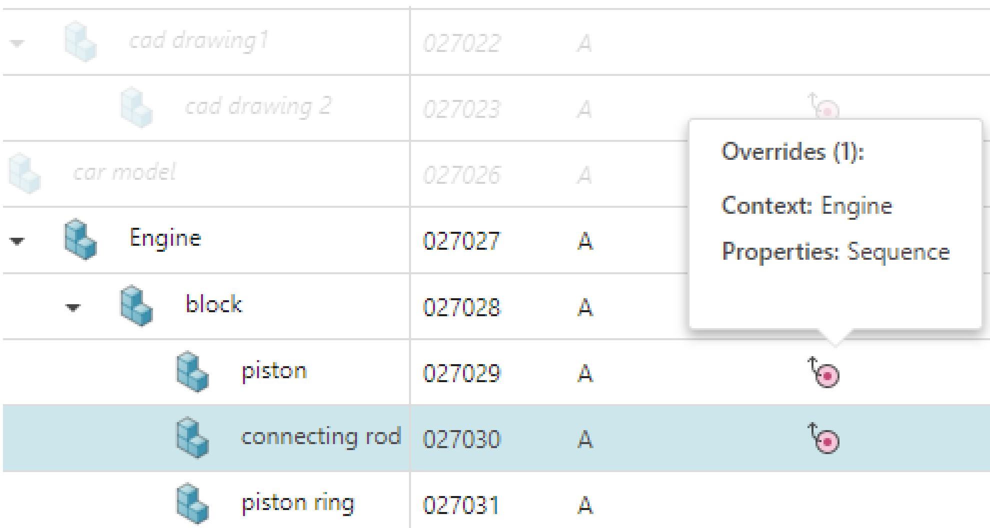
1. Search for and open the structure.
2. Select the grandparent assembly, two-levels above the component, to be defined as the override context.
3. Select **Edit Structure** > **In-Context** .
   The **Override Context** in the header is updated to reflect the selected assembly. Only the selected assembly remains active. The rest of the structure is made unavailable.
4. Click **Edit** .

5. Change the properties you want to edit in the context of the selected assembly.

   You can override the **Is Element Suppressed?**, **Absolute Transform**, trace link, and JT Override properties for a usage part in the Usage BOM.

   When you click **Create Trace Link**, you can create a trace link for occurrence to occurrence, occurrence to revision, and revision to occurrence. Right-click a usage part and click **Existing Trace Link** to view the existing trace links for the selected usage part.

6. Save your changes by clicking **Save Edits**. Alternatively, to cancel your edits, click **Cancel Edits**. The override icon appears against the selected component in the **Overrides** column of the table. On mouse hover, it shows information about overridden properties with the context.

   The override icons are also shown along with the individual overridden properties.

   **Tip**

   To close the **Set In-Context** view, select the same line where you set the in-context override, click **Configuration**, and select **Set In-Context**.

# 11. Create and maintain Assembly BOM

## About creating Assembly BOM

Create Assembly BOM in Teamcenter either automatically from existing BOMs, manually, by duplicating existing structures, or by importing them from Excel or other Teamcenter sites.

You can create a:

- Assembly BOM **automatically** from an existing design structure (design BOM). You can also create a Assembly BOM **manually**.
- Structure by importing it from **Excel**. You can also **import structures along with their partitions** from other Teamcenter sites.

## Generate an Assembly BOM automatically

You can automatically generate an Assembly BOM from a design structure and then align it. When a design structure is updated, the alignment data is also carried over to the Assembly BOM. You can validate the alignment. In case of alignment mismatch, you can manually realign the parts with designs.

You can automatically generate an aligned Assembly BOM from a design BOM (design structure) that already exists in Teamcenter. The variability scope associated with the design BOM is carried over to the generated Assembly BOM.

If the aligned design structure is updated, you must also update the Assembly BOM. The alignment data is carried over to the updated structure. If the design structure is released, you must first revise it and then update it. In this case, the alignment data is not carried over to the updated structure.

After generating or updating the aligned Assembly BOM, you can validate if the alignment is done correctly. In case of an alignment mismatch, you can realign the parts and designs and the part occurrences and design occurrences manually.

## Create an Assembly BOM manually

You can manually create an Assembly BOM by adding child items.

For a **Configurable Assembly** part, you create an Assembly BOM manually by adding child and sibling parts to it. After creating the Assembly BOM, you can generate its aligned design BOM.

### Procedure

1. Open the part and go to the **Content** tab.
2. Click **Add** ⊕ > **Child** to add a new part.

   You can add an existing part or solution variant from **Search** or **Palette**.

   > **Note**
   >
   > You can also reuse revision-controlled parts and assemblies. To do so, ensure that these parts are qualified as engineering Parts by the administrator.

3. If you choose to add a new part, in the **New** tab:
   a. Enter the required information.
   b. Select the **Design Required** check box if you need a corresponding design.
   c. Select the **Automatically Create Design** check box if you want the design to be automatically created and aligned to the part.

      OR

      Click **Add Design** to manually create a design. In the **Add Design** panel, enter the required information, and click **Add**.

      After manually adding a design, if you select the **Automatically Create Design** check box, the manually added design is ignored, and a new design is automatically created by Teamcenter and aligned to the part.

      If the selected parent part is a supplier part, the child part that you add is treated as a spare part.

4. Follow the above steps to build the Assembly BOM structure by adding parts as required. To add parts that are at the same level as the selected part, click **Add** ⊕ > **Sibling**.

   You can include embedded software artifacts in the structure. For example, the part *Tire Pressure Sensor* in the product *Crosskart* can be programmed such that when the air pressure drops below the product's recommended level, the sensor captures this information, and the dashboard indicator lights up.

   Software developers create and maintain the software artifacts using the Embedded Software Management functionality.

   Additionally, you can include ECAD parts created and managed in integrated Electronic Design Automation (EDA) applications.

   ECAD parts are represented both as designs (in the design BOM) and as parts (in the engineering BOM), hence they are shared between design structure and Assembly BOM.

   > **Note**
   >
   > It is recommended that you edit these parts in the integrated EDA application, and not directly in the BOM.

   Additionally, you can **carry over** parts from one Assembly BOM to another.

To remove a part from the structure, right-click it and select **Remove**.

If you are not able to remove an aligned occurrence, contact your administrator.

The element is removed and it may split due to **effectivity cutback**.

#### What to do next

After creating a Assembly BOM, you must **set maturity levels** for the newly added parts. You can also set the assembly indicator. If the newly added part is a piece part, bought out part, purchased assembly, or a foreign part, you can edit it and select the **Supplier Part** in the **Assembly Indicator** list. While setting up the value in the **Assembly Indicator** list, if the part is not a **Generic Part**, **Configurable Assembly**, or a **Supplier Part**, leave the value blank.

If your administrator has enabled effectivity propagation on your structure, you can view the **Occurrence ECN In** and **Occurrence ECN Out** properties for your parts. They show the responsible ECN names from which the part has derived its element effectivity start and end dates.

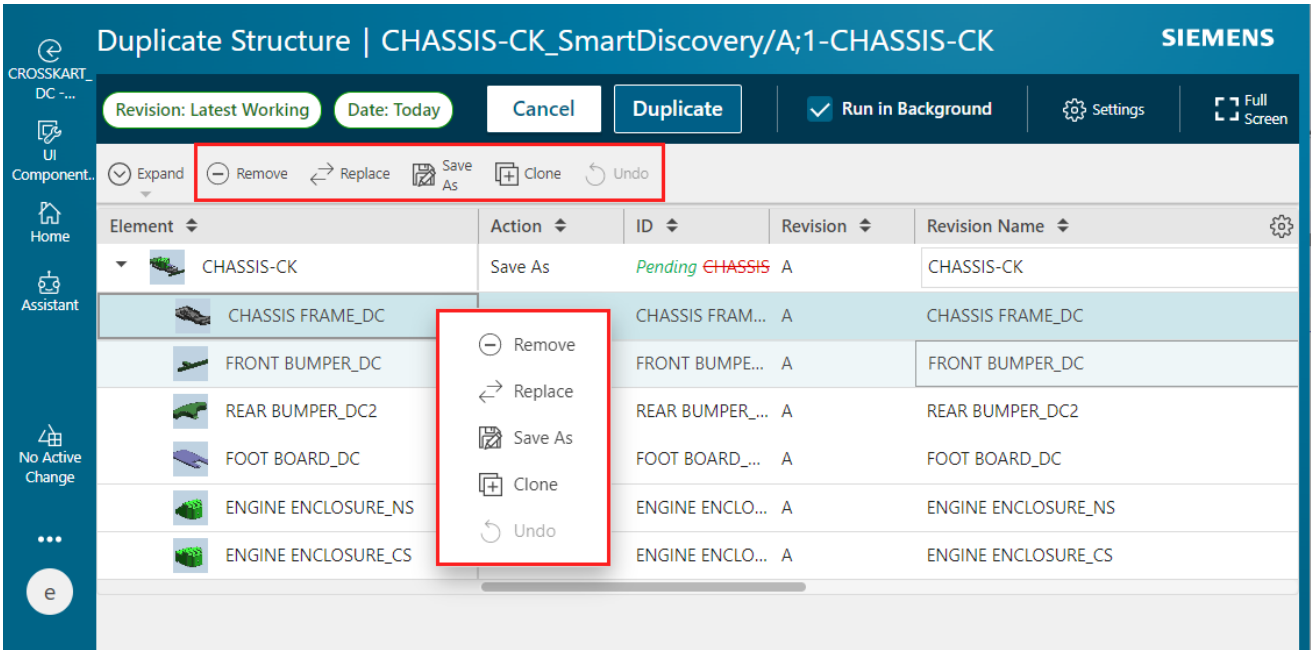
## Duplicate a structure

You can create a new structure by duplicating an existing structure. You can also duplicate structures that you do not own. When you duplicate a structure that you do not own, you become the owner of the new structure.

For an Assembly BOM or Usage BOM, duplicate the child items instead of the top-level structure. Design structures can be duplicated in their entirety.

#### Procedure

1. For an Assembly BOM or Usage BOM, select the child items. For a design structure, select the top line of an assembly in the structure that you want to duplicate.
2. Click **Duplicate**.
3. Right-click each element and select one of the following actions. You can also select these actions from the work area toolbar.




| Action | Description |
|---|---|
| **Remove** | The element is not included in the new structure. |
| **Replace** | The element is replaced with the replacement element that you specified in the **Replace** panel. |
| **Save As** | A copy of the element is added to the new structure. You own this new element.<br><br>If the element is a structure, all its child elements are only referenced in the new structure. These are still owned by the user who created the source structure.<br><br>You can edit the **Revision Name** and **Description** of the element. You can edit the custom properties if your administrator has configured those custom properties accordingly. |
| **Clone** | A copy of the element is added to the new structure. You own this new element.<br><br>If the element is a structure, copies of its child elements are used in the new structure.<br><br>You can edit the **Revision Name** and **Description** of each element. You can edit the custom properties if your administrator has configured those custom properties accordingly. |

**Note**

If any child element is a structure, all its children are also marked with the same action. However, if you change the action of one of its child elements later, the action of the parent element changes accordingly.

If you do not select an action, and the **Action** column is blank, the element is only referenced in the new structure. The element continues to be owned by the user who created the source structure.

4. To undo an action set on an element, select the element, and click **Undo**.

5. (Optional) Click **Settings**, and then perform these additional actions on the duplicated structure:

   a. Choose an option for the ID naming in the Settings panel:

      - Click the **Default IDs** option to retain the original element IDs for the cloned elements and for any elements that are saved as new elements.
      - Alternatively, click the **ID Naming Rule** option and enter the following details.

| ID naming rule | Description |
|---|---|
| **Prefix** | The text is prefixed at the start of the existing IDs. |
| **Suffix** | The text is appended at the end of the existing IDs. |
| **Replace/With** | The text specified in **Replace** is replaced with the text specified in **With**.<br>The text specified in **Replace** must be text that is part of the existing ID, and it is case-sensitive. |

   b. To retain the original ownership for cloned elements and the elements that are saved as new elements, click the **Retain Original Ownership** check box.

      The elements on which no action is taken are referenced in the duplicated structure and their original ownership is retained.

   c. Ensure that the **Owning Project** section displays the project that is a part of your active Teamcenter session.

      You can change your project so that the duplicated structure becomes a part of a different project. If no project is selected in the active Teamcenter session, you do not see the **Owning Project** section.

   d. To include the Designcenter NX dependencies, if any, that are associated with the existing elements in the newly duplicated structure, click the **Include all Designcenter Dependencies** check box.

   e. Click **Close**.

6. To run the duplicate process in the background or foreground, select or clear the **Run in Background** check box. If you choose to run the duplicate process in the background, you receive a notification in **Alerts** once the structure is duplicated.

> **Note**
>
> If your administrator has enabled the parallel duplication of large structures, the duplication process always runs in the background. Contact your administrator to adjust this setting.

7. Click **Duplicate**.

### Results

The alignment data is carried over from the original structure to the new structure. The element effectivity for the new structure is derived from the active change.

By default, your active Teamcenter session project is assigned to the duplicated structure and its elements. If the administrator has enabled the project assignments in the Teamcenter environment, the following actions are performed on the duplicate structure:

- If an item in the original structure is associated to a project, that project is replaced with the project selected in the active Teamcenter session for the corresponding duplicated item.
- If no project is associated to an item in the original structure, no project is assigned to the corresponding duplicated item, even though a project is selected in the Teamcenter session.
- For referenced items that are not cloned but are reused in the new structure, if the original item is associated to a project, the project that is selected in the active Teamcenter session is appended to its existing project list.

## Create a new structure from an existing structure

You can create a new structure from an existing structure by saving the existing structure as a new structure.

### Procedure

1. Search for a structure.
2. Click **More Commands** ••• > **New** > **Save As**.
3. In the **Save As** panel, enter the required information.
4. If you are creating a part, select the **Copy Design Alignment** check box. If you select this check box, another revision of the existing aligned design is automatically created and aligned with the part. If you do not select this check box, a new revision of the existing aligned design is created at the time of alignment.

   If you are creating a design, select the **Copy Part Alignment** check box. If you select this check box, another revision of the existing aligned part is automatically created and aligned with the design. If you do not select this check box, a new revision of the existing aligned part is created at the time of alignment.
5. Click **Save**.

### Results

A new structure is created from the existing structure. If the existing structure contains any partitions, the partitions are also carried over to the new structure.

# Associate a variability scope with a structure

The variability data is managed outside of a structure using Product Configurator. To include the variability data within the structure, you associate it with an appropriate variability scope.

### Procedure

1. Search for a structure of type **Configurable Assembly** and open it.
2. Click **Edit** .
3. In **Variability Scope**, click **Add** .
4. Search for variability scope, select it, and click **Add**.
5. Click **Save** .

# Carry over parts from one Assembly BOM to another

Engineering parts can be reused in several structures. You can quickly populate a Assembly BOM by carrying over parts from another structure.

### Procedure

1. Open the Assembly BOM from which you want to carry over the parts.
2. Click the **Split Context** .
   The Assembly BOM is split and shown within two views displayed side-by-side.
3. In one view, click **Open in View**
4. In the **Open in View** panel, add the Assembly BOM to which you want to carry over the parts. You can either add the structure from the palette or search for it.
5. Click **Open**.
6. Drag the parts from one Assembly BOM to the other in order to carry over the parts.
   As the aligned designs are also carried over to the new Assembly BOM, you can immediately visualize the carried over parts in the new structure.

# Create standalone parts

Each component that makes up a product is identified as a part in Teamcenter. You can create standalone parts and later add these parts to an engineering BOM.

#### Procedure

1. Open the folder where you want to create a part and click **Add to** ⊕.
2. In the **Add** panel, filter to select **Part**.
3. Enter the required information and click **Add**.

   By default:

   - **Finish Type** is set to **None**.
   - **Part Maturity** is set to **Concept**.

   You can update the part properties later as needed.

   While updating the **Assembly Indicator** for the newly added part, if the part is not a *Generic Part*, *Configurable Assembly*, or a *Supplier Part*, leave it blank. If the newly added part is a *bought out* part (that is, a piece part, purchased assembly, or a foreign part), you can edit it and select **Supplier Part** in the **Assembly Indicator** list. If the selected parent part is a supplier part, the child part that you add to it is always treated as a spare part and is not visible in Teamcenter. The spare part does not participate in alignment, structure configuration, and visualization.

## Replace a part

You can replace a part with another part. You can also replace an existing part or a sub-assembly in an assembly with its copy using the **Save As And Replace** command. If the **Save As And Replace** option is not visible, you may not have the required privileges.

#### Procedure

1. Search for the element that you want to replace.
2. Select the occurrence and click **Edit Structure** > **Replace**.
3. In the **Replace** pane, select the replacement and click **Replace**.

   Teamcenter replaces the element and refreshes the display.

   In case of a released Usage BOM, these modifications followed by alignment update might result in the element split due to effectivity cutback.

## Replace a structure component with its copy

A copy is a new item with identical properties but a new ID. You can copy a part or a subassembly, with or without children, save it with a new name, and replace the existing subassembly in the structure with its copy. If the subassembly is copied with children, the child parts are renamed based on the naming convention specified using the naming rule.

### Procedure

1. Select the structure from which you want to copy the element, and click **Open**.
2. Choose **More Commands** > **New** > **Save As and Replace**.

   The **Save As And Replace** pane is displayed.
3. (Optional) Select the **Clone Children** check box.

   If you choose the **Clone Children** option, the child elements in the assembly are replaced along with the top element.
4. Select **ID Naming Rules** to automatically assign new IDs to the copied elements. The **ID Naming Rules** option is visible only when the **Clone Children** check box is selected.

   - **Prefix**

     The text string provided is added to the start of the existing IDs, and new IDs are assigned.
   - **Suffix**

     The text string provided is added to the end of the existing IDs, and new IDs are assigned.
   - **Replace/With**

     The text string specified in the **Replace** field is replaced with the text string specified in the **With** field in the new IDs.
5. (Optional) Select the **Run in Background** check box.

   **Note**

   For the save and replace operation to run in background, your system administrator must configure Dispatcher services.
6. If you are replacing a part, select the **Copy Design Alignment** check box. If you select this check box, another revision of the existing aligned design is automatically created and aligned with the part. If you do not select this check box, a new revision of the existing aligned design is created at the time of alignment.

   If you are replacing a design, select the **Copy Part Alignment** check box. If you select this check box, another revision of the existing aligned part is automatically created and aligned with the design. If you do not select this check box, a new revision of the existing aligned part is created at the time of alignment.
7. Click **Save As And Replace** at the bottom of the **Save As And Replace** pane.

   Teamcenter replaces the element and refreshes the display.

   If you choose to run the replace operation in the background, a notification is generated once it is complete. Click the notification to view the details of the replacement.

   In case of a released Usage BOM, these modifications followed by alignment update might result in the element split due to **effectivity cutback**.

# Set the maturity level of a part

The maturity of a part indicates its progression in its development cycle. For example, the maturity of a part can indicate if it is being conceptualized, if its prototype is created, or if it has entered production.

**Procedure**

1. Select the part for which you want to set the maturity level.
2. Click **Edit**.
3. Set **Part Maturity** in the **Part Properties** section.
4. Click > **Save**.

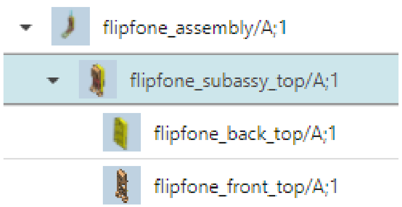
# Insert or remove levels in a structure

You must have write access to the structure to insert or remove a level. The insert action adds a level between the selected object and its parent object. The remove action removes the selected element's parent object one level above and attaches the element and all its siblings to its current grandparent object two levels above.

Search for the structure you want to edit and click **Open**.

| To | Do this | Result |
|---|---|---|
| Insert a level | 1. Click the element above which you want to insert a level and click **Edit Structure** >**Insert Level**.<br>2. Fill in the details and click **Insert**. | A new level is added above, as the new parent of the selected element. |
| Remove a level | 1. (Optional) If any child of the level you want to remove has effectivity or variant condition set on it or is suppressed, enable the following options:<br>• **Show Excluded by Effectivity**<br>• **Show Excluded by Variants**<br>• **Show Suppressed**<br>2. Click the element that you want to remove from the assembly and select **Edit Structure** >**Remove Level**. | The selected element is removed from the structure, and its child elements are attached to what was previously their grandparent elements. |

The following example shows how the remove-action works in the three-level assembly in the graphic:




In both cases, removing or adding levels, the element may split due to **effectivity cutback**.

If you are not able to remove an aligned occurrence, contact your administrator.

# Revise a structure

You can revise a released structure to update it. When you revise a structure, you can retain existing aligned parts or create and use new ones. You can also address the alignment mismatches by setting a primary aligned part.

### Procedure

1. Open the structure that you want to revise.
2. Click **More Commands** > **New** > **Revise**.
3. In the **Revise** panel, enter the required information.

   If the selected structure is a design structure, additionally select the **Keep Existing Part** check box to retain the part that is already aligned with the selected design. The existing part and design alignment will be automatically carried over to the revised structure.

   You can also choose to create a new part or use another existing part by clicking **Add Part**. In this case, the existing part and design alignment will not be carried over to the revised structure. After choosing to create or use another part, if you select the **Keep Existing Part** check box, the new part or the part that you chose is ignored.

   If you see any error while previous alignment is getting removed, contact your administrator.
4. Click **Revise**.

   To align the design and part again to fix the mismatched alignments that are indicated as :

   a. Switch to the **Tree with Summary** view.
   b. Select a design in the design structure.
   c. Go to the **Aligned Parts** section in the **Overview** tab, select the required part, and click **Set Primary**.

# Discontinue a part

Sometimes a part is unavailable due to logistic issues or raw material shortages. In such cases, you can remove that part for a finite amount of time and reintroduce into the product later.

**Example**

Imagine that you have a released car GPS module which is effective from January 1, 2023 to December 31, 2024. Due to a semiconductor shortage, you want to discontinue the part for 6 months, starting June 1, 2023 until December 31, 2023. After December 31, 2023, you want to reintroduce the part into to the product.

### Procedure

1. Create a change notice with a **release effectivity** range for the duration of the part discontinuation, and **set it as an active change**.
2. Open the engineering BOM and select the released part.
   a. Click **More Commands** > **New** > **Revise** to first revise the part.
   b. Next, in the **Part Properties** section of the **Overview** tab, select the **Discontinued** check box.

   OR

   a. Click **More Commands** > **Manage** > **Submit to Workflow**.
   b. In the **Submit to Workflow** panel, select **Part Discontinue** in the **Template** section.
   c. Click **Submit**.

   You can continue this part again by revising it and clearing the **Discontinue** check box.

   **Note**

   Discontinuing a part does not discontinue its corresponding part usage.

# Update a part

You might be required to update a part for many reasons, including to update its finish type and find number. If the part is an assembly part, you can update its structure by adding, removing, or replacing parts.

### Procedure

1. Open the part.
2. Click **Edit**.

In case of parts having a released current or older revisions and no design-part alignments, you can edit following properties without revising the part. If an alignment exists, you can first undo this alignment and then edit the properties.

- **Part Required**
- **Design Required**
- **Skip alignment**

3. Update the required information, and click **Save** .

   If the part is a *piece* part, *bought out part*, purchased assembly, or a foreign part, in the **Assembly Indicator** list, select **Supplier Part**.

   If the **Assembly Indicator** value for the part is already set as **Supplier Part**, the part contains an aligned design, and you want to change its assembly indicator value, you must first remove the alignment of the part with its design and then update the assembly indicator.

### What to do next

After updating the structure, you must update its aligned design BOM.

## Split the occurrences of a part based on quantity

Considering your business requirement, you can split the occurrences of an element based on quantity.

Suppose a *Light Duty conveyor* for small packages needs only four *M8 Hex Bolts* per *Support Leg Assembly*, whereas a *Heavy Duty conveyor* for bulk materials requires 8 *M8 Hex Bolts* for increased structural integrity. In this situation, you can split the *M8 Hex Bolt* occurrence to define different quantities, with each tied to a specific conveyor load capacity.

You can split an occurrence based on quantity if the quantity of the element is at least two. You cannot split the occurrence of an element if the quantity is one.

### Procedure

1. Search for the occurrence and click **Edit Structure** > **Split by Quantity**.
   The Split panel appears.
2. In the **Number of Splits** box, type a number to divide the element by that many splits. Teamcenter suggests the default quantity for each split.
3. Modify the default quantity for the suggested splits in the **Quantity** box as needed.
4. Click **Split**.

### Results

The specified number of occurrences for the element are created with the defined quantity. If you split the occurrences of the element as a part of an active change, the changes, including the creation of new occurrernces and the modification of existing occurrences, are tracked.

> **Note**
>
> After splitting the occurrences of an element, the alignment of designs are retained. However, you must review if the alignments are valid considering your business requirement. Do not use guided or automated updates for adjusting the alignment of these elements.

## Split the occurrences of a part based on variant criteria

Considering your business requirement, you can split the occurrences of an element based on variant criteria.

Suppose the *Front Exterior Assembly* contains a single, generic *Headlight Assembly* occurrence. However, the bus model is sold in different regions, including North America and Europe, with distinct regulatory requirements for headlights. For example, the headlight requirements are amber turn signals in North America, clear in Europe, and different beam patterns. In this situation, you split the *Headlight Assembly* occurrence into multiple variant-specific occurrences. Each new occurrence contains a region-specific headlight part.

### Prerequisites

You can split an occurrence based on variant criteria if the variability scope is defined for the element.

### Procedure

1. Search for the occurrence and click **Edit Structure** > **Split by Variant**.
   The Split panel appears and displays that the element is split into two occurrences and it also displays the variant formula.

2. Click **Split**.

### Results

The two occurrences of the element are created with the same variant condition. You can adjust the variant formula. If you split the occurrences of the element as a part of an active change, the changes, including the creation of the new occurrences and the modifications of the existing occurrences, are tracked.

**Note**

After splitting the occurrences of an element, the alignment of designs are retained. However, you must review if the alignments are valid considering your business requirement. Do not use guided or automated updates for adjusting the alignment of these elements.

**What to do next**

You must edit the **variant condition** for the element.

# Classify parts

Classifying parts makes finding and reusing existing parts easier. You classify parts using the **Classification** tab.

**Example**

You classify the part *Wheel* as follows:

| Property | Value |
|---|---|
| Fabric Carcass | Radial |
| If Tyres With Tube | No Tube Allowed |
| Season | Studded Winter Tyres |
| Rim Diameter | 15 Inches |

Later, when you want to use a 15-inch diameter wheel in your product, instead of searching for all the wheels available in the database, you can search for wheels that are 15 inches wide. Based on the other classification properties, you can choose the appropriate wheel for your product.

# Editing occurrence properties in the context of an assembly

## Setting in-context overrides

You can edit the properties of an occurrence in the context of a selected higher level assembly. This in-context override is not defined in the context of the immediate parent but in the context of the next higher level (grandparent level) component.

When a component or sub-assembly is reused in another assembly, some information might have to be overridden to adapt it to fit the context of the new assembly.

> **Example**
>
> Properties such as **Sequence** or **Quantity** can be different for a component when the same component is used in the context of two different assemblies.
>
> When component **C1** is used in the context of the assembly **A1**, the quantity could be **4** and Sequence value could be **10**.
>
> However, it is possible that when the same component **C1** is used in another assembly, namely **A2**, the quantity required could change to **6** instead of **4** and the **Sequence** value could change to **20** from **10**.
>
> In case of such requirements, for the same component **C1**, you can use overrides to define the properties **Quantity** and **Sequence** differently in the context of assembly **A1** and **A2**.

For more information about in-context override, refer to *Structure Manager* in the Teamcenter documentation.

## Set in-context overrides

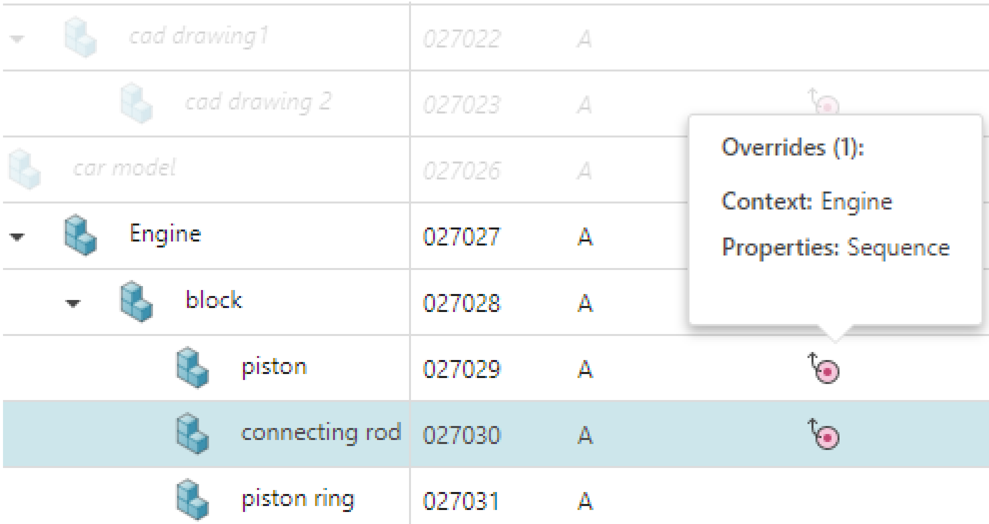
You can set in-context overrides to modify or override certain properties and attributes of elements within a specific structure context without affecting the original item. This helps you add context-specific information, like position and orientation, different mounting positions of an element.

### Procedure

1. Search for and open the structure.
2. Select the grandparent assembly, two-levels above the component, to be defined as the override context.
3. Select **Edit Structure** > **In-Context**.
   The **Override Context** in the header is updated to reflect the selected assembly. Only the selected assembly remains active. The rest of the structure is made unavailable.
4. Click **Edit**.
5. Change the properties you want to edit in the context of the selected assembly.

   You can override the **Is Element Suppressed?**, **Absolute Transform**, trace link, and JT Override properties for a usage part in the Usage BOM.

   When you click **Create Trace Link**, you can create a trace link for occurrence to occurrence, occurrence to revision, and revision to occurrence. Right-click a usage part and click **Existing Trace Link** to view the existing trace links for the selected usage part.
6. Save your changes by clicking **Save Edits**. Alternatively, to cancel your edits, click **Cancel Edits**.
   The override icon appears against the selected component in the **Overrides** column of the table. On mouse hover, it shows information about overridden properties with the context.




The override icons are also shown along with the individual overridden properties.

**Tip**

To close the **Set In-Context** view, select the same line where you set the in-context override, click **Configuration**, and select **Set In-Context**.

# 12. Create and maintain design structure

## About creating design structures

Create design structures in Teamcenter either automatically from existing BOMs, manually, by duplicating existing structures, or by importing them from Excel or other Teamcenter sites.

You can create a:

- Design structure **automatically** from an existing engineering BOM. You can also create a design structure **manually**.
- Design structure by **duplicating** an existing structure.
- Structure by importing it from **Excel**. You can also **import structures along with their partitions** from other Teamcenter sites.

## Generate a design structure automatically

You can generate a design structure automatically from a Usage BOM or from an Assembly BOM. After the design structure is generated, you can validate its alignment.

You can automatically generate an aligned design structure from a Usage BOM or an Assembly BOM that already exists in Teamcenter.

If the aligned engineering BOM is updated, you must also update the design structure. The alignment data is carried over to the updated structure.

After generating or updating the aligned design structure, you can validate if the alignment is done correctly. In case of an alignment mismatch, you can realign the parts and designs and the part occurrences and design occurrences manually.

## Create a design structure manually

You can create a design structure manually by adding child items to it.

**Procedure**

1. Open the folder in which you want to create a new structure.
2. In the **Overview** tab, click **Add to** ⊕ in the **Contents** section.
3. In the **Add** panel, select the **Type** as **Design**, and enter the required information.

If you want to add further information for an item or assembly, add a secondary item. The **Occurrence Type** of the secondary item is **Reference Geometry**. The secondary components are displayed as a part of the structure. You can hide the secondary components from being displayed into the structure.

4. Select the **Part Required** check box if you need a corresponding part and then choose one of the following options:

   - Click the **Auto-Create** option to automatically create a new part and align it with the design.
   - Click the **Add Part** option to add a new part or search for an existing part and align that part with the design.
   - Click the **Assign Part ID** option to assign only the part number corresponding to the design. The part is neither created nor aligned to the design at this point. You can then choose to create a part with the assigned part number and align it to the design.
   - Click the **Defer for later** option to create a part and align it with the design later.

5. Open the design that you added, go to the **Content** tab, and click **Add** ⊕ > **Child**.

6. In the **Add Child** panel, enter the required details, and click **Add**.

   Add more children to build the structure. To add children that are at the same level, click **Add** ⊕ > **Sibling**.

   You can drag a design from one place to another to reorder the structure. To remove a design, right-click it and choose **Remove** ⊖.

   If you are not able to remove an aligned occurrence, contact your administrator.

#### What to do next

After the design is created, you might want to update its assembly indicator considering your business requirement.

## Duplicate a structure

You can create a new structure by duplicating an existing structure. You can also duplicate structures that you do not own. When you duplicate a structure that you do not own, you become the owner of the new structure.

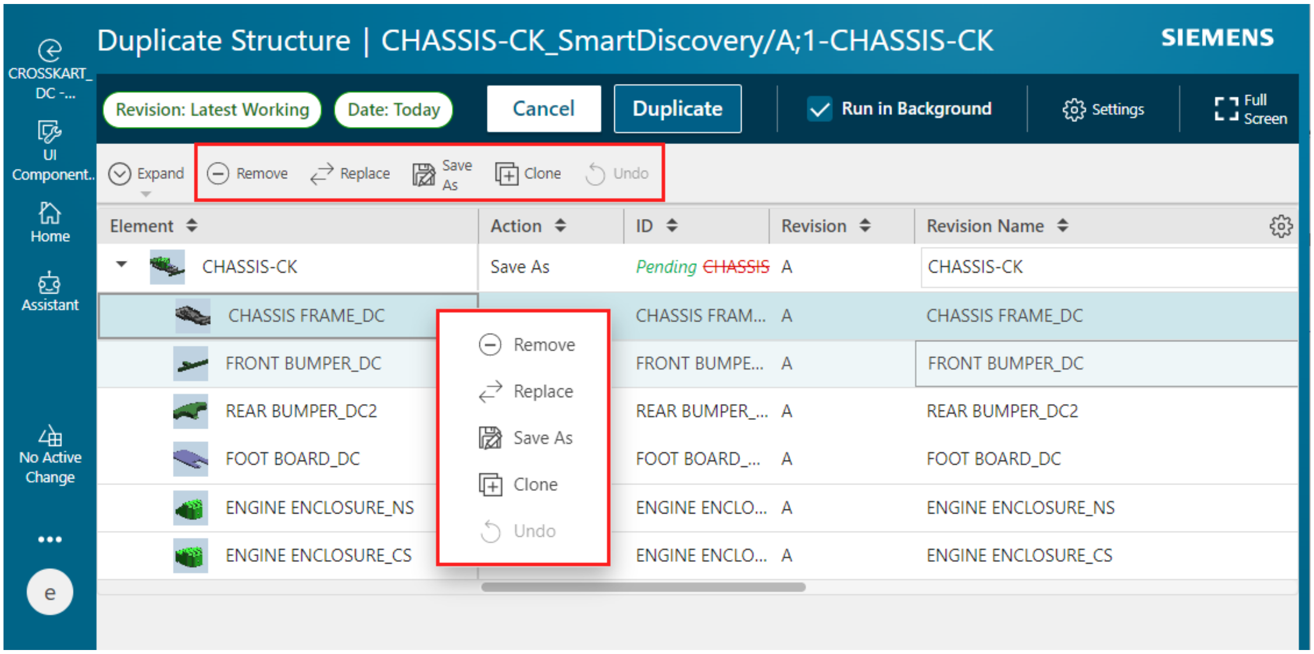
For an Assembly BOM or Usage BOM, duplicate the child items instead of the top-level structure. Design structures can be duplicated in their entirety.

#### Procedure

1. For an Assembly BOM or Usage BOM, select the child items. For a design structure, select the top line of an assembly in the structure that you want to duplicate.

2. Click **Duplicate**.

3. Right-click each element and select one of the following actions. You can also select these actions from the work area toolbar.




| Action | Description |
|---|---|
| **Remove** | The element is not included in the new structure. |
| **Replace** | The element is replaced with the replacement element that you specified in the **Replace** panel. |
| **Save As** | A copy of the element is added to the new structure. You own this new element.<br><br>If the element is a structure, all its child elements are only referenced in the new structure. These are still owned by the user who created the source structure.<br><br>You can edit the **Revision Name** and **Description** of the element. You can edit the custom properties if your administrator has configured those custom properties accordingly. |
| **Clone** | A copy of the element is added to the new structure. You own this new element.<br><br>If the element is a structure, copies of its child elements are used in the new structure.<br><br>You can edit the **Revision Name** and **Description** of each element. You can edit the custom properties if your administrator has configured those custom properties accordingly. |

> **Note**
>
> If any child element is a structure, all its children are also marked with the same action. However, if you change the action of one of its child elements later, the action of the parent element changes accordingly.

If you do not select an action, and the **Action** column is blank, the element is only referenced in the new structure. The element continues to be owned by the user who created the source structure.

4. To undo an action set on an element, select the element, and click **Undo**.

5. (Optional) Click **Settings**, and then perform these additional actions on the duplicated structure:

   a. Choose an option for the ID naming in the Settings panel:

      - Click the **Default IDs** option to retain the original element IDs for the cloned elements and for any elements that are saved as new elements.
      - Alternatively, click the **ID Naming Rule** option and enter the following details.

      | ID naming rule | Description |
      | --- | --- |
      | **Prefix** | The text is prefixed at the start of the existing IDs. |
      | **Suffix** | The text is appended at the end of the existing IDs. |
      | **Replace/With** | The text specified in **Replace** is replaced with the text specified in **With**.<br>The text specified in **Replace** must be text that is part of the existing ID, and it is case-sensitive. |

   b. To retain the original ownership for cloned elements and the elements that are saved as new elements, click the **Retain Original Ownership** check box.

      The elements on which no action is taken are referenced in the duplicated structure and their original ownership is retained.

   c. Ensure that the **Owning Project** section displays the project that is a part of your active Teamcenter session.

      You can change your project so that the duplicated structure becomes a part of a different project. If no project is selected in the active Teamcenter session, you do not see the **Owning Project** section.

   d. To include the Designcenter NX dependencies, if any, that are associated with the existing elements in the newly duplicated structure, click the **Include all Designcenter Dependencies** check box.

   e. Click **Close**.

6. To run the duplicate process in the background or foreground, select or clear the **Run in Background** check box. If you choose to run the duplicate process in the background, you receive a notification in **Alerts** once the structure is duplicated.

**Note**

If your administrator has enabled the parallel duplication of large structures, the duplication process always runs in the background. Contact your administrator to adjust this setting.

7. Click **Duplicate**.

### Results

The alignment data is carried over from the original structure to the new structure. The element effectivity for the new structure is derived from the active change.

By default, your active Teamcenter session project is assigned to the duplicated structure and its elements. If the administrator has enabled the project assignments in the Teamcenter environment, the following actions are performed on the duplicate structure:

- If an item in the original structure is associated to a project, that project is replaced with the project selected in the active Teamcenter session for the corresponding duplicated item.
- If no project is associated to an item in the original structure, no project is assigned to the corresponding duplicated item, even though a project is selected in the Teamcenter session.
- For referenced items that are not cloned but are reused in the new structure, if the original item is associated to a project, the project that is selected in the active Teamcenter session is appended to its existing project list.

## Create a new structure from an existing structure

You can create a new structure from an existing structure by saving the existing structure as a new structure.

### Procedure

1. Search for a structure.
2. Click **More Commands** ••• > **New** > **Save As**.
3. In the **Save As** panel, enter the required information.
4. If you are creating a part, select the **Copy Design Alignment** check box. If you select this check box, another revision of the existing aligned design is automatically created and aligned with the part. If you do not select this check box, a new revision of the existing aligned design is created at the time of alignment.

   If you are creating a design, select the **Copy Part Alignment** check box. If you select this check box, another revision of the existing aligned part is automatically created and aligned with the design. If you do not select this check box, a new revision of the existing aligned part is created at the time of alignment.
5. Click **Save**.

### Results

A new structure is created from the existing structure. If the existing structure contains any partitions, the partitions are also carried over to the new structure.

## Associate a variability scope with a structure

The variability data is managed outside of a structure using Product Configurator. To include the variability data within the structure, you associate it with an appropriate variability scope.

### Procedure

1. Search for a structure of type **Configurable Assembly** and open it.
2. Click **Edit** .
3. In **Variability Scope**, click **Add** .
4. Search for variability scope, select it, and click **Add**.
5. Click **Save** .

## Insert or remove levels in a structure

You must have write access to the structure to insert or remove a level. The insert action adds a level between the selected object and its parent object. The remove action removes the selected element's parent object one level above and attaches the element and all its siblings to its current grandparent object two levels above.

Search for the structure you want to edit and click **Open**.

| To | Do this | Result |
|---|---|---|
| Insert a level | 1. Click the element above which you want to insert a level and click **Edit Structure** >**Insert Level** .<br>2. Fill in the details and click **Insert**. | A new level is added above, as the new parent of the selected element. |
| Remove a level | 1. (Optional) If any child of the level you want to remove has effectivity or variant condition set on it or is suppressed, enable the following options:<br>• **Show Excluded by Effectivity** | The selected element is removed from the structure, and its child elements are attached to what was previously their grandparent elements. |

| To | Do this | Result |
|---|---|---|
| | • **Show Excluded by Variants**<br>• **Show Suppressed**<br>2. Click the element that you want to remove from the assembly and select **Edit Structure** >**Remove Level** . | |

The following example shows how the remove-action works in the three-level assembly in the graphic:


In both cases, removing or adding levels, the element may split due to **effectivity cutback**.

If you are not able to remove an aligned occurrence, contact your administrator.

# Revise a structure

You can revise a released structure to update it. When you revise a structure, you can retain existing aligned parts or create and use new ones. You can also address the alignment mismatches by setting a primary aligned part.

**Procedure**

1. Open the structure that you want to revise.
2. Click **More Commands** > **New** > **Revise** .
3. In the **Revise** panel, enter the required information.

   If the selected structure is a design structure, additionally select the **Keep Existing Part** check box to retain the part that is already aligned with the selected design. The existing part and design alignment will be automatically carried over to the revised structure.

   You can also choose to create a new part or use another existing part by clicking **Add Part**. In this case, the existing part and design alignment will not be carried over to the revised structure. After choosing to create or use another part, if you select the **Keep Existing Part** check box, the new part or the part that you chose is ignored.

   If you see any error while previous alignment is getting removed, contact your administrator.

4. Click **Revise**.

   To align the design and part again to fix the mismatched alignments that are indicated as :

   a. Switch to the **Tree with Summary** view.

   b. Select a design in the design structure.

   c. Go to the **Aligned Parts** section in the **Overview** tab, select the required part, and click **Set Primary** .

## Update properties of a design

You might want to update the properties of a design for many reasons, including to update its finish type and find number. By updating the properties of a design ensures that the design information is accurate, complete, and effectively communicated for downstream processes.

### Procedure

1. Open the design.

2. Click **Edit** .

   In case of designs having a released current or older revisions and no design-part alignments, you can edit following properties without revising the design. If an alignment exists, you can first undo this alignment and then edit the properties.

   - **Part Required**
   - **Skip alignment**

3. Update the required information, and click **Save** .

## Specify mirror designs

Specify the designs that are mirror of each other to establish design dependency and to enable synchronized change management. In Teamcenter, you can define two designs as mirror, or left-handed and right-handed copies of each other if not already set in the CAD software, such as NX.

In Teamcenter, mirror designs are geometrically identical designs, oriented differently. For example, the left and right versions of headlights, doors, or suspension components. Specifying these mirror relationships in Teamcenter creates design dependency, supports searching for such related designs, and facilitates synchronized change management.

If any form-fit changes are made to one of the designs making it different than the counter design, you can remove the link and the design acts as a separate and standalone design.

### Prerequisites

The design that you want to add as a mirror design must already exist in Teamcenter.

### Procedure

Specify a master design.

2. Open the first design and click **Edit**.
3. In the **Mirror Type** list, select **Master** and then click **Save**.

   The design is saved as a master design in Teamcenter.

Specify the mirror design.

5. Open the other design that you want to specify as a mirror design and click **Edit**.
6. In the **Mirror Type** list, select **Mirror**, and click **Save**.

Link the master and mirror designs.

**Note**

You can link the master and mirror designs only from the mirror design.

8. In **Mirror Link** section of the mirror design, click **Add**.

   A design can be a master design for only one mirror design. Similarly, a design can be set as a mirror design only for one master design.
9. Search for the master design and click **Add**.
10. Click **Save**.

    The designs are stored as mirror designs. If an engineering BOM is created for the design structure that already contained the mirror designs, the equivalent aligned parts of the engineering BOM also contain the mirror information. If an engineering BOM already existed for the design structure, you must then update the alignments for the mirror information to be added to the aligned parts.
    If a design is revised due to a business requirement and is no longer a mirror design, you can remove the link. To do so, open the mirror design, and in the **Mirror Link**, click **Delete**.

# 13. Create and manage designs and parts within a single structure

## Creating designs and parts within a single structure

You can create a single structure using the **Design Part** business object and store both design and part information on the same item. You can manually create structures, duplicate existing ones, or import them from Excel. You can manage items within a structure by adding, editing, removing, or replacing them as needed.

You can create a structure using the **Design Part** business object by:

- **Manually** adding child items to the parent item.
- **Duplicating** an existing structure.
- **Creating a new structure from an existing structure**.
- Importing a structure.

  You can import a structure from **Excel**. You can also **import structures along with their partitions** from other sites.

## Create a structure manually

You can create a structure manually by adding the child items to it.

**Procedure**

1. Open the folder in which you want to create a new structure.
2. In the **Overview** tab, click **Add to** ⊕ in the **Contents** section.
3. In the **Add** panel, select the **Type** as **Design Part**, and enter the required information.
4. Open the item that you added, go to the **Content** tab, and click **Add** ⊕ > **Child**.
5. In the **Add Child** panel, enter the required details, and click **Add**.

   Add more children to build the structure. To add children that are at the same level, click **Add** ⊕ > **Sibling**.

   You can drag an item cross the structure to reorder the structure. To remove an item, right-click it and choose **Remove** ⊖

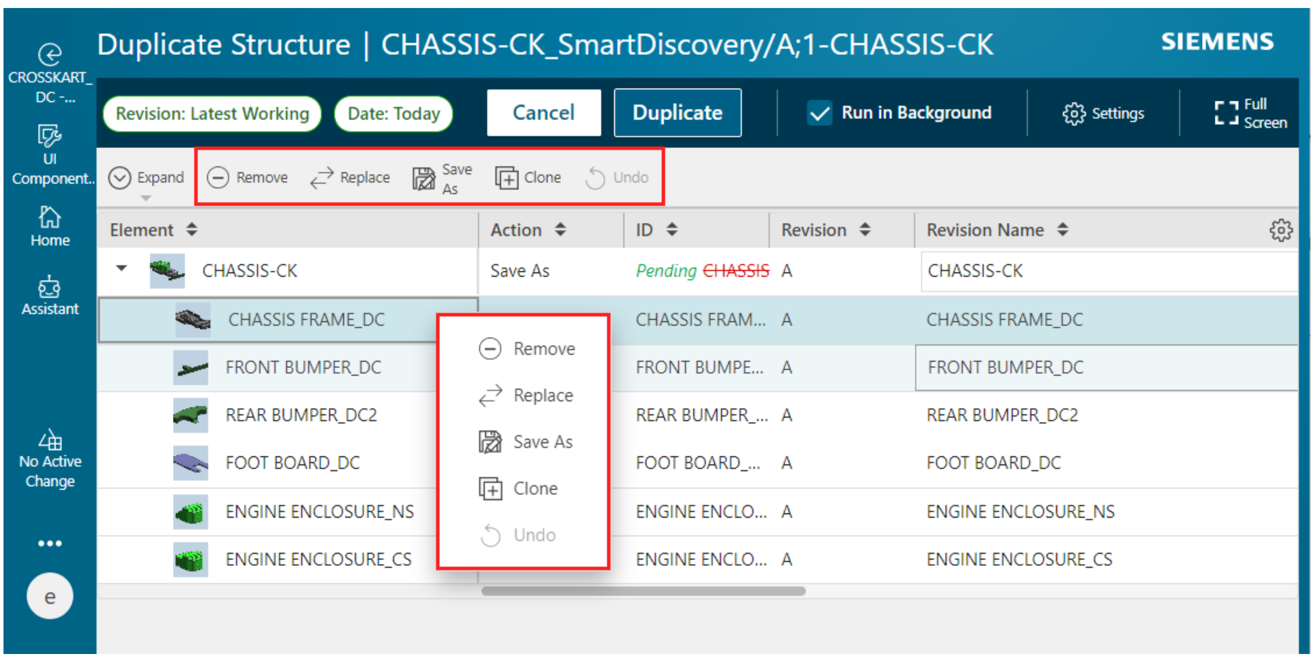
# Duplicate a structure

You can create a new structure by duplicating an existing structure. You can also duplicate structures that you do not own. When you duplicate a structure that you do not own, you become the owner of the new structure.

For an Assembly BOM or Usage BOM, duplicate the child items instead of the top-level structure. Design structures can be duplicated in their entirety.

### Procedure

1. For an Assembly BOM or Usage BOM, select the child items. For a design structure, select the top line of an assembly in the structure that you want to duplicate.
2. Click **Duplicate** .
3. Right-click each element and select one of the following actions. You can also select these actions from the work area toolbar.




| Action | Description |
|---|---|
| **Remove** | The element is not included in the new structure. |
| **Replace** | The element is replaced with the replacement element that you specified in the **Replace** panel. |
| **Save As** | A copy of the element is added to the new structure. You own this new element.<br>If the element is a structure, all its child elements are only referenced in the new structure. These are still owned by the user who created the source structure. |

| Action | Description |
|---|---|
| | You can edit the **Revision Name** and **Description** of the element. You can edit the custom properties if your administrator has configured those custom properties accordingly. |
| **Clone** | A copy of the element is added to the new structure. You own this new element.<br><br>If the element is a structure, copies of its child elements are used in the new structure.<br><br>You can edit the **Revision Name** and **Description** of each element. You can edit the custom properties if your administrator has configured those custom properties accordingly. |

**Note**

If any child element is a structure, all its children are also marked with the same action. However, if you change the action of one of its child elements later, the action of the parent element changes accordingly.

If you do not select an action, and the **Action** column is blank, the element is only referenced in the new structure. The element continues to be owned by the user who created the source structure.

4. To undo an action set on an element, select the element, and click **Undo**.

5. (Optional) Click **Settings** ⚙, and then perform these additional actions on the duplicated structure:

   a. Choose an option for the ID naming in the Settings panel:

   - Click the **Default IDs** option to retain the original element IDs for the cloned elements and for any elements that are saved as new elements.
   - Alternatively, click the **ID Naming Rule** option and enter the following details.

   | ID naming rule | Description |
   |---|---|
   | **Prefix** | The text is prefixed at the start of the existing IDs. |
   | **Suffix** | The text is appended at the end of the existing IDs. |
   | **Replace/With** | The text specified in **Replace** is replaced with the text specified in **With**.<br><br>The text specified in **Replace** must be text that is part of the existing ID, and it is case-sensitive. |

   b. To retain the original ownership for cloned elements and the elements that are saved as new elements, click the **Retain Original Ownership** check box.

   The elements on which no action is taken are referenced in the duplicated structure and their original ownership is retained.

c. Ensure that the **Owning Project** section displays the project that is a part of your active Teamcenter session.

   You can change your project so that the duplicated structure becomes a part of a different project. If no project is selected in the active Teamcenter session, you do not see the **Owning Project** section.

d. To include the Designcenter NX dependencies, if any, that are associated with the existing elements in the newly duplicated structure, click the **Include all Designcenter Dependencies** check box.

e. Click **Close**.

6. To run the duplicate process in the background or foreground, select or clear the **Run in Background** check box. If you choose to run the duplicate process in the background, you receive a notification in **Alerts** once the structure is duplicated.

> **Note**
>
> If your administrator has enabled the parallel duplication of large structures, the duplication process always runs in the background. Contact your administrator to adjust this setting.

7. Click **Duplicate**.

#### Results

The alignment data is carried over from the original structure to the new structure. The element effectivity for the new structure is derived from the active change.

By default, your active Teamcenter session project is assigned to the duplicated structure and its elements. If the administrator has enabled the project assignments in the Teamcenter environment, the following actions are performed on the duplicate structure:

- If an item in the original structure is associated to a project, that project is replaced with the project selected in the active Teamcenter session for the corresponding duplicated item.
- If no project is associated to an item in the original structure, no project is assigned to the corresponding duplicated item, even though a project is selected in the Teamcenter session.
- For referenced items that are not cloned but are reused in the new structure, if the original item is associated to a project, the project that is selected in the active Teamcenter session is appended to its existing project list.

## Create a new structure from an existing structure

You can create a new structure from an existing structure by saving the existing structure as a new structure.

### Procedure

1. Search for a structure.
2. Click **More Commands** ••• > **New** > **Save As**.
3. In the **Save As** panel, enter the required information.
4. Click **Save**.

### Results

A new structure is created from the existing structure. If the existing structure contains any partitions, the partitions are also carried over to the new structure.

## Associate a variability scope with a structure

The variability data is managed outside of a structure using Product Configurator. To include the variability data within the structure, you associate it with an appropriate variability scope.

### Procedure

1. Search for a structure of type **Configurable Assembly** and open it.
2. Click **Edit**.
3. In **Variability Scope**, click **Add**.
4. Search for variability scope, select it, and click **Add**.
5. Click **Save**.

## Revise a structure

You can revise a released structure to update it. When you revise a structure, you can retain existing aligned parts or create and use new ones. You can also address the alignment mismatches by setting a primary aligned part.

### Procedure

1. Open the structure that you want to revise.
2. Click **More Commands** ••• > **New** > **Revise**.
3. In the **Revise** panel, enter the required information.

If the selected structure is a design structure, additionally select the **Keep Existing Part** check box to retain the part that is already aligned with the selected design. The existing part and design alignment will be automatically carried over to the revised structure.

You can also choose to create a new part or use another existing part by clicking **Add Part**. In this case, the existing part and design alignment will not be carried over to the revised structure. After choosing to create or use another part, if you select the **Keep Existing Part** check box, the new part or the part that you chose is ignored.

If you see any error while previous alignment is getting removed, contact your administrator.

4. Click **Revise**.

   To align the design and part again to fix the mismatched alignments that are indicated as :

   a. Switch to the **Tree with Summary** view.

   b. Select a design in the design structure.

   c. Go to the **Aligned Parts** section in the **Overview** tab, select the required part, and click **Set Primary** .

# 14. Map engineering BOM with requirements

You can map an engineering BOM with requirements using trace links for impact analysis and quick navigation. This mapping helps you manage and trace evolving designs and requirements.

A part is always created or added inside the structure due to a requirement. A requirement, as well as a part, can evolve over time. When either the requirement or the part changes, there might be an impact on some other requirement or a part.

**Example**

A ball bearing was designed for a load requirement of 100 kg, but now the load requirement has changed to support 200 kg. This results in a change to the part design of the bearing, and it can also impact other parts, such as the suspension spring.

Therefore, it is important to have a mapping between the part and its corresponding requirements for impact analysis and quick navigation. You can create this mapping using trace links.

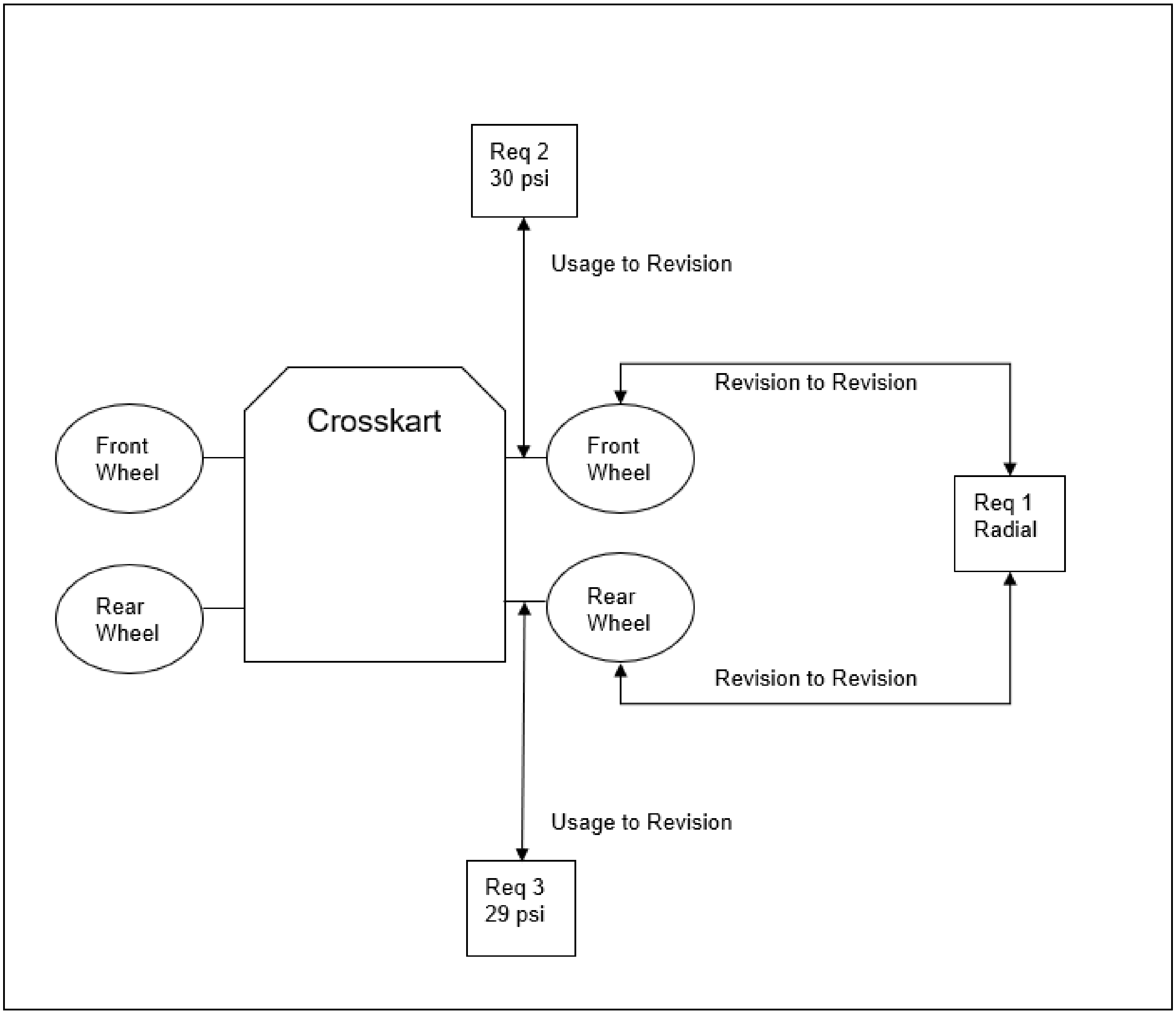
When you work with Usage BOM, you have a part and the usage of that part in the product. Both the part and its part usage can have their own unique set of requirements. The part requirements are independent of the product where it is used, and the part usage requirement is only valid in the context of the product where it is used.

**Example**

The part tire is used in the Crosskart, and it has a tire requirement for a radial type of tire, independent of the tire location. However, when the tire is used as a front tire, it should have a pressure of 30 psi, and when used as a rear tire, it should have a pressure of 29 psi. This means the part usage of the front tire has a requirement of 30 psi and the rear tire has a requirement of 29 psi.

You can create a trace link between the part tire and the requirement of the radial tire, making a revision to revision trace link.

Then, you can create the part usage to revision trace links between the part usage of the front tire and the requirement of 30 psi, and between the part usage of the rear tire and the requirement of 29 psi.




After creation, you can also view the existing trace links.

# 15. Specify alternates and substitutes for parts

## About global alternates and substitutes

Understand the difference between global alternates and substitutes in Teamcenter, including their definitions, examples, and key characteristics, to effectively manage interchangeable parts within product structures.

Parts are the different components that make up a product. In certain cases, one part can be interchanged with another part. There are two types of interchangeable parts, namely, global alternates (also known as alternates) and substitute components (also known as substitutes).

### Alternates

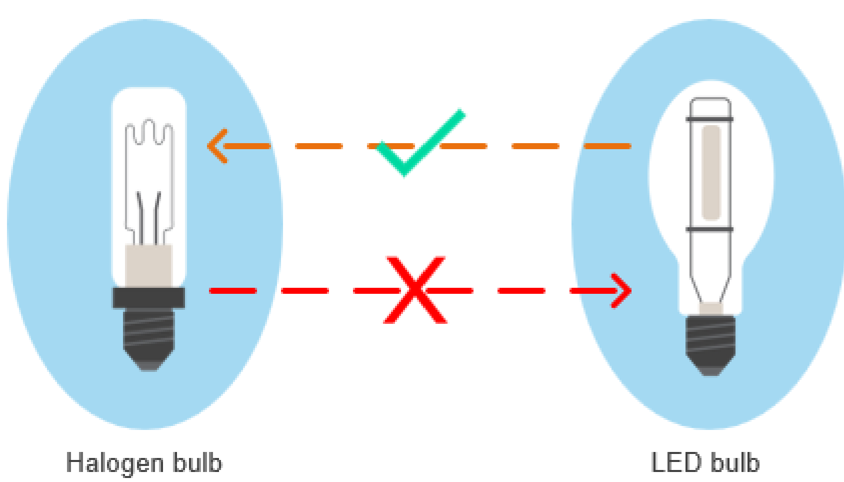
An alternate is a part that can be interchanged with another part in all circumstances, irrespective of where the other part is used in a product structure. Alternates are not specific to a variant or a product. You use alternates to specify that a part is interchangeable in any assembly.

**Example**

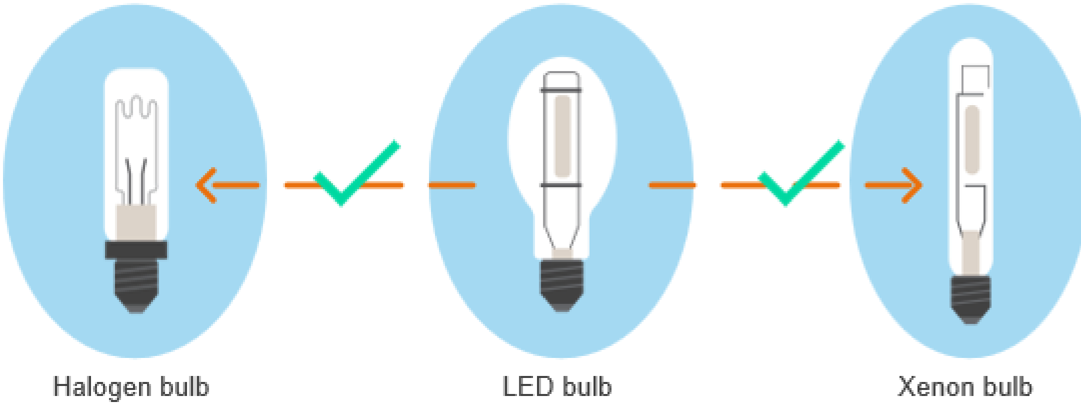
Consider a Halogen bulb that is used in several assemblies across products. You can specify an LED bulb as an alternate for the Halogen bulb. This means that in all assemblies and products, in place of the Halogen bulb, the LED bulb can be used.

Note the following important points about alternates:

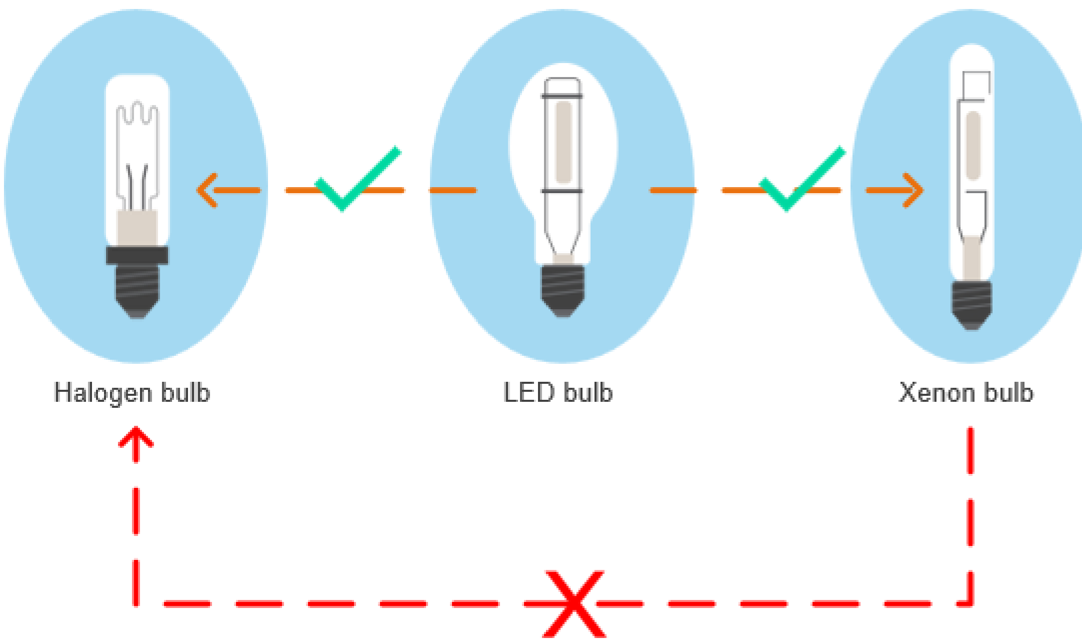
- If you reuse a part in some other assembly, the alternates of the part are carried over.
- Parts and their alternates are related in one way only. For example, the LED bulb is an alternate of the Halogen bulb. However, the Halogen bulb is not an alternate of the LED bulb.



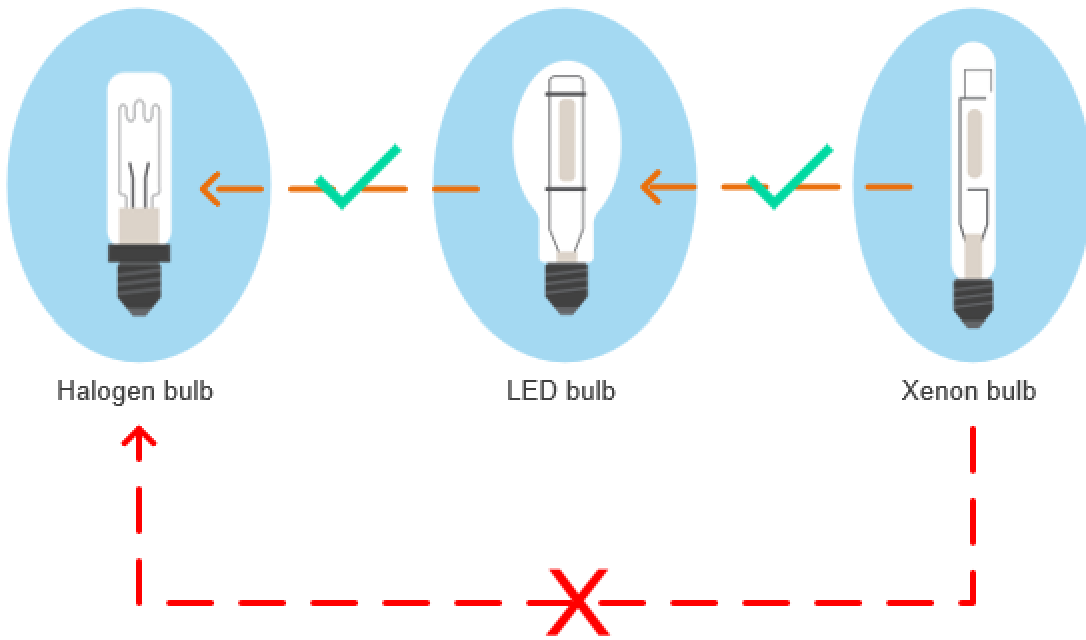
- One part can be an alternate of multiple other parts. For example, the LED bulb is an alternate of the Halogen bulb and of the Xenon bulb.




- Alternate relationships are not shared. For example, the Xenon bulb is not an alternate of the Halogen bulb, even though the LED bulb is an alternate of both bulbs.




- Alternate relationships are not chained. For example, the LED bulb is an alternate of the Halogen bulb. And the Xenon bulb is an alternate of the LED bulb. However, the Xenon bulb is not an alternate of the Halogen bulb.




## Substitutes

Substitutes are parts that can be interchanged only with a particular occurrence of a component within a single assembly. You define a substitute for a single occurrence in an assembly and not for an item. You use substitutes to specify that a part is interchangeable only in a specific assembly.

> **Example**
>
> Consider that Crosskart has a lighting assembly that uses the Halogen bulb, by default. You set the Xenon bulb as a substitute for the Halogen bulb within the lighting assembly. A substitute can be used in place of another part but only in a specific assembly. So, you cannot use the Xenon bulb in place of the Halogen bulb in another assembly.

Note the following important points about substitutes:

- You define a substitute for a single occurrence in an assembly and not for an item.
- You can add multiple substitutes for an occurrence of a part.
- You can replace a part in an assembly with a substitute.
- All substitutes of one occurrence share the same occurrence attributes. The occurrence attributes can be a find number, quantity, and notes.

# Set global alternates for a part

An *alternate part* is a part that can be used in place of another part in any product. You can set and manage global alternates in Teamcenter, including adding new alternates, selecting from existing ones, and removing alternates for a part, to define interchangeable components across all product structures.

You can set an alternate for only those parts whose **Assembly Indicator** is not set as **Generic Part** and **Configurable Assembly**.

**Procedure**

1. Select the part for which you want to set a global alternate.
2. Click the **Overview** tab.

   The alternates available for a part are listed under the **Global Alternates** section.

## Add an alternate

**Procedure**

1. In the **Global Alternates** section, click **Add Alternate** ⊕.
2. In the **Add Alternate** panel, click one of the following tabs:

   | Tab | Description |
   |---|---|
   | **New** | On this tab, you can add a new occurrence. |
   | **Palette** | On this tab, you can paste an occurrence from the clipboard or select one from your **Favorites** or **Recent** list. |
   | **Search** | On this tab, you can search for an occurrence to add. |

3. In the **Add Alternate** panel, select the part that you want to add as a global alternate. You can select multiple parts.
4. Click **Add**.

   The selected parts are added to the **Global Alternates** list.

## Remove an alternate

**Procedure**

1. From the **Global Alternates** list, select the alternate that you want to remove. You can select multiple alternates.
2. Click **Remove** ⊖.

The selected parts are removed from the **Global Alternates** list.

# Set substitutes for a part in a structure

A *substitute part* is a part that can be used in place of another part in a specific product configuration. In other words, substitutes are specific to an assembly. A substitute part can be added in the context of an engineering BOM. You can set, remove, and replace parts with substitutes in a structure.

You can set a substitute for only those parts whose **Assembly Indicator** is not set as **Generic Part** and **Configurable Assembly**.

### Procedure

1. Select the part for which you want to set a substitute.
2. Click the **Overview** tab.

   The substitutes available for a part are listed under the **Substitutes** section.

## Add a substitute

### Procedure

1. In the **Substitutes** section, click **Add Substitute** ⊕.
2. In the **Add Substitute** panel, click one of the following tabs:

   | Tab | Description |
   |---|---|
   | **New** | On this tab, you can add a new occurrence. |
   | **Palette** | On this tab, you can paste an occurrence from the clipboard or select one from your **Favorites** or **Recent** list. |
   | **Search** | On this tab, you can search for an occurrence to add. |

3. In the **Add Substitute** panel, select the part that you want to add as a substitute. You can select multiple parts.
4. Click **Add**.

   The selected parts are added to the **Substitutes** list.

## Remove a substitute

### Procedure

1. From the **Substitutes** list, select the substitute that you want to remove. You can select multiple substitutes.

2. Click **Remove** ⊖.

   The selected parts are removed from the **Substitutes** list.

## Replace an element with a substitute

### Procedure

1. From the **Substitutes** list, select the substitute that you want to use in place of an element.

2. Click **Use**.

   In the structure, the element is replaced with the selected part. The element that is replaced in the structure is added to the **Substitutes** list.

# Add and manage occurrence substitutes for a part in a structure

A *substitute part* is a part that can be used in place of another part in a specific product configuration. In a structure, you can add a substitute for a part, remove a substitute, and set a substitute as primary item. If you add and manage occurrence substitutes in an active change notice, the changes are indicated by redlining.

### Restrictions and limitations

You can add and manage occurrence substitutes for a part in a structure if your Teamcenter setup includes the BOM Structure Management license.

### Procedure

1. Select the part for which you want to add a substitute.

2. Click **Add** ⊕ > **Substitute**.

   > **Note**
   >
   > You can use the **Substitutes** section in the **Overview** tab to add a substitute only when at least one substitute is already added for the item.

3. In the **Add Substitute** panel, click one of the following tabs:

| Tab | Description |
|---|---|
| **New** | On this tab, you can add a new occurrence. |
| **Palette** | On this tab, you can paste an occurrence from the clipboard or select one from your **Favorites** or **Recent** list. |
| **Search** | On this tab, you can search for an occurrence to add. |

4. In the **Add Substitute** panel, select the part that you want to add as a substitute. You can select multiple parts.

5. Click **Add**.
   The selected occurrence of the parts are added to the structure. The **Substitute Indicator** column displays for the primary part and for the substitutes. The **Substitutes column** shows the substitute name for the primary item. The occurrence attributes of the primary part, including find number, are also set for the substitutes. The **Substitutes** section of the **Overview** tab for the selected primary part displays the information for the substitutes.

   Click **Configuration Settings** > **Show Substitute Components** to show or hide the substitutes in the structure.

   If you copy or cut a primary part to the same structure or to a different structure, all occurrences of the substitutes along with the attribute information are also copied. If you insert a level on the primary part, all substitutes are also moved along with the primary part. If you remove a primary part from the structure, all its substitutes are also removed.

## Remove a substitute

If a substitute is no more a valid substitute for any reason, you can remove it.

### Procedure

1. Select the substitute that you want to remove and click **Edit Structure** > **Remove** .
2. Click **Remove** on the confirmation message.

   The selected substitute along with its information is removed from the structure.

## Set a substitute as a primary part

At times, you might want to mark a substitute as a primary part in the structure. After you mark the substitute as a primary part, the original primary part becomes substitute for it.

### Procedure

1. Select a substitute that you want to set as primary part in the structure.

   If you select multiple substitutes, the last substitute that you selected is chosen as primary.

2. Right-click and select **Set Primary**.

   The selected substitute is set as a primary part and the primary part is set as a substitute in the structure.

# Add and manage a substitute group for parts in a structure

When two or more parts in a structure have the same set of substitutes, you can define these substitutes by using a substitute group. For example, a structure contains different caps for the 12V, USB-A, and USB-C charging ports in the car and you want to define similar substitute items for them. In such a situation, you can select these caps as the primary items in the structure, create a substitute group, and then define the substitute parts. If you add and manage occurrence-based substitute groups in an active change notice, the changes are indicated by redlining.

A substitute group can contain multiple sets of substitutes. After a substitute group is created, you can add or remove primary parts, substitutes, and substitute sets to or from it. In addition, you can map a primary part to a substitute part in a substitute group.

### Restrictions and limitations

You can add and manage a substitute group for parts in a structure only if your Teamcenter setup includes the BOM Structure Management license.

### Procedure

1. Select two or more parts in the structure for which you want to add a substitute group.
2. Click **Add** > **Substitute Group**.
3. In the **Add Substitute Group** panel, ensure that the **Occurrence Substitute Group** type is selected, and then perform the following tasks:
   a. In the **Properties** section, specify a name and description for the substitute group.
   b. In the **Primary Elements** section, ensure that the required parts are listed.
   c. In the **Substitutes** section, specify a name for **Substitute Set ID** and then click **Add** to add substitute elements.

      **Tip**

      Using the **Add Substitute Elements** panel, you can now add one or more substitute elements. All the substitutes that you add become a part of the substitute set.

      After you click **Add**,

      For example, in the first substitute set, you can add three caps as substitute parts for the 12V, USB-A, and USB-C charging ports.
   d. To add another substitute set, in the **Substitutes** section, click **Add**.

You can add multiple substitute sets.

4. Click **Create**.
All the substitutes that you selected are added to the structure. The **Substitute Indicator** column displays for the primary parts and for the substitutes. The **Substitutes column** shows the substitute names for the primary parts. The occurrence attributes of the primary part, including find number, are set for the substitutes. The **Substitutes** section of the **Overview** tab for the primary part displays the primary parts, the substitute sets, and the substitutes.

5. To remove a substitute, substitute set, or a primary part, select the item in the **Substitutes** section, and click **Remove** ⊖.

## Add a primary part to the substitute group

You can add another part from the structure as a primary part to the existing substitute group. For example, suppose that the USB-B port is added to the newer model of the car. In such a situation, you might want to add the cap for the USB-B port as a primary part to an existing substitute group.

**Procedure**

1. In the **Substitutes** section, select the **Elements** node, and click **Add Elements** ⊕.

   The **Add Elements** panel appears and displays the remaining parts from the structure.

2. Select a part and click **Add**.
The selected part is added as a primary part to the substitute group and the information about the substitutes is also added for it.

## Add a substitute set to the substitute group

You might want to add another set of items as substitutes for the primary parts. In such a situation, add a substitute set to an existing substitute group.

**Procedure**

1. In the **Substitutes** section, select the substitute group, and click **Add Substitute** ⊕.
A substitute set is added.

2. Select the newly added substitute set and click **Add Substitute** ⊕.

3. In the **Add Substitute Elements** panel, search for an existing substitute or add a new substitute and click **Add**.
The substitute is added to the selected substitute set.

# Map a substitute to a primary part

You can map a substitute to a primary part in the group. For example, map the newly added cap as a substitute for the primary cap for USB-C port.

### Procedure

In the **Substitutes** section, select a primary part and a substitute, and then click **Map** .
In the **Substitutes** section, the **Mapping** column displays the name of the primary part.

# 16. Manage structure effectivity

## About structure effectivity

A product structure goes through many changes during the evolution of its product definition. Using structure effectivity, you can capture how the product structure has evolved over a period. You can use element and release effectivity to capture the evolution of product structures over time, manage product variability, and configure different revisions or elements based on date or unit ranges.

An effectivity can be of the following types:

| | |
|---|---|
| **Element effectivity** | It denotes from which date or for which unit (and end item) an element is effective. |
| **Release effectivity** | It denotes from which date or for which unit (and end item) an element revision is effective. It is necessary to have a release status associated with an element to author release effectivity on it. |

Release effectivity helps in configuring different revisions of the same element, wherein with the help of element effectivity you can decide if to show the element itself or not inside the structure.

For example, in an engine cooling block, revision *A* has 5 cooling fins and in revision *B* it is changed to 6, now to decide which revision to use, release effectivity is used. Now suppose we decide to use another engine block itself which is liquid-cooled, to decide between the air-cooled or liquid-cooled engine block element effectivity is used.

An effectivity can be expressed as a date range or a range of units or both. When you edit the effectivity range for one occurrence, the change is applied to all occurrences.

Element effectivity can be automatically derived from the release effectivity of the current active change notice using **effectivity propagation**. In case of overlapping element effectivities when you modify the structure, the element effectivities are adjusted using the **cut back** mechanism.

If an element does not have an associated effectivity object, it is assumed to be always effective. It is not constrained by any effectivity.

Effectivity can be used for the following purposes:

You can use **group effectivity** to combine multiple end items and range of units for each end item or multiple date ranges to configure multiple occurrences at the same time which have different effectivities on them.

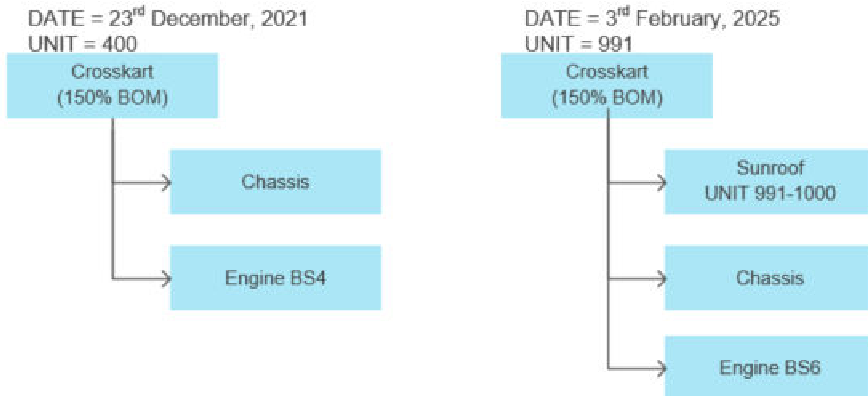
**Note**

Videos are not included in PDF. To access videos, use HTML.

- To reflect changes to the structure over time when new parts replace old ones.

  For example, due to an engineering change, from 1st January 2025 onwards, Engine BS6 must be used instead of Engine BS4. For this, you specify date effectivity on both the engines.

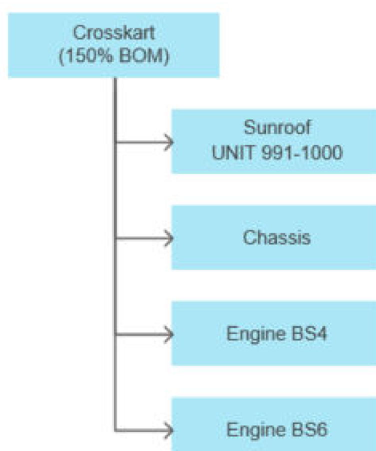
  When you configure Crosskart on 23rd December 2021, Engine BS4 is used. However, when you configure Crosskart on 3rd February 2025, Engine BS6 is used.



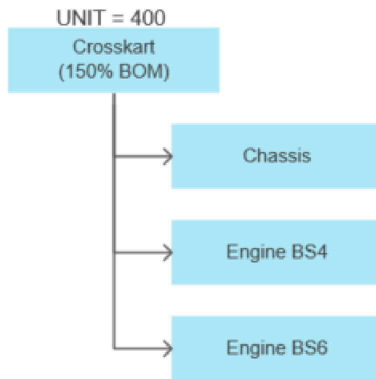

- To state the content of a unit or range of units as a means of managing the variability of the product.

  For example, consider that your company manufactures a product, Crosskart. As per the company's business requirement, a total of 1000 units of Crosskart must be manufactured.

  However, 10 customers want Crosskart with Sunroof. Due to such special customer requests, there is a slight deviation from the standard configuration of the product. It is inconvenient to maintain unique identifiers for several such small deviations. But with occurrence effectivity, it is easier to accommodate such deviations as the existing structure can be modified to accommodate the changes.




  If you specify a unit between 1 and 990, or a range of units between 1 and 990, Sunroof is not included in the configured structure. If you specify a value between 991 and 1000, Sunroof is included.




## About effectivity propagation

The *effectivity propagation* is the method to propagate or transfer the effectivity of an active change notice to the different elements of a Usage BOM structure.

When you set an engineering change notice sent to the **EBOM Release Process workflow** as an **active change**, its **release effectivity** is inherited as the element effectivity by the part usages that are added to or updated in a Usage BOM.

For example, if the release effectivity of an active change notice is from 1st April 2022 to 30th September 2022, any part usage added, replaced, or revised in the context of this change notice inherits the release effectivity as its element effectivity.

If a part usage has element effectivity inherited from some previous change, which overlaps with the release effectivity of the active change, an ***effectivity cutback*** is performed on the original part usage. The new part usage inherits the release effectivity of the active change.

In case, the active change notice is not yet released, and its release effectivity is updated after it is propagated to part usages, you must verify the data first and then **repropagate the effectivity**.

The following example explains why you must verify the data before repropagating the effectivity.

**Example**

Consider that revision *A* of the engine assembly is effective from 1st Jan 2022 to UP and contains *piston, rubber gasket*, and *stud*.

Revision *B* of the engine assembly is effective from 1st March 2022 to UP and contains *piston, rubber-gasket*, and *bolt*.

The original change effectivity is set for 1st March 2022, so revision *B* of the engine assembly is the basis for the change.

At this point, the engine assembly is modified, and its revision *C* contains *piston, silicon gasket*, and *bolt*.

Now if the effectivity is changed to 1st February 2022, revision *A* of the engine assembly becomes the basis for the change. Revision *A* of the assembly does not contain the *bolt*. Therefore, you must work on revision *A* of the engine assembly again before repropagating the effectivity.

# About effectivity cutback

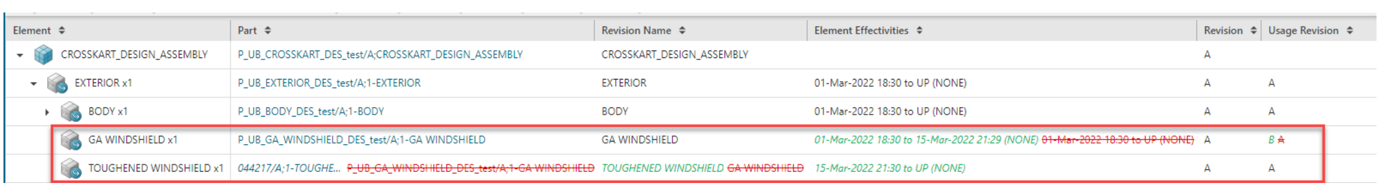
*Effectivity cutback* is the result of modifying an effectivity range when you introduce a new part to replace an old part in a released Usage BOM structure. In this case, it is necessary to reduce (cut back) or make room for the effectivity range of the old part so that it does not overlap with the effectivity of the new part.

For example, Engine BS4 has effectivity ranging from days 01–20 and Engine BS6 has effectivity ranging from days 12–20. If Engine BS4 is replaced by Engine BS6, both engines are effective from days 12–20. To resolve this, the effectivity of Engine BS4 must be cut back or adjusted to end before Engine BS6 becomes effective, that is, days 01–11. This ensures that Engine BS4 is effective from days 01–11, and Engine BS6 is effective from days 12–20.

Consider the following scenario:

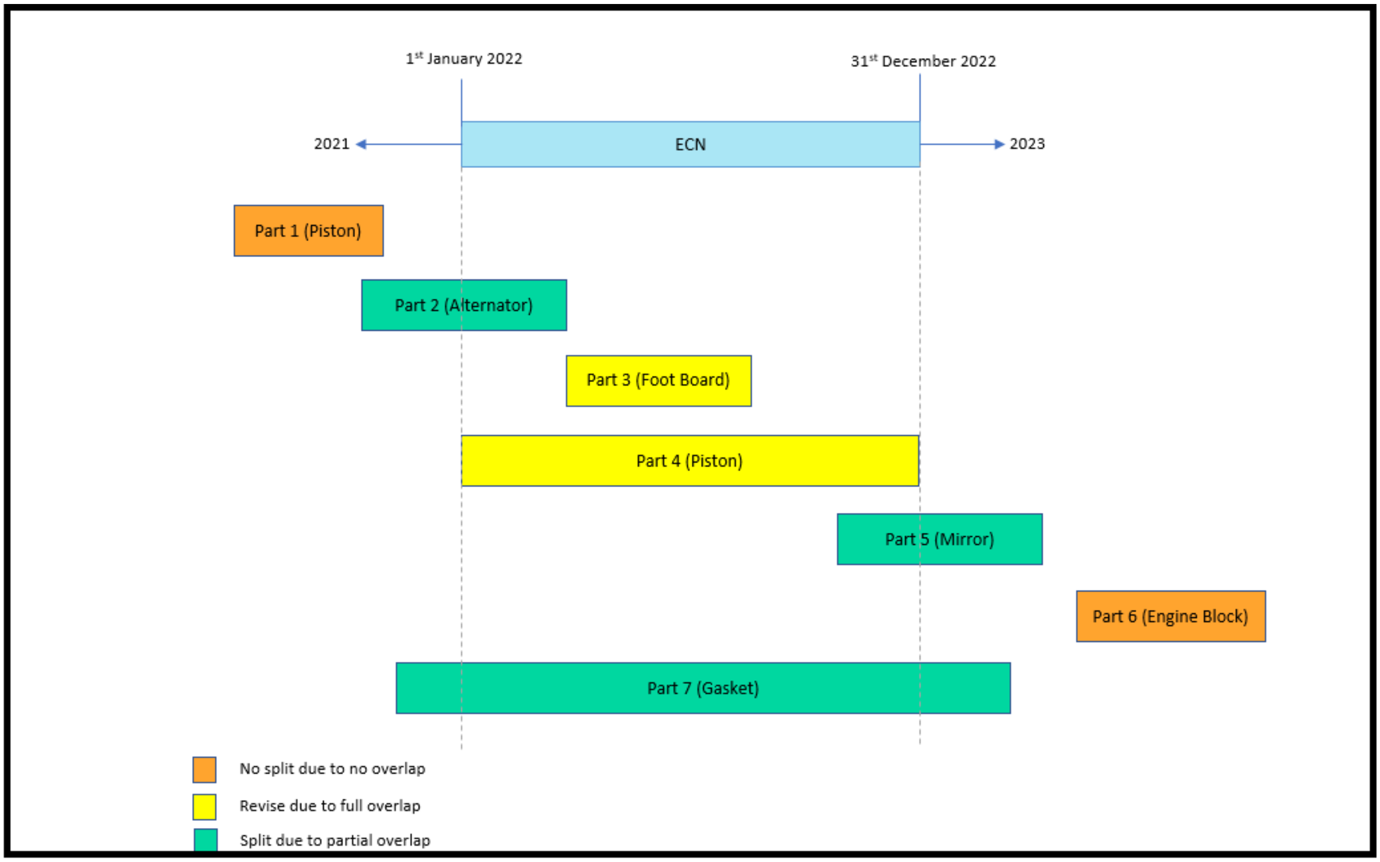
You have a released structure with **effectivity propagation** resulting from a change notice. You modify this structure in context of a new change notice by replacing an existing part. This existing part, the impacted item, has a partial effectivity overlap with the new part, the solution item. In this scenario, the new part, or the solution item, inherits its element effectivity from the active change.

In the following example, the part GA WINDSHIELD (created in change notice 1, with a release effectivity starting March 1, 2022) is replaced with the part TOUGHENED WINDSHIELD (created in change notice 2, with a release effectivity starting March 15, 2022). The element effectivity of the part GA WINDSHIELD (now revised to **Usage Revision** B) is cut back to a date before the part TOUGHENED WINDSHIELD (having **Usage Revision** A) is effective.



| Element | Part | Revision Name | Element Effectivities | Revision | Usage Revision |
|---|---|---|---|---|---|
| CROSSKART_DESIGN_ASSEMBLY | P_UB_CROSSKART_DES_test/A;CROSSKART_DESIGN_ASSEMBLY | CROSSKART_DESIGN_ASSEMBLY | | A | |
| EXTERIOR x1 | P_UB_EXTERIOR_DES_test/A;1-EXTERIOR | EXTERIOR | 01-Mar-2022 18:30 to UP (NONE) | A | A |
| BODY x1 | P_UB_BODY_DES_test/A;1-BODY | BODY | 01-Mar-2022 18:30 to UP (NONE) | A | A |
| GA WINDSHIELD x1 | P_UB_GA_WINDSHIELD_DES_test/A;1-GA WINDSHIELD | GA WINDSHIELD | 01-Mar-2022 18:30 to 15-Mar-2022 21:29 (NONE) ~~01-Mar-2022 18:30 to UP (NONE)~~ | A | B ~~A~~ |
| TOUGHENED WINDSHIELD x1 | 044217/A;1-TOUGHE... ~~P_UB_GA_WINDSHIELD_DES_test/A;1-GA WINDSHIELD~~ | TOUGHENED WINDSHIELD ~~GA WINDSHIELD~~ | 15-Mar-2022 21:30 to UP (NONE) | A | A |

The following graphic explains various use cases for effectivity cutback, where each block represents the effective period of the part and its overlap with the release effectivity of the change notice.




Sometimes, when a new engineering change notice is created, its effectivity completely overlaps with the effectivity of the released part (Part 3 and Part 4 in the image). If you want to modify this released part, its part usage is first revised and then updated.

For example, consider that the part Foot Board with quantity 2 was created and released in the context of change notice 1. This part is effective starting March 1, 2002. Now, the part Foot Board is updated to change its quantity to 4 in the context of change notice 2, a new change notice. The release effectivity of change notice 2 is the same as that of change notice 1. Therefore, before updating Foot Board, it is first revised and then updated, and the quantity of its part usage is set as 4.

Now, after the quantity is updated, the effectivity of change notice 2 is changed such that it is effective starting May 1, 2002, to have a partial overlap instead of full overlap. In such a scenario, you must repropagate the effectivity so that Foot Board gets the new effectivity date. On doing so, Foot Board is split into two usages due to a partial overlap of effectivity dates. The old, revised part usage is effective from March 1, 2022, to April 30, 2022, while the new part usage created is effective starting May 1, 2022.

## Repropagate effectivity

If the release effectivity of an unreleased change notice is updated after it is already propagated to the part usages as their element effectivity, you must repropagate the effectivity from the change notice to the respective part usages.

### Procedure

1. From **Launcher**, click **Changes**.

   If this option is unavailable, search for it. You can pin it from the search results for future access.

2. Open the required change for which its release effectivity has been modified.
   The **Affected Items** tab shows the **IMPACTED ITEMS** and **SOLUTION ITEMS** for the change notice.
   The **CHANGE SUMMARY** section in the **Overview** tab shows the modifications made in the context of this change notice.

3. Verify the data first and then click the **Repropagate Effectivity** button in the **Overview** tab.

   The following example explains why you must verify the data before repropagating the effectivity.

   **Example**

   Consider that revision *A* of the engine assembly is effective from 1st Jan 2022 to UP and contains *piston, rubber gasket*, and *stud*.

   Revision *B* of the engine assembly is effective from 1st March 2022 to UP and contains *piston, rubber-gasket*, and *bolt*.

   The original change effectivity is set for 1st March 2022, so revision *B* of the engine assembly is the basis for the change.

   At this point, the engine assembly is modified, and its revision *C* contains *piston, silicon gasket*, and *bolt*.

   Now if the effectivity is changed to 1st February 2022, revision *A* of the engine assembly becomes the basis for the change. Revision *A* of the assembly does not contain the *bolt*. Therefore, you must work on revision *A* of the engine assembly again before repropagating the effectivity.

### Results

The element effectivity of the part usages is updated as per the new release effectivity of the change notice. In certain cases, the element effectivity of a part usage is displayed as **Not Effective**, which implies that the part is no longer effective in the structure. On hovering the mouse over the element effectivity, Teamcenter displays:

- 0 if a unit effectivity was applied on the part.
- 5 Jan 1900 to 5 Jan 1900 in UTC (or a subsequent date in your time zone considering the UTC time) if a date effectivity was applied on the part.

**Example**

Consider an engine assembly that includes a piston, rubber gasket, and stud, created in the context of *ECN1* with a release effectivity from 1st Jan 2022 to UP. After this, *ECN1* was released.

Later, due to business requirements, it was decided that the stud from XYZ company should be used from 1st Jan 2022 to 31st Dec 2022. Consequently, *ECN2*, effective from 1st Jan 2022 to 31st Dec 2022, was introduced. With *ECN2* set as the active change, Teamcenter splits the part usage of the stud into two in the engine assembly. The effectivity of the stud procured from XYZ company is set to 1st Jan 2022 to 31st Dec 2022, while the effectivity of the other stud is set to 1st Jan 2023 to UP.

At this stage, due to further changes in business requirements, it was decided that the stud from XYZ company will always be used. Therefore, the effectivity of *ECN2* is modified to 1st Jan 2022 to UP, and this change is repropagated to the engine assembly. As a result of this repropagation, the element effectivity of both part usages of the stud is now 1st Jan 2022 to UP. In such a situation, Teamcenter automatically sets the part usage of the older stud created under *ECN1* as **Not Effective** and sets the effectivity of the stud from XYZ company to 1st Jan 2022 to UP.

# Repropagate effectivity from a closed change notice using a workflow

You can edit and repropagate the effectivity for a closed change notice that you own.

### Prerequisites

You must be the owner of the closed change notice for which you want to repropagate the effectivity.

### Procedure

1. Select the closed change notice and click **Release Effectivity** .
2. In the **Release Effectivity** panel, edit the effectivity values and click **Save**.
3. Click **Submit to Workflow** .
4. In the **Submit to Workflow** panel, select the **Effectivity Update Workflow** template, and click **Submit**.
   The modified effectivity value is repropagated to **Solution Items** in the change notice.

   **Note**

   This repropagation on a closed change notice works only when the solution items of the current change notice are not part of one or more other change notices as impacted items.

# Add or modify an element effectivity

You can add and modify element effectivities using date ranges or unit numbers. You can also create new shared effectivities, and understand how effectivity changes can impact the element visibility.

You can **create** an element effectivity using a range of dates, a unit number or a range of units. You can also **update** an existing element effectivity.

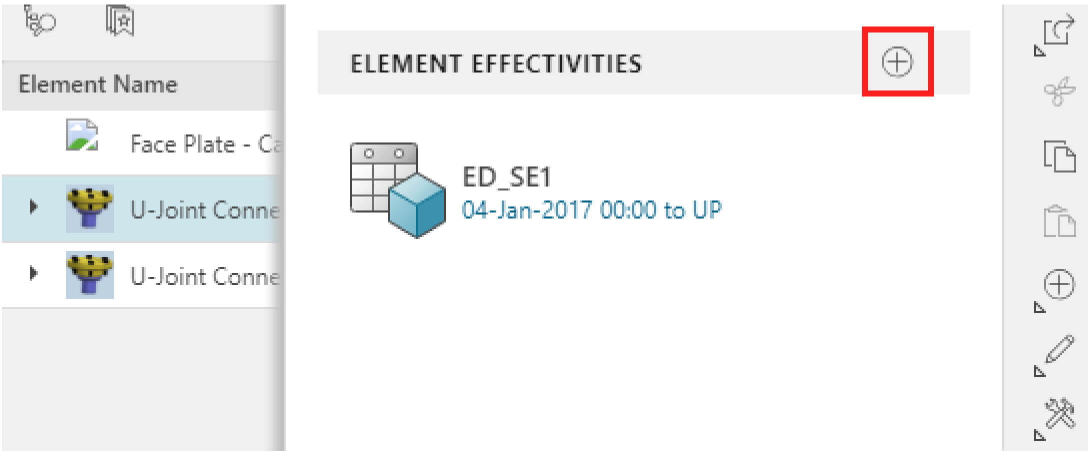
# Add an element effectivity

### Procedure

1. Set an active change notice to track a Usage BOM.

2. Select one or more elements and click **Add** ⊕ > **Element Effectivity**.

3. In the **Element Effectivity** panel, click **Add Effectivities** ⊕.




4. You can add multiple effectivities to the selected element.

> **Note**
> Add a new effectivity or locate existing effectivities from the **Search** tab.

To author a new date effectivity:

a. In the **New** tab, select **Date**.

b. (Optional) Select the **Share** check box to create a shared effectivity.

   The **Name** field is displayed only for shared effectivities, so that users can search for the effectivity by name.

c. Select the **Start** date from the calendar.

d. Select the **End** date from the calendar. If applicable, you can select **UP (all future dates)** or **SO (stock out)**.

e. (Optional) Click **Replace** to add a new **End Item**, or search for an existing **End Item**.

f. (Optional) Select the **Protect** check box if you do not want the effectivity to be edited.

g. Click **Add** to create the effectivity.

   The element effectivity is applied to the selected element. If the applied **Revision Rule** or **Date** effectivity is not applicable to this element, then it is excluded from the structure. Select the Show Excluded By Effectivity toggle to show this element.

> **Note**
>
> You cannot author multiple date ranges inside a single date effectivity on a single occurrence.

OR

To author a new unit effectivity:

**a.** In the **New** tab, select **Unit**.

**b.** (Optional) Select the **Share** check box to create a shared effectivity.

The **Name** field is displayed only for shared effectivities, so that users can search for the effectivity by name.

**c.** Specify the desired unit or a range of units in the **Unit** field.

**d.** Click **Replace** to add a new **End Item**, or search for an existing **End Item**.

**e.** (Optional) Select the **Protect** check box if you do not want the effectivity to be edited

**f.** Click **Add** to create the effectivity.

The element effectivity is applied to the selected element. If the applied **Revision Rule** or **Unit** effectivity is not applicable to this element, then it is excluded from the structure. Select the **Show Excluded By Effectivity** toggle to show this element.

> **Note**
>
> You cannot edit multiple date ranges inside a single date effectivity on a single occurrence.

## Edit or remove an element effectivity

Before updating an element effectivity, ensure that you have **set an active change notice to track a Usage BOM**.

- To update an element effectivity, select it and click **Edit**.
- To remove an element effectivity, select it and click **Remove**.

  After the element effectivity is updated for an element, if the applied **Revision Rule** or *effectivity criteria* is not applicable to this element, then it is excluded from the structure. Select the **Show Excluded By Effectivity** toggle to show this element.

# Add or modify release effectivity

You can add and modify release effectivities using date ranges or unit numbers for elements with an associated release status. You can create new shared effectivities and understand how these changes impact the element configuration.

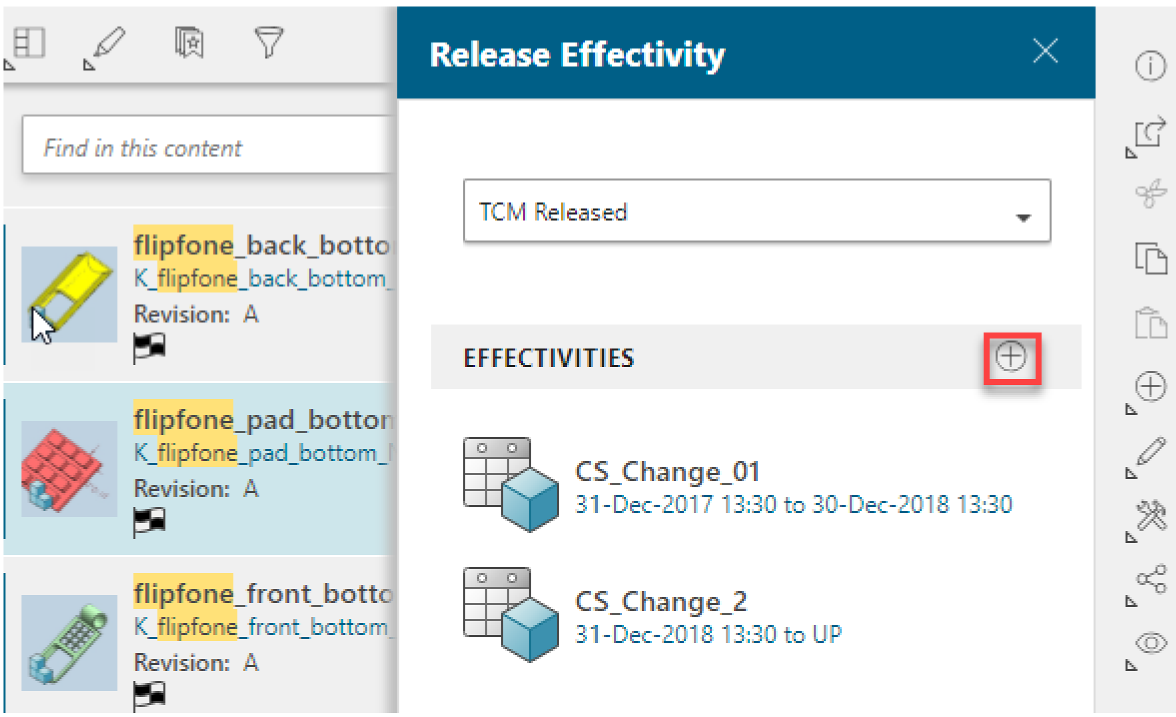
You can **create** a release effectivity using a range of dates, a unit number or a range of units. You can also **update** existing release effectivity.

# Add release effectivity

### Procedure

1. Set an active change notice to track an engineering BOM.
2. Select the element (having some release status associated with it) from the structure and click **Add** ⊕ > **Release Effectivity**.
3. You can also search the element (having some release status associated with it) and click **Manage** > **Release Effectivity**.
4. In the **Release Effectivity** panel, select the appropriate release status.
5. In the **Release Effectivity** panel, click **Add Effectivities** ⊕.




6. Add a new effectivity or from the **Search** tab locate existing effectivities. You can add multiple effectivities to the selected element.

   To specify the dates for release effectivity:

   a. In the **New** tab, select **Date**.
   b. Select the **Share** check box to create a shared effectivity. Specify the effectivity **Name**.

      The **Name** field is displayed only for shared effectivities. You can search for the effectivity by name.
   c. Select the **Start** date from the calendar.
   d. Select the **End** date from the calendar. If applicable, you can select **UP (all future dates)** or **SO (stock out)**.

e. (Optional) Click the **Replace** icon to add a new **End Item** or search for an existing **End Item**.

f. Select the **Protect** check box if you do not want the effectivity to be edited

g. Click **Add** to create the effectivity.

   After release effectivity is added, the occurrence is configured based on the current revision rules.

OR

To specify units for release effectivity:

a. In the **New** tab, select **Unit**.

b. Select the **Share** check box to create a shared effectivity. Specify the effectivity **Name**.

   The **Name** field is displayed only for shared effectivities. You can search for the effectivity by name.

c. Specify the desired unit or a range of units in the **Unit** field.

d. Click the **Replace** icon to add a new **End Item** or search for an existing **End Item**.

e. Select the **Protect** check box if you do not want the effectivity to be edited

f. Click **Add** to create the effectivity.

   After the release effectivity is added, the element is configured based on the current revision rules.

## Edit or remove the release effectivity

Before you do this, **set an active change notice to track a collaborative product engineering BOM** , if it is not done already.

To update the newly added release effectivity, select it and click **Edit** .

To remove the release effectivity, select it and click **Remove** .

After release effectivity is updated, the part revision is configured based on the current revision rules.

# Configure structure using group effectivity

You can configure product structures using group effectivity, which combines multiple end items, unit ranges, or date ranges to simultaneously manage the effectivity of various occurrences. You can also add, modify, or remove these group effectivities.

A group effectivity is a combination of multiple end items and range of units for each end item or multiple date ranges. When you want to configure multiple occurrences at the same time which have different effectivities on them, you can configure using group effectivity.

You can configure the structure by adding group effectivities if your administrator has configured it for you at your site.

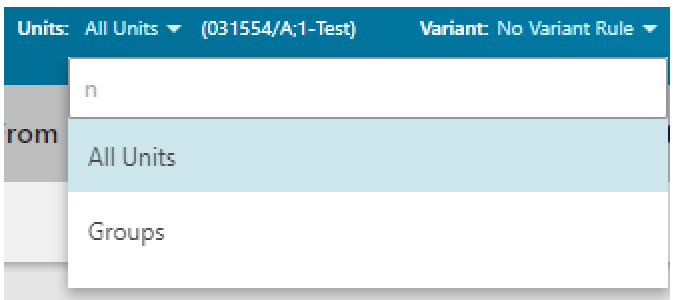
Group effectivity is used to configure product structure occurrences of an assembly by:

- Specifying multiple end items
- Specifying the unit effectivity ranges for each of those end items
- Specifying multiple date ranges

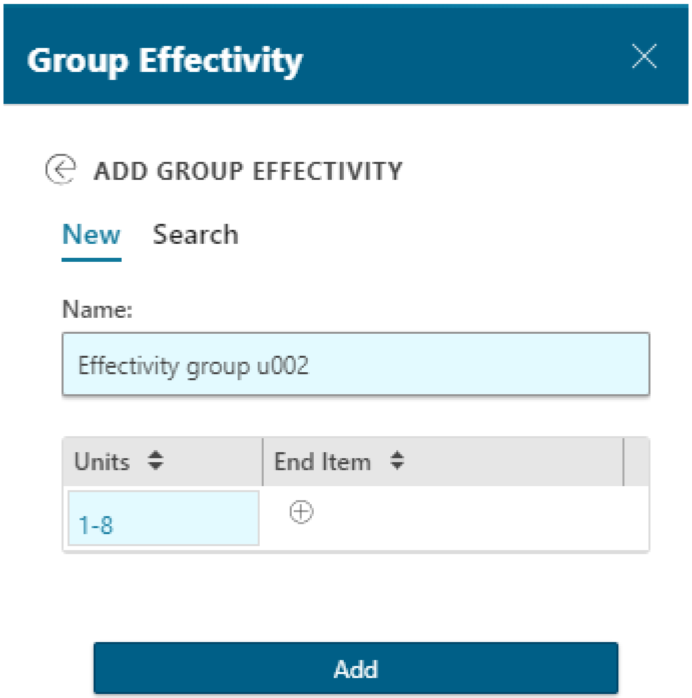
## Configure a structure using a group of unit effectivities and end items

### Procedure

1. Search and open the structure that you want to configure with group effectivity.
2. Select the structure and click **Units** > **Groups**.




3. In the **Group Effectivity** pane, click the **Add Group Effectivity** ⊕ icon.
4. Click **New**.




5. Specify the effectivity **Name**.
6. Specify the desired unit or a range of units in the **Unit** field.
7. In the **End Item** field, click **Add** ⊕ to add a new **End Item** or search for an existing **End Item**.

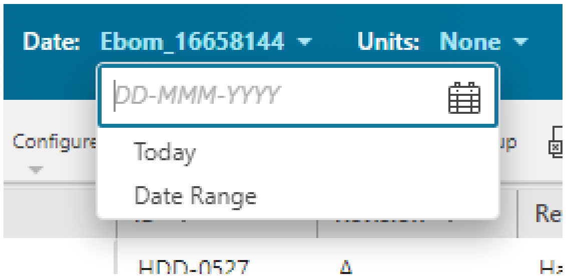
   Once the unit and end item information is entered, a new row is added to the table.
8. Add more units and unique end items as required.
9. Click **Add**.

   A group effectivity is created and applied to the currently displayed structure. As the effectivity criteria set in the header, and in the revision rule are in sync, you may lose any earlier **configuration done using the revision rule**.

## Configure a structure using a range of date effectivities containing an end item

### Procedure

1. Search for and open the structure that you want to configure with date range effectivity.
2. Select the structure and click **Date** > **Date Range**.




3. In the **Date Range Effectivity** panel, click **New**.
4. In the **Name** box, specify a name for the date range effectivity.
5. In the **Start** box, specify a start date for the date range effectivity.
6. In the **End** box, choose one of the following values:
   - **Date**
   - **UP - All Future Dates**
   - **SO - Stock Out**
7. In the **End Item** field, click **Add**.
8. Select an end item from the **Palette** tab or search for and select an end item from the **Search** tab.
9. Click **Add**.
10. Click **Apply**.

## Edit or remove group effectivity of units and end items

### Procedure

1. Select the structure which is configured with group effectivity and click **Units** > **Groups**.
2. In the **Group Effectivity** pane, select the effectivity that you want to edit or remove.

   The **Remove Group Effectivity** and **Edit** icons are displayed.

   a. To remove the group effectivity, click the **Remove Group Effectivity** icon to remove the effectivity.

   b. To modify the group effectivity, click **Edit** and then change the properties you want to edit.
3. Click **Save**.

## Edit or remove range of dates

### Procedure

1. Select the structure which is configured with group effectivity and click **Date** > **Date Range**.
2. In the **Date Range Effectivity** pane, select the date range from the **Search** tab.

3. To edit the date range, click **Edit** and modify the dates and click **Apply**.
4. To remove the applied date range, select **Today** from the header.

# 17. Filter structures (Smart Discovery)

## About filtering structures

You filter the product structures to work with the specific product definitions. You can filter a product structure by spatial proximity and volume, creation or modification period, properties (like owner or type), and partitions. You can also filter the product structure using Copilot if Teamcenter Artificial Intelligence Services are installed in your Teamcenter environment

You **filter a structure** so that you can work with a specific product definition, which is comparatively smaller in size than the entire structure.

You can filter structures if your Teamcenter setup has the Context Management User and Smart Discovery licenses.. Additionally, you can filter only those structures that are indexed using Smart Discovery Indexing. Such structures can be identified by the *Indexed* indicator . Hovering over the indicator shows the start time of the last successful index update indicating whether the modified structure is available for filter. You can further visualize the 3D design data of those structures if they are indexed for massive model visualization (MMV) using Smart Discovery Indexing. Such structures can be identified by the *Massive model visualization indexed* indicator .

The different ways to filter a product structure are as follows:

- **Filter by spatial proximity and volume**

  Filter a structure by locating parts that are at a certain proximity from a specific part. You can also filter a structure by locating parts that are inside or outside a certain spatial zone. The spatial zone is represented by a bounding box. You can also choose to locate parts that are on the edges of the bounding box.

- **Filter by period**

  Filter a structure by selecting a date range. You can select the date range when the structure was created or modified.

- **Filter by properties**

  Filter a structure by specifying certain properties such as *owner*, *type*, and *occurrence notes*.

  You cannot find an element within a structure by specifying occurrence properties. You can do so by specifying item revision properties. However, you can filter a structure by specifying both occurrence properties and item revision properties.

- **Filter by partitions**

  Filter a structure by selecting partitions.

## Filter a structure using Copilot

Apart from using the regular **filters**, you can also **filter your currently open structure** using Copilot if Teamcenter Artificial Intelligence Services are installed in your Teamcenter environment. If the structure is indexed using Smart Discovery Indexing, you can filter the structure in Copilot:

- **by spatial proximity and volume**

  If the spatial properties exist for the structure, you can use Copilot to filter the structure by locating parts that are at a certain proximity from a specific part. You can also filter a structure by locating parts that are inside or outside a certain spatial zone. The spatial zone is represented by a bounding box. You can also choose to locate parts that are on the edges of the bounding box.

- **by properties**

  You can use the following properties only in Copilot to filter the structure: *name*, *last modified*, *owning group*, *owning user*, *quantity*, *sequence number*, *description*, and *item ID*.

> **Tip**
>
> When you filter a structure using Copilot, provide a single instruction at a time.

You cannot filter the structure in Copilot by period and partitions.

# Create hierarchy for freeform zones

Create a hierarchy for freeform zones to allow for efficient filtering and searching within your product structure. Engineers can use this hierarchy to identify components that intersect with specific spatial zones, such as maintenance access areas.

Freeform zones are spatial regions within a product structure that help engineers visualize, filter, search, and collaborate. You can obtain the JT files that contain the freeform zones already defined in CAD software, for example, NX, and then create a hierarchy to associate zones with relevant product components and assemblies.

In Teamcenter, a freeform zone hierarchy is created using a secondary component of type **Spatial Zone Container** at the top, **Item** or **Design** for middle nodes, and **Spatial Zone** for the lowest-level (leaf) child items. Each assembly can contain one such freeform zone hierarchy.

> **Example**
>
> A coffee vending machine product structure can have separate freeform zone hierarchies for Brewing Unit Assembly, Water System Assembly, and Waste Management Assembly.

### Prerequisites

- JT files containing freeform zones are available.
- You are the administrator of the product structure and your role has the permission to create the freeform zone hierarchy for the structure.
- The Teamcenter environment has the Smart Discovery Indexing license.

### Procedure

1. Obtain the JT files that contain the freeform zones already defined in CAD software, for example, NX.
2. Search for a structure, and click **Open**.
3. Expand the structure and select an assembly or sub-assembly for creating the freeform zone hierarchy.
4. Click **Add** > **Secondary Component**.
5. In the Add Secondary Component panel, perform the following steps:
   a. In the **Secondary Component Type** list, select **Spatial Zone Container**.
   b. Enter the name and description for the freeform zone container and click **Add**.
6. Select the newly added spatial zone container, and click **Add** > **Child** to create the hierarchy.
7. In the Add Child panel, perform the following steps:
   a. In the **Type** list, perform one of the following actions: .
      - If you want to create a lowest-level node in the hierarchy, select the **Spatial Zone**
      - If you want to create different levels in the hierarchy, select **Item** or **Design** as per your requirement.

   b. Enter the required details and click **Add**.

      Add more child items to create the freeform zone hierarchy, considering your business requirement.
8. Perform the following steps to attach the JT file to each lowest-level child of the hierarchy.
   a. Select the lowest-level child of the hierarchy and click the **Attachments** tab.
   b. In the **Files** section, click **Add**.
   c. In the Add panel, upload the JT file and click **Add**.

### Results

The freeform zone hierarchy is available in the **Spatial** section of the **Filter** panel for engineers, allowing them to filter and search the product structure content.

# Filter a structure

You filter a structure so that you can work with a specific product definition, which is smaller in size than the entire structure.

### Restrictions and limitations

You can filter structures if your Teamcenter setup has the Context Management User and Smart Discovery licenses.. Additionally, you can filter only those structures that are indexed using Smart Discovery Indexing. Such structures can be identified by the *Indexed* indicator. Hovering over the indicator shows the start time of the last successful index update indicating whether the modified structure is available for filtering. You can further visualize the 3D design data of those structures if they are indexed for massive model visualization (MMV) using Smart Discovery Indexing. Such structures can be identified by the *Massive model visualization indexed* indicator.

### Procedure

1. Search for a structure, and click **Open** to open it.

2. Click **Filter**.

3. In the **Filter** panel, click **Settings**, enable or disable the following options, and click **Apply**:

   - **Auto-update filters**: By default, this option is enabled, and the structure is filtered in real-time as you apply each filter. When not enabled, you select all filters first and then apply them on the structure.

   - **Filter/Append Elements with Children**: By default, this option is enabled, and the children of the structure element selected as a filter are automatically added to the filter criteria. When not enabled, the children are not added to the filter criteria.

4. In the work area, right-click a structure element to choose a filter. The filters that you can choose and how these filters are applied on the structure depends on whether the administrator has configured the advanced search or not.

   - *SCENARIO 1:* The administrator has not configured the advanced search.

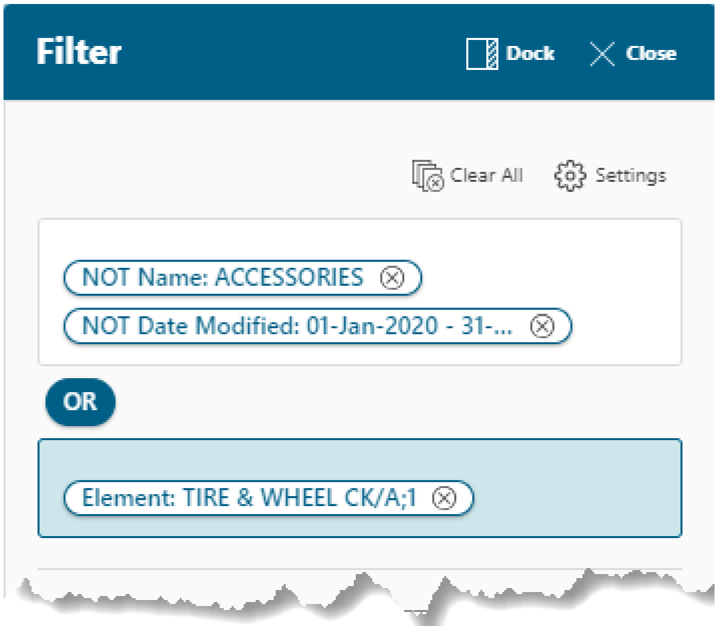
     If the administrator has not configured the advanced search options, you can use the following filters:

| Filter | Description |
|---|---|
| **Filter Selected Elements with children** | Use this filter to narrow down the structure by including the selected elements along with the children. The elements and the children are added to the filter criteria using the **AND** operator. |

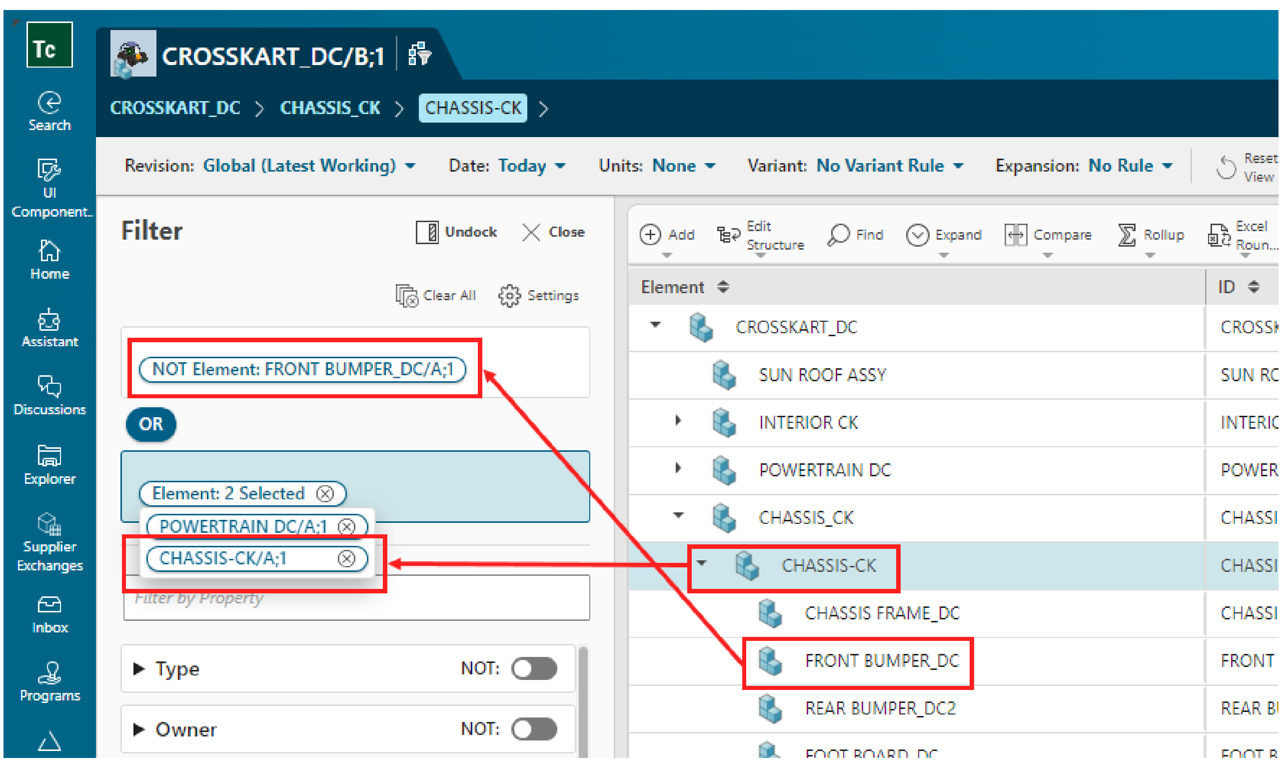
| Filter | Description |
|---|---|
| | This filter option is displayed when the **Filter/ Append Elements with Children** setting is enabled. |
| **Filter Selected Elements** | Use this filter to narrow down the structure by including only the selected elements and not their children. The selected elements are added to the filter criteria using the **AND** operator. |
| **Append Selected Elements with children** | Use this filter to broaden the filtered structure by including the selected elements along with the children. The elements and the children are added to the filter criteria using the **OR** operator.<br><br>This filter option is displayed when the **Filter/ Append Elements with Children** setting is enabled. Additionally, this option is displayed only when after you first apply another filter. |
| **Append Selected Elements** | Use this filter to broaden the filtered structure by including only the selected elements and not their children. The selected elements are added to the filter criteria using the **OR** operator.<br><br>This filter option is displayed when the **Filter/ Append Elements with Children** setting is disabled. Additionally, this option is displayed only when after you first apply another filter. |
| **Exclude Selected Elements** | Use this filter to narrow down the structure by excluding the selected elements along with the children. The elements and children are added to the filter criteria using the **NOT** operator. For example, if you want to exclude an assembly and its children parts, select the assembly and click **Exclude Selected Elements**. |

The filters are applied in groups. If the advanced search options are not configured, a maximum of three groups with the **OR** operator can be created and you can use the same filter expression in NX.

The filters applied using the **OR** operator are always placed last in the filter criteria.




Sometimes, when you append selected elements with children, a filter applied using the **OR** operator overrides a filter applied using the **NOT** operator. For example, if you want to exclude **FRONT_BUMPER_DC** from the structure but want to append its parent **CHASSIS-CK** with children, **FRONT_BUMPER_DC** is automatically included in the structure.



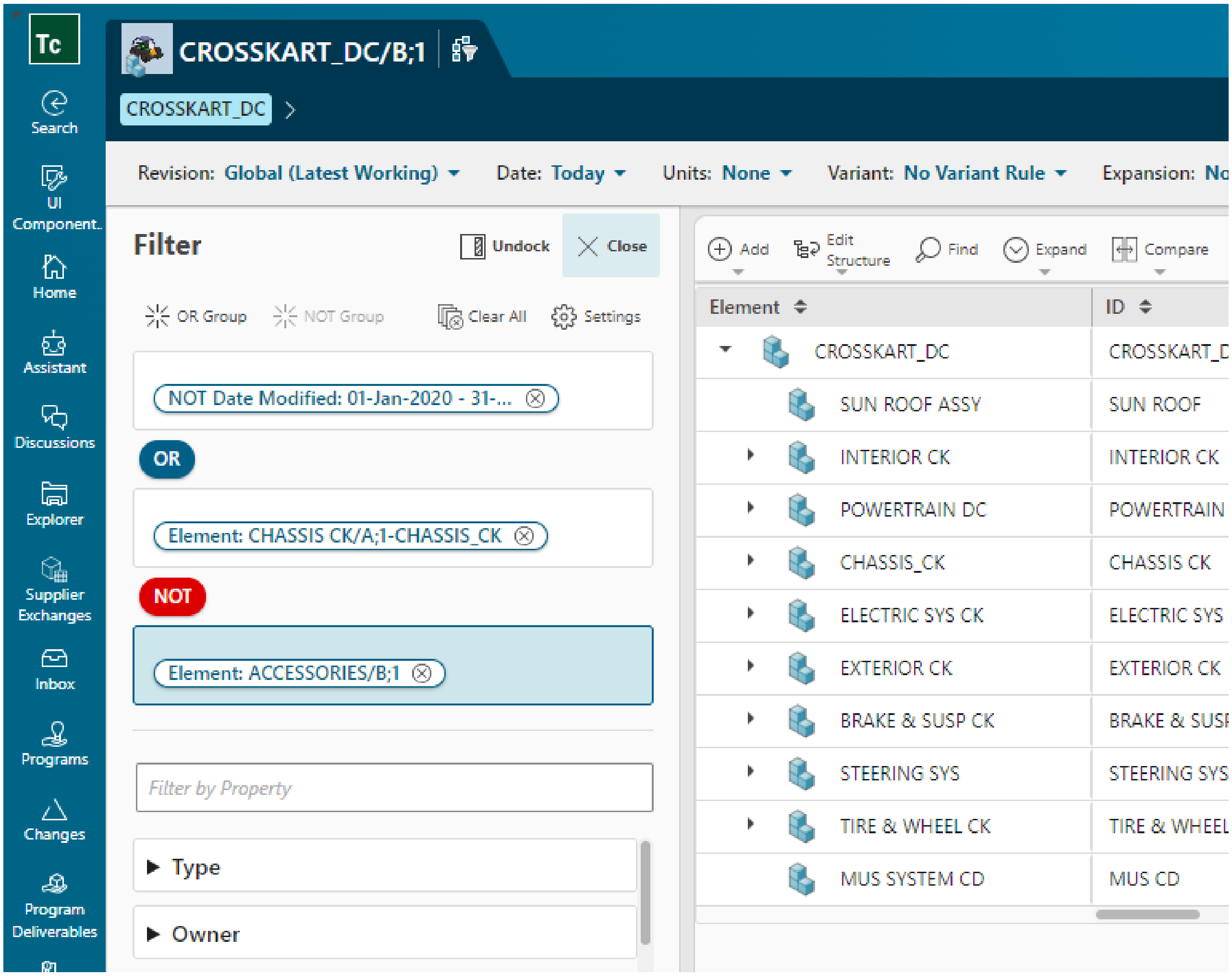

- *SCENARIO 2:* The administrator has configured the advanced search.

  If the administrator has configured the advanced search options, you can use the following filters:

| Filter | Description |
|---|---|
| **Filter Selected Elements with children** | Use this filter to narrow down the structure by including the selected elements along with the children. The elements and the children are added to the filter criteria using the **AND** operator.<br><br>This filter option is displayed when the **Filter/ Append Elements with Children** setting is enabled. |
| **Filter Selected Elements** | Use this filter to narrow down the structure by including only the selected elements and not their children. The selected elements are added to the filter criteria using the **AND** operator. |
| **Exclude Selected Elements** | Use this filter to narrow down the structure by excluding the selected elements along with the children. The elements and children are added to the filter criteria using the **NOT** operator. For example, if you want to exclude an assembly and its children parts, select the assembly and click **Exclude Selected Elements**. |

If the advanced search options are configured, the **OR Group** and **NOT Group** buttons are available in the **Filter** panel. You can use these buttons to create the filter expression which is a combination of the **OR** and **NOT** groups. You can create multiple **OR** groups in the expression but only one **NOT** group can exist in the filter expression.

When the filter expression is a combination of **OR** groups and a **NOT** group, the filter shows all results from the **OR** groups by excluding the result from the **NOT** group. For example, suppose that you want to filter the structure that was not created between 1 January 2020 through 31 December 2020, contains **CHASSIS_CK**, and does not contain **ACCESSORIES**. In such a case, you can use the filters shown in the following image.




If the administrator has configured a wild card search in the Teamcenter environment, in a text-based filter, you can use operators like **Contains**, **Does not contain**, **Equals**, **Does not equal**, **Begins with**, **Ends with**, and **Like** operators. For example, suppose that you want to search for the parts in the structure that are owned by John. In such a case, use the **Contains** operator and then type **John** in the **Owner** filter. Similarly, if you want to search for all the parts that are not owned by John, enter **John** and then change the operator to **Does not contain** in the **Owner** filter.

**Tip**

To use a negative operator, use its positive operator first and then change to its negative one. For example, in order to use **Does not contain** operator, first select the **Contains** operator, enter a value in the filter, and then change the operator to **Does not contain**.

In a numeric filter, you can use operators like **Greater than or equals**, **Greater than**, **Less than or equals**, **Less than**, **Equals**, **Does not equal**, **Range inclusive**, and **Range exclusive**.

After you select a value in a filter, the values in the remaining filters are dynamically populated. For example, imagine that the structure contains parts that are created by John who belongs to the Engineering group. In such case, if you select **John** in the **Owner** filter, the values in the **Group ID** filter are dynamically populated and only show **Engineering**. If advanced filters are configured, the values in the filters are dynamically populated even when you switch between groups.

You can use an asterisk (*) to filter the parts in the structure. For example, consider that the structure contains parts owned by different engineers whose names are John, Johan, and Johnnie. In such a case, if you want to search the parts that are owned by all these engineers, use the **Like** operator, and then type **Joh*** in the **Owner** filter.

The advanced search might not provide you with the intended results.

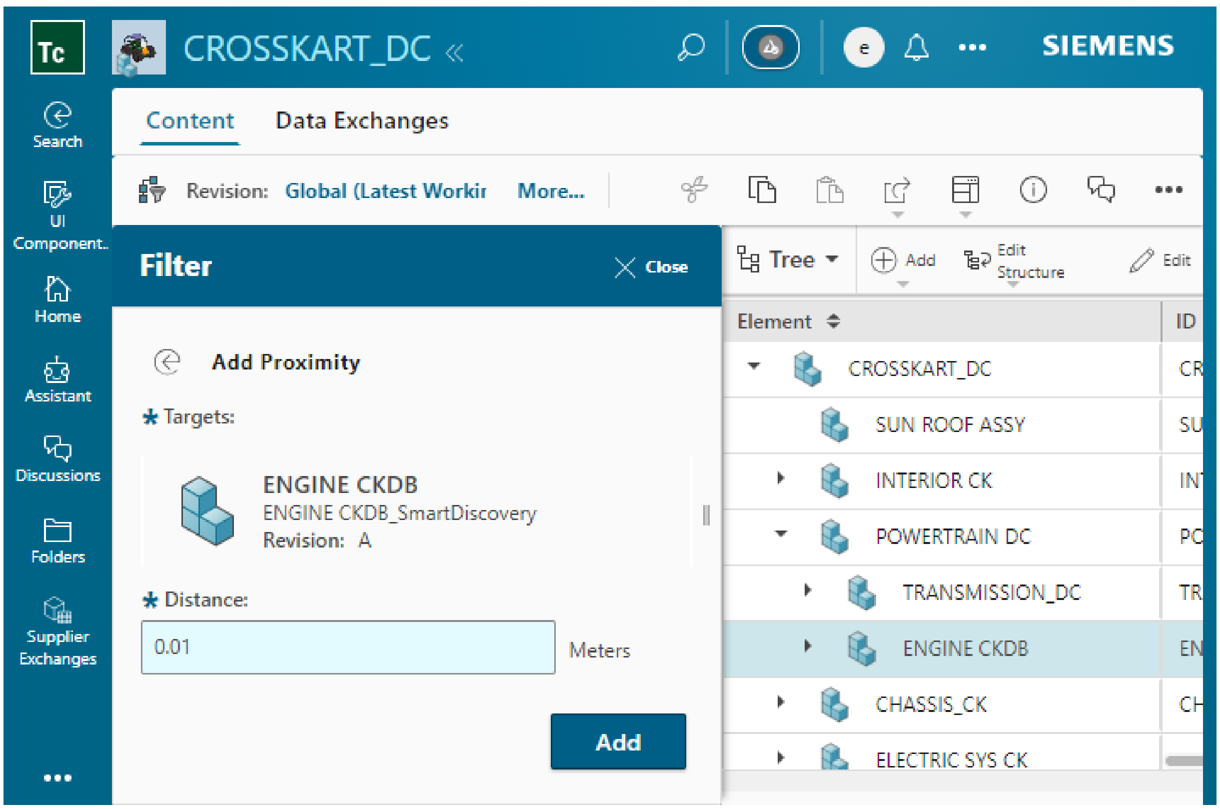
5. In the **Filter** panel, apply additional filters, as required:

   **To filter by spatial proximity:**
   You require the Smart Discovery license to filter by proximity.

   a. From the work area, select a structure element to locate other structure elements that are at a certain proximity from it.

   b. In the **Filter** panel, select **Proximity** from the **Spatial** section.

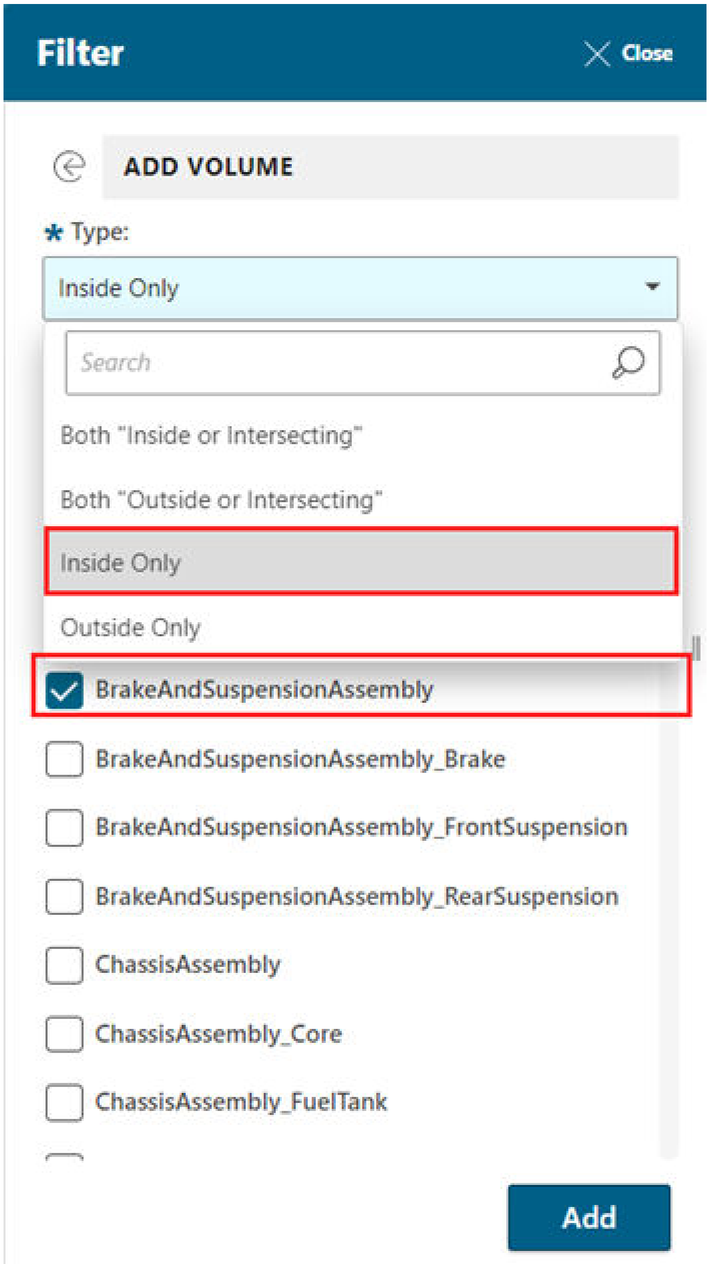
   c. Enter the proximity in **Distance** and click **Add**.




   **To filter by spatial volume:**
   You require the Smart Discovery license to filter by volume.

   a. In the **Filter** panel, select **Volume** from the **Spatial** section.

   b. Select the volume for which you want to perform the search. For example, you can choose to locate parts that are inside the brake and suspension assembly.




c. Click **Add**.

**To filter the structure by spatial freeform zone:**

If the structure administrator has set up the freeform zone hierarchy and configured its display in the **Filter** pane, use the following steps:

a. In the **Filter** panel, select **Freeform Zone** from the **Spatial** section.

b. In the **Add Freeform Zone** section, select a type from the list.

For example, **Inside or Intersects**.

c. In the freeform zone hierarchy, search for and select the zones for filtering.

For example, you can choose to locate parts that are inside the brake and suspension assembly.

d. Click **Add**.

**To filter the structure by properties:**

**Note**

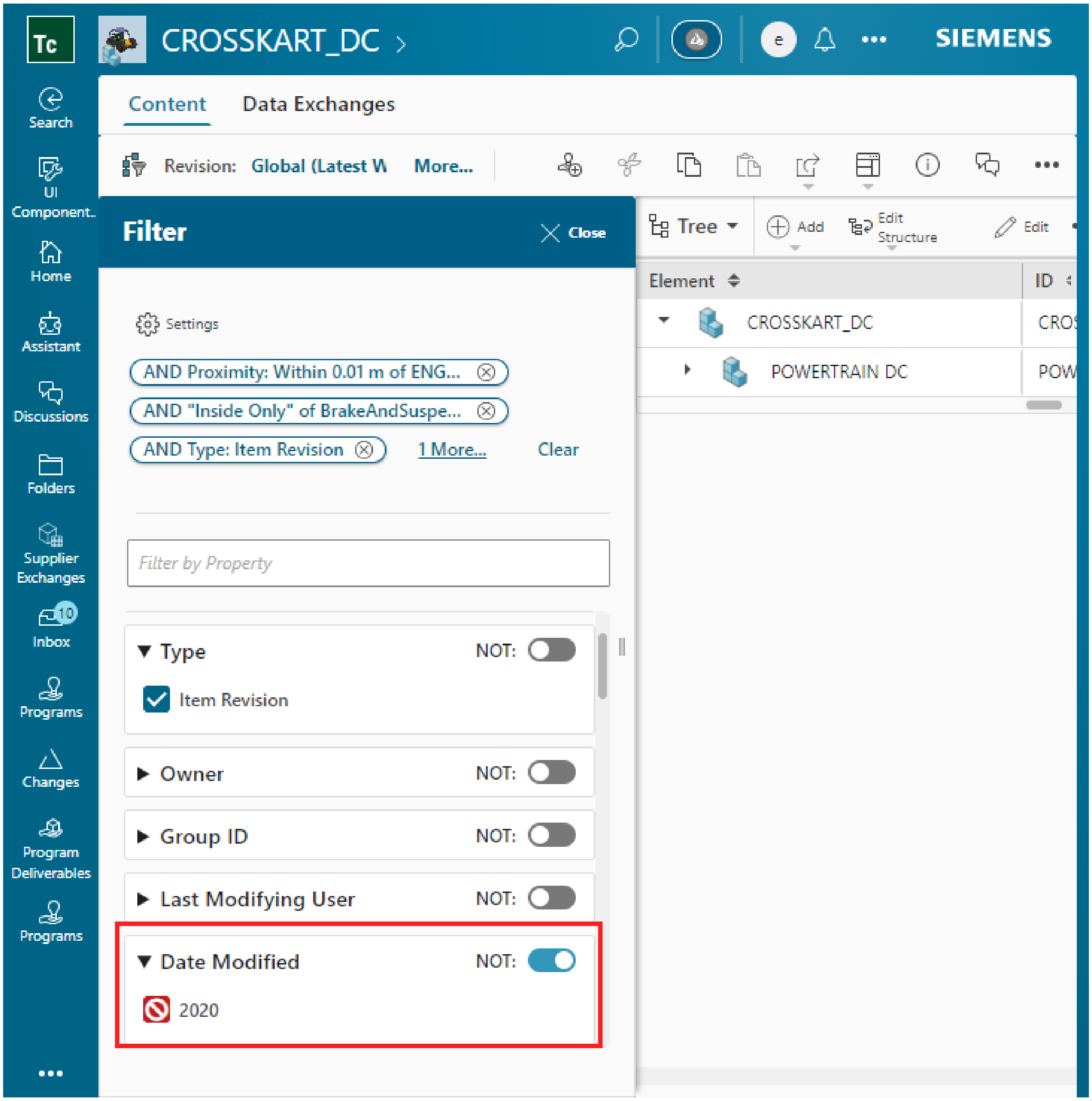
The top element in a structure or BOM is not considered for the **Find** or **Filter** actions. Therefore, the properties of the top element are not available while you perform these actions.

**Note**

If you do not have the Read access privilege for the structure element selected in the work area, you are not be able to view the value of its indexed occurrence properties in the **Filter** panel.

a. Select the attributes, such as **ID**, **Occurrence Type**, and **Date Modified**, by which you want to further filter the structure In the **Filter** panel. For example, include all elements that are of the type, **Item Revision**.

   To exclude a property as a filter, select the property and enable the **NOT** option. For example, to exclude all elements that were not modified in 2020, select **2020** and enable the **NOT** option in the **Date Modified** section.




**To filter by partitions:**

You require the Smart Discovery license to apply a proximity filter.

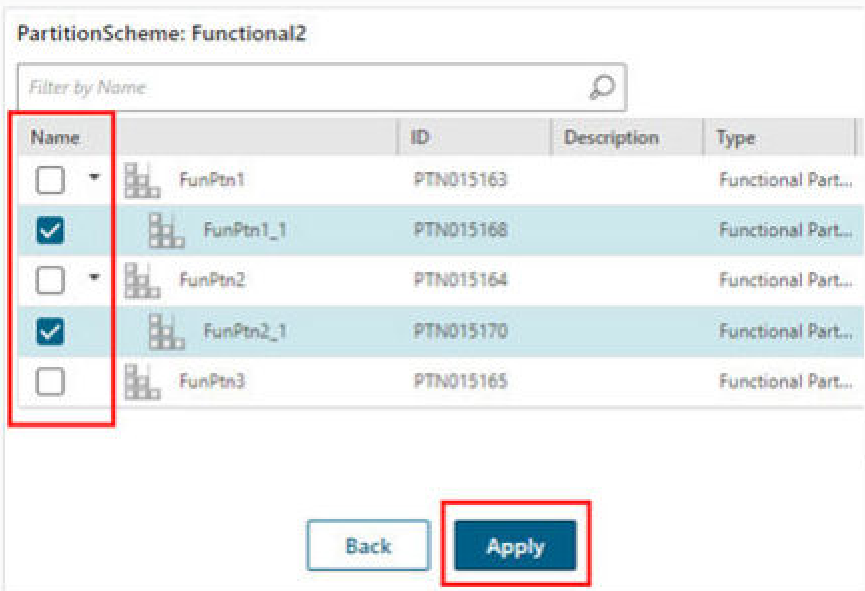
**Tip**

To quickly filter a structure by a partition using a shortcut menu, right-click the partition and then click **Filter Selected Elements with children**. However, to filter a structure by a partition using the **Filter** panel, follow step **a** and later in this section.

If you use the shortcut menu option, the structure is narrowed and the selections are included with children as filters. The selections are added to the filter criteria using the AND operator. This action is synchronized with the method of filtering by using the **Filter** panel.

a. In the **Filter** panel, select a partition scheme.

**b.** Select the partitions that contain the required structure elements and click **Apply**.




If a selected partition has child partitions, the elements of the child partitions are also considered.

As you select different filters in the **Filter** panel, an expression is built at the top.

To remove any filter from the expression, click ⊗ next to it.

To remove all filters, click **Clear All**.

6. If you disabled auto-update, click **Filter** to filter the structure.

> **Note**
>
> Filtering by spatial proximity and volume supports one-to-one, one-to-many, and many-to-one alignments.

### What to do next

To view all the structure elements including the ones that are not displayed due to the applied filters, click **Configure** > **Show Excluded by Filtering**.

You can save the filtered structure in a ***workset*** or ***session*** for easy retrieval.

## Reset a filtered structure

Reset removes all filters and sets the configuration criteria to its default settings.

### Procedure

1. Search for the structure.
2. Click **Open** to open the structure.
3. Select **Configure** > **Reset View** to remove the filters applied to the structure.

# 18. Configure structures

## About configuring structures

Understand the various methods for configuring and viewing specific product structure configurations, including using selection, proximity, effectivities, closure rules, revision rules, variant rules, and custom configurations.

To view a specific configuration of the structures, you can configure a structure using:

- **Selection**
- **Proximity**
- **Effectivities**
- **A closure rule for expansion**
- **Revision rules**
- **Variant rules**
- **A custom configuration**

## Configure a structure by selection

You can configure a structure based on the Product Configurator-authored variants used in a structure element. This method of configuring a structure is called *configure by selection*. Based on the variability of the selected elements, other elements are configured in or out of the structure, which enables you to validate all possible and valid variant combinations according to your product configurator-authored variant definitions.

### Restrictions and limitations

You can configure a structure by selection only if the structure is indexed by using Smart Discovery Indexing and your Teamcenter setup has the Context Management User license.

### Procedure

1. Search for a product structure on which variability is set by using Product Configurator variants.
2. Open the structure.
3. Locate a structure element on which variability is set. You cannot configure the structure by selecting its root element.
4. Click **Configuration Settings** > **Configure by selection**.

# Configure a structure by proximity

You can configure a structure based on the legacy classic variants used in a structure element. This method of configuring a structure is called *configure by proximity*. Based on the variability of the selected elements, other elements are configured in or out of the structure.

### Restrictions and limitations

You can configure a structure by proximity only if your Teamcenter setup has the Context Management User license.

### Procedure

1. Search for a structure on which variability is set by using legacy variants.
2. Click **Open** to open the structure.
3. Click **Filter** .
4. Select a structure element from the work area. You cannot configure the structure by selecting its root element.
5. In the **Filter** panel, click **Proximity**.
6. Enter the proximity in **Distance** and click **Add**.

# Configure a structure by effectivity

You can configure product structures using either date effectivity (specifying a single date, a date range, or an end item) or unit effectivity (entering a specific unit or using groups). You can understand how these configurations impact the displayed structure.

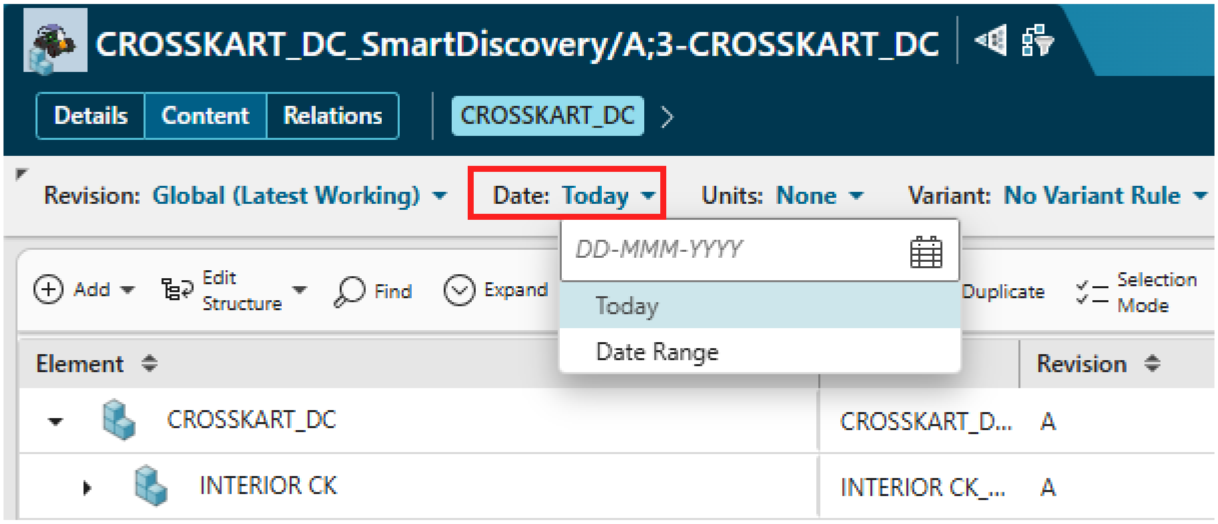
You can configure a structure by **date** effectivity and **unit** effectivity.

## Configure a structure by date effectivity

### Procedure

1. Search for and open the product to be configured.
2. Perform one of the following actions:
   - Select **Date** in the header.
   - Click **Configure** to display the **Configure** panel.

In the **Date** section, you can select **Today** or a specific date from the calendar. You can also add a range of dates.




To add a new range of dates:

a. Select **Date Range**.

b. In the **Date Range Effectivity** panel, you can add a new range of dates or locate the existing date effectivities from the **Search** tab.

c. To author a new range of dates inside the **New** tab:

   i. Give a suitable name to the new date range.

   ii. Select the **Start** date from the calendar.

   iii. Select the **End** date from the calendar. If applicable, you can select **UP** (all future dates) or **SO** (stock out).

   iv. Click **Apply**.

(Optional) To add an end item in the Configure panel:

a. In the **End Item** section of the **Configure** panel, click ⊕ to add the end item. By default, the top-level element is displayed in this section.

b. Search for an end item and select it or choose an end item from the **Palette** tab.

   The end item that you selected is displayed in the **End Item** section on the Configure panel.

3. If you are using the Configure panel, click **Configure** to apply the date effectivity.

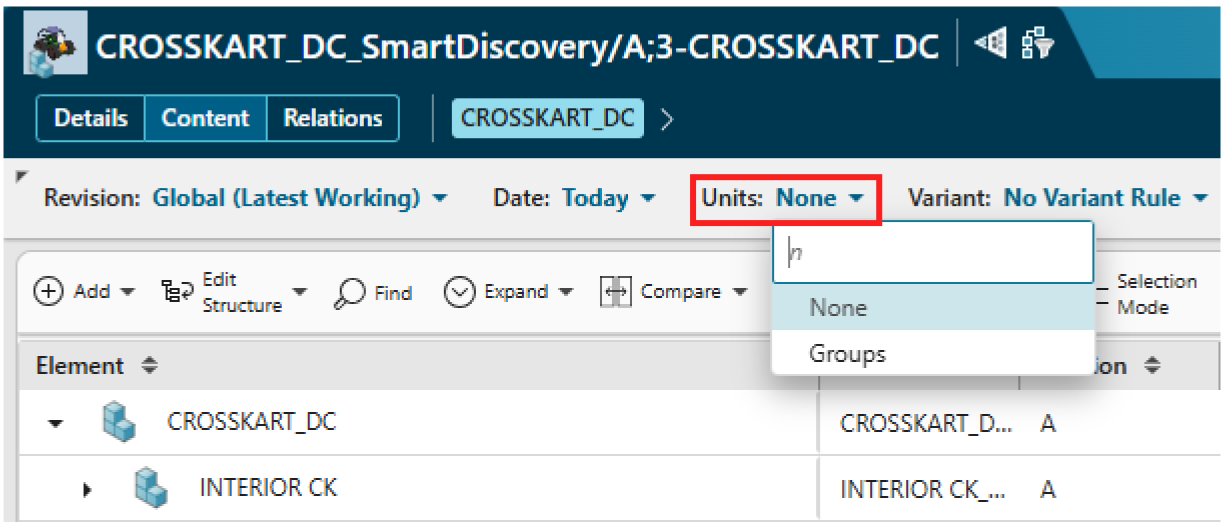
> **Note**
>
> In the Configure panel, if you select two configurations that are of the same type or nature but with different values, the latest configuration is applied. For example, suppose that you first selected a date range from January 1, 2025 through June 30, 2025 for applying the date effectivity, and then selected a revision rule that already contained an effectivity date from February 1, 2025. In such a case, the structure is configured using the revision rule that contained the effectivity date from

February 1, 2025 because the revision rule with the effectivity date is the latest configuration that is applied to the structure.

## Configure a structure by unit effectivity

**Procedure**

1. Search for and open the product to be configured.
2. Perform one of the following actions:
   - Select **Unit** in the header.
   - Click **Configure** to display the **Configure** panel.




You can enter a specific integer or select the following options in unit effectivity:

| | |
|---|---|
| **None** | To display all the elements without effectivity. |
| **Groups** | To author new group effectivity and apply it. |

3. If you are using the Configure panel, click **Configure** to apply the unit effectivity.

## Configure a structure with a closure rule for expansion

A closure rule holds subsidiary rules that define the objects of interest in a given structure. The rules determine if an object is included in a given structure based on their types, classes, and the relationship between them.

Typically, a Teamcenter administrator creates and adds closure rules.

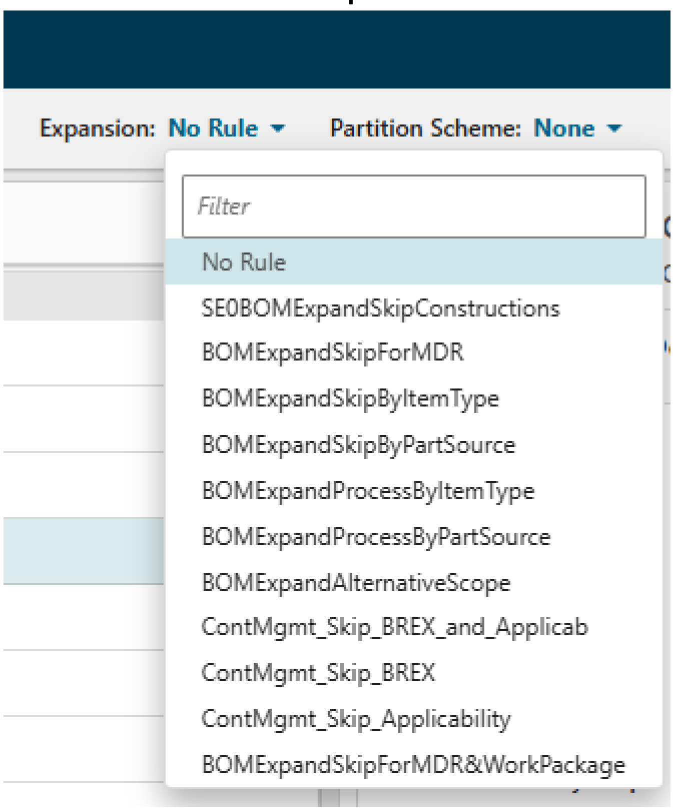
For more information about closure rules, see *Managing closure rules* in *PLM XML/TC XML Export Import Administration* in the Teamcenter documentation.

In Teamcenter:

- You can configure a non-indexed structure with a closure rule to apply an expansion or filtering logic to the structure.
- You cannot create or edit closure rules.
- You can use only static closure rules.

### Procedure

1. Search for and open the structure (only non-indexed structures) to be configured.
2. Perform one of the following actions:
   - Select the rule from the **Expansion** list in the header.

   - Use the Configure panel:
     a. Click **Configure** to display the Configure panel.
     b. In the Configure panel, select a rule from the **Expansion** list and click **Configure**.

### Results

Teamcenter refreshes and displays the configured content based on the applied closure rule.

# Configure a structure with revision rules

## Understanding revision rules

You can use revision rules to select specific revisions of parts and assemblies in a product structure by defining criteria such as working revisions, release status, effectivity (date or unit), and override folders. You can understand how these revision rule entries are evaluated to configure the structure.

You can create and apply revision rules that select the appropriate revision of parts and assemblies in a product structure. A revision rule:

- Selects the working revisions and (optionally) specifies the owning user or group.
- Selects revisions by status (according to status precedence) or the latest revision with any based on the using release date.
- Optionally specifies the effectivity against which the revisions are configured. Effectivity may be specified by date or by unit number.
- Selects revisions in a specified override folder.
- Selects the latest revisions according to the revision ID in the following order: alphanumeric, numeric, or creation date. This selection does not depend on whether the revisions are in the working or released state.

You define each of these criteria with a revision rule entry. A revision rule may contain any number of rule entries, each of which attempts to select a revision according to the specified criteria, for example, the status that the revision should have or the user or group that owns the revision.

Teamcenter evaluates rule entries in the order of precedence until a revision is successfully configured. You can include some entries more than once to define the order of precedence. You can modify the order of the rule entries to change the precedence Teamcenter uses when evaluating the revision rule. Certain rule entries can also be grouped so that they are evaluated with equal precedence.

## Viewing and updating a revision rule

You can view and update revision rule clauses using Date, End item, Latest, Override, Precise, Status, Unit Number, and Working and then configure product structures.

You can configure a structure with a revision rule. You can also view or update revision rule clauses in the context of a structure.

The following revision rule clauses are supported for viewing and updating in Active Workspace:

| Revision | Description |
|---|---|
| **Date** | Loads item revision with a specific date. |
| **End item** | Configures a structure by effectivity with respect to a product, system, or module. For example, you can configure the structure of unit number **110** in product **X400**, where **X400** is the end item. |
| **Latest** | Loads item revisions regardless of whether they are released or not. |
| **Override** | Overrides all item revisions available in an override folder. The revision rule is not evaluated for these item revisions. |
| **Precise** | Loads item revisions in a precise product structure. |
| **Status** | Loads item revisions that are released with a specific status. |
| **Unit Number** | Loads item revisions with a unit number as specified by the user. It is used in combination with the status rule entry with unit number effectivity.<br><br>Typically, a unit number is a property of the end product or a major module of a product. As Teamcenter may manage many units, you typically qualify a unit number entry with an end item entry. |
| **Working** | Loads the working item revision that does not have a release status. |

The **Branch** clause is not supported in Active Workspace, while the **Nested Effectivity** clause is read only.

## Updating a revision rule

You can view and modify the revision rule clauses. When you modify the clauses of an existing revision rule, a new, temporary instance of that rule is created. This modified instance of the revision rule is session-specific and accessible only to the user who modified it.

You can create multiple modified versions of a revision rule within a single user session. However, these temporary instances are not saved permanently and will be lost upon logging off. The original revision rule remains unaffected by these modifications.

If the original revision rule includes a nested effectivity clause, this clause will be visible and usable within the modified temporary revision rule, but it cannot be updated.

# Configure structures with a revision rule

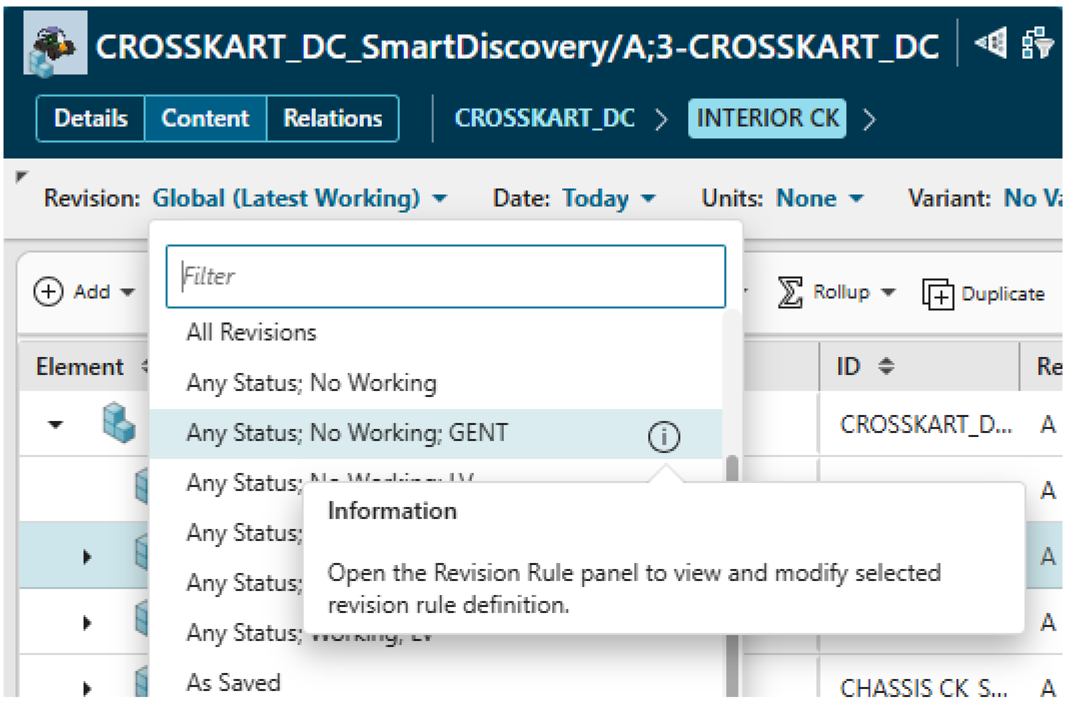
You can configure a structure by using an existing revision rule and by using the **Configuration** panel.

## Configure a structure by using an existing revision rule

### Procedure

1. Search for and open the product to be configured.
2. To apply a revision rule, select the rule from the **Revision** list in the header.

   Depending on how your administrator has configured it, you might see a limited list of revision rules. Click **Show All** to view all revision rules.




   Teamcenter refreshes and displays the configured content based on the new configuration.

## Configure a structure by using the Configure panel

### Procedure

1. Search for and open the product to be configured.
2. Click **Configure** to display the **Configure** panel.
3. Select the rule from the list in the **Revision** section.

   After a revision rule is selected, the Configure panel displays the unit, date, and end item information for the clause, if they exist. You can override any clause information if required.

4. Click **Configure** to apply the revision rule and close the Configure panel.
   Teamcenter refreshes and displays the configured content based on the new configuration. Because the effectivity criteria set in the revision rule and in the header are in sync, you may lose any earlier configuration done using the **effectivity group configuration**.

## Configure a structure with a modified revision rule

You can modify an existing revision rule to configure a structure as per your business requirement.

You must also modify a revision rule to include released part usages in a Usage BOM. In the modified revision rule, you must include the following clauses in the given sequence:

Working

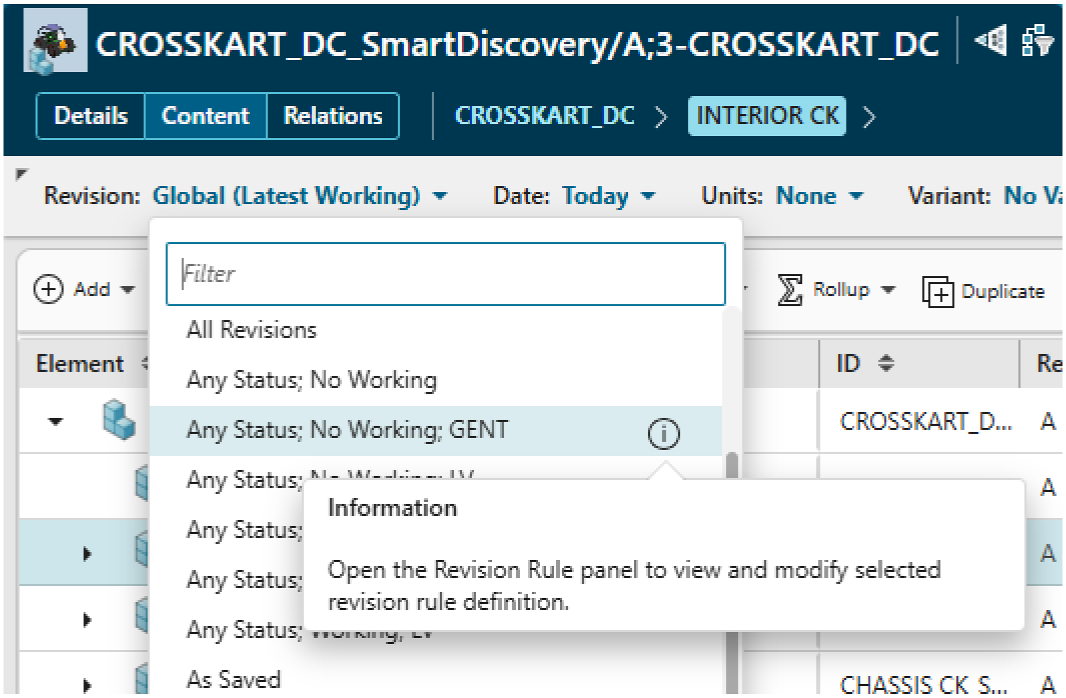
Has (Specific Release Status, Configure by Effective Date)

Has (Any Status, Configure by Release Date)

On applying this revision rule, part usages that are released but are effective as per the active change notice are configured in the BOM irrespective of when the part usage is released. If a part usage has older revisions, the latest released revision is always configured in the BOM. This is because Teamcenter considers the current date (Today) while configuring the part usages in BOM.

### Procedure

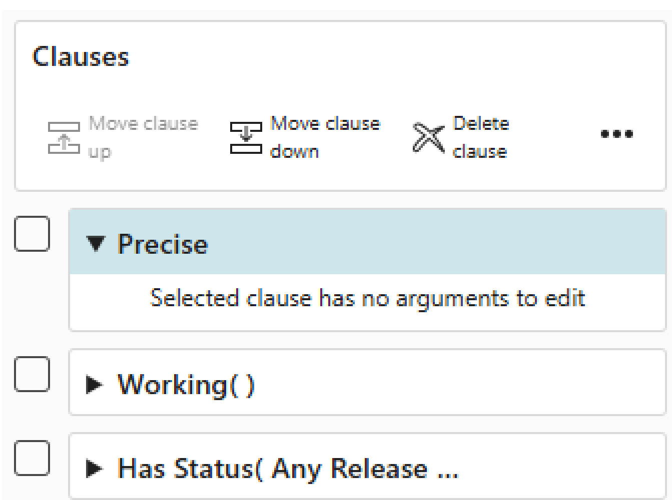
1. Search for and open the product to be configured.
2. To modify a revision rule, perform one of the following actions:
   - In the **Revision** list, go to the rule that you want to modify and click ⓘ **Information**.



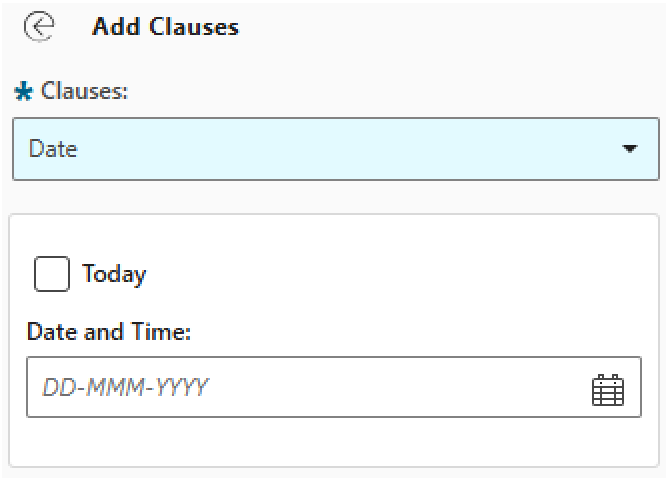

- Click **Configure** to display the **Configure** panel and then click **Information** for a revision that you want to modify.

The revision rule details are displayed with all the clauses.

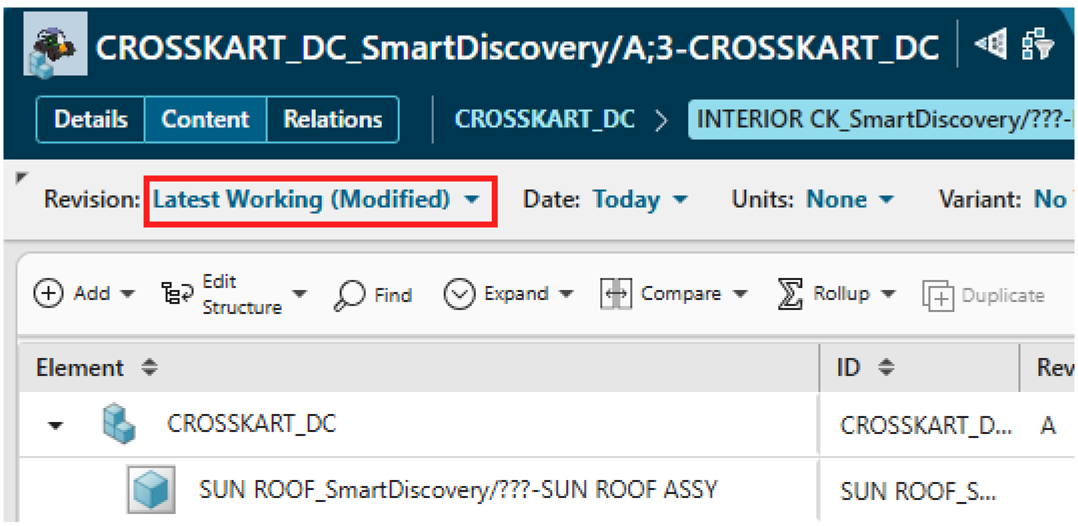
3. To modify the rule, you can add a new clause, delete a clause, or change the order of precedence by moving a clause up or down.




4. To edit a clause, select the clause.
The editable attributes for the clause are displayed. For example, if the **Date** clause is selected, the date-related fields are displayed.




5. To apply the modified revision rule, click **Modify and Configure**.
Teamcenter refreshes and displays the content configured by the modified revision rule. The **Revision** in the header area shows the modified revision rule.




### Results

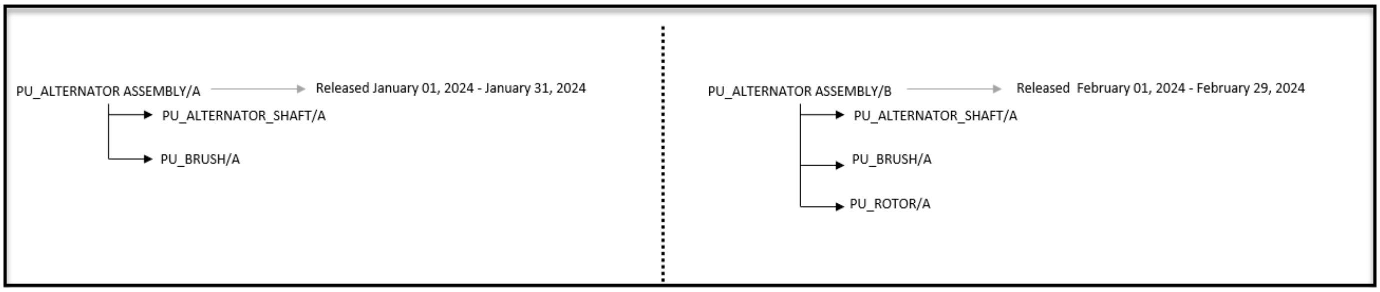
The modified rule is available to you only if you modified the rule. Alternatively, you can use the Configure panel to modify the revision rule clauses and update the **Date** and **Unit** values.

As the effectivity criteria set in the revision rule and in the header are synchronized, you might lose any configuration done earlier using the effectivity group configuration.

## Configuration of a fixed assembly in a Usage BOM

A fixed assembly is a part that has a fixed set of child parts. A fixed part cannot have a variability associated with it. A fixed assembly in a Usage BOM can have multiple revisions released over a period, and each revision can have various parts.

For example, consider the following revisions of the assembly **Alternator**that differ in their part breakdown, with different parts and components.




You open revision A of this assembly for the first time on a future date, namely, April 1, 2024.

The revision rule applied is as follows:

Latest Working (Any status, Configured by Release date)

Date = Today (April 1, 2024)

In this case, as the date for the configuration is later than the valid range for revision A, which is from January 1, 2024, to January 31, 2024, the revision rule date is automatically modified to match the start date already specified for revision A.

The modified revision rule appears as follows:

Latest Working (modified) (Any status, Configured by Release date)

Date = January 1, 2024

It is possible to manually override this revision rule criteria to different values.

If you have set an active change notice, the release date set on this change notice supersedes all other dates for configuration.

## Configure a structure with a revision rule that contains an override clause

You can configure a product structure by using a revision rule with an override clause for various tasks, such as marking the structure as precise, checking temporary changes, performing what-if analyses, and analyzing changes to the structures based on business requirements.

### Procedure

1. Identify the folder that is used as an override folder in the revision rule.

If you cannot, contact your administrator.

2. Search for and copy the required items to the folder.

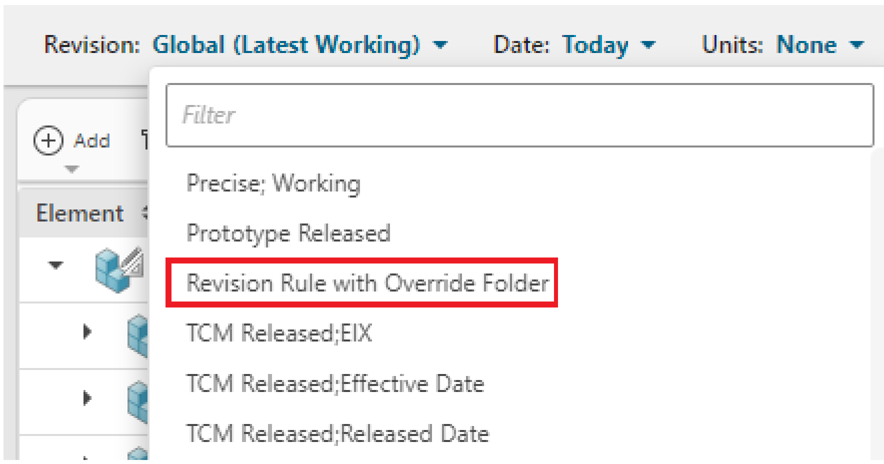
   For example, copy item revisions.

3. Additionally for a Usage BOM, search for and copy the required part usages to the folder.

4. Search for and open the product to be configured.

5. To apply the revision rule with an override clause, select the rule from the **Revision** list.

   The following screenshot shows an example of a revision rule that the administrator has created with an override clause.




   Teamcenter refreshes and displays the configured content using the selected revision rule.

# Configure a structure with variant rules

## Configuring structures with variant rules

You can configure product structures using variant rules, which allow you to define options, their allowed values, and variant conditions to create a single generic product structure that can be tailored for different product variants. You can understand how to manage default options, derived defaults, and variant rule checks.

Manufacturers often want to develop a range of products based on the same generic platform, offering their customers choice, but at the least engineering cost. One approach is to create a single generic product structure that can be configured for each variant of the product offered.

Using Teamcenter, you can define options and the corresponding allowed values and attach them to an item, typically the top-level item in the structure. For example, you can define a Gearbox option with the allowed values of Manual and Automatic. You then attach a logical expression, referred to as a variant condition, to any occurrences of the components that are configurable, for example, the automatic and manual gearboxes. The expression refers to the defined options and can be as complex as necessary.

You choose the desired option values for a configuration and set them in a variant rule. Teamcenter evaluates the variant conditions on the occurrences in the structure against the set option values, and components are configured accordingly. Unconfigured components can be hidden.

A designer can preset the option values in the variant rule. The preset value may be a default option or a derived default option.

- **Default option**

  A default value is a value that you preset for an option. For example, the option **aerial** may have a default value set to **standard**.

- **Derived default option**

  A derived default is a value that is set to a value that depends on a condition. For example, the option **radio** may have a value **stereo** if **car type = GLX**.

For more information about default and derived default options, see the *Working with option defaults* in *Structure Management on Rich Client —Usage*.

Designers can define combinations of option values that are not allowed using the variant rule checks. A variant rule check consists of a condition (for example, **engine = 1200 AND gearbox = automatic**) and an error message (for example, **Incompatible engine and gearbox**). An error message containing the condition and message is displayed if the rule check fails when you configure a structure with the variant rule.

You can view existing variant rules and variant configuration data. Additionally, you can change the configuration data and save the updated rule as a new variant rule. You can apply any existing or updated variant rule configuration to your structure.

To view or update saved variant rules, ensure that the **PSEVariantsMode** preference is set to **`legacy`**.

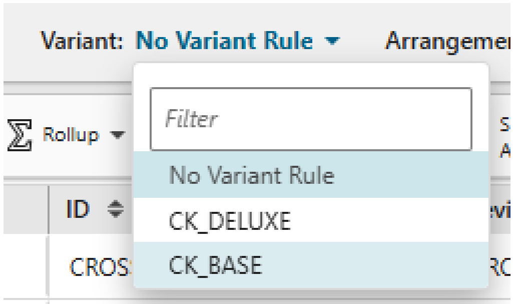
## Configure a structure with a variant rule

In Teamcenter, you can configure a structure with a variant rule. You can view the existing variant rules and variant configuration data. You can also change the configuration data and save the updated rule as a new variant rule. If you have **Partitions for Structure** deployed in your Teamcenter setup, you can set variability on partitions so that you see only those partitions that are relevant to your configuration.

> **Example**
>
> In a structure where no variant rule or no partition scheme is specified, you choose the variant *Sedan* and then choose the partition scheme *Physical*. Now, you see only the partitions that are relevant to this configuration. Specifically, with the variability applied on the partitions, the partitions *Truck* and *Hatchback* are not displayed in the configured structure.

**Procedure**

1. Search for and open the product to be configured.
2. To apply a variant rule, select the rule from the **Variant** list in the header.

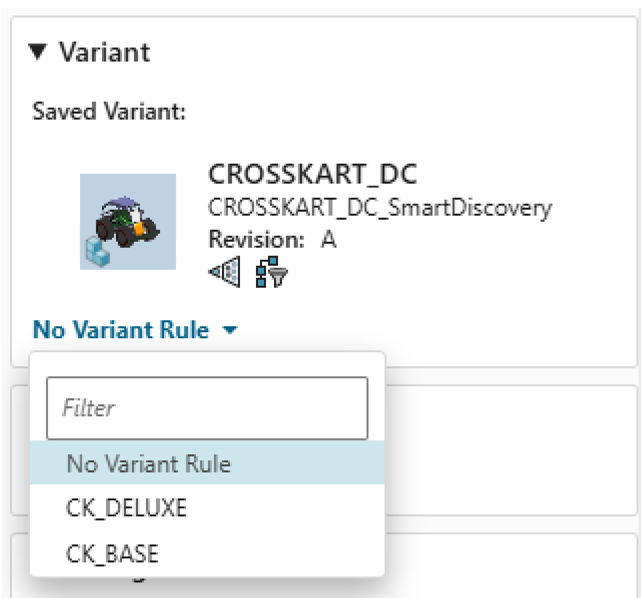



Teamcenter refreshes and displays the configured content based on the new configuration.

## Configure a structure with a variant rule by using the Configure panel

**Procedure**

1. Click **Configure** to display the **Configure** panel.
2. To apply a variant rule, select the rule from the list in the **Variant** section in the **Configure** panel.




3. Click **Configure** to select the variant and close the Configure panel.
Teamcenter refreshes and displays the configured content based on the new configuration.

### Configure a sub-assembly by using the end item's configurator context

Consider that a structure has a configurator context set to it, you open a sub-assembly from that structure, and the sub-assembly does not have its own configurator context set to it. In such a case, the sub-assembly does not carry the configurator context of the parent assembly. However, you can set the end item to the sub-assembly and then configure the sub-assembly by using the end item's configurator context.

#### Procedure

1. Search for and open the sub-assembly to be configured.
2. Click **Configure** .
3. In the **Configure** panel, select the parent structure as the end item.
   The configurator context of the parent structure is selected as the configurator context for the sub-assembly.

#### What to do next

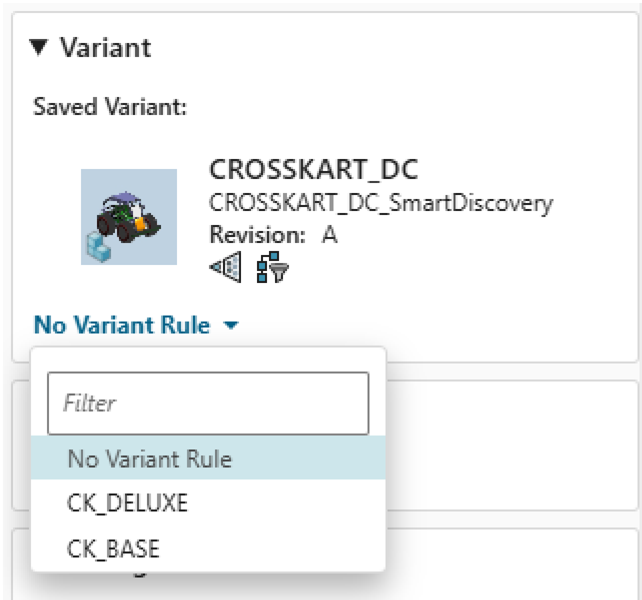
Click the **Variant Configuration** tab and the **Variant Conditions** tab to configure the variants for the sub-assembly.

## View variant rule details

You can view the details of an applied variant rule in the Configure panel, including the default values and the derived default values.

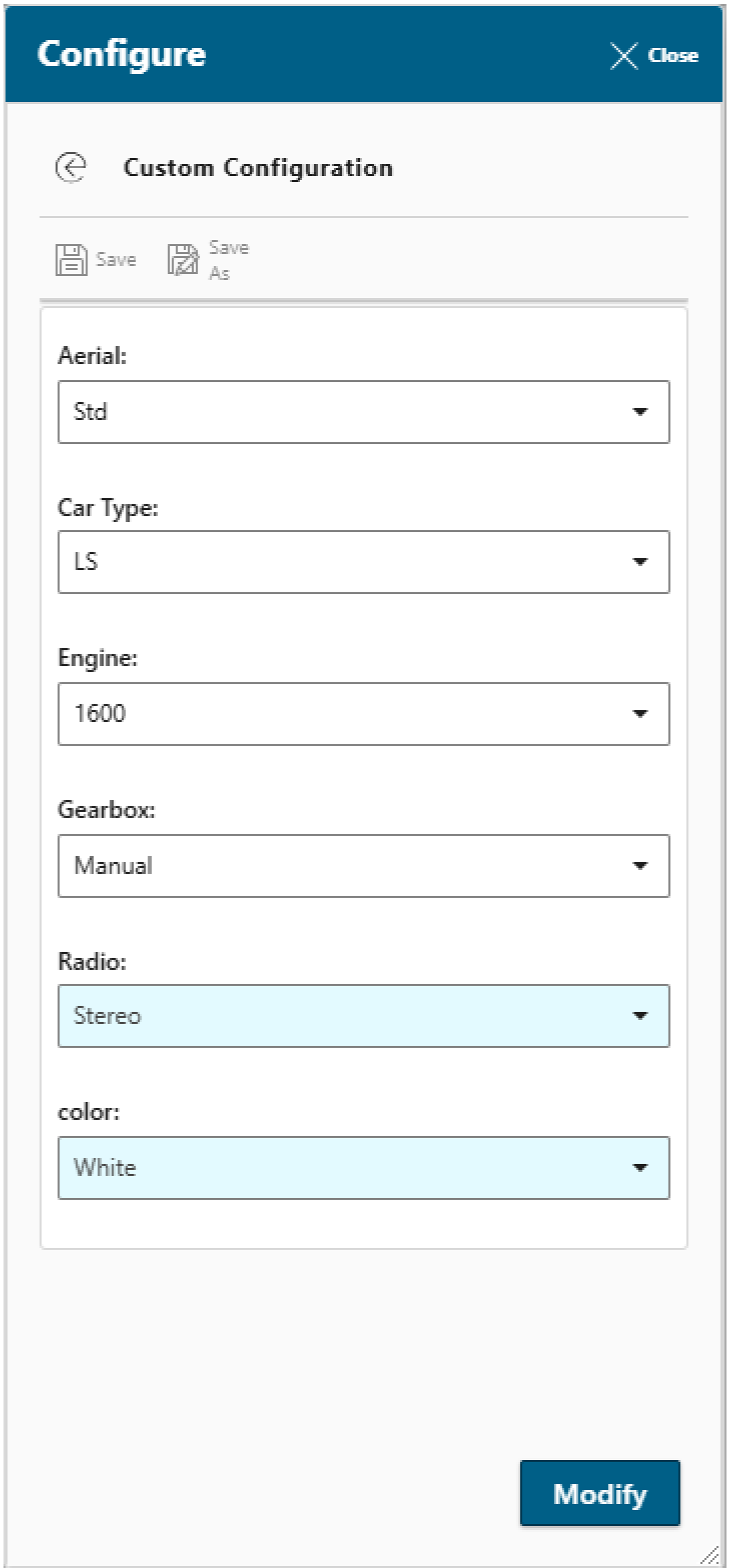
#### Procedure

1. Search for and open the structure to be configured.
2. Click **Configure**  to display the **Configure** panel.
3. In the **Variant** section, select the rule you want to view.




4. To view the details of the applied variant rule, select **Custom** from the variant rule list. The **Configure** panel shows the options and values defined in the variant rule.

   For the options in the variant rule that do not have a defined value, the default values and derived default values are shown.




5. Change a default value and click **Modify** to view the derived default values for the selected default value.

> **Note**
>
> Unless the **Modify** button is clicked, the derived default value is not updated, even after the default value is changed.

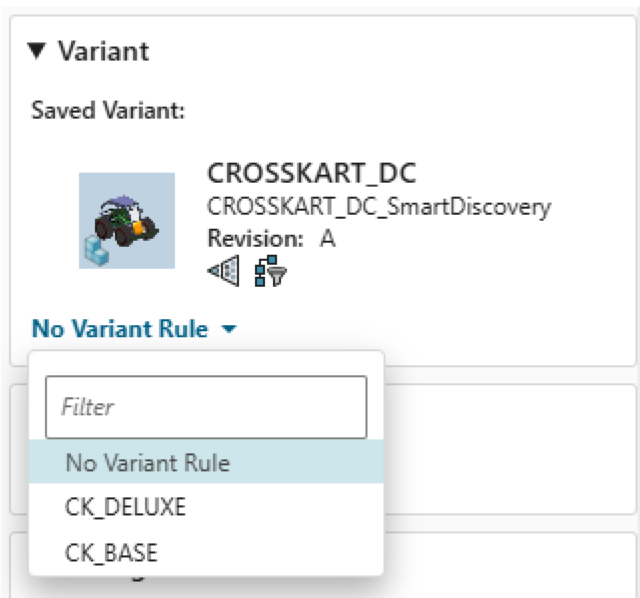
6. Click **Configure** to apply the configuration and close the Configure panel.
   The rule checks for the variant rule only after the **Configure** button is clicked.

## Update a variant rule

Only users with the requisite access privileges can modify a variant rule.

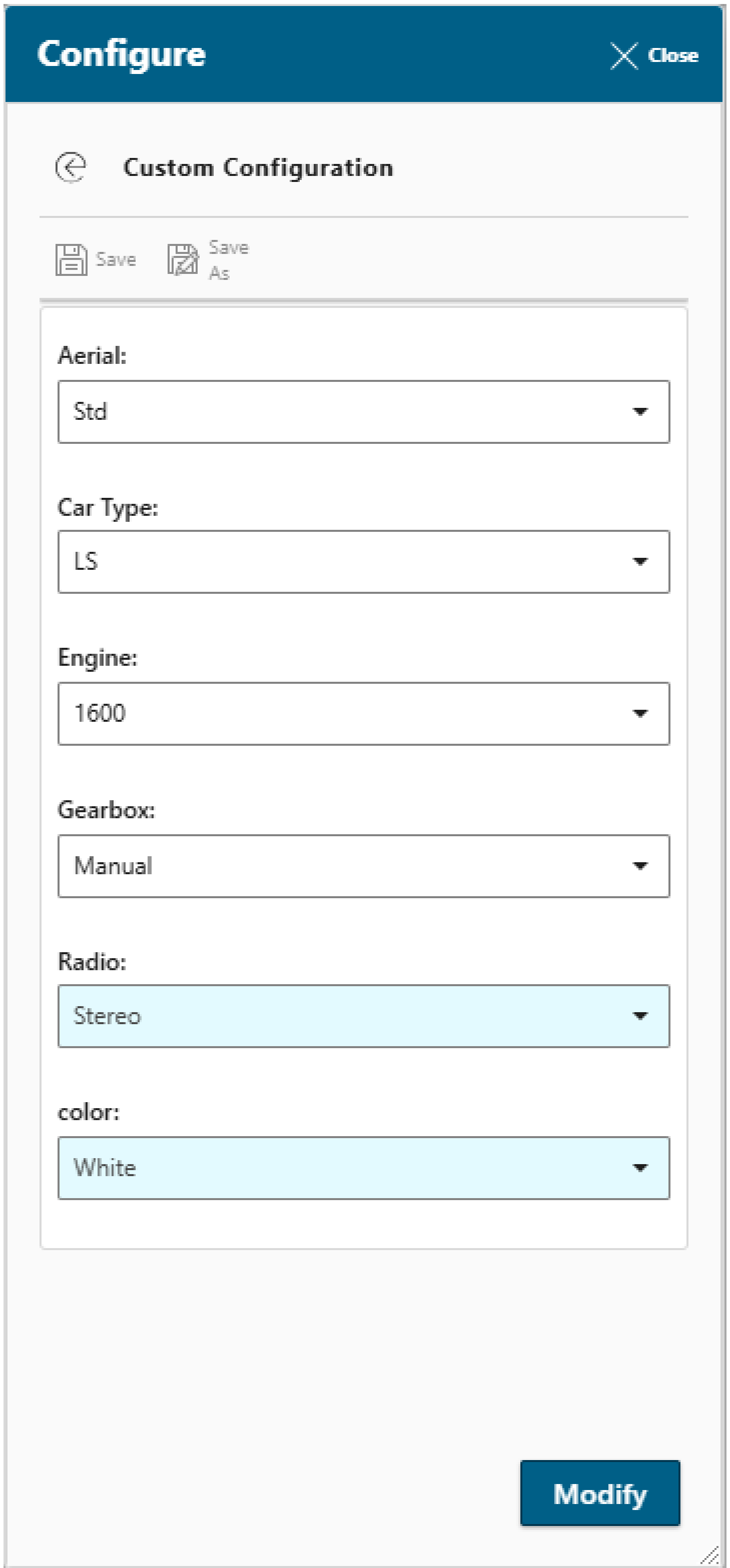
### Procedure

1. Click **Configure** to display the **Configure** panel.
2. Select the variant rule you want to modify from the list in the **Variant** section in the **Configure** panel.




3. To update the variant rule, select **Custom** from the list.
   The **Configure** panel shows the options and values defined in the variant rule.

   For the options in the variant rule that do not have a defined value, the default values and derived default values are shown.




4. Make the changes to the option values as required and click the **Save** icon.
   The variant rule is saved with the updates. You can use it to configure structures.

## Save a modified variant rule as new

You can save a modified variant rule as a new rule. When you save the variant rule as a new rule, you can attach this new rule to your current structure or configurator context for future use in configuring structures.

#### Procedure

1. After **updating a variant rule**, to save it as a new rule, click the **Save As** icon.
2. Enter a **Name** and **Description** and click the **Save** button.
3. To attach this variant rule to the current structure or the configurator context, select the appropriate option.

#### Results

A new variant rule is created that you can use to configure structures.

## Associate a variability scope to a structure

When you associate a variability scope to a structure, you apply the configurator context information to the structure. The configurator context, which contains information such as inclusion and exclusion rules, is used for creating product variants.

#### Procedure

1. Search for the required structure element such as item, part, or product design, and open it.

   Alternatively, from the item revision, part revision, or design revision, navigate to the respective item, part, or product design using the **Relations** tab.
2. In the **Overview** tab, click **Edit** > **Start Edit**.
3. In the **Variability Scope** section, click **Add**.
4. Select the required configurator context using **Search** or **Palette** and click **Add**.
5. Click **Edit** > **Save Edits**.

## Create variant conditions for a part

You can create and update variant conditions for a part through the **Variant Conditions** tab and also inline within a structure.

## Create and update variant conditions using the Variant Conditions tab

### Procedure

1. Search for the product structure that contains the part for which you want to add a variant condition.
2. **Associate a configurator context with the product structure**, if this is not already done.
3. Select the product design and click **Open**.
4. Select the relevant part and click the **Variant Conditions** tab.
5. Click **Show All Families** to view all options.
6. To expand each option family, click **Show Children**.
7. Click **Edit**.
8. Click a cell in the grid next to a variant option to set the variant condition for the part:
   - Click once to display a check mark ✓ to include the variant option as a variant condition.
   - Click twice to display a circle backslash ⊘ to exclude the variant option as a variant condition.
   - Click three times to display a blank cell to indicate that the variant option is not used as a variant condition.
9. Click **Save Edits** to update the variant conditions.

   Multiple variant conditions are connected using a logical **AND** operation to create the final variant condition.

## Create and update variant conditions inline

### Procedure

1. Search for the product structure that contains the part for which you want to add a variant condition.
2. **Associate a configurator context with the product structure**, if this is not already done.
3. Select the product design and click **Open**.
4. Select the relevant part and hover over the **Variant Formula** column and click.
5. Click **Edit**.
6. Type the syntax for the variant condition for the selected part.

   **Tip**

   When you are authoring the variant condition, click **Intellisense** to receive the syntax suggestions to author the variant condition.

Multiple variant conditions are connected using a logical **AND** operation to create the final variant condition.

7. If you want to clear the typed variant condition, click **Clear Text** ×.
8. If you want to validate the syntax and the variant condition, click **Validate Syntax** ✓.
9. Click **Save**.

# Configure a structure using baselines

## Overview of capturing the structure configuration using baselines

Baseline is a persistent record of all or part of a product's definition data that can be accessed at any time during or after a product's life. Baselines help you capture and preserve the state of the structure and its sub-assemblies at a particular checkpoint in the product development.

For instance, if you are working on an aircraft structure and have nearly completed the engine assembly while still progressing on the stabilizer and wings assemblies, you can create a configuration baseline for the engine assembly. After the stabilizer assembly is finished, you can establish another baseline for it.

You can create a precise baseline and a configuration baseline. When you create a baseline, you can create a precise baseline and capture the 150% structure of the BOM. After creating a precise baseline, when you open it, you must configure it again to view the exact structure that you want to view. For example, when you open a precise baseline, you must select the revision rule to view the exact configuration of the structure. After a precise baseline is created, you cannot further modify it.

When you create a configuration baseline, you can capture the exact same configuration of the structure. If your Teamcenter environment includes the required license and when you create a configuration baseline, you can choose the required type of the configuration baseline that you want to create.

Using baselines, you can perform the following tasks:

- **Create precise baselines** and capture the 150% structure of the BOM.

  **Note**

  You cannot create a precise baseline for a Collaborative Product EBOM.

- **Create configuration baselines** for the complete structure or for an individual sub-assembly.
- Considering multiple milestones involved in the product development, generate multiple baselines to track different development milestones for the same sub-assembly.

  For example, you might first create an ad hoc design review baseline and later a system design review baseline for the engine assembly.

- When multiple baselines are created for the same sub-assembly, **compare configuration baselines** to identify changes between the milestones.

- And finally, view the entire structure using the baseline configurations.

## Create a precise baseline

You can create a baseline of the item revision that represents a structure at any stage. You create precise baselines to preserve the state of a structure at a particular checkpoint. A baseline revision is created for all the elements in the structure, irrespective of the element selected. After a precise baseline is created, you cannot further modify it.

After creating a precise baseline, when you open it, you must configure it again to view the exact structure that you want to view. For example, when you open a precise baseline, you must select the revision rule and variant to view the exact configuration of the structure.

### Restrictions and limitations

You cannot create a precise baseline for a collaborative product EBOM.

### Procedure

1. Locate the structure that you want to create a baseline for and click **Open**.
2. Click **New** > **Baseline**.
3. In the Baseline panel, click the **Precise** check box.
4. (Optional) In the **Baseline Workflow** list, modify the default baseline workflow.

   By default, the **TC Default Baseline Process** workflow is selected.
5. (Optional) In the **Description** box, enter a description for the precise baseline.
6. Do one of the following:

   | To create a revision baseline | Do this |
   |---|---|
   | Asynchronously (in the background) | Select the **Run in Background** check box. |
   | Synchronously (in the foreground) | Clear the **Run in Background** check box. |

   When creating a precise baseline, the **Run in Background** check box is selected by default. If you clear the **Run in Background** check box, Teamcenter remembers your choice. The next time you log on, the check box is not selected.

   **Note**

   If you want to create the revision baseline asynchronously, you must have the AsyncService translator installed, configured, and running for the baseline feature to operate properly.
7. If you want to open the baseline immediately after it is created, click the **Open on Create** check box.

8. Click **Create**.

   The precise baseline is created and is available in the **Revision History** table of the **History** tab.

9. If you selected the **Run in Background** check box, a notification is displayed in the **Alerts** panel. Click **Alerts** and click the baselined structure in the **Target Object** section.

## Create a configuration baseline

You can create a configuration baseline for the entire structure. The configuration baseline helps you preserve the state of the structure at a particular checkpoint and visualize it later. You can create a configuration baseline for a structure even if it is released.

### Restrictions and limitations

You can create different types of configuration baselines only if your Teamcenter setup has the required license.

### Procedure

1. Locate the structure that you want to create a baseline for and click **Open** .
2. Click **New** > **Baseline** .
3. In the Baseline panel, perform the following tasks:
   a. Clear the **Precise** check box if it is already selected.

      **Note**

      If you are creating a configuration baseline for Collaborative Product EBOM, the **Precise** check box is not available.

   b. In the **Configuration Baseline Type** list, select a value.

      If you are creating the configuration baseline for performing an ad hoc design review, select **Ad-hoc Design Review**.

      If you are creating the configuration baseline for performing the design review of the entire selected assembly system, select **System Design Review**.

      If you are almost done with the system design and want to get the structure reviewed before finalizing it, select **Preliminary Design Review**.

      If you are creating the configuration baseline for evaluating whether the structure is complete, meeting all requirements, and is ready to proceed the formal release, select **Critical Design Review**.

      If the selected structure is ready and no more changes are expected to it, select **Formal Release Baseline**.

> **Note**
>
> Use the values in the **Configuration Baseline Type** list considering your product development stage. No Teamcenter action is performed on the configuration baseline.

c. In the **Name** and **Description** boxes, enter a name and description for the configuration baseline.

d. In the **Action on Working Revisions** list, select a value.

   If you want to capture the baseline revisions of the working revisions and apply the workflow, select **Capture and Apply Status**.

   If you want to freeze the baseline revisions of the working revisions and apply the workflow, select **Freeze and Apply Status**.

   If you want to capture the working revisions as is without applying any workflow, select **Capture Configuration As Is**.

e. In the **Baseline Workflow** list, select a workflow to apply to the configuration baseline.

4. If you want to create the configuration baseline in the background, select **Run in Background** checkbox.

5. Click **Create**.

### Results

The configuration baseline is created for the selected structure. If the structure contains substitutes and remote checked out objects, the configuration baseline is not supported for them. If the selected part structure contains an aligned design and a configuration baseline is created for it, the configuration baseline is created for its aligned design structure too. However, if the configuration baseline is created for a design structure, the configuration baseline is not created for its aligned part structure. Click the **Baselines** tab to view the configuration baselines that are already created for the selected structure. If your Teamcenter environment is set up with smart baseline and you close the configuration baseline, the new revisions are created for the associated items whose work in progress revisions are modified since their last baseline was created.

## View a configuration baseline

You can view configuration baselines for structures or subassemblies and further configure them by using effectivity and variant configurations.

### Procedure

1. Select the structure or the subassembly for which the configuration baseline is created and click **Baselines** tab.
   The **Baselines** tab displays all the configuration baselines created for the selected structure.

2. Select a configuration baseline and click **Open**.

The configuration baseline opens. You can identify it with the **Configuration Baseline** icon.
The configuration baseline displays the same configuration of the structure that it was created in, including the revision rules, partitions, and effectivities. You can also use the Filters panel to view the required content of the configuration baseline if your Teamcenter setup has the Context Management User license.

## View the content of the configuration baseline in other applications

You can open and view the content of the configuration baseline in other applications, such as NX and CATIA.

| Application | Action |
|---|---|
| Teamcenter Lifecycle Visualization | Select **Open** > **Open in Visualization**. |
| NX | Select **Open** > **Open in** NX. |
| CATIA | Select **Open** > **Open in CATIA**. |

## Compare configuration baselines

You can compare two configuration baselines within the same structure to view their differences. These configuration baselines can have different types, states, and release statuses.

### Procedure

1. Select the structure for which the configuration baselines are created and click the **Baselines** tab.
The **Baselines** tab displays all configuration baselines created for the selected structure.

2. Select two or more configuration baselines and click **Compare Structures**.
The **Compare Structures** view appears and displays the comparison between the selected configuration baselines. The **Compare** panel displays the elements that are unique to both the configuration baselines with color bars under the Results. This comparison is performed with the default comparison options.

3. (Optional) Change the comparison level:

    a. Specify the comparison level by selecting one of the following from **Options**.

| Comparison option | Description |
|---|---|
| **All Levels** | Compares the configuration baselines at every level in the two structures. If this option is selected for very large structures, performance may be adversely affected. |
| **Components Only** | Compares only the parts, or components that are at the lowest level. |
| **Current Level** | Compares only the parts that are the immediate children of the selections in the two structures. This is the default level of comparison. |
| **Linked Assemblies or Components** | Compares the two configuration baselines by using an accountability check. This comparison level is displayed only if Multi-Structure Foundation is available in your Teamcenter environment. Accountability check is a verification mechanism to check if all the parts in one structure have equivalent parts in the other structure. You can define your own properties for equivalence. |
| **All Levels Below Selected** | Compares the configuration baselines below the levels that you selected. |

b. Specify what you want to display in the comparison results by selecting one or more of the following options from **Options** > **Display**.

| Comparison option | Description |
|---|---|
| **Matched** | Displays the parts that are a match in both of the configuration baselines. |
| **Different** | Displays the parts that differ between the two configuration baselines. |
| **Multiple Matches** | Displays the different numbers of occurrences of the same part in the comparison as a difference. |
| **Unique in Left View** | Displays the parts that are only listed in the configuration baseline that is displayed on the left. |
| **Unique in Right View** | Displays the parts that are only listed in the configuration baseline that is displayed on the right. |

c. For **Linked Assemblies or Components**, select the **Dynamic Equivalence** check box.

In some cases, the occurrences that are slightly different in the two configuration baselines might not be reported as equivalent. The dynamic equivalence check reports such occurrences.

This check box is displayed only if Multi-Structure Foundation is available in your Teamcenter environment.

d. (Optional) Select the **Generate Report** check box to generate the comparison report.

e. Click **Compare**.

The comparison results are listed under **Results**. Clicking a part in the list highlights that part in the configuration baselines.

Comparison results are retained for a specific period. If a comparison result is older than the retention period or if a more recent result exists, the old result is automatically deleted by the system.

> **Tip**
>
> If you closed the **Compare** panel and want to perform any further comparison, click **Compare** on the task bar to view the **Compare** panel again.

When you are comparing the two configuration baselines, you can configure them by unit effectivity, date effectivity, and by variants.

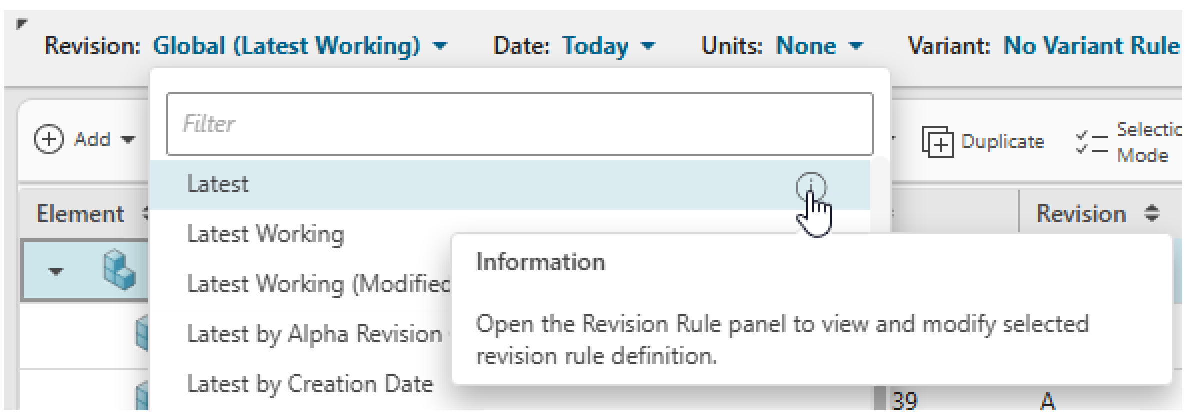
4. Click **Exit Compare** to close the comparison.

## View a structure using a configuration baseline

You can modify the revision rule to configure a structure with a configuration baseline and then view it. You can also use effectivity and variants for further analysis.

### Procedure

1. Search for and open the structure.
2. Modify a revision rule to include the configuration baseline in the structure.
   a. In the revision rule list, go to the rule that you want to modify and click **Information**.




   The revision rule details are displayed with all the clauses.

   b. On the **Clauses** section of the **Revision Rule** panel, click **More** > **Add clause** to add the clause to the configuration baseline.

   c. Perform one of the following tasks:

- If configuration baselines are created for multiple subassemblies and you want to configure the structure for all of the subassemblies that satisfy the clause you define, in the **Clauses** list, select **`Baseline Type`** and then select the **Configuration Baseline Type**, and a date.
- To analyze the structure for a specific configuration baseline, in the **Clauses** list, select **`Baseline Object`** and then add the required configuration baseline.

d. Click **Add** to modify the selected revision rule.

e. If required, add a new clause, delete a clause, or change the order of precedence by moving a clause up or down.

f. Click **Modify and Configure**.
The revision rule is modified and configured to display the content of the configuration baseline.

3. If required, use effectivity and variants to further configure the structure.

## Using smart baselines

The standard baseline process may be time-consuming and use significant disk space as it may create many new revisions and (consequently) copies of all the associated data and CAD designs. To avoid this, you can create a *smart* baseline, reusing a baseline revision whose work in progress revision is unmodified since its last baseline was created. You can only create smart baselines of assemblies.

If you try to create a smart baseline of an assembly that has not changed since the previous baseline, a new baseline is not created. No informational message is displayed in Structure Manager to indicate this.

**Note**

If you make changes to an occurrence property (for example, quantity) the last modified date of the parent BVR changes and Teamcenter therefore considers the parent item revision as a candidate for a new baseline. However, the item revision of the BOM line on which changes were made does not qualify for the new baseline.

# Perform a solve using a modular configuration

## Why use a modular configuration?

Learn about the benefits of using modular configurations to minimize the engineering and production costs. The configurator administrator or a designated user creates *configurator modules*. The BOM engineers can use these modules to configure products, such as a generic door assembly for refrigerators and freezers, by setting parameters and variant conditions.

The configurator modules are self-contained, plug-compatible units that can be *reused*.

Manufactured goods are often designed and assembled from configurator modules. For example, consider a company that produces a range of refrigerators and freezers in different sizes and colors. The door assemblies are developed in a department that designs a modular door suitable for use in any refrigerator or freezer. They design a generic door assembly that has all possible components for any use—a sheet steel outer door and two internal covers, one for a freezer and one for a refrigerator.

You can then configure the door assembly for a particular use in a refrigerator or freezer by setting various parameters or variant conditions that describe it, for example, **door width**, **door height**, **application** (refrigerator or freezer), and **color** (white or stainless steel). This intelligent door assembly is called a *configurator module*.

The door configurator module is completely self-contained and can be reused in any product by setting its modular configuration. The modular configuration settings only control the features and components contained in the door configurator module itself and make no reference to features in higher- or lower-level configurator modules.

## Solve the modular configuration

As a BOM engineer, you can perform a solve using the configurator module.

### Procedure

1. Search and open the modular configuration.
2. Click **Details** > **Variant Configuration**.
3. Choose one of the following modes:
   - In the **Guided** mode, only valid features are displayed.

     This mode allows you to navigate only valid selections and configurations. If no configuration is applied, a model opens in guided mode, and the system guides you to select a valid group of features in which to configure a product. Every time a selection is made, the system reevaluates the rules and refreshes the list of features.

     Guided mode supports both complete and partial configurations.
   - In the **Manual** mode, all available features, including variants that are not valid in the current context, are displayed.

     In this mode, you can look at the entire variability, choose a variety of features, and assess if those selections are valid based on the configurator rules. You can validate your selections and save your configuration. If there are any conflicting constraints, the system reports violations as error messages and specifies the conflicting selections.

   While radio buttons are displayed for making selections in **Guided** mode, check boxes are displayed for **Manual** mode.
4. To **filter configurator data by revision rule, effectivity, and rule date**, and rule date, click ⚙ to view the **Settings** panel and change them as appropriate.

5. Set the validation mode.

   a. Click ⚙ to view the **Settings** panel.

   b. In the **Filter Criteria** section, choose the product hierarchy depth.

      The default value is **1** and the maximum is **5**.

      > **Note**
      >
      > By default, five levels of structures are supported. However, this example provides procedures to configure only three levels of structures.

   c. In the **VALIDATION MODE** section, **choose a mode from the drop-down list**.

      - **Order** (default profile)

        This mode is intended for a user who wants to quickly arrive at a 100% BOM. You can use a 100% BOM to create a prototype, perform simulations, compute the price, calculate the weight of the product, or visualize the structure.

      - **Overlay**

        This mode does not expect you to specify the complete configuration. It is intended for developing a new product or making changes to an existing product.

6. To configure the modular configuration, make the appropriate selections in the **Variant Configuration** view.

7. Select a product model with which you would like to work.

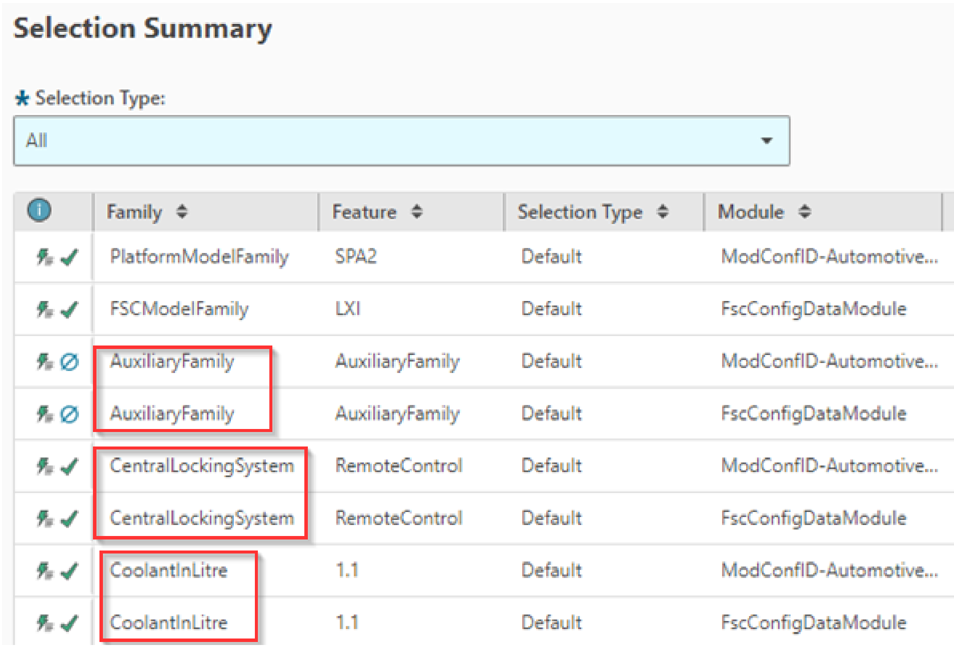
8. Select each group from the list in the left panel and make the necessary feature selections in the work area.

9. (Optional) To find the required selections, in the **Variant Configuration** view, choose **More Commands** > **Next Required** or **Previous Required** as appropriate.

10. (Optional) To find user and system selections, in the **Variant Configuration** view, click **Full Screen**, choose **Selection Summary**, and select **All** from the **Selection Type** list.

    When the product hierarchy depth is greater than **1**, the **Selection Summary** panel shows an additional column, **Module**. This column displays where the selection has been made, that is, the module, family, feature, and type of selection.

    The **Selection Summary** panel sometimes displays the same feature in two rows. This happens when either the user or the system (based on constraints) has made a selection on a feature that exists both in the parent module as well as the child module. You can see the name of the parent and the child module for that repeated feature in the **Module** column for that feature.




11. Make all the required selections.

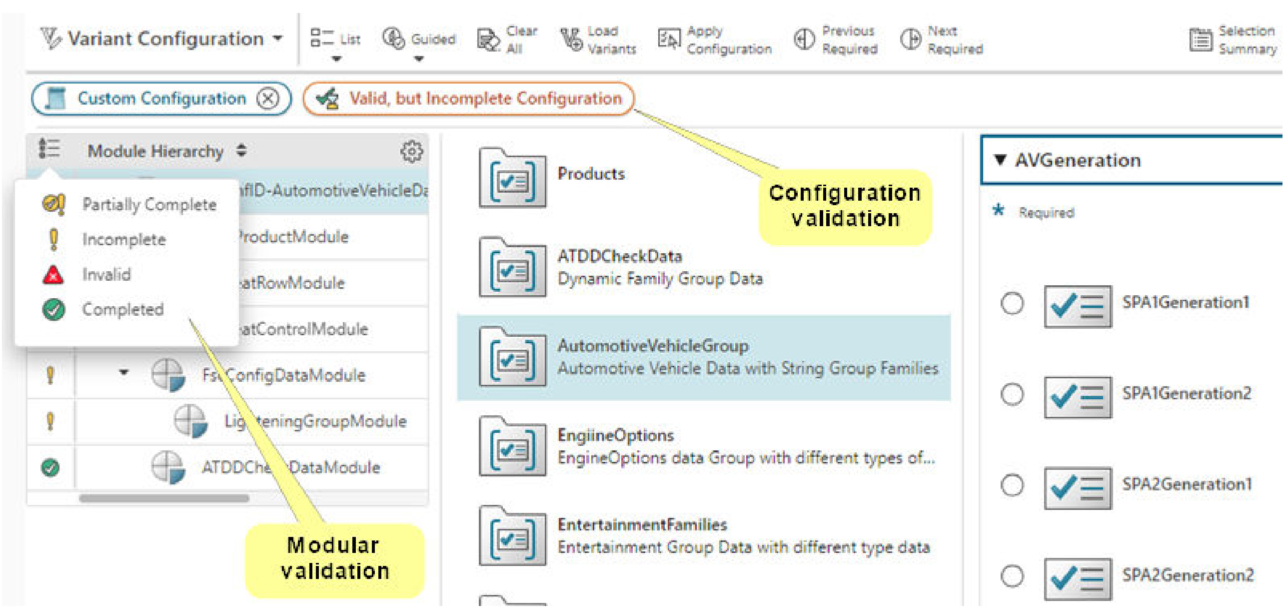
    You can make selections at the root level and child level modules (configurator context levels).

    In the **Guided** mode, the validation happens for each of your selections, and its configuration completeness is displayed automatically.

    If you are in the **Manual** mode, when you click **Expand** or **Validate**, the current selections in the UI are validated or expanded, respectively.

    With modular configuration, along with configuration completeness, a module completeness check is also done.

    The configuration completeness is standard for all product structures, irrespective of the modular configuration.



The completeness indicator at the top of the view displays one of three states: **Valid and Complete**, **Valid and Incomplete**, and **Invalid**.

- **Valid and Complete** (applicable for both guided mode and manual mode)

  Valid and complete means that the BOM is targeting a complete structure, that is, a 100% BOM.

  When a configuration is valid and complete, you could take this configuration input and execute an order entry in the order system. When a configuration is valid and complete, you can take this configuration input and perform a cost or a weight rollup, compute the center of gravity, or perform a digital analysis of that product.

  On the contrary, if you do not see this completeness indicator, you should not typically perform the above action. It means that you have less content or more content.

  When a configuration is valid and complete, it assigns a value to *every* family. This is irrespective of whether the family is mandatory or discretionary.

- **Valid and Incomplete** (applicable for both guided mode and manual mode)

  In guided mode, you can select **Next Required** or **Previous Required** to navigate to the next or the previous family or the feature that requires a value to be assigned to complete the configuration.

  In the case of modular configuration, **Next Required** or **Previous Required** takes you to the next module after reaching the last incomplete family of the last group.

- **Invalid** (applicable for manual mode only)

  It means that the input criteria is invalid based on the configurator rules set in the current configurator context.

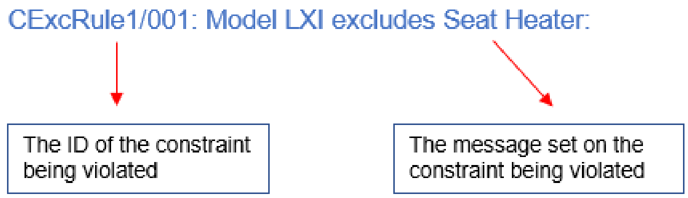
The modular validation indicates the following states:

| State | Description |
|---|---|
| **Partially Complete** | The selections for the parent are complete, but the selections for some of the children are not. |
| **Incomplete** | Some required families do not have complete selections. |
| **Invalid** | Some violations have occurred due to some constraints. |
| **Completed** | All the required selections are complete for the module, and they do not have any conflicts with constraints. |

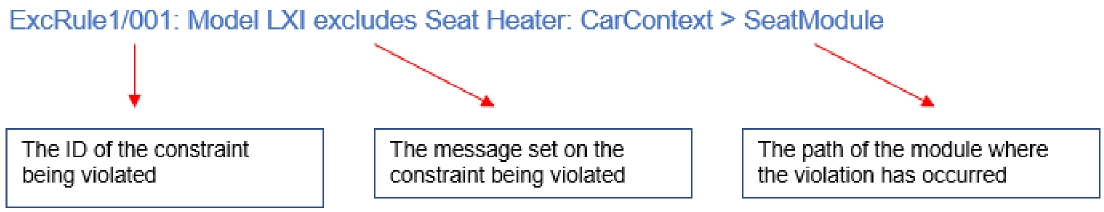
12. In the **Manual** mode, when you click on **Expand** or **Validate**, you might enter into an invalid state because of some constraint being violated. In such cases the system reports violations.

    The following are some examples of violation messages in the **Manual** mode:

    Example of violation message without modular configuration:




Example of violation message with modular configuration:




> **Note**
>
> Violation indicators are available at the child level also.

13. Record your selections by clicking the **Save** button.

    If you have not loaded a variant rule or a variant condition, the system prompts you to save the selections as a new variant rule or variant condition. In such cases, specify the name in the **Save** dialog box and select one of the following options:

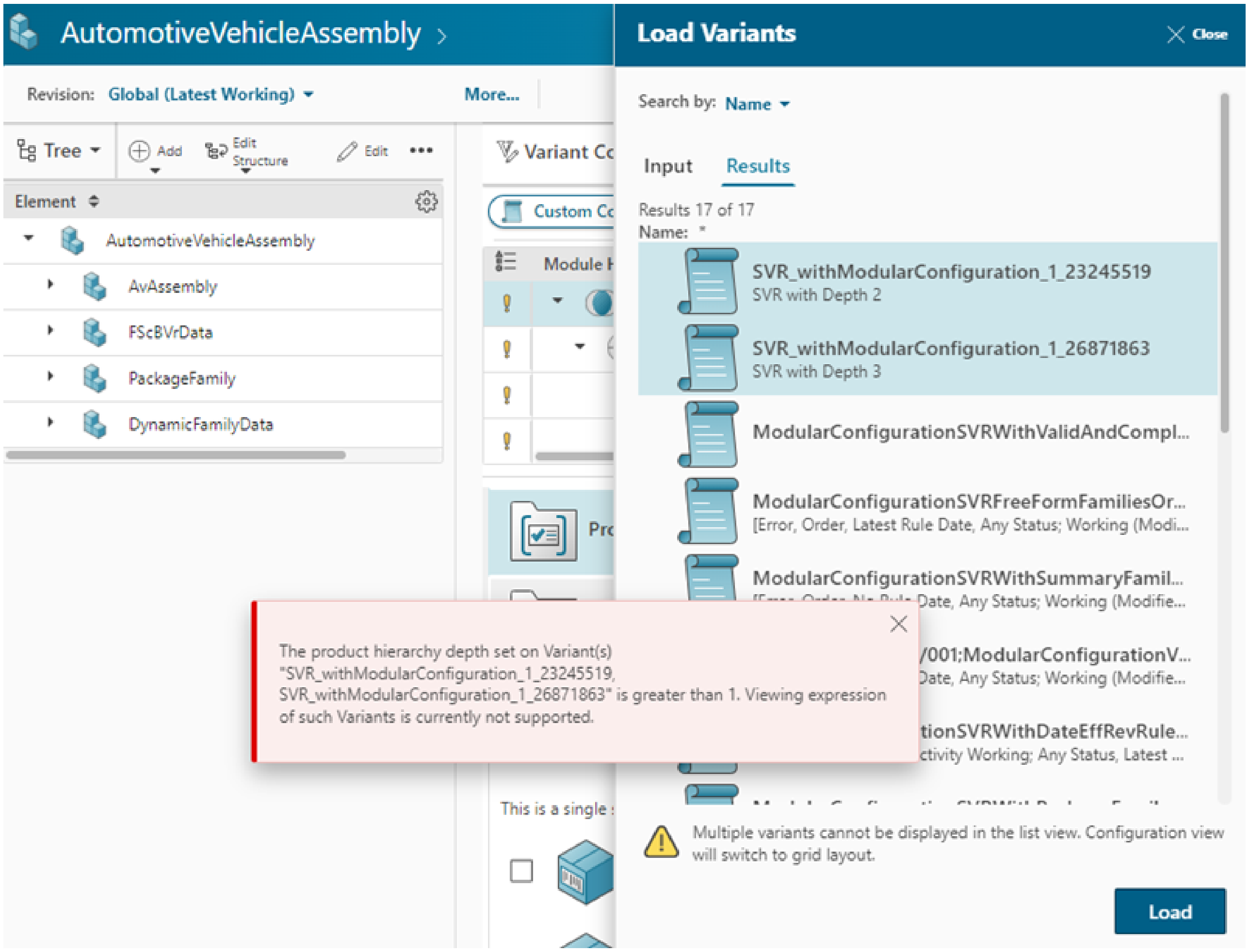
    - **Current Structure Revision** to save the variant rule or the variant condition to the current structure.
    - **Configurator Context** to save the variant rule or the variant condition to the configurator context. This saved variant rule or variant condition is available in all structures to which the configurator context is attached.

14. To load a variant rule or variant condition, choose **More Commands** > **Load Variants**.

    Currently, when you load multiple variants using the **Load Variant** dialog, the system switches to the variant configuration's **Grid Mode**.

    With a modular configuration:

    - If the selected variants have a product hierarchy depth greater than **1**, the system does not switch to **Grid Mode** and an error message is displayed. This is currently not supported.
    - Any variant with a depth greater than **1** can be viewed only in the variant configuration's **List** view.
    - If the selected variants have a mix of depth greater than **1** and depth as **1**, the system switches to the variant configuration's **Grid Mode**. It then displays only the variants with a depth of **1**. The rest of the variants with a depth greater than **1** are skipped with a warning message.




15. Save selections after loading a variant rule or variant condition.
    - Click the **Save** button to save the changes to the existing variant rule or variant condition; OR
    - Click the **Save As** button to save the changes to a new variant rule or variant condition.
16. To apply the selected variant configuration to the current content and to refresh the content in the product, choose **More Commands** > **Apply Configuration**.

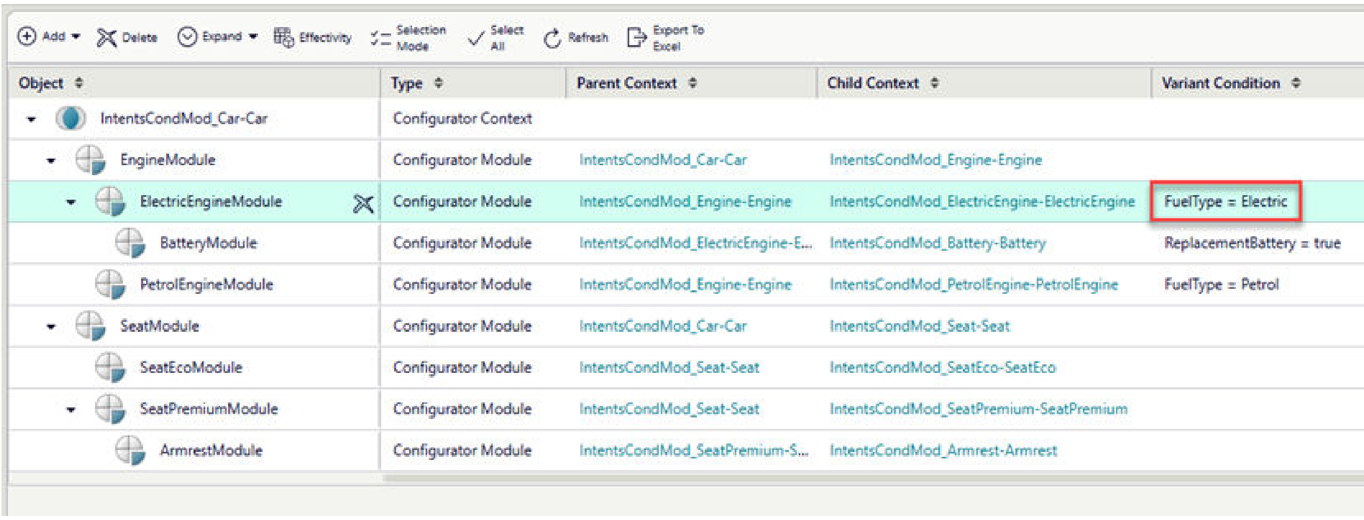
## Solve the modular configuration using intents and conditional module

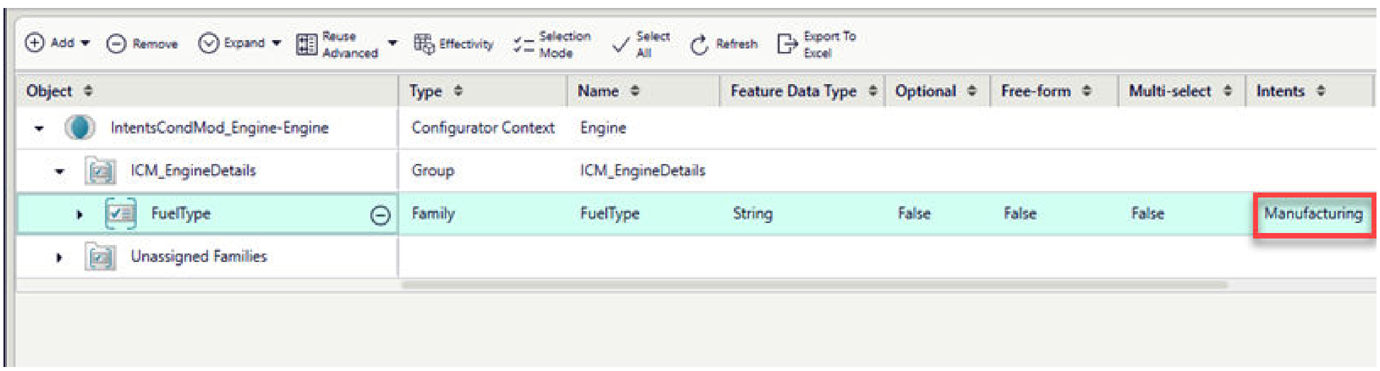
You can use intents to scope the modular configuration before performing a solve. Teamcenter configures out modules and their features based on intents specified by the configurator administrator.

You can refine the modular configuration by using variant conditions on modules. For example, when configuring a marketing-specific solution, you can filter out modules not intended for marketing by authoring a variant condition on the module and setting the appropriate intent.

### Prerequisites

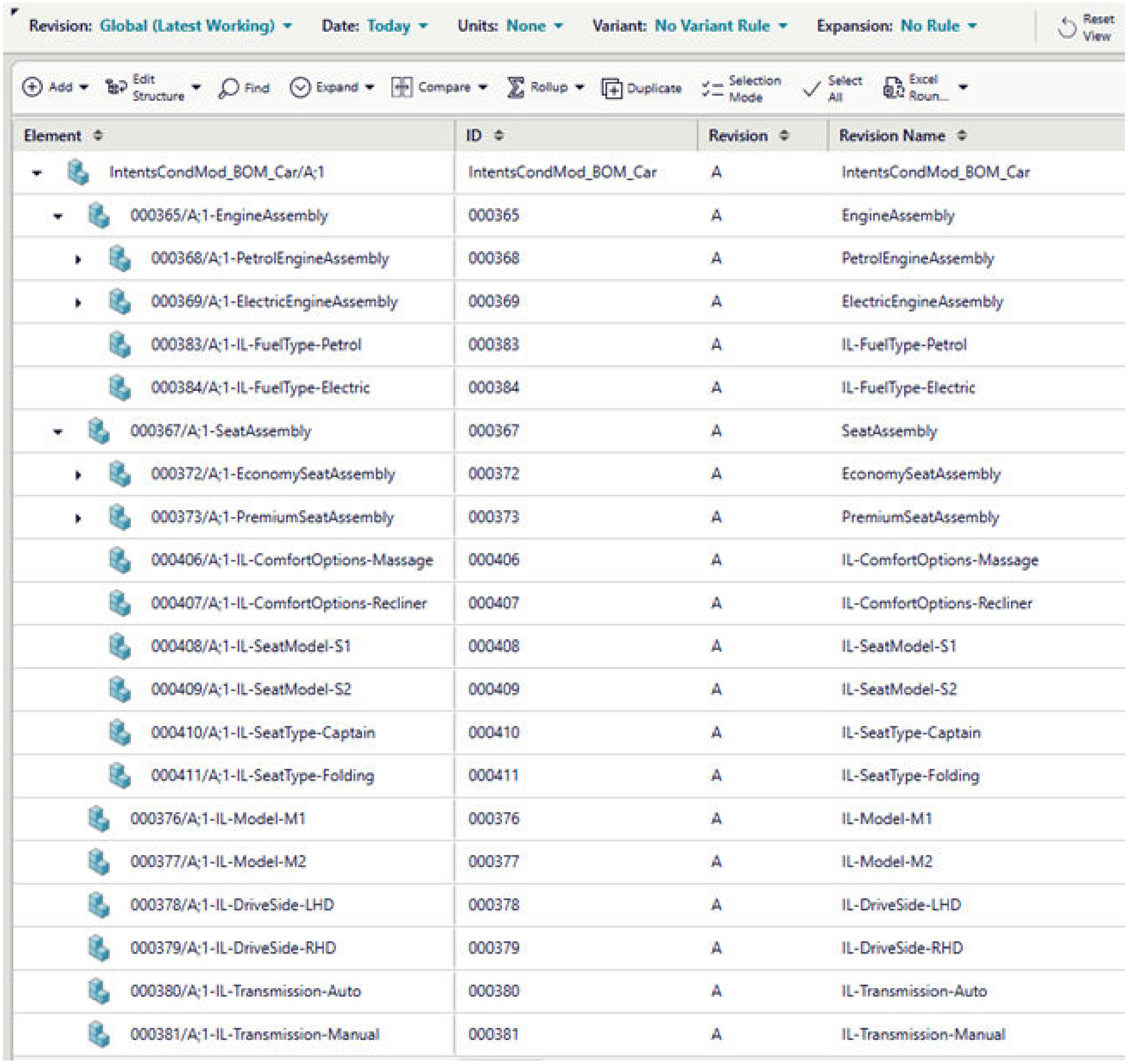
This topic assumes that the configurator administrator has created a conditional module with **FuelType** family and **Petrol** and **Electric** as its features. Additionally, a variant condition has been created with **FuelType=Electric** on the configurator module and the **Manufacturing** intent selected for the **FuelType** family.




### Procedure

1. In the search box, type the item revision name, open the item revision, and switch from **Details** > **Variant Configuration**.




2. Choose one of the following modes:

   - In the **Guided** mode, only valid features are displayed.

     This mode allows you to navigate only valid selections and configurations. If no configuration is applied, a model opens in guided mode, and the system guides you to select a valid group of features in which to configure a product. Every time a selection is made, the system reevaluates the rules and refreshes the list of features.

     Guided mode supports both complete and partial configurations.

   - In the **Manual** mode, all available features, including variants that are not valid in the current context, are displayed.

In this mode, you can look at the entire variability, choose a variety of features, and assess if those selections are valid based on the configurator rules. You can validate your selections and save your configuration. If there are any conflicting constraints, the system reports violations as error messages and specifies the conflicting selections.

While radio buttons are displayed for making selections in **Guided** mode, check boxes are displayed for **Manual** mode.

3. Apply filter criteria and validation mode.

   a. Click **Settings** to view the **Profile Settings** panel.

   b. In the **Filter Criteria** section, choose the product hierarchy depth.

      The default value is **1** and the maximum is **5**.

   c. Change the snapshot revision rule, effectivity, and rule date policy as appropriate.

      For more information, see **filter configurator data by revision rule, effectivity, rule date, and intent**.

   d. In the **VALIDATION MODE** section, **choose a mode from the drop-down list**.

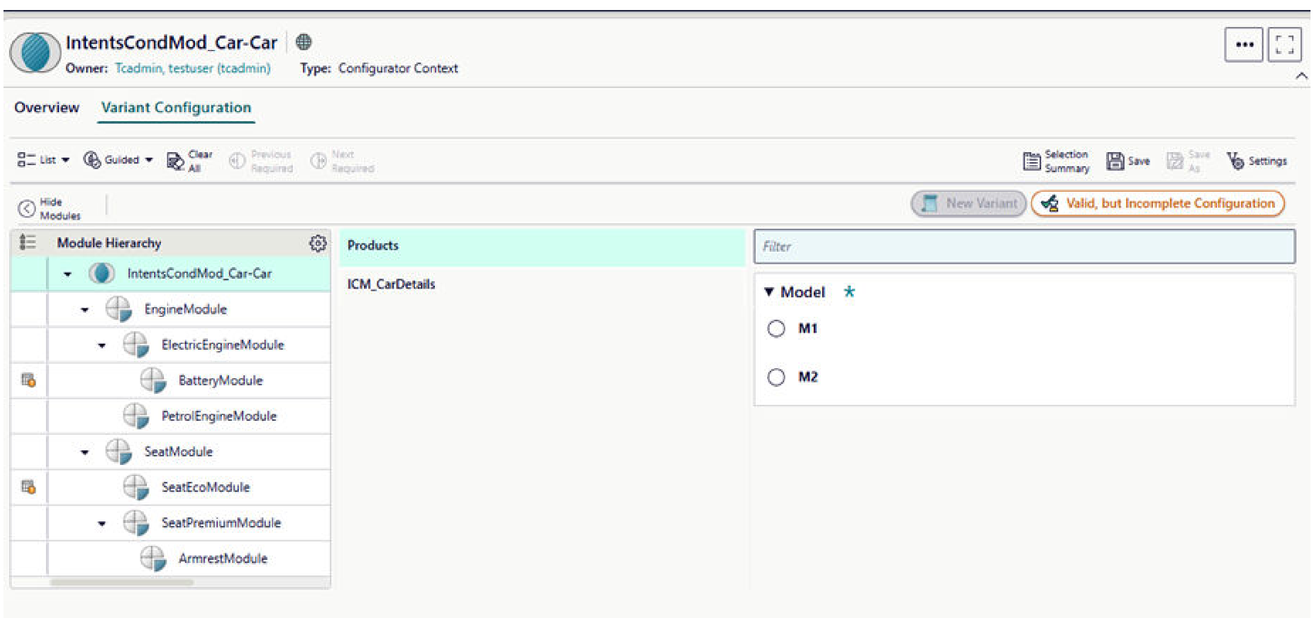
      - **Order** (default profile)

        This mode is intended for a user who wants to quickly arrive at a 100% BOM. You can use a 100% BOM to create a prototype, perform simulations, compute the price, calculate the weight of the product, or visualize the structure.

      - **Overlay**

        This mode does not expect you to specify the complete configuration. It is intended for developing a new product or making changes to an existing product.

4. In the **Variant Configuration** view, observe the module hierarchy.



Note that two modules, **BatteryModule** and **SeatEcoModule** are configured out due to some constraints or rules.

5. Select a product model.

6. Select each group from the list in the left panel and make the necessary feature selections in the work area.

7. (Optional) To find the required selections, choose **More Commands** > **Next Required** or **Previous Required** as appropriate.

8. (Optional) To find user and system selections, click **Full Screen**, choose **Selection Summary**, and select **All** from the **Selection Type** list.

   When the product hierarchy depth is greater than **1**, the **Selection Summary** panel shows an additional column, **Module**. This column displays where the selection has been made, that is, the module, family, feature, and type of selection.

9. Make all the required selections.

   You can make selections at the root level and child level modules (configurator context levels).

   In the **Guided** mode, the validation happens for each of your selections, and its configuration completeness is displayed automatically.

   If you are in the **Manual** mode, when you click **Expand** or **Validate**, the current selections in the UI are validated or expanded, respectively.

   With modular configuration, along with configuration completeness, a module completeness check is also done.

   The configuration completeness is standard for all product structures, irrespective of the modular configuration.

   The completeness indicator at the top of the view displays one of three states: **Valid and Complete**, **Valid and Incomplete**, and **Invalid**.

   - **Valid and Complete** (applicable for both guided mode and manual mode)

     Valid and complete means that the BOM is targeting a complete structure, that is, a 100% BOM.

     When a configuration is valid and complete, you could take this configuration input and execute an order entry in the order system. When a configuration is valid and complete, you can take this configuration input and perform a cost or a weight rollup, compute the center of gravity, or perform a digital analysis of that product.

     On the contrary, if you do not see this completeness indicator, you should not typically perform the above action. It means that you have less content or more content.

     When a configuration is valid and complete, it assigns a value to *every* family. This is irrespective of whether the family is mandatory or discretionary.

   - **Valid and Incomplete** (applicable for both guided mode and manual mode)

     In guided mode, you can select **Next Required** or **Previous Required** to navigate to the next or the previous family or the feature that requires a value to be assigned to complete the configuration.

In the case of modular configuration, **Next Required** or **Previous Required** takes you to the next module after reaching the last incomplete family of the last group.

- **Invalid** (applicable for manual mode only)

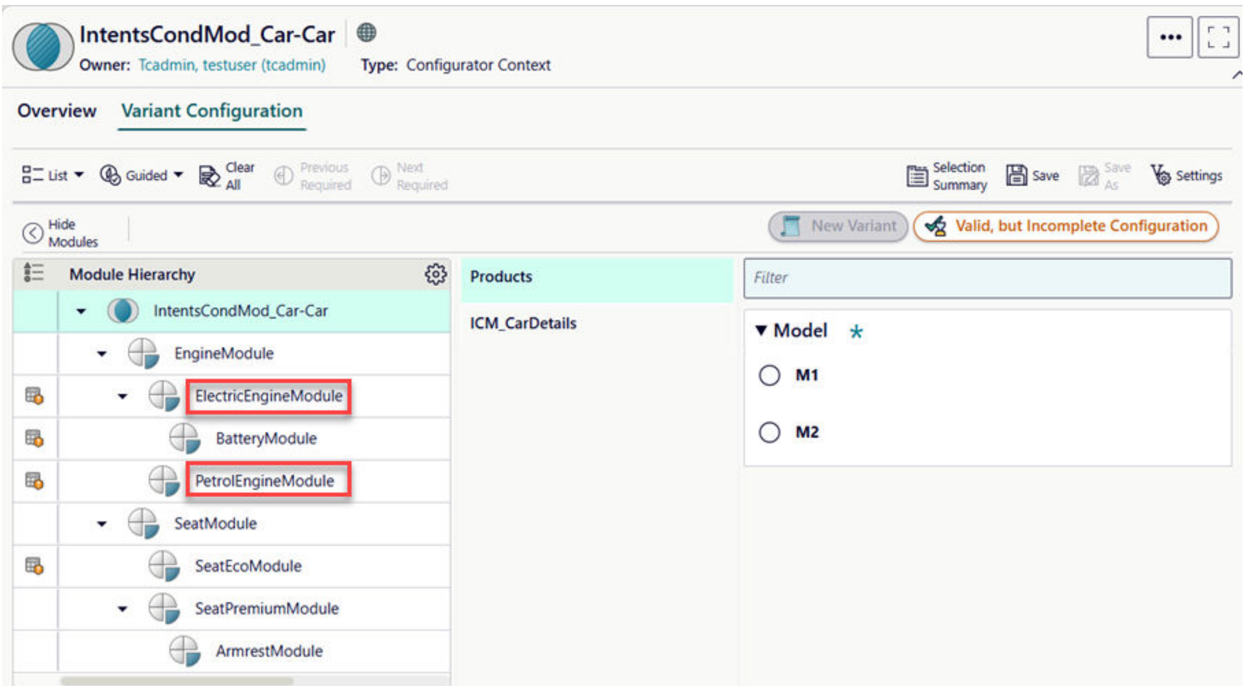
  It means that the input criteria is invalid based on the configurator rules set in the current configurator context.

10. Click **Clear All** to clear all your selections.
11. Select an intent to scope down the structure.

> **Note**
>
> Modules that are already configured out due to variant conditions remain configured out when intents are applied.

a. Click **Settings** and select **Marketing** from **Snapshot Intents**.

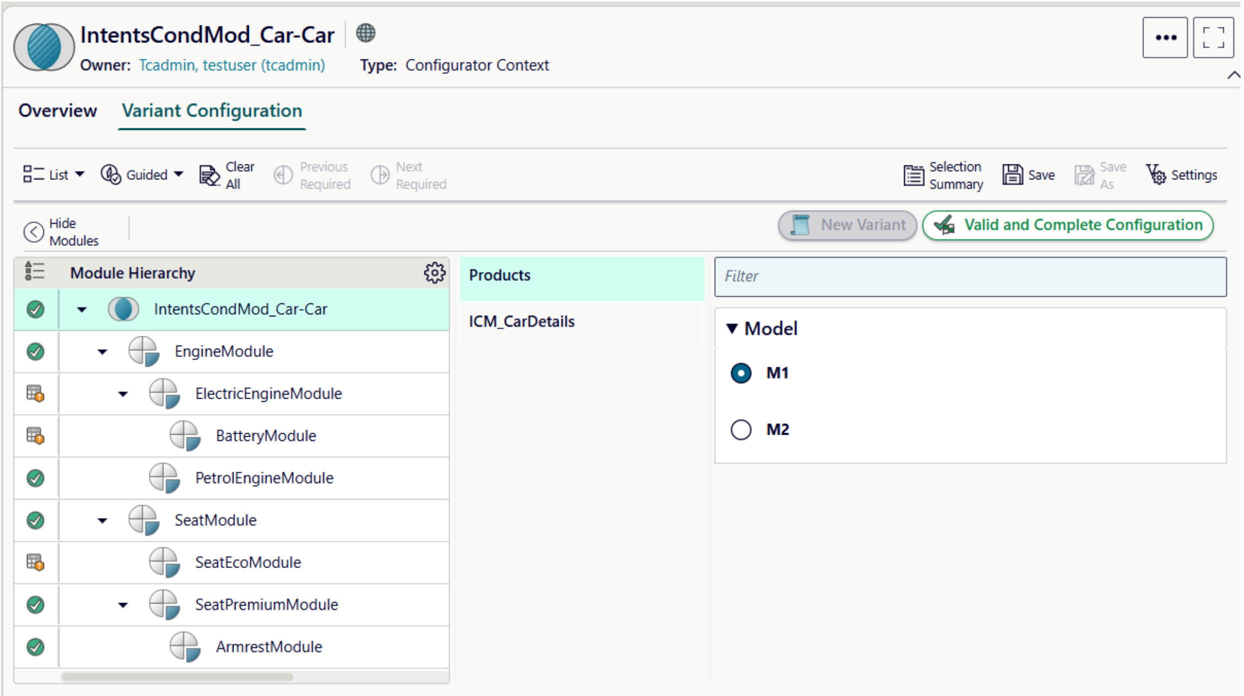
b. In the **Variant Configuration** view, observe the module hierarchy.




Note that two additional modules, **ElectricEngineModule** and **PetrolEngineModule** are configured out. This is because you have selected the **Marketing** snapshot intent to scope the configuration before performing a solve.

c. Select a product model such as **M1**.

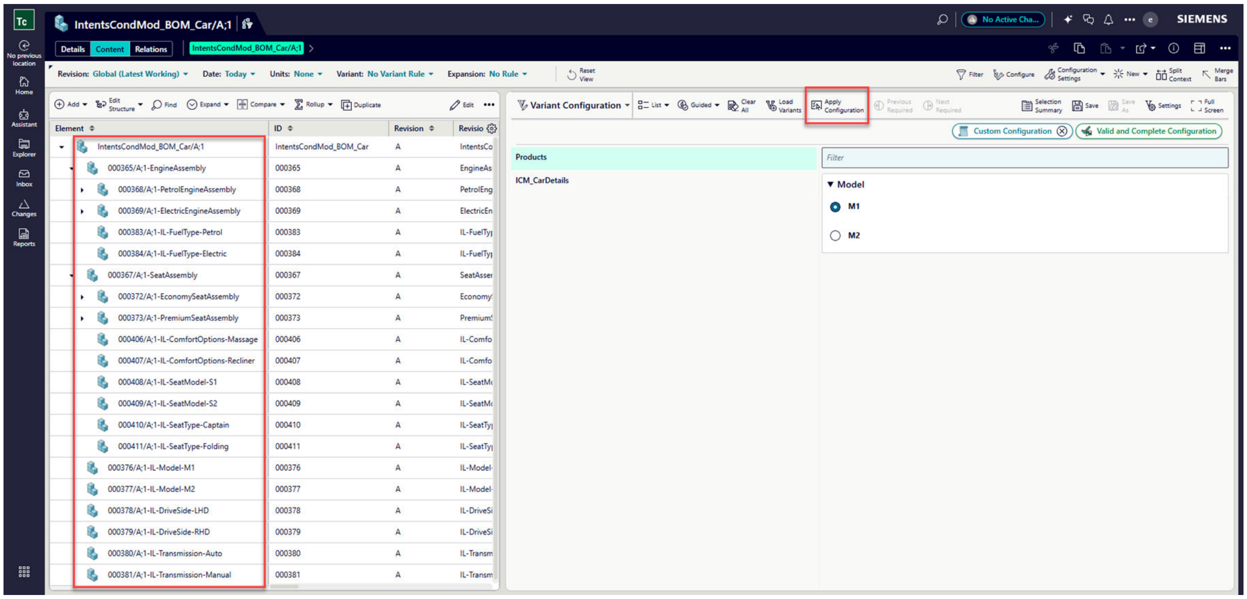
The system makes all the required selections based on constraints and the configuration becomes valid and complete.



d. Click **Apply Configuration**.

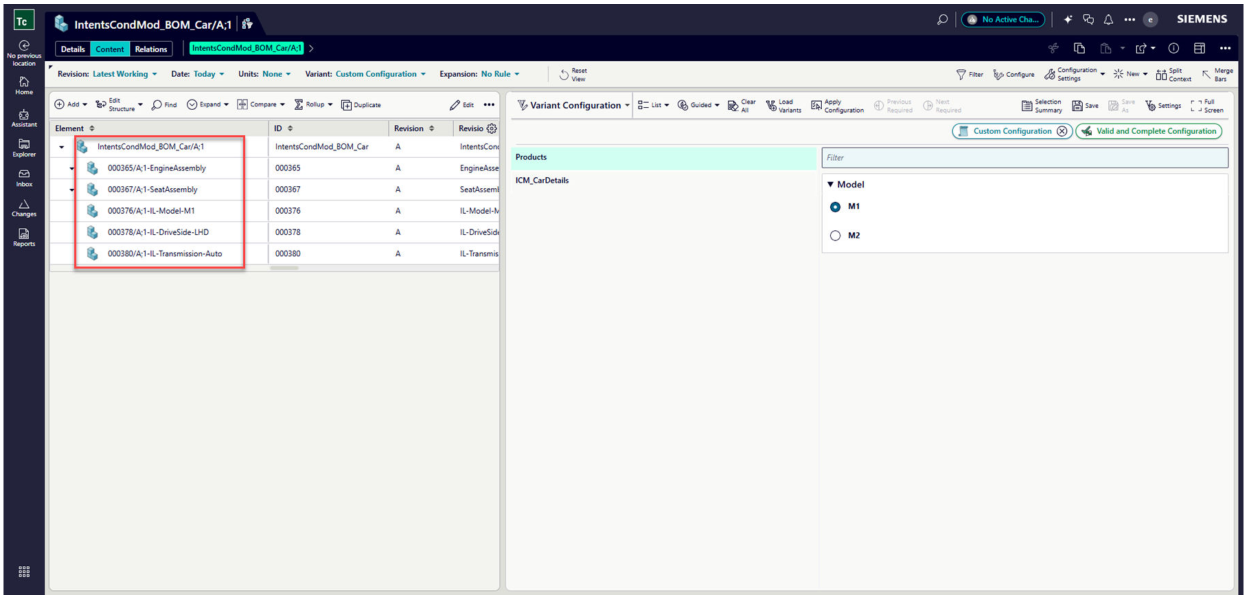
The BOM lines are configured according to the applied configuration.

Before applying the configuration:



After applying the configuration:



# Perform a solve using a custom configuration

## Different modes to analyze configurations

You can use **Order** and **Overlay** modes for configuring and analyzing variant configurations. The **Order** mode helps you achieve a 100% Bill of Material (BOM) for shippable products by automatically completing configurations. The **Overlay** mode is used during product development to show compatible features and content without requiring a complete configuration.

You can use the **Order** mode or the **Overlay** mode to configure and analyze the variant configuration.

- **Order** mode

  This mode is intended when the product is ready to be shipped. It is for a user who wants to quickly arrive at a 100% BOM. The 100% BOM is an *as sold* product configuration. An example of this is the configuration of a car that is going to be built and shipped to the dealer.

  You can use a 100% BOM to create a prototype, perform simulations, compute the price, calculate the weight of the product, or visualize the structure.

  In this mode, the system considers all the information that it can get to autocomplete the configuration. You can select a certain model and select the country of sale, for example, UK or India, and the system automatically selects the right-hand drive. The system scans the entire knowledge base of the configurator in order to predict what the final 100% product looks like. You can select another color or a different engine. The configurator attempts to consider your inputs and complete the configuration as much as possible as per the default rules.

  In cases where your inputs and the default rules are insufficient, the configurator does not achieve the original goal of completing a 100% configuration. When some user options are possibly left open, it achieves only a 90% configuration.

- **Overlay** mode

  This mode is intended when the product is in the development stage. It does not expect you to specify the complete configuration. You can, but it is not necessary. It configures all the features and the product content that are compatible with your inputs. For example, a new fuel filter, that you plan to introduce might have collision or interference problems within the existing space. However, at this stage, your focus is in determining all the environments in which the fuel filter can be assembled. The filter can be assembled in the 1.5 liter, 2.0 liter, and 3.0 liter gasoline engine options with or without an automatic transmission. However, the filter cannot be used in a diesel engine. Therefore, in this mode, when you select a fuel filter, you see a large overlay with various types of gasoline engines, but not a diesel engine. You can view only the options that are compatible with your selection.

Therefore, both modes converge for complete configurations, even if they might begin with very different starting positions, based on more inputs provided by the user. While the **Order** mode might automatically complete the empty input configurations to come up with a complete, default product, the **Overlay** mode leaves the same input configuration empty. When the input expression reaches a complete product configuration, the system behavior of both modes is identical.

## Set the validation mode

The **Order** mode or the **Overlay** mode is used to configure and analyze the variant configuration. This can set the validation model for the current session.

### Procedure

1. Find and open the product to be configured and choose **Details** > **Variant Configuration**.
2. Click ⚙ to view the **Settings** panel.
3. In the **VALIDATION MODE** section of the **Settings** panel, choose a mode from the drop-down list.

   - **Order** (default profile)

     This mode is intended for a user who wants to quickly arrive at a 100% BOM. You can use a 100% BOM to create a prototype, perform simulations, compute the price, calculate the weight of the product, or visualize the structure.

   - **Overlay**

     This mode does not expect you to specify the complete configuration. It is intended for developing a new product or making changes to an existing product.

   For more information about these modes, see ***Different modes to analyze configurations***.

   The validation severity, expansion severity, and selection behavior settings are updated based on the selected mode. These settings are defined by your administrator and are therefore, unavailable and read-only.

4. Click **Apply**.
   All user selections are retained, but the system selections and violations are cleared.

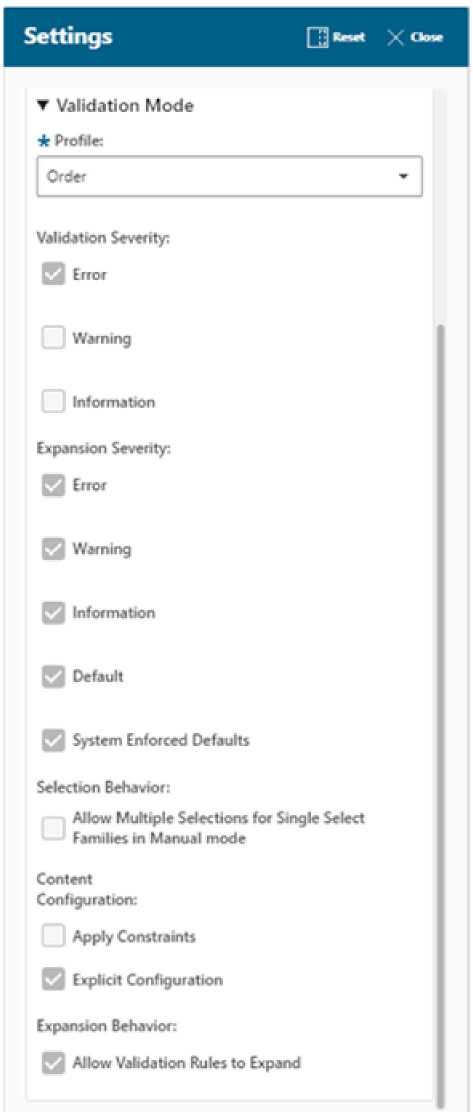
**Tip**

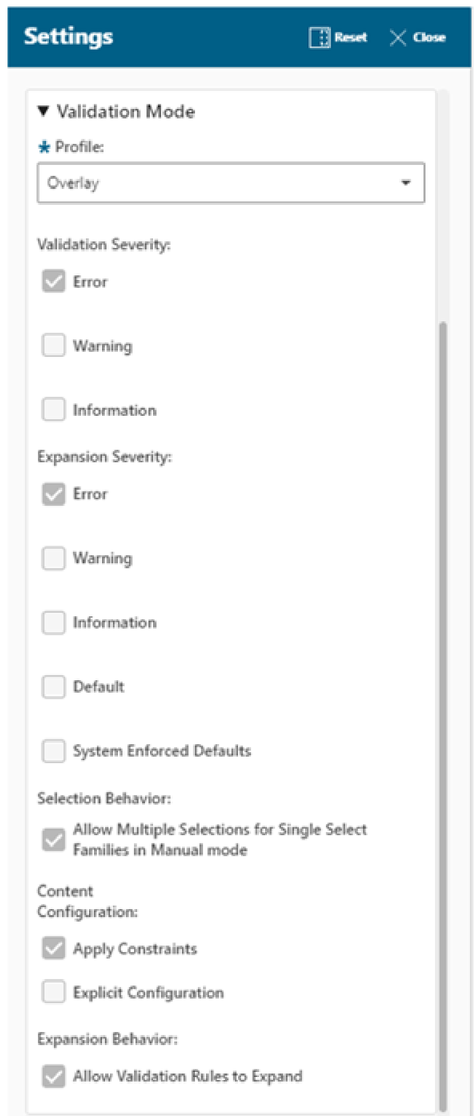
Once you change tabs or otherwise start a new session, the validation mode settings will return to the defaults. A **saved variant** stores this setting information, however, so loading the appropriate SVR is a quick way to return to your desired settings.

The individual settings that make up each mode definition are defined by your administrator and are read-only. If you change from one mode to the other, the settings change to reflect the selected mode.

When applied, all user selections are retained, but the system selections and violations are cleared.

Choosing the validation mode sets the severity for reporting and expansion in order to control how the solver behaves and how violations are reported. When a mode is selected, the definition of that mode is shown, including its validation severity, expansion severity, and its selection behavior. These settings determine how strictly you want to evaluate the constraints in the system.



- The **Validation Severity** applies when the configuration is validated and is always higher than the expansion severity.
- The **Expansion Severity** is relevant when an order is expanded by clicking **Apply System Selections**, and the user wants to communicate the severity level to the solver.
- (Not applicable for **Order** mode by default.) The **Selection Behavior** consists of a checkbox which determines whether multiple selections for a single select family will be considered valid or not (in manual mode, where single select families are represented with check boxes for each feature).

  **Note**

  The **Selection Behavior** setting is ignored in guided mode.

- (Not applicable for **Order** mode by default.) The **Apply Constraints** check box, if selected, determines if all constraints should be considered while applying the variant expression on the content. If deselected, only user selections are applied.
- (Not applicable for **Overlay** mode by default.) The **Explicit Configuration** check box, if selected, returns only BOM lines where the features have been explicitly selected in **Order** mode. This leads to a configuration that is 100% or less. Conversely, if the **Overlay** mode is selected, the **Explicit Configuration** check box is not selected by default. This leads to a configuration that is more than 100%, for example, a 120% BOM. This happens unless a complete configuration is specified that results in a 100% BOM.

  For more information, see *Example: Solve behavior using explicit configuration*.

- The **Allow Validation Rules to Expand** check box, if selected, determines if validation rules should participate during criteria expansion. If deselected, only violations, if present, are reported. The system selection, based on validation rules, does not occur in the expanded criteria.

  By default, this check box is selected.

  **Example**

  When the validation severity is set to **Error** (as it is for both **Order** and **Overlay** modes), the configuration is considered invalid only when the **Error** severity rules are in conflict. When there are conflicts between **Warning** and **Information** constraints, the configuration is considered valid.

## Filter configurator data by revision rule, effectivity, rule date, and intent

You can filter configurator data by applying revision rules, effectivity (date or unit ranges), rule dates (like Latest or Date), and intents (such as Manufacturing or Marketing) to view specific configurations and manage product variations effectively.

You can filter configurator data by **revision rule**, **effectivity**, **rule date**, and intent.

### Procedure

1. Find and open the product to be configured and choose **Details** > **Variant Configuration**.

> **Note**
>
> If your configuration analyst has not associated a configurator context with the product, **Variant Configuration** is not displayed.

2. Click **Settings** to view the **Settings** panel.

3. Filter configurator data by revision rule.

   Content changes can be managed by revising product data. Changes that may necessitate a new revision include addition or removal of components, relationships to other products or assemblies, and changes to properties. You can then apply a revision rule to view the correct revisions of the product data.

   Similarly, you can apply revision rules to configure the families and features for a particular configuration. The revision rule (as well as effectivity and rule date) in the **Settings** panel of **Variant Configuration** *only* applies to the configurator families and features. For example, a feature may have a specific rule date, and the **Settings** panel controls whether that feature is available for selection in a configuration.

   > **Note**
   >
   > The revision rule and effectivity for configuration is separate and independent of the revision rule and effectivity for product data.
   >
   > Rule date is unique to configuration.

   You cannot view multiple configurations at the same time; if you want to see another configuration, you must apply a different revision rule.

   Revision rules are predefined by your system administrator, and you select a rule from the list of available rules.

   To override the default revision rule for the current session:

   a. In the **FILTER CRITERIA** section of the **Settings** panel, click the current revision and select a revision rule from the resulting **Revision Rule** list.

      > **Note**
      >
      > An invalid revision rule is a rule that contains a clause that is not valid for a particular type of object. While both valid and invalid revision rules appear in this list, the system will not let you apply an invalid revision rule.

   b. Click **Apply**.
      The system refreshes the displayed content to show only content configured in by the selected revision rule.

4. Filter configurator data by effectivity.

   You can configure configurator data with effectivity. When you apply the effectivity, the features are configured according to the unit range or the date range you specify. The system then shows the features that are in effect for the specified date, unit number or range.

You can edit the **Cfg0DefaultEffectivity** user preference to set the default effectivity. While editing this preference, you can specify date ranges or units.

**Note**

This preference is not supported in rich client.

Examples:

- *Date Range*: *<YYYY-MM-dd..YYYY-MM-dd>* (Example: **2023-11-08..2023-11-30** or **2023-11-30..UP**)
- *Today*: The present date. (Example: If **Today** is **2023-11-08**, it is displayed in the user interface as **2023-11-08..2023-11-09**)

For more information about setting this preference, see the description of this preference.

For information about retrieving a list of preferences, see *Where can I get a list of preferences?* in *Teamcenter Administration*.

Only the valid features are displayed in the structure views when you set the default effectivity.

**Note**

Your administrator configures the effectivity scheme used by your business by setting the **PCA_effectivity_shown_columns** preference. This preference can be set to show **Effectivity Date**, **Unit**, or **Both**.

You must ensure that the values specified for both **Cfg0DefaultEffectivity** and **PCA_effectivity_shown_columns** preferences are in sync. For example, if you set the default value as date range, then the effectivity shown column preference should also be set to date. Otherwise, the system displays a warning and the default value is set.

If your site supports date and unit effectivity, both selections are displayed in the **Settings** panel.

To override the default or change the existing dates:

a. In the **FILTER CRITERIA** section of the **Settings** panel, click **All Dates** or the displayed date under **Effectivity**.

b. Use the calendar to select the desired start date.

   To set a date range, use the calendar to select the desired end date.

   Using the pull-down menu, you can select **UP** (any date after the one specified) or **SO** (stock out), rather than a specific date.

c. Click **Set**.

d. To reconfigure the content to match the new date, click **Apply**.

5. Filter configurator data by rule date.

   BOM engineers who develop new products typically prefer to use the **Latest** rule date. That is because the configurator engineer is continuously improving configurator rules or adding new ones, and, therefore, the BOM engineer wants to validate the product against the **Latest** rule date.

Order engineers need to validate the variant rules or variant criteria against the rules that were valid at the time of the order. If the order is already in production, they might not want to apply the **Latest** rule date. In such cases, **Date** that is set on the order is used to validate variant rules or variant criteria.

The **Cfg0DefaultRuleDateForContent** user preference specifies the default rule date mode value for the configurator views. By default, it is set to **Latest**.

For information about retrieving a list of preferences, see *Where can I get a list of preferences?* in *Teamcenter Administration*.

a. Select a rule date as appropriate and click **Set**.

| Rule date | Description |
|---|---|
| **Date** | Sets the rule date selected from the widget. |
| **Latest** | Sets the rule date as the current time on the clock. |
| **System Default** | Sets the rule date as the system default. |
| **No Rule Date** | Sets the rule date as null. |

b. To apply the settings, click **Apply**.

6. Filter configurator data by intent.

You can filter out configurator data by specific intent while authoring variant conditions. For example, if you are authoring a bill of materials (BOM), you can focus on features with technical intent. Conversely, if you are looking at a bill of process (BOP), you can focus on features with marketing intent. The configurator administrator defines a business purpose or intent, such as manufacturing, marketing, or technical, to categorize families and constraints.

a. In the **Snapshot Intent** drop-down list, select the intents you want to filter the configurator data.

If configured, you can select the default values such as Manufacturing, Marketing, or Technical or select any custom intents defined by your administrator. The intents selected by default in the Snapshot Intents list may vary according to the configuration set by your administrator. The displayed content refreshes to reflect the filtering for the selected intent. Filtering is applied to all configurator contexts associated with the product.

b. Click **Apply**.

**Note**

If a family is used in multiple configurator contexts, it is recommended that you apply the same intents to that family across all contexts. For example, if the **Fuel Type** family has **Manufacturing** and **Technical** intents in the **Car** context, ensure that it also has **Manufacturing** and **Technical** intents in the **Engine** context. This consistency helps maintain predictable behavior and simplifies management across your product structures.

c. Make your selections to create a valid and complete configuration and click **Save**.

When you filter configurator data by applying **Snapshot Intents**, and then save the valid and complete SVR, the system records the specific intent that was applied when the valid and complete SVR was created and saved. This allows you to create multiple SVRs for the same product configuration, with each considered complete but also under different intents. For example, one SVR might be complete for a Technical intent, while another is complete for a Marketing intent. The variant rule text for each SVR reflects only the families and features that participated in the configuration under its specific snapshot intent.

The **Snapshot Intent** column in the **Variants** tab displays the intent that was used during configuration. If the column is hidden, click **Table Settings** > **Arrange**, and add **Snapshot Intents** from the **Available Columns** to the **Displayed Columns**, and then click **Save and Arrange**.

## Filter configurator objects using intents

You can dynamically filter families and constraints based on their assigned intents and allocation intents. This filtering helps you focus on specific subsets of variability relevant to your current task or domain.

### Prerequisites

Your administrator has enabled filtering by intents.

### Procedure

1. Open the configurator context and click the **Models**, **Features**, or **Constraints** tab.
2. In the **Intents** list, select the intent that you want to specify use for filtering.

   You can select multiple values as required. The objects in the view are refreshed based on the selected intents.

### Results

- For dynamic families, the standalone features are filtered as specified.
- If a family has an allocation intent assigned, the system uses this allocation intent for filtering. For example, if a family has an allocation intent of **Marketing** and a standard intent of **Manufacturing**, and you select **Marketing** in the breadcrumb, the family is filtered in. If you select **Manufacturing**, the family is filtered out.
- If a family does not have an allocation intent, the system uses its standard intent for filtering.
- If a family has no specified intents, neither standard nor allocation, it is always filtered in, which means that it is displayed regardless of the intents selected in the breadcrumb.

## Configure variants in guided mode

You can configure variants using the guided mode. The guided mode simplifies the configuration process by displaying only valid features and helps you create customized variant configurations. It also supports both complete and partial configurations while providing real-time feedback on configuration validity and completeness.

#### Procedure

1. Find and open the product to be configured and choose **Details** > **Variant Configuration**.
   The system displays a list of available product lines, product models, and groups. If the product has a variant configuration defined, the selections in the configuration are shown with the variability in the context.

   > **Note**
   >
   > Your administrator can deploy the system with the **Cfg0PrimaryBusinessRelevantAttribute** global constant that allows you to use either **Name** or **ID** as the primary business-relevant attribute. When the system is configured with the global constant that points to **Name** as a primary business property, all Product Configurator views display **Name** instead of **ID** in the corresponding areas.

2. Select the **Guided** mode. This is the default mode.

   In this mode, only valid features are displayed.

   Guided mode allows you to navigate only valid selections and configurations. If no configuration is applied, a model opens in guided mode, and the system guides you to select a valid group of features in which to configure a product. Every time a selection is made, the system reevaluates the rules and refreshes the list of features.

   Guided mode supports both complete and partial configurations.

   While radio buttons are displayed for making selections in **Guided** mode, check boxes are displayed for **Manual** mode.

   > **Note**
   >
   > If you have an invalid configuration, the system does not allow you to switch from manual to guided mode.

3. To begin creating a customized variant configuration, do one of the following:

   - Select the product line containing the product model with which you would like to work.

     The model families in the selected product line are selected by the system.

   - Select a product model with which you would like to work.

     The system automatically selects the product line containing the selected product model.

     When you make some selection changes and revert them, the system continues to display the **Save** icon.

> **Note**
>
> Product lines are only listed if they are defined in the configurator model.

4. After you have selected the appropriate product model in the work area, select each group from the list in the left panel and make the necessary feature selections in the work area.

   - Use the **Filter** box to help locate the appropriate feature. The text filter is not case-sensitive and is based on the family and feature properties shown (**Name** and **Description**, for example).

     > **Example**
     >
     > You type **mfv** into the **Filter** box.
     >
     > - The filtered list shows the families containing **mfv** in the family name, along with all the features in that family.
     > - The filtered list shows the features containing **mfv** in the feature name, along with its parent family.

   - An option button displays when a single-select feature is selected. Click the feature again to deselect.
   - In multiselect and Boolean families, you can select as follows:
     - indicates include.
     - indicates unselected.
     - indicates exclude.

   To exclude a feature, hover over it and select the **Exclude** icon.

5. Select one feature in a single-select family or one or more features in a multiselect family.
   The completeness indicator at the top displays one of three states: **Valid and Complete**, **Valid and Incomplete**, and **Invalid**.

   - **Valid and Complete** (applicable for both guided mode and manual mode)

     Valid and complete means that the BOM is targeting a complete structure, that is, a 100% BOM.

     When a configuration is valid and complete, you could take this configuration input and execute an order entry in the order system.

     On the contrary, if you do not see this completeness indicator, you should not typically perform the above action. It means that you have less content or more content.

     When a configuration is valid and complete, it assigns a value to *every* family. This is irrespective of whether the family is mandatory or discretionary.

   - **Valid and Incomplete** (applicable for both guided mode and manual mode)

     In guided mode, you can select the **Next Required** or the **Previous Required** option to navigate to the next or the previous family or feature that requires a value to be assigned to complete the configuration.

   - **Invalid** (applicable for manual mode only)

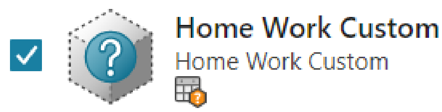
It means that the input criteria is invalid based on the configurator rules set in the current configurator context.

It may also refresh features displayed for other families as it applies the relevant rules by hiding invalid features. If none of the features in a family is still valid, the family itself is hidden.

Default features are selected by the system if they do not already have a value and designated with the indicator. Constraints are applied, validating the expression and adding any additional expression terms.

6. Select a family or feature that has a specified revision rule, effectivity, or rule date. Apply a different revision rule, effectivity, or rule date. This becomes an unconfigured family or feature.

   Such families and features are displayed with a different icon and have a question mark and an indicator that shows it as *configured out*. In such cases, the system switches to the manual mode. To proceed in guided mode, you can clear your selection of the unconfigured family or feature and select another family or feature. The system switches to the guided mode and you can continue to configure the structure.



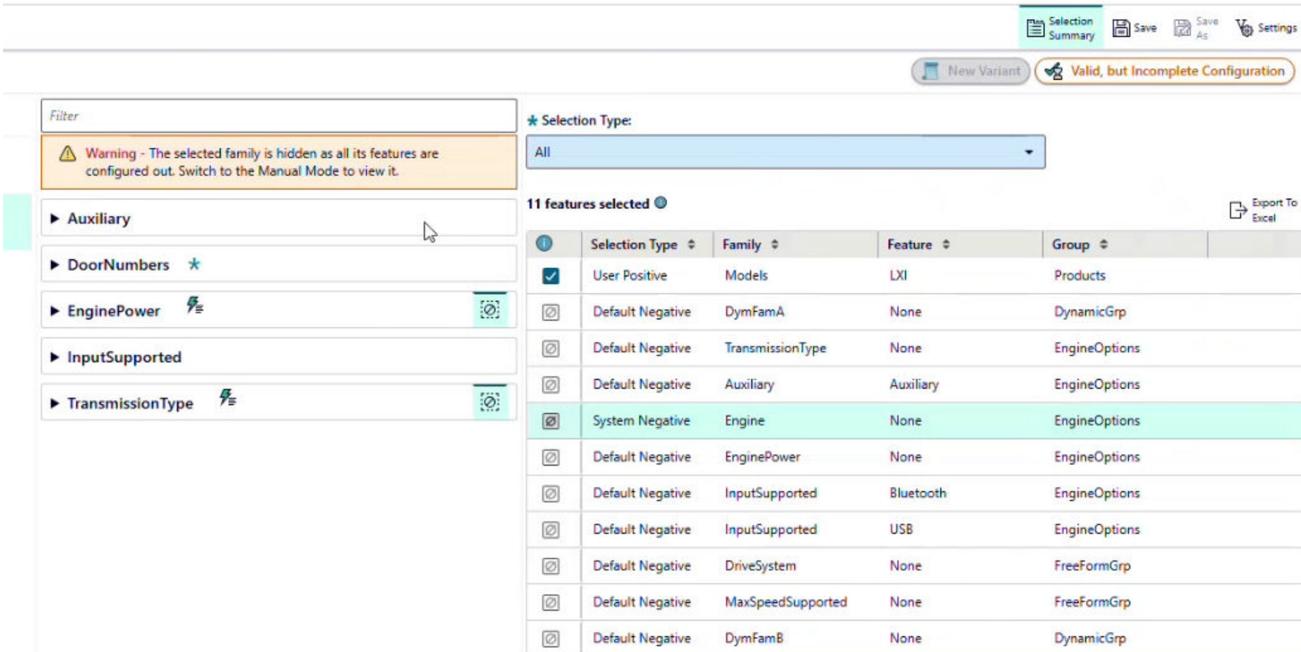

   - If a family is unconfigured and you select a feature within it, then both the family and feature are shown as unconfigured.
   - If you deselect an unconfigured feature or family and navigate to a different group, both the family and feature are not displayed by the system. Similarly, if you deselect an unconfigured feature or family, save it, and reopen it, neither the family nor the feature are displayed by the system.

   You can optionally save the content with an unconfigured family or feature as an SVR.

7. An optional feature family contains product configurations where users are not required to select a feature for the family. For example, you may have a family with features for optional equipment.

   If optional feature families are defined in a product model, an additional option is displayed in the family section header. Click if you want *none* of the optional family features to appear in the structure.

   When Teamcenter configures out all features from an optional family based on constraints, the family itself is hidden from the **Variant Configuration** view. This does not impact the configuration. If you select **Selection Summary**, and **All** from the **Selection Type**, the optional family that is hidden from the GUI is displayed as a **System Negative** type. When you click the optional family that is hidden from the GUI, the system displays a message as shown below.

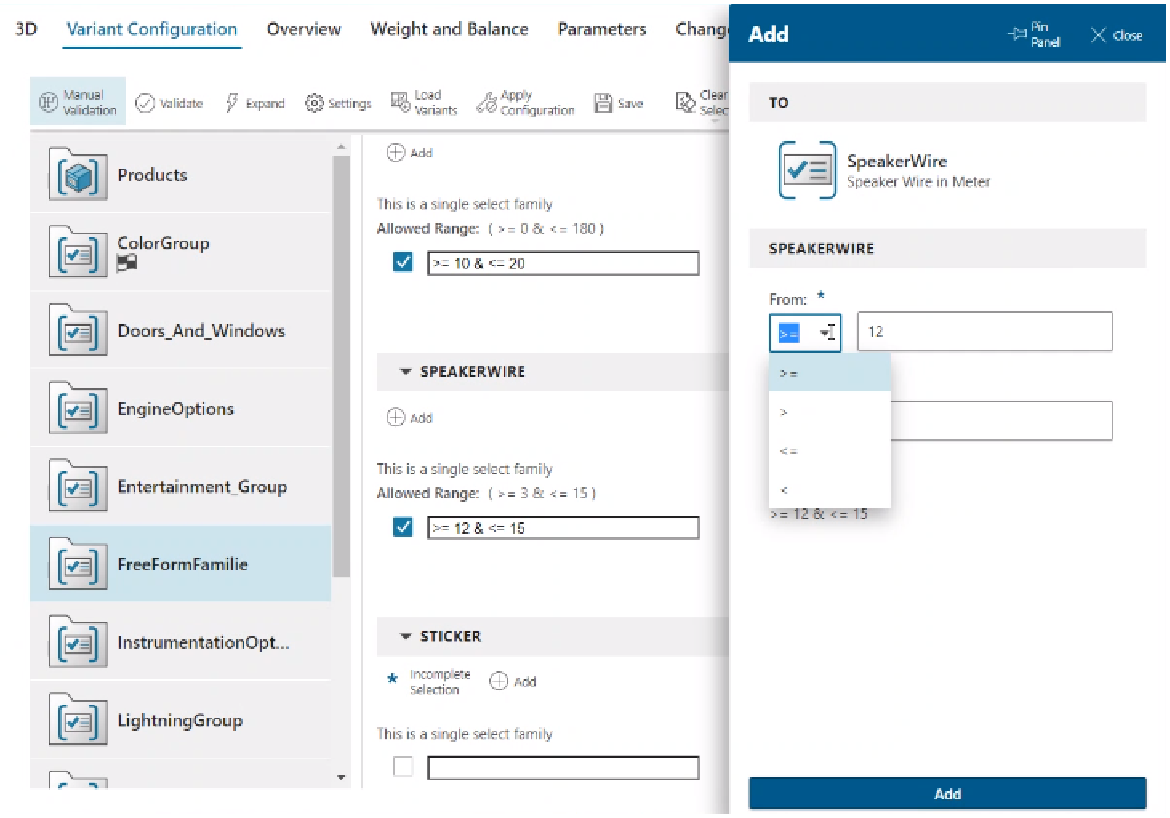



8. Typically, features are predefined for each family, but free-form families allow you to enter a value for a feature at any time. If a family is specified as free-form, you can enter numbers, strings, or dates, depending on the data type defined.

   Select an assembly that contains free-form families in the **Variant Conditions** tab, and click the **Add Feature** option. You can use operators such as **>=**, **>**, **<=**, and **<**. If you select **<=** or **<** in the **From** field, the **To** field is disabled.

   If you have a **SpeakerWire** free form family in your configuration context, then you can enter the free form value as a discreet value or a range in the text box or use the **Add** panel to author the range. According to the range or the discreet value you specified, the configuration context displays only the correct BOM lines. For combining the values, you can use the AND (**&**) operator. The OR (**|**) operator is not supported.

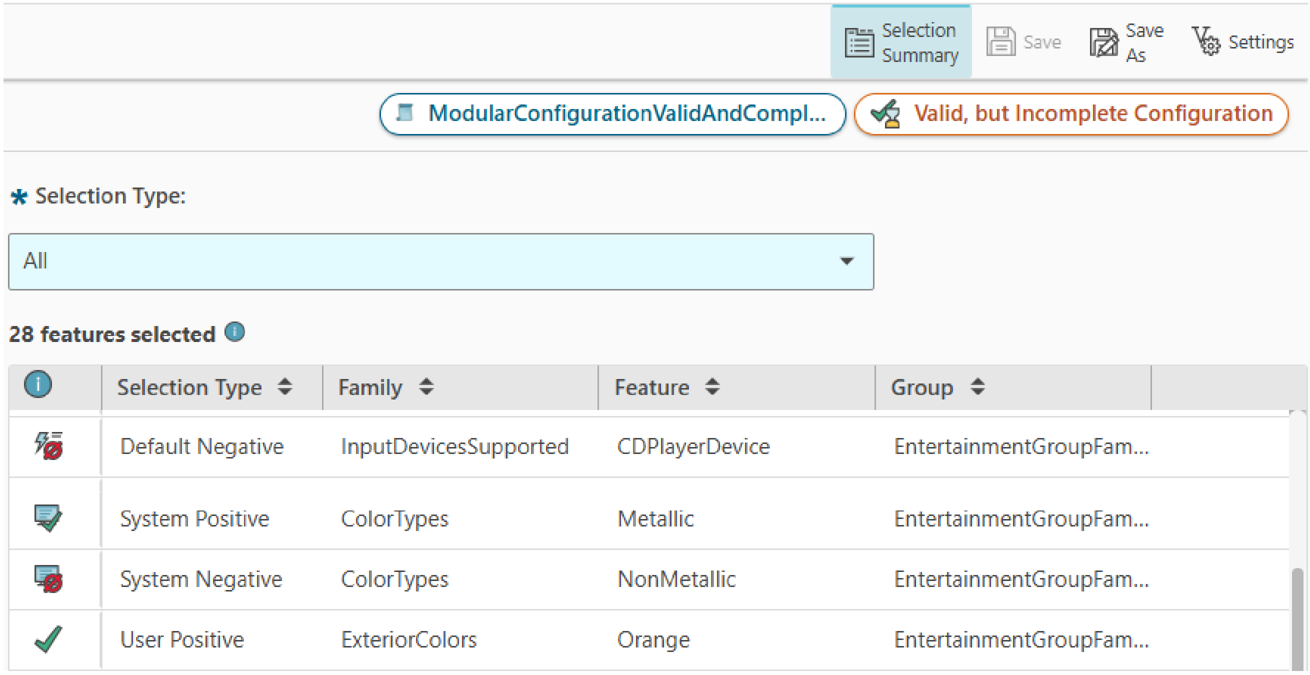
 




When you apply the configuration to the structure, the system filters the structure to show only the items that meet the condition for the feature.

9. As you make selections for the custom configuration, in the **Selection Summary**, filter your selections using one of the following options from the **Selection Type** list.

| Selection Type | Description |
|---|---|
| **User** | Displays the features selected by the user. |
| **All** | Displays all features that are selected. These include user and system selections. |
| **Incomplete Families** | Displays the families from which you have not made any selections yet. |




In the selection summary table, you can:

- Identify the type of selection by viewing the icon in the first column. For example, user positive selections are indicated by ✓. To know more about the icons and the type of selection they indicate, hover over the ⓘ icon in the column header.
- View information about the **Family**, **Feature**, and **Group** of the selected features.
- Sort the columns in ascending or descending order as required. To do this, click the header cell of the specific column and select the required option.
- Filter the column contents. To do this, click the header cell of the specific column, select the option from the **Contains** list, enter the filter criteria in the search box and click **Filter**.

The selection summary options help you cross-probe to narrow down and select specific features to complete your configuration.

10. At any time, click **Clear All** to clear all user and system selections as well as any reported violations across all groups.
    This provides a blank configuration from which to start.
11. (Optional) **Save the custom configuration**.
12. Click **Apply Configuration** to apply the selected variant configuration to the current content and to refresh the content in the product.

## Configure variants in manual mode

You can configure variants using the manual mode. The manual mode help you view all the available features, including potentially invalid ones, and manually select and validate configurations. You can identify and resolve conflicting constraints, understand system indicators for violations, and apply configurations to customize product structures.

You can view all the product lines and product models in a configurator context. Product lines are sometimes referred to as *series* and provide a high-level organization of product models.

### Procedure

1. Find and open the product to be configured and choose **Details** > **Variant Configuration**.
   The system displays a list of available product lines, product models, and groups. If the product has a variant configuration defined, the selections in the configuration show with the variability in the context.

   > **Note**
   >
   > Your administrator can deploy the system with the **Cfg0PrimaryBusinessRelevantAttribute** global constant that allows you to use either **Name** or **ID** as the primary business-relevant attribute. When the system is configured with the global constant that points to **Name** as a primary business property, all Product Configurator views display **Name** instead of **ID** in the corresponding areas.

2. Switch to **Manual** mode by selecting **Guided** > **Manual**.
   You are now in manual mode.

   - You see all available features, including features that are not valid in the current context.
   - Summary models are listed below the models in a particular product group and act as a dependency chain to other places in the model. For example, one particular car model requires gasoline, while another requires diesel. Choosing one of these fuel categories triggers changes in other parts of the model, such as the engine itself, the need for spark plugs, the need for a fuel receptacle, and so on.
   - You can look at entire variability, choose variety of features, and assess if those selections are valid based on the configurator rules. You can validate your selections and save your configuration. If there are any conflicting constraints, the system reports violations as error messages and specifies the conflicting selections.
   - Single select groups are designated as such to show the intent of the groups. However, when you select a feature, other features in the same family remain displayed, allowing you to select additional features.
   - Configurator rules and defaults are *not* automatically applied. You must manually apply configurations by clicking .
   - Automatic selection, filtering, and indications of invalid selections are *not* automatically provided. You must manually validate and apply system selections .

   > **Note**
   >
   > If you have an invalid configuration, the system does not allow you to switch from manual to guided mode.

3. Select the appropriate model and the summary model.

When you make some selection changes and revert them, the system continues to display the **Save** icon.

4. Select the desired group to see the features.

   You can only view one group at a time.

   - Use the **Filter** box to help locate the appropriate feature. The text filter is not case-sensitive and is based on both the family and the feature names.

     **Example**

     You type **mfv** into the **Filter** box.

     - The filtered list shows the families containing **mfv** in the family name, along with all the features in that family.
     - The filtered list shows the features containing **mfv** in the feature name, along with its parent family.

   - Because manual mode allows multiple selections for multiselect and Boolean families, you can select as follows:

     - ☑ indicates include.
     - ☐ indicates unselected.
     - ∅ indicates exclude.

   To exclude a feature, hover over it and select the **Exclude** icon.

5. Select the appropriate features for this custom variant configuration.

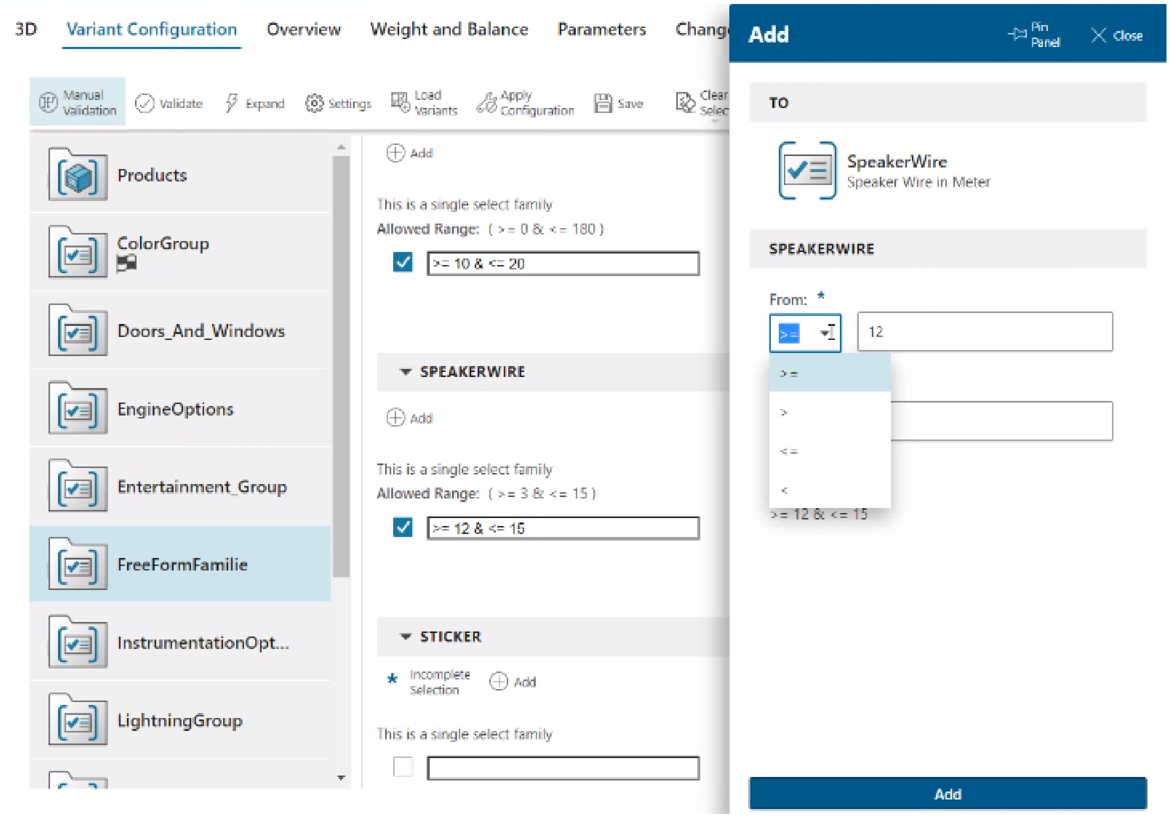
6. An optional feature family contains product configurations where users are not required to select a feature for the family. For example, you may have a family with features for optional equipment.

   If optional feature families are defined in a product model, two options display in the family section header. Click to allow *any* of the optional family features to appear in the structure. Click if you want *none* of the optional family features to appear in the structure.

7. Typically, features are predefined for each family, but free-form families allow you to enter a value for a feature at any time. If a family is specified as free-form, you can enter numbers, strings, or dates, depending on the data type defined.

   Select an assembly that contains free-form families in the **Variant Conditions** tab, and click the **Add Feature** option. You can use operators such as **>=**, **>**, **<=**, and **<**. If you select **<=** or **<** in the **From** field, the **To** field is disabled.

   If you have a **SpeakerWire** free form family in your configuration context, then you can enter the free form value as a discreet value or a range in the text box or use the **Add** panel to author the range. According to the range or the discreet value you specified, the configuration context displays only the correct BOM lines. For combining the values, you can use the AND (**&**) operator. The OR (**|**) operator is not supported.




When you apply the configuration to the structure, the system filters the structure to show only the items that meet the condition for the feature.

**8.** At any time, click **Validate** to validate the configuration.
Each violation is identified using the error indicator.

> **Note**
>
> The type of violation assigned to a feature or feature group is based on the severity assigned by your configuration analyst. You cannot change the severity level in the current Active Workspace release. The severity is currently set to *error* by default.

Hover over the indicator icon next to a feature to view the reason for the violation.

- Next to a feature to view the reason for the violation.
- Next to a group to view the features within that group that have a violation.

The system identifies violations in the configuration using the following indicators:

- error
- warning
- information

**Note**

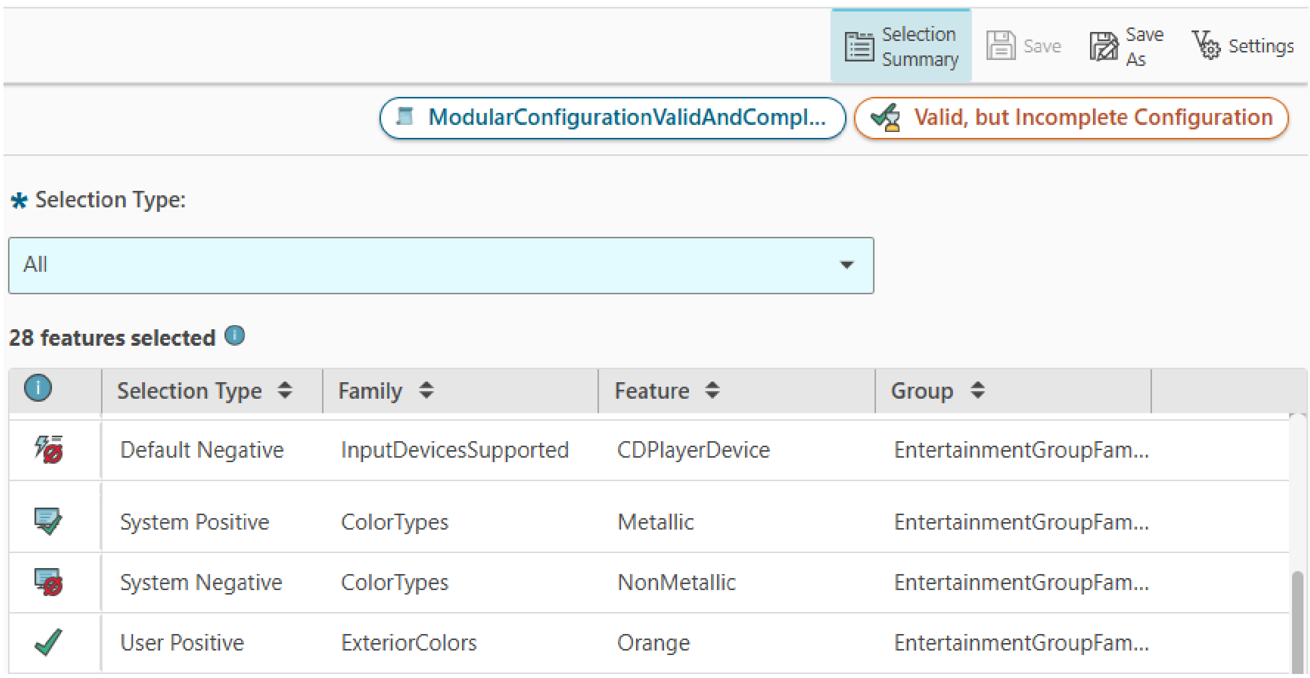
The type of violation assigned to a feature or feature group is based on the severity assigned by your configuration analyst. You cannot change the severity level in the current web client release. The severity is currently set to *error* by default.

Hover over the indicator icon:

- Next to a feature to view the severity and reason for the violation.
- Next to a group to view the features within that group that have a violation.

9. As you make selections for the custom configuration, in the **Selection Summary**, filter your selections using one of the following options from the **Selection Type** list.

| Selection Type | Description |
|---|---|
| **User** | Displays the features selected by the user. |
| **All** | Displays all features that are selected. These include user and system selections. |
| **Incomplete Families** | Displays the families from which you have not made any selections yet. |




In the selection summary table, you can:

- Identify the type of selection by viewing the icon in the first column. For example, user positive selections are indicated by ✓. To know more about the icons and the type of selection they indicate, hover over the ⓘ icon in the column header.
- View information about the **Family**, **Feature**, and **Group** of the selected features.

- Sort the columns in ascending or descending order as required. To do this, click the header cell of the specific column and select the required option.
- Filter the column contents. To do this, click the header cell of the specific column, select the option from the **Contains** list, enter the filter criteria in the search box and click **Filter**.

The selection summary options help you cross-probe to narrow down and select specific features to complete your configuration.

**Note**

You can export the contents of the selection summary if required. To do this, in the **Variant Configuration** tab, in the **Selection Summary** view, click **Export to Excel**.

10. At any time, click **Expand** to apply system selections.

    The system makes selections based on configurator rules or constraints and displays the indicator to designate the system selections.

    The completeness indicator at the top displays one of three states: **Valid and Complete**, **Valid and Incomplete**, and **Invalid**.

    - **Valid and Complete** (applicable for both guided mode and manual mode)

      Valid and complete means that the BOM is targeting a complete structure, that is, a 100% BOM.

      When a configuration is valid and complete, you could take this configuration input and execute an order entry in the order system.

      On the contrary, if you do not see this completeness indicator, you should not typically perform the above action. It means that you have less content or more content.

      When a configuration is valid and complete, it assigns a value to *every* family. This is irrespective of whether the family is mandatory or discretionary.

    - **Valid and Incomplete** (applicable for both guided mode and manual mode)

      In guided mode, you can select the **Next Required** or the **Previous Required** option to navigate to the next or the previous family or feature that requires a value to be assigned to complete the configuration.

    - **Invalid** (applicable for manual mode only)

      It means that the input criteria is invalid based on the configurator rules set in the current configurator context.

11. At any time, click **Clear Selections** > **Clear System Selections** to clear any system selections that may have been previously calculated, while preserving your user selections.

    Alternatively, click **Clear Selections** > **Clear All** to clear all user and system selections, as well as any reported violations across all groups.

12. (Optional) **Save the custom configuration**.

13. Click **Apply Configuration** to apply the selected variant configuration to the current content.

> **Note**
>
> If you save the current context (for example, the subset definition or workset), the variant configuration displays in the list of saved variants.

## Configure by saved variant or variant criteria

You can configure a specific product variant by applying the predefined saved variants or variant criteria. You can load these stored configurations that consist of families and features (for example, color = red).

Variants allow you to select and apply feature families (for example, color) and allow features of those families (for example, red and blue) to configure a part or assembly.

To configure a particular variant of a product, apply the appropriate variant (a group of families and features such as **color = red**, **material = cotton**) or variant criteria (a revisable version of a saved variant). These items are predefined by the configuration analyst and are stored in Teamcenter for users to retrieve later.

To configure content by a saved variant or variant criteria (VC):

### Procedure

1. Find and open the product to be configured and choose **Details** > **Variant Configuration**.

    > **Note**
    >
    > If your configuration analyst has not associated a configurator context with the product, **Variant Configuration** is not displayed.

2. To open the SVR or VC, click **Load Variants**.

3. Search for the appropriate configuration by typing the name (or part of the name) of the SVR or VC to be loaded.

    The available SVRs and VCs is obtained from the configurator context associated with the product or from the product itself. Only SVRs and VCs directly associated with the product can be loaded. A variant that is created for a product is not available for other products, even if the products share a configurator context.

    The SVRs and VCs matching the search criteria are listed.

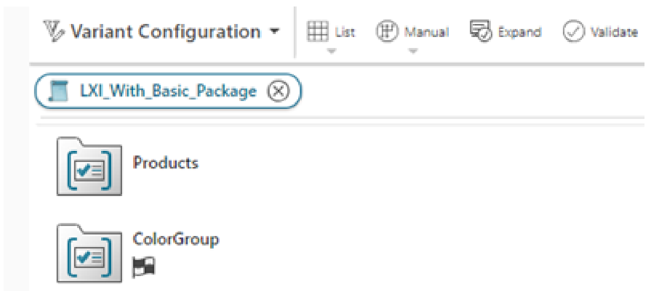
4. Select an SVR or VC and click **Load**. The system loads the selected configuration.

    > **Note**
    >
    > In this example, a single variant configuration is applied. Multiple variants can be applied to *4G structures* from the **Variant** section of the **Configuration** panel. Access the **Configuration** panel by clicking **Configure**.

- If no variant is applied to the structure, the model opens in guided mode.
- If a configuration is applied, the model displays in manual mode.

5. Remove the variant rule or variant condition applied to the structure.

**Example**




## Solve behavior for default rules

Default rules anticipate common settings that align with most users' preferences. When conflicts arise, Teamcenter resolves them by determining which default rules to suppress based on their sequence number and affiliation with the input configuration.

Default rules, created by the configuration specialist, enhance usability by assisting users in forming a complete product configuration. These rules anticipate common settings that align with the preferences of most users.

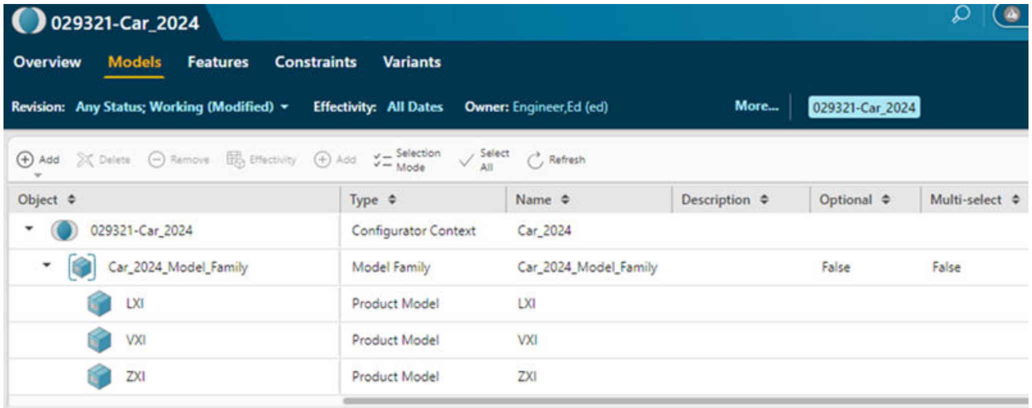
Default rules that are in conflict with the user input or constraint rules with a higher severity are automatically skipped when expanding the configuration. Teamcenter automatically resolves all conflicts that involve default rules by determining the optimum set of default rules to be suppressed based on their affiliation with the input configuration and their ranking based on sequence number. Default rules with a higher sequence number are evaluated later and therefore have a lower ranking. Those with the same sequence number have the same ranking. The system infers the rank of default rules based on site preferences set by the administrator.

### Prerequisites

This topic assumes that the administrator has set the required site preferences for default rules as described in Setting preferences for default rules in *Product Configurator on Rich Client — Usage*.

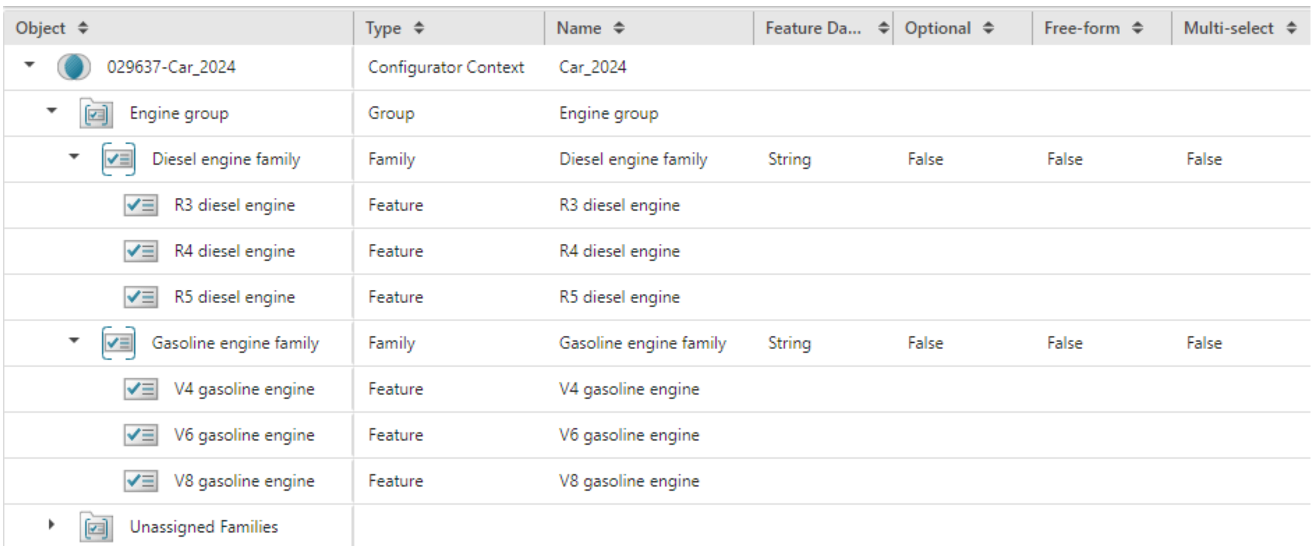
### Procedure

1. Create a **Car_2024** configurator context with a positive or negative bias.
2. Create a model family and product models as follows:




| Object | Type | Name | Description | Optional | Multi-select |
|---|---|---|---|---|---|
| 029321-Car_2024 | Configurator Context | Car_2024 | | | |
| Car_2024_Model_Family | Model Family | Car_2024_Model_Family | | False | False |
| LXI | Product Model | LXI | | | |
| VXI | Product Model | VXI | | | |
| ZXI | Product Model | ZXI | | | |

3. Create a group, family, and features as follows:



| Object | Type | Name | Feature Da... | Optional | Free-form | Multi-select |
|---|---|---|---|---|---|---|
| 029637-Car_2024 | Configurator Context | Car_2024 | | | | |
| Engine group | Group | Engine group | | | | |
| Diesel engine family | Family | Diesel engine family | String | False | False | False |
| R3 diesel engine | Feature | R3 diesel engine | | | | |
| R4 diesel engine | Feature | R4 diesel engine | | | | |
| R5 diesel engine | Feature | R5 diesel engine | | | | |
| Gasoline engine family | Family | Gasoline engine family | String | False | False | False |
| V4 gasoline engine | Feature | V4 gasoline engine | | | | |
| V6 gasoline engine | Feature | V6 gasoline engine | | | | |
| V8 gasoline engine | Feature | V8 gasoline engine | | | | |
| Unassigned Families | | | | | | |

4. Verify the value of the **Cfg0DefaultRuleSortSequence** site preference. The default value of this preference is **cfg0Sequence:ASC**. This means that only sequence numbers are considered for default rules and the lower sequence takes priority.

   Override this preference by setting the value as **Cfg0DefaultRule:object_name:ASC**.

5. Create availability rules as follows in the **Grid Editor**.

   Availability rules are required only if you create a configurator context with a negative bias.

| Variability Content | 1964/001 Availability Rule | 1973/001 Availability Rule | 1974/001 Availability Rule |
|---|---|---|---|
| Subject | | | |
| Diesel engine family | | | |
| R3 diesel engine | | | ✓ |
| Gasoline engine family | | | |
| V4 gasoline engine | | ✓ | |
| V6 gasoline engine | ✓ | | |
| Condition | | | |
| Car_2024_Model_Family | | | |
| LXI | ✓ | ✓ | ✓ |

6. Create default rules as follows in the **Grid Editor**.

| Variability Content | 1976/001 Default Rule | 1977/001 Default Rule |
|---|---|---|
| Subject | | |
| Gasoline engine family | | |
| V4 gasoline engine | ✓ | |
| V6 gasoline engine | | ✓ |
| Condition | | |
| Car_2024_Model_Family | | |
| LXI | ✓ | ✓ |

7. Click **Settings** > **Arrange**. In **Available** columns, search for **object_name**, select the **Name** column, and move it to **Displayed Columns**. Ensure that this is next to the **Sequence Number**. Click **Save and Arrange**.

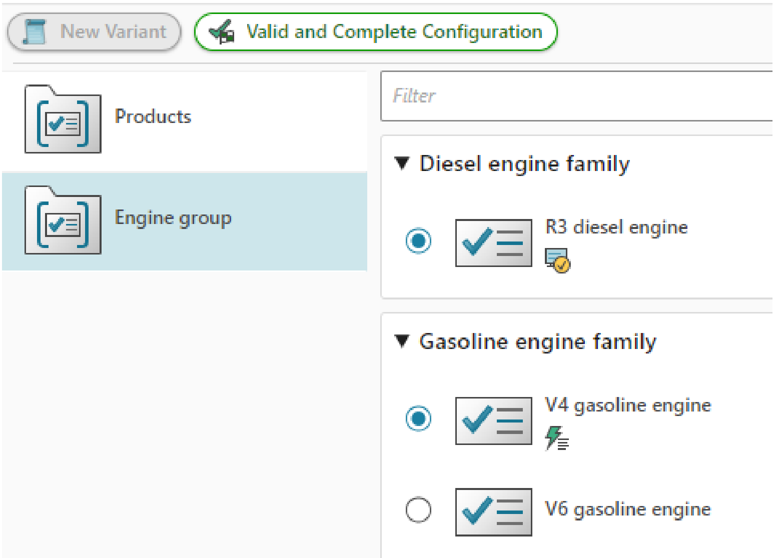
   **object_name** is the internal name for the **Name** column.

8. Edit the default rules to specify the **V4 gasoline engine** and **V6 gasoline engine names** as **V4_gasoline** and **V6_gasoline**, respectively.

| Severity | Name | Subject | Condition |
|---|---|---|---|
| Error | 1964 | 'Gasoline engine family' = 'V6 gasoline engine' | Car_2024_Model_Family = LXI |
| Error | 1973 | 'Gasoline engine family' = 'V4 gasoline engine' | Car_2024_Model_Family = LXI |
| Error | 1974 | 'Diesel engine family' = 'R3 diesel engine' | Car_2024_Model_Family = LXI |
| | V4_gasoline | 'Gasoline engine family' = 'V4 gasoline engine' | Car_2024_Model_Family = LXI |
| | V6_gasoline | 'Gasoline engine family' = 'V6 gasoline engine' | Car_2024_Model_Family = LXI |

9. Open the **Variants** tab and click **Variant Configuration**. It shows a valid and complete configuration.

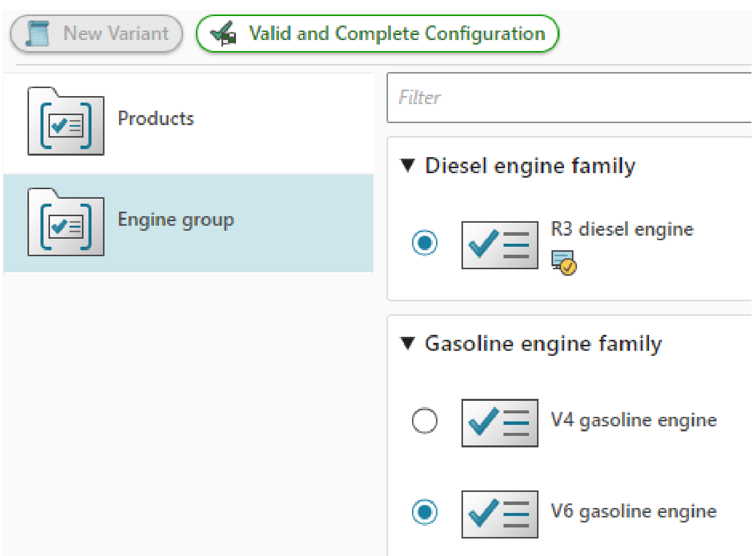
10. Navigate to the Engine group.




The **R3 diesel engine** is a system selection because of an availability rule. The **V4 gasoline engine** is the default selection. This is because the default rule name was **V4_gasoline** for **V4 gasoline engine** and **V6_gasoline** for **V6 gasoline engine**. The alphabetical order of the default rule name takes priority for the default rules since you have overridden the **Cfg0DefaultRuleSortSequence** site preference and set the value to **Cfg0DefaultRule:object_name:ASC**.

11. Edit the names of the default rules to specify **y_gasoline** and **x_gasoline** for **V4 gasoline engine** and **V6 gasoline engine**, respectively.

12. In the **Variant Configuration** view, navigate to the Engine group.




The **V6 gasoline engine** is the default selection now. This is because the default rule name was **y_gasoline** for **V4 gasoline engine** and **x_gasoline** for **V6 gasoline engine**. The alphabetical order of the default rule name takes priority for the default rules since you have overridden the **Cfg0DefaultRuleSortSequence** site preference and set the value to **Cfg0DefaultRule:object_name:ASC**.

13. Override the **Cfg0DefaultRuleSortSequence** site preference and set the value as follows:

    **Cfg0DefaultRule:object_name:ASC**

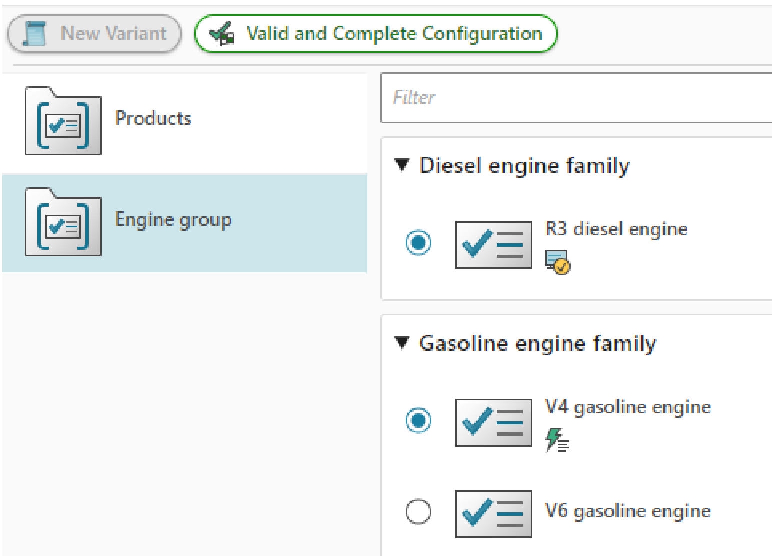
    **Cfg0DefaultRule:cfg0Sequence:ASC**

14. Edit the names of both the default rules and specify **gasoline** as the value.

15. Edit the **Sequence Number** property of the default rule and specify **10** as the value for **V6 gasoline engine**. The sequence number of **V4 gasoline engine** remains unchanged as **0**.

16. Save your changes.

17. In the **Variant Configuration** view, navigate to the **Engine** group.




The **V4 gasoline engine** is selected. This is because the names (**gasoline**) are the same for both the default rules. However, the sequence number was **0** for **V4 gasoline engine** and **10** for **V6 gasoline engine**. The system, based on the preference value, first considers the names of the default rules and since they are the same, it considers the sequence number next to prioritize the rules. In this case, the lower sequence number is prioritized.

## Guidelines for selecting search methods

The example in this topic provides search guidelines while configuring product structures.

While configuring a product structure, you can search for features using the **Feature** or **Expression** option in the **Search by** list. Understanding when to use each method is crucial for efficient variant configuration.

To illustrate the various scenarios for using these search methods, consider the following families in a vehicle configuration:

| Engine Type Family | Transmission Type Family | Drivetrain Type family |
|---|---|---|
| Petrol Engine | Manual Transmission | FWD Drivetrain |
| Diesel Engine | Automatic Transmission | RWD Drivetrain |
| Hybrid Engine | CVT Transmission | AWD Drivetrain |
| Electric Engine | - | - |

For searching using basic feature combinations that span across different families, such as Petrol Engine AND Manual Transmission, selecting either **Feature** or **Expression** will generate identical results. Similarly, when searching within a single feature family, such as all Petrol Engine configurations, you can choose either search method based on your preference, as both approaches will provide the same comprehensive results.

When you use expression-based filtering, the system performs solving operations in the background. As a result, searching by expression is slower compared to using the standard search by feature option.

The following table suggests the recommended guidelines for searching using the **Expression** or **Feature** option.

| Search method | When to use | Example | Special considerations |
|---|---|---|---|
| Search by **Feature** | When features are connected by OR across different families. | Petrol Engine OR Manual Transmission | Recommended for inclusive searches. |
| | When dealing with potentially conflicting criteria. | Electric Engine with Petrol Engine AND Manual Transmission | Provides more predictable results. |
| | When searching across multiple families with OR conditions. | Petrol Engine OR Manual Transmission OR AWD Drivetrain | More user-friendly for multiple selections. |
| Search by **Expression** | When using nested conditions or multiple logical operators. | Petrol Engine AND (Manual Transmission | Provides more precise control when you |

| Search method | When to use | Example | Special considerations |
|---|---|---|---|
| | | OR Automatic Transmission) | specify the search criteria. |
| | When excluding specific features or combinations. | Petrol Engine AND Not Automatic Transmission | Required for NOT conditions. |
| | When using specific combinations of multiple features. | (Petrol Engine AND Manual Transmission) OR (Electric Engine AND Automatic Transmission) | Allows precise logical grouping of the search criteria. |

## Filter variability data by intent while authoring variant conditions

You can filter out configurator objects by specific intent while authoring variant conditions. For example, if you are authoring a bill of materials (BOM), you can focus on objects with technical intent. Conversely, if you are looking at a bill of process (BOP), you can focus on objects with marketing intent.

The configurator administrator defines a business purpose or intent such as manufacturing, marketing, or technical to categorize families and constraints. For more information, see Categorize and filter families and constraints by business purpose or intent.

### Procedure

1. Find and open the product or item revision to be configured.
2. Click the **Models**, **Features**, or **Constraints** tab.
3. To filter the variability data by intents, in the breadcrumb, select the required option from the **Intents** drop-down list. For example, you can filter using the **Technical**, **Manufacturing**, or **Marketing** intent.

   When you apply intent filters for a Bill of Materials (BOMs) with its standard variant conditions, the **Variant Configuration** view displays the specific intent filters that are applied.

   Any BOM line falling outside of the applied intent filter is treated as if it is configured out, and this status influences how other constraints are evaluated. For example, if a family A has an intent X that is outside of the current intent filter, then Family A will be configured out, and subsequent constraint evaluations will proceed based on this exclusion.

   > **Note**
   >
   > You can filter using intents if your administrator has included this ability in your configuration.

## Author variant formula in the grid

You can author variant formula by selecting features from the grid. This is the recommended method.

You can also **author variant formula by typing the formula in the text editor**. This is recommended only for complex formula or combinations that cannot be selected on the grid.

**Procedure**

1. Find and open the product for which you want to author variant formula.
2. Select the BOM lines for which you want to apply variant conditions and click the **Variant Conditions** tab.

   The BOM lines you selected are displayed as columns at the top. You can hover over the column header to view the complete name of the BOM line.
3. (Optional) Enable **Variant Formula** column in your work area, if it is not available.

   This column displays the variant formula for each BOM line, and you can edit this in the grid.

   a. In your work area, click **Settings** .
   b. Click **Arrange** and search for the **Variant Formula** column.
   c. Select the **Variant Formula** column and click **Move up**  to bring it to the top of the **Arrange** panel.
4. Click **Start Edit**  on the top of the grid
5. Make the required selections in the grid.
6. Click **Save Edits** .

   **Note**

   You can export the contents of the Variant Conditions grid view to Microsoft Excel if required. To do this, click **Export to Excel** . The **Export to Excel** panel appears if you select **Expand All**. The icons for the user positive, user negative, and configured out objects, and the corresponding text is displayed in the panel. You modify the default names if required, and click **Export**. When you specify names for the icons, other than the default ones, you can include spaces in the names. However, the text box cannot be blank, you must specify a name or retain the default values.

   If there are no icons, a message which indicates that the contents are exported to Microsoft Excel is displayed.

## Author variant formula in the editor

You can author variant formula by typing the formula in the text editor. This is recommended only for complex formula or combinations that cannot be selected on the grid.

You can also **author variant formula by selecting features from the grid**. This is the recommended method.

In the **Variant Formula Editor**, you can author complex variant conditions that cannot be authored in the grid. This view is used by expert users who want to write the expression directly in the editor. When you save such a variant condition, it is not available in the grid.

### Procedure

1. Find and open the product for which you want to author variant formula.
2. Select the BOM lines for which you want to apply variant conditions and click the **Variant Conditions** tab.

   The BOM lines you selected are displayed as columns at the top. You can hover over the column header to view the complete name of the BOM line.
3. (Optional) Enable **Variant Formula** column in your work area, if it is not available.

   This column displays the variant formula for each BOM line, and you can edit this in the grid.

   a. In your work area, click **Settings** .
   b. Click **Arrange** and search for the **Variant Formula** column.
   c. Select the **Variant Formula** column and click **Move up**  to bring it to the top of the **Arrange** panel.
4. Select one of the BOM lines, click on the top of the column, and click **Show Variant Formula Editor**.

   Even if you select multiple BOM lines in the **Variant Conditions** view, you can edit only one BOM line at a time.
5. To edit the variant condition, in the **VARIANT FORMULA EDITOR**, click **Start Edit** .
6. Click inside the box to start creating your formula.

   The **Intellisense** autocompletion feature is turned on by default. When it is turned on, you can get suggestions by pressing either Ctrl + space bar or space bar and then make a selection. After making the selection, press the space bar again to get the appropriate logical operators. Continue to create a complete formula and click **Validate Syntax** to validate the syntax of the formula you have created manually.

   Example: `ExteriorColor = NeptuneBlue`

   When the autocompletion feature is turned on, the system honors the name or the ID as the primary business attribute depending on how the attribute is set at your site by your administrator. This means that if name is set as the primary attribute, you must type the name in the formula editor and not the ID to complete the formula. Additionally, the administrator at your site can set the visibility of option families and the display of family namespace. By default, option families are visible, and the family namespace is not displayed. In such cases, you can type the option families in the formula editor, but not the family namespace to complete the formula. The system honors the configurations set by the administrator and makes suggestions based on these configurations. Consult the administrator at your site for more information.

> **Note**
>
> If you have families or features with identical names in the configurator context, they create ambiguity when creating an expression in the formula editor. In such cases, you should use the object IDs instead of the names to create the expressions. In case the object IDs are also identical, then contact your administrator to enable the necessary preferences, allowing you to create the expression by prefixing the namespace.

When you type the suggestions, the system performs validations for each suggestion when the feature is turned on. However, when the feature is turned off, the system does not perform any validation.

When the autocompletion feature is turned off, you must author the formula using the internal Teamcenter formula format.

You can also use custom formula in the formula editor. By default, custom formula is not enabled. Consult your administrator to enable it.

7. Make the required changes to the variant formula for the selected BOM line.

   The formula displayed in the **Variant Formula** column is the display formula whereas it is an internal formula for the editor. The server converts the internal formula to the display formula.

   Examples where the logical operators are different:

| Variant Formula column | Formula in editor |
|---|---|
| **Engine = V4 OR Engine =V8** | **(Engine = V4 \| Engine =V8)** |
| **Music Player = Radio AND Music Player = CD Player** | **(Music Player = Radio & Music Player = CD Player)** |

8. Click **Save Edits** .

   The system validates the expression syntax only after you save the changes.

   If you specify a wrong formula in the editor and click **Save Edits**, the system displays a syntax error or invalid error as appropriate.

   You cannot save the changes if they are invalid.

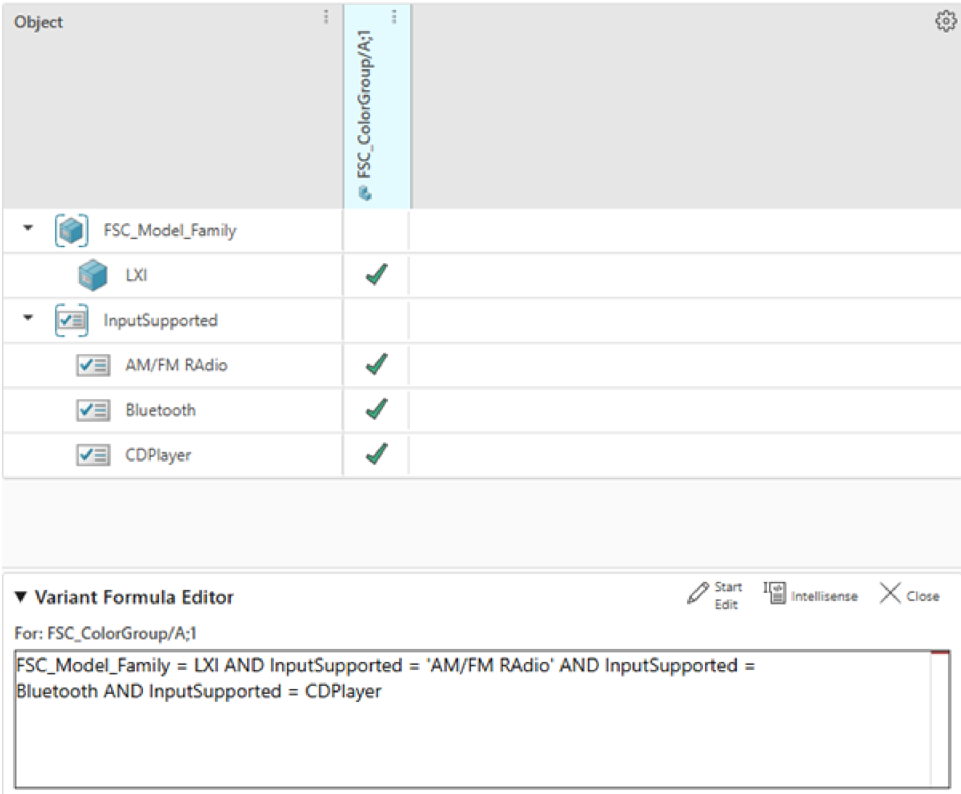
9. (Optional) To validate the expression against the constraints associated with the configurator context, after saving multiple changes, click **Validate All**  on the top of the grid.

### Examples

The following are some examples of the differences when selections are made for single select and multiselect families.

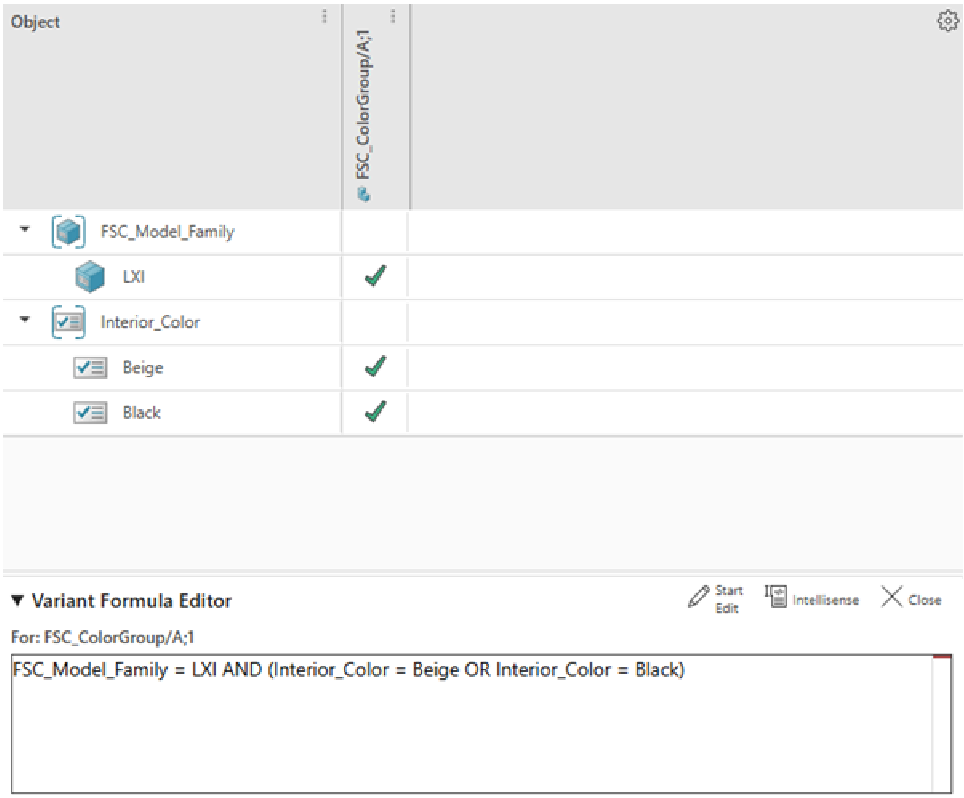
1. Open a structure, select a BOM line, and make selections in the **Variant Conditions** tab as follows. Additionally, select the BOM line, click on the top of the column, and click **Show Variant Formula Editor**.
2. In a multiselect family, feature selections are connected using **AND**.

Example:



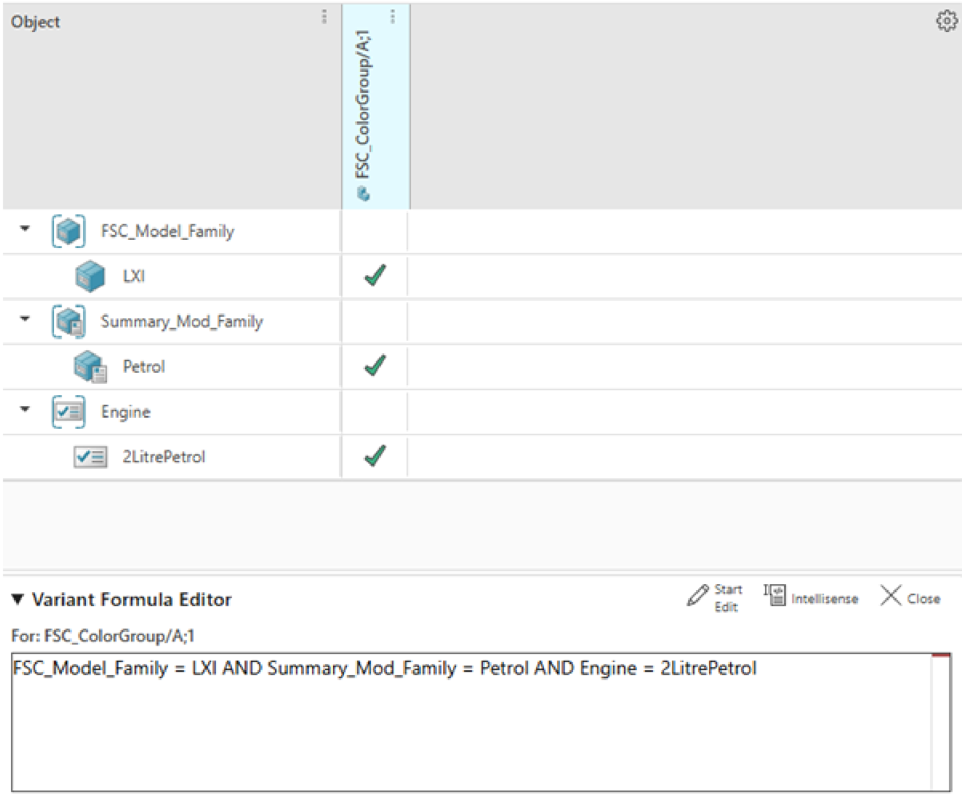

3. In a single select family, feature selections are connected using **OR**.

   Example:



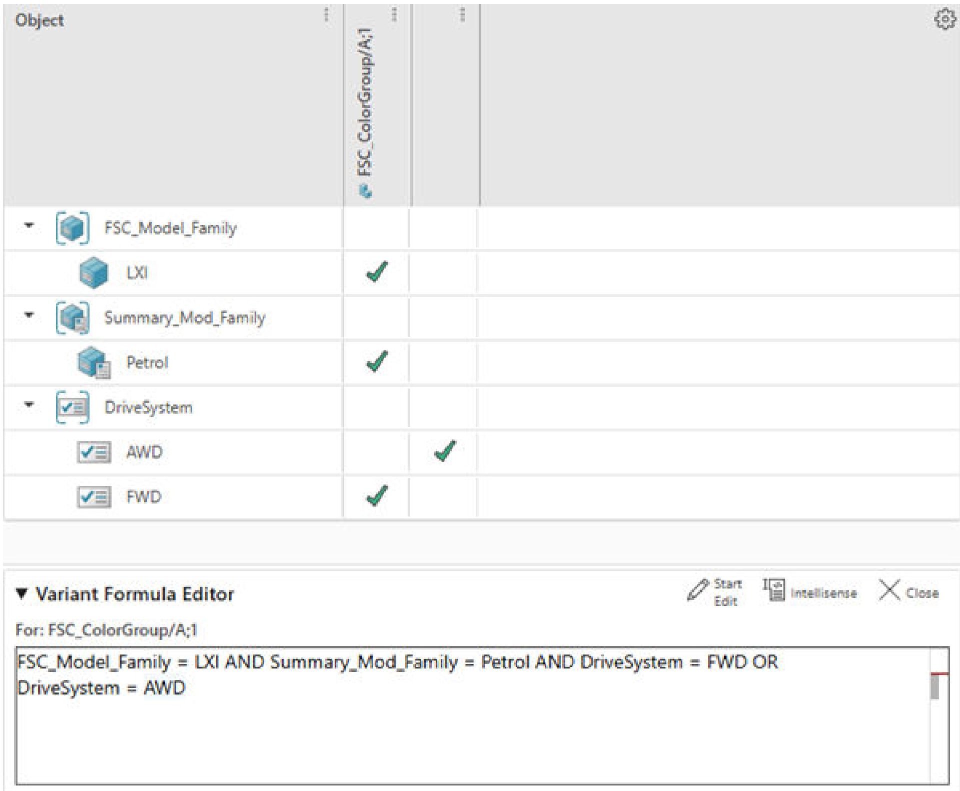

4. Feature selections from different families are connected using **AND** in the same column.

   Example:




5. For split columns, the expression generated by feature selections across columns are connected using **OR**.

   Example:




## Save changes in a configuration

You can save changes in your existing variant or save your custom configuration as a new variant rule. Alternatively, you can save them as a variant criteria. This depends on how the administrator configures your Teamcenter environment.

When an SVR or VC is saved, the user and system selections are saved, along with everything in the **Settings** panel, including the validation severity, expansion severity, and selection behavior settings, as well as the filter criteria information (revision rule, effectivity, and rule date).

In earlier versions, the system only saved the user selections when saving an SVR. If you load an SVR that was saved in a previous version, it loads the saved user selections but keeps the current settings in the **Settings** panel.

To save changes in your configuration,

### Procedure

1. To open the **Save** panel, choose **More Commands** ••• > **Save** .
2. Enter a name.
3. (Optional) Type a description.

4. Select one of the following options in **Attach to**:

   - **Current Structure Revision** to save your changes to the structure you have opened. These changes are available only for the specific structure.
   - **Configurator Context** to save your changes to the configurator context. These changes are available to all the structures to which this configurator context is associated.

   The default selection for this option is determined by how the administrator configures your Teamcenter environment. By default, it is **Configurator Context**.

5. Click **Save**.

# 19. Save filtered and configured structures within a workset

## About worksets

A workset is your personal working context. You can save your product definition in a workset. It is a container that holds one or more structures together. These structures can be the different views of the same structure or two entirely different structures. You can perform all BOM operations for a structure within a workset even when a partition scheme is applied to the structure.

You can create and use worksets only if your Teamcenter setup has the Smart Discovery license.

> **Note**
>
> Videos are not included in PDF. To access videos, use HTML.

## Create a workset

A workset is your personal working context in which you can save your product definition. You can create a workset to bring separate structures together. A workset can be created for all structures, including those on which a partition scheme is applied.

**Procedure**

1. Open the structure that has variability data created using Product Configurator.

   You cannot create a workset for a structure that has variability data defined using classic variants.

2. Click **New** > **Workset**.
3. In the **Workset** panel, do the following:
   a. In the **Type** list, select either **`Workset`** or **`Workset Custom`** depending on whether you want to create a workset or a custom workset.
   b. In the **ID** box, enter an ID for the new workset.
   c. In the **Revision** box, enter a revision for the new workset.
   d. In the **Name** box, enter a name for the new workset.
   e. (Optional) In the **Description** box, enter a description for the new workset.
   f. (Optional) Click **Add Project** to assign a project to the new workset.
4. Click **Add**.

### Results

A workset is created for your structure.

## Update a workset

When you change the filter parameters of a configuration in a workset, the updates you made to the workset and the structures within it are saved automatically. You can also update a structure (within the workset) for which a partition scheme is applied. However, the tree structure does not reflect the change. Therefore, you must perform a recalculation. Subsequently, the partition scheme displays the configured BOM lines under a partition, and this is based on the applied configuration settings and filter criteria.

### Procedure

1. Search for a workset using the global search or **Quick Access**.
2. Click **Open** to open the workset.

   The workset is always loaded with static data.
3. To update the workset with the latest applied filters and configurations, select the workset and then click **Replay**. All modified structures within the workset are updated.

   OR

   To update just a structure within the workset, select the structure and then click **Replay**.
4. Select a product within a workset and then click **Filter** or select **Configure** to apply filters or configurations to the workset.
5. Select the workset and click **Replay** to update it.

## Save a workset as a new workset

When you change the filtering aspects of a configuration in a workset, the updates you made to the workset and the structures within it are saved automatically. When you save a workset as a new workset, the name of the workset is changed, but its configurations are retained. You can also save a structure (within the workset) for which a partition scheme is applied as a new structure.

### Procedure

1. Search for a workset using the global search or **Quick Access**.
2. Click **Open** to open the workset.
3. Select a product within a workset and then click **Filter** or select **Configure** to modify the filters or the configurations applied on the workset.

4. Click the workset and then select **More Commands** ••• > **New** > **Save As**.
5. In the **Save As** panel, enter a name for the new workset.
6. (Optional) In the **Description** box, enter a description for the new workset.
7. (Optional) Click **Add Project** to assign a project to the new workset.
8. Click **Save**.

# Revise a workset

When you change the filter parameters of a configuration in a workset, the updates you made to the workset and the structures within it are saved automatically. When you update or revise a workset, the revision of the workset is changed, but its configurations are retained. You can also revise a structure (within the workset) for which a partition scheme is applied.

### Procedure

1. Search for a workset using the global search or **Quick Access**.
2. Click **Open** to open the workset.

   The workset is always loaded with static data.
3. To update the workset with the latest applied filters and configurations, select the workset and then click **Replay**. All modified structures within the workset are updated.

   OR

   To update just a structure within the workset, select the structure and then click **Replay**.
4. Select a product within a workset and then click **Filter** or select **Configure** to modify the filters or the configurations applied on the workset.
5. Click the workset and then select **More Commands** ••• > **New** > **Revise**.
6. On the **Revise** panel, do the following:
   a. In the **Revision** box, enter a revision for the workset.
   b. In the **Name** box, enter a name for the workset.
   c. (Optional) Enter a description for the workset in the **Description** box.
   d. (Optional) Click **Add Project** to assign a project to the new workset.
   e. Click **Revise**.

# Find elements within a workset

You can find elements across the different product structures (indexed using Smart Discovery Indexing) saved within a workset.

### Procedure

1. Open a workset from the search results.

2. Click **Find**.

3. In the **Find** panel, enter the search keywords and click **Search**.

   The search result displays elements that match the search keywords. These elements can belong to the different structures within the workset.

   You can also search within a specific structure. However, this structure must be indexed using Smart Discovery Indexing. To search within a structure in the workset:

   - Select the structure and click **Find**.

   - In the **Find** panel, select the **Find within <Product Name>** check box, enter the search keywords, and click **Search**.

     You can also find an element in a single structure (within the workset) for which a partition scheme is applied.

# Compare worksets

You can compare two different worksets to view their differences. You cannot open the same workset for comparison.

### Procedure

1. Search for and select two different worksets.

2. Click **Open**.
   The two worksets are opened in the Split View.

3. Click **Compare**.

4. (Optional) Change the comparison level:

   a. In the **Options** list, select a comparison level. The following table explains the different options.

| Comparison option | Description |
|---|---|
| **All Levels** | Compares the elements at every level in the two worksets. |
| **All Levels Below Selected** | Compares all levels below the selected level in the two worksets. |

| Comparison option | Description |
|---|---|
| **Components Only** | Compares only the parts that are at the lowest level, that is, components. |
| **Current Level** | Compares only the parts that are immediate children of the selections in the two worksets. This is the default level of comparison. |
| **Linked Assemblies or Components** | Compares two worksets by using an accountability check. This comparison level is displayed only if Multi-Structure Foundation is available in your Teamcenter environment. Accountability check is a verification mechanism to check if all the parts in one structure have equivalent parts in the other structure. You can define your own properties for equivalence. |

b. Specify what you want to display in the comparison results by selecting one or more of the following options from **Options** > **Display**:

| Comparison option | Description |
|---|---|
| **Matched** | Displays the parts that are a match in both worksets. |
| **Different** | Displays the parts that differ between the two worksets. |
| **Multiple Matches** | Displays the different numbers of occurrences of the same part in the comparison as a difference. |
| **Unique in Left View** | Displays the parts that are only listed in the workset displayed on the left. |
| **Unique in Right View** | Displays the parts that are only listed in the workset displayed on the right. |

c. For **Linked Assemblies or Components**, select the **Dynamic Equivalence** check box.

In some cases, the occurrences that are slightly different in the two structures might not be reported as equivalent. The dynamic equivalence check reports such occurrences.

This check box is displayed only if Multi-Structure Foundation is available in your Teamcenter environment.

d. If required, select the **Generate Report** check box to generate the comparison report.

e. Click **Compare**.

The comparison results are listed under **Results**. Clicking a part in the list highlights that part in the workset.

Comparison results are retained for a specific period. If a comparison result is older than the retention period, or if a more recent result exists, the old result is automatically deleted.

If you selected the **Generate Report** check box, the comparison report is generated in Excel. If you have the permissions to edit the target engineering BOM, the report is attached to it. You can click the **Attachments** tab of the target engineering BOM to download the report. If you do not have permissions to edit the target engineering BOM, the report is saved in the **Newstuff** folder. The file name for the comparison report is in the format *Compare_Report_<Source_ID>_AND_<Target_ID>_YYYYMMDD_HH_MM_SS*, where *Source_ID* is the ID of the first engineering BOM that you want to compare and *Target_ID* is the ID of the engineering BOM with which you want to compare the first engineering BOM. You can also click **Alerts** to understand the location of the comparison report if the BOMs are compared in the background.

> **Tip**
>
> If you closed the **Compare** panel and want to perform further comparison, click **Compare** on the task bar to view the **Compare** panel again.

> **Note**
>
> If the content of one of the worksets is modified, compare the worksets again.

5. When you are comparing two worksets, if the content of a workset is modified, click **Replay** to bring the latest content to the workset.

## Create a snapshot of a workset

You can create a snapshot of all worksets, irrespective of whether a partition scheme applied to the structures within the worksets.

Snapshots capture the 3D data associated with your product. When you load the product in the 3D viewer and take a snapshot, you capture the current 3D view, including camera, visibility, selection, view port (pan, zoom, and rotate modes), orientation, sections, measurements, queries, and markups. You also capture the configuration and filtering criteria.

When working with worksets, you can capture snapshots of a workset to preserve the working context and save it for later use.

## Working with Product Configurator data in a workset

You can work with the Product Configurator data within a workset. A workset helps you manage multiple structures, enabling you to author variant conditions and load Saved Variant Rules (SVRs) directly from the workset or its included structures, while noting that classic variants are not supported in this context.

A workset is a container that holds two or more structures together. Therefore, to view configuration information, select the relevant structure within the workset.

You can perform Product Configurator-related tasks, such as authoring variant conditions and loading a Saved Variant Rule (SVR), from within the workset and from the structures included in the workset. Classic variants are not supported in a workset.

## Modifying the data in a workset

You can modify data within a workset, which initially loads with the static data. The changes like adding or deleting elements are immediately visible. However, the changes like property edits, edits made to variant condition, or authoring occurrence effectivity require replaying the workset or the structure within the workset to fully take effect.

After you open a workset, it is always loaded with static data. You can modify the data in a workset, for example, you can add a structure element, remove an element, and edit the properties of an element. When you do this, you must replay the workset or a structure within the workset for the changes to take effect.

When you insert an element in a structure within a workset, the element is immediately displayed in the tree structure. However, when you reopen the workset or refresh the browser page, the element is not displayed. Now, when you replay the workset or a structure within the workset, the element might be displayed in the tree structure based on the applied filter criteria.

When you delete an element from a structure within a workset, the element is removed from the tree structure.

When you edit the properties of an element or edit a variant condition or author occurrence effectivity, the updates are immediately done. However, to display the changes, you must replay the workset or the structure within the workset. Now, when you reopen the workset or refresh the browser page, the updates are maintained even though the changed property does not satisfy the filtering or configuration criteria.

## Export and import structures along with worksets

You can export and import structures along with worksets to and from other Teamcenter sites.

A workset is a context collector for other structures. If your organization has very large structures with a multitude of occurrences, these occurrences can be collated within a workset to do a *what-if* analysis. But when you send the workset to another site by using a Multi-Site environment or a Briefcase, you might not want to export such a large number of occurrences along with the workset. Therefore, while exporting, the **Include entire BOM** option is not enabled by default. Consequently, the TCXML-based Multi-Site functionality works to share workset and related objects but not the product item or item revisions or their content.

Worksets also support site-consolidation activities through the TCXML-based Multi-Site functionality.

To export and import structures along with worksets, you can use:

- Briefcase files
- Multi-Site Collaboration
- PLM XML

# 20. Save filtered and configured structures within a session

## About sessions

Sessions help you save and retrieve specific product structures, along with any applied filters and configurations. You can store either static or dynamic data, enable 3D visualization and snapshots in the sessions. You can share the sessions with other users.

To easily retrieve a structure that you are currently working with, you save it within a ***session***. You can further apply filters to the structure or configure it to derive a smaller structure that is more relevant to you. The session retains the filters and configurations applied to the structure.

When you open a session, you view the structure as it was previously saved retaining its applied filtering and configuration criteria. Depending on how your application administrator has configured the system, the session displays:

- Static data already stored in the session.

  OR

- Dynamic data where the stored data is re-evaluated against the current data and the latest data is loaded.

You can override the value of this preference by creating a new instance of the preference at a higher-precedence location.

You can make **changes** to the session by applying new filters and configuration, and can also **save it as a new session**. Additionally, you can capture a snapshot of the product within the session to include the 3D measurement data applied on the structure. You can also **share a session** with other users.

Depending on how structures are indexed, you can perform the following tasks:

| Scenarios | | Tasks |
|---|---|---|
| Structures are indexed using Smart Discovery Indexing. | Structures are filtered. | Create a session.<br><br>Visualize large structures and capture snapshots. |
| | Structures are not filtered. | Create a session.<br><br>Visualize large structures and capture snapshots. |

> **Note**
>
> When applying the **Global Latest Working** revision rule, the system will reload the entire structure to ensure the components are the latest working versions.

# Create a session

You can create a session for a regular structure and also for a structure within a workset with a partition scheme applied. You can save this session and share it with other applications.

### Procedure

1. Search for and **Open** the desired structure.
2. Filter the structure as per your requirement.
3. Click **New** > **Session** .
4. In the **Session** panel, specify the access level for other users:
   a. Select the **Allow others to view** check box to provide *read* access to other users. Clear the check box to deny read access.
   b. Select the **Allow others to edit** check box to provide *write* access to other users. Clear the check box to deny write access.

   > **Note**
   >
   > The **Allow others to edit** check box is displayed only when the **Allow others to view** check box is selected.

5. Click **Create** to create the session.

# Search for a session

You can search for a session based on the properties associated with the top-structure node in that session. To quickly search for a session, you can further filter sessions by their properties such as **Name**, **ID**, and **Type**.

### Procedure

1. Search for sessions by performing a global search.

   You can search with a keyword or the wildcards characters * or ?.

   The **Filters** panel is displayed.
2. In the **Filter** search box under the **Type** list, select the **Session** check box.

The session that you are searching for should be displayed in the panel to the right of the **Filters** panel.

You can further refine your search by applying more filters, such as **ID Contained in**, **Name Contained in**, and **Type Contained in** and many other filters.

# Save a session as a new session

You can save an existing session as a new session. thereby preserving the original session while creating an updated version.

### Procedure

1. Search for a session.
2. Click **Open** to open the session.
3. Click **Filter** or select **Configure** > **Configuration** to modify the filters or the configurations applied on the session.
4. Click the session and then select **Session** > **Save As** .
5. In the **Save As Session** panel, enter the required details and click **Save**.

# View and update a session

When you view a session, the session is loaded either with the static or dynamic data depending upon how the administrator has configured it for you. You can restore the sessions after unexpected logouts, replay to see the latest filters and configurations, modify applied settings, and save changes—either by overwriting the original session (if permitted) or saving it as a new one. You can also reset a session to its last saved state.

### Procedure

1. Search for a session.
2. Click **Open** to open the session. Your site administrator sets whether to load a session with the static data stored in the session or with the latest data.

   Sometimes, when a session is opened, you get a message asking you to choose to restore the session. Due to some issues, such as getting logged out while making updates to a session, results in this message. Click **Restore** if you want to autorecover the updates that you earlier made to the session. If you do not want to restore, you can ignore the displayed message, and continue with your updates to the session.

   Session information that gets restored includes snapshots, quick measurements, part orientation, queries, sections, and workspace settings such as floor options and material.

3. To view the latest filters and configurations applied on the structure within the session, click **Replay**.

   If you opened the structure in another window and edited it, the updates are not reflected in the session when you replay it. You must refresh the browser window to reload the changes in the session.

4. Click **Filter** or select **Configure** > **Configuration** to modify the filters or the configurations applied on the session.

5. Click **Save Session**.

   If the session is opened by another user at the same time and the other user made some changes to the session, you can choose to overwrite the session or save the session as a new one. You can overwrite the session only if your administrator has allowed overwriting of the session.

   If another user removes some structure elements from the session, you receive a message with the list of the structure elements that are no longer available in the session.

6. To save a session as a new one:

   - Click **Session** > **Save As**.
   - In the **Save As Session** panel, enter the required details and click **Save**.

7. To undo the updates to the session, click **Session** > **Reset View**.

   On resetting, the session goes back to its previously saved state.

# Configure a session

You can configure a session to save specific filtering and configuration criteria for the product structures. You can retain the applied configurations (like variants and effectivity) and filters, and control the display of excluded or suppressed occurrences.

Sessions are used to save the filtering and configuration criteria applied to a structure. When you open a session, you view the structure as it was previously saved retaining its applied filtering and configuration criteria.

When a new session is created for a structure, objects in the structure are displayed based on the configurations applied prior to creating the session. The status of the following **Show...** sets a preference. The value of the preference is not saved when the session is saved.

- Show Excluded by Effectivity
- Show Excluded by Variants
- Show Suppressed

You can use these **Show...** toggles in the **Configuration Settings** menu to control the display of the structure elements. You can apply any filters on the configured structures as required. The configuration is retained with the filters applied.

When the session is saved, the applied configuration (variant, effectivity, filtering, etc.) is also saved. However, the status of the **Show...** commands is not saved.

> **Note**
>
> The system will reload the entire structure to reflect the changes in the configuration settings. This is due to the dependency on the BOM and the need to reprocess the entire structure.

### Show or hide occurrences excluded by effectivity

When a structure is loaded, all objects including those which are not effective for the specified revision rule are displayed by default. However, you can choose to display only those occurrences that are effective for a specified date, unit, range of dates, or unit numbers.

To show or hide the occurrences that are excluded by the currently applied effectivity, in the work area toolbar, click **Configuration Settings** > **Show Excluded By Effectivity**.

### Show or hide occurrences excluded by variant

When a structure is loaded, all occurrences including those occurrences which are not configured by the variant are displayed by default. However, a user can choose to hide or show the that are not configured.

To show or hide the occurrences that are excluded by the variant configuration, in the work area toolbar, click **Configuration Settings** > **Show Excluded By Variants**.

### Show suppressed occurrences

Occurrences in a structure can be hidden by setting the suppress property to *True*. When a structure is loaded, all occurrences including the suppressed occurrences are displayed by default.

To show or hide the suppressed occurrences, in the work area toolbar, click **Configuration Settings** > **Show Suppressed**.

## Share a session with other users

You can share a session with other users. As a prerequisite, the system administrator must set the **Has Class( Fnd0AppSession )** Access Manager rule. To share the session, you must be the creator of the session. After you share the session, other users can view or edit the session that is shared by you.

# Make a session shareable while saving the session

### Procedure

1. Search for and open the structure.
2. Configure the structure as required.
3. Click **More Commands** > **New** > **Create Session**.
4. In the **Create Session** pane, specify the access level for other users:
   a. Select the **Allow others to view** check box to grant *read* access to other users. Clear the check box to deny read access.
   b. Select the **Allow others to edit** check box to grant *write* access to other users. Clear the check box to deny write access.

   > **Note**
   >
   > The **Allow others to edit** check box is displayed only when **Allow others to view** check box is selected.
5. To create the session, click **Create**.

# Initiate a workflow to share a session

When you share a session, you must inform the person you have shared the session with. You can communicate using any method such as emailing the session URL. You can also use a workflow to share sessions and assign tasks to other users.

### Procedure

1. Search for and select the session that you want to share.
2. Click **More Commands** > **Manage** > **Submit to Workflow** .

   The **Submit to Workflow** panel with a list of workflow templates is displayed.
3. Select the **Session Collaboration Workflow** template.
4. Enter a description for the workflow participants, and select the appropriate workflow template.

   A default workflow is defined for sharing the session. It is automatically selected as the workflow template.
5. Click **Assignments** to assign users and resource pools to the workflow.

   > **Note**
   >
   > You must have selected the **Allow Others to View** check box while creating a session in order for the assigned user to receive an email about the shared session.
6. Click **Submit**.

# Release a session

You can release a session by submitting it to a workflow. When you release a session, you ensure that the proper release maturity is set for your session.

### Procedure

1. Search for the session you want to release.
2. Click **Open** to open the session.
3. Click **More Commands** ••• > **Manage** > **Submit to Workflow**.

   The **Submit to Workflow** panel with a list of workflow templates is displayed.
4. Select the **Session Release Workflow** template.
5. Enter the required details and click **Submit**.

# Find an element within a session

You may want to search for a structure element within a session. Searching for an element is not limited to the filtered structure displayed in the session. The element is searched for across the unfiltered structure.

### Procedure

1. Open a session.
2. Click **Find**.
3. Enter the search keyword.
4. (Optional) To narrow your search, select **Find within** to find the element within a certain assembly.

# Capture a snapshot of a session

Snapshots capture the 3D data associated with your product. When you load the product in the 3D viewer and take a snapshot, you capture the current 3D view, including camera, visibility, selection, view port (pan, zoom, and rotate modes), orientation, sections, measurements, queries, and markups. You also capture the configuration and filtering criteria.

You can capture a snapshot of the current view of an assembly as an in-session snapshot. In-session snapshots are contained within the session. You cannot share these snapshots independently with other users.

# View a session in other applications

You can a session and its associated structures in other applications like Teamcenter lifecycle visualization, NX, or CATIA so that you can leverage saved filters and configurations for 3D visualization, design modifications.

To easily retrieve a structure that you are currently working with, you save it within a *session*. The session also retains any filters and configurations applied to the structure. You can view a saved session and update the structure in applications such as Teamcenter lifecycle visualization, NX, or CATIA.

When you make changes to a session, and then open the unsaved session in other applications such as Lifecycle Visualization, NX, or CATIA, the changes must be saved or discarded before opening the structure in the other application. You can save or discard the unsaved changes.

### Procedure

1. Search for a session that you want to view.
2. Click **Open** to open the session.
3. To work with the 3D data associated with the structure, open the session in Teamcenter lifecycle visualization by clicking **Open** > **Open in Visualization**.

   Before opening the session in Teamcenter lifecycle visualization, you must first replay and save the session in Active Workspace.

4. To make design modifications to the engineering BOM structure, open the session in CATIA or NX by clicking **Open** > **Open in CATIA** or **Open in NX**. The aligned design BOM is opened in the selected application.

# 21. Create and maintain solution variants

## About solution variants

The solution variants in Teamcenter are 100% configured product variants generated from a single variable structure (150% BOM). The solution variants are linked to the source structure. Teamcenter helps you ensure their uniqueness and also facilitates updates.

For effective product definition and management, a range of product variants are managed as a single variable structure (150% BOM) by using a configurator instead of managing them as discrete product variants. You can configure this variable structure to generate a 100% variant by applying a valid variant configuration to create a *solution variant*. The solution variant has a unique item ID, and it is linked to the source structure.

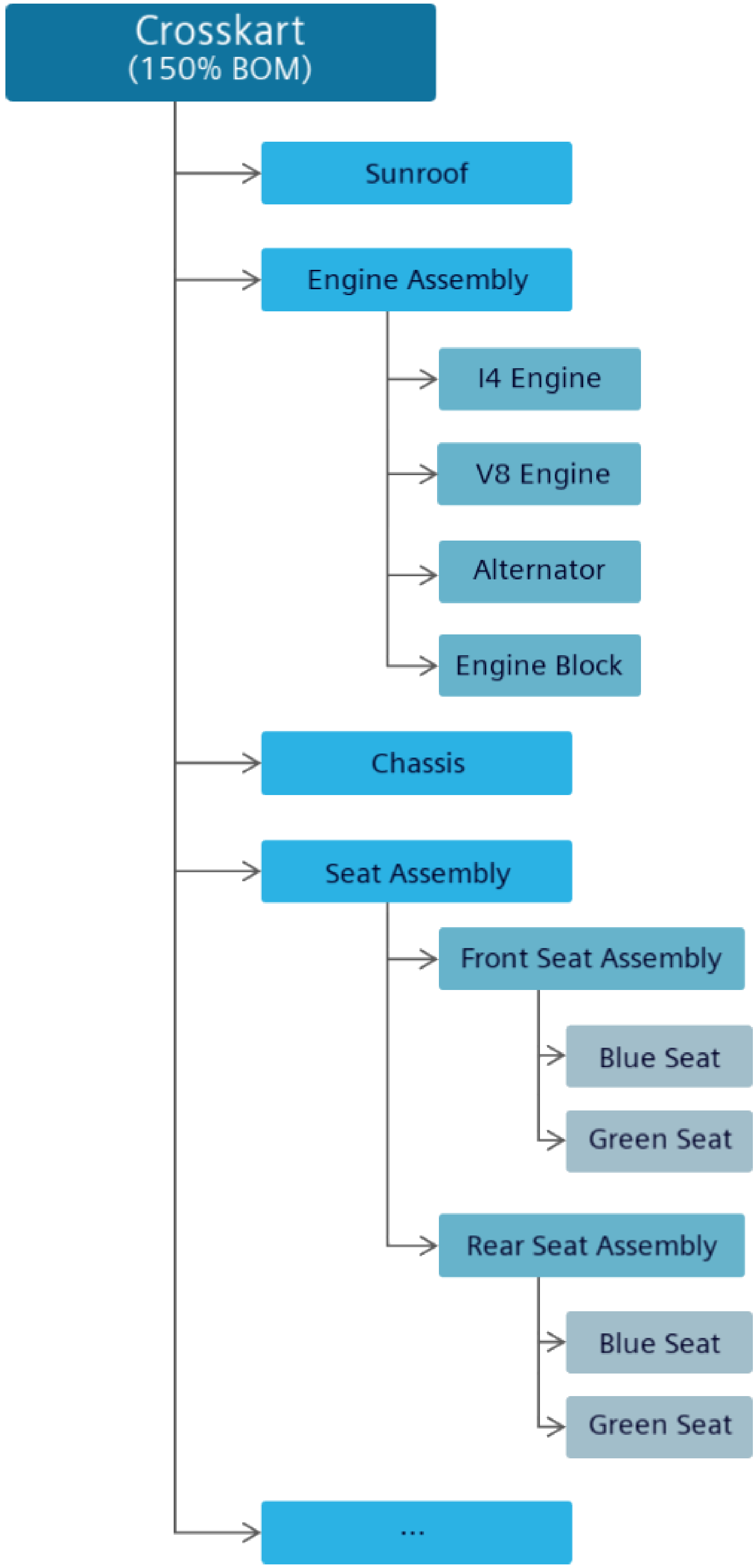
**Note**

Videos are not included in PDF. To access videos, use HTML.

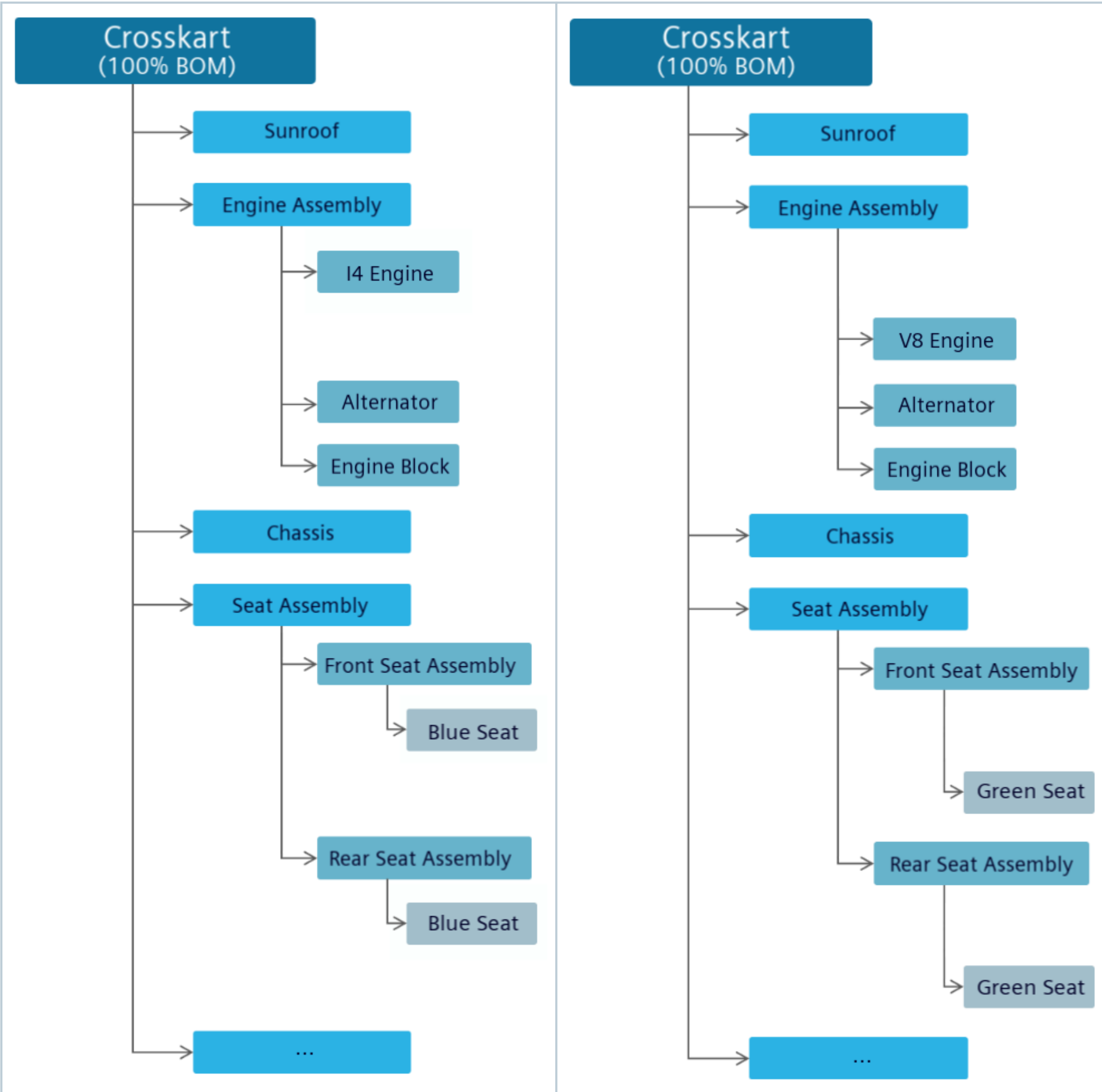
Multiple solution variants can be created for each valid variant configuration of the 150% variable structure. Teamcenter ensures the uniqueness of each solution variant by comparing the content with a list of existing solution variants before creating a new one. As the solution variants are linked to the source variable structure, the solution variants can be updated to easily incorporate the changes made to the source structure.

**Example**

Consider a product, *Crosskart*, that has two variants, *Base* and *Deluxe*. Both these variants are managed within a single variable structure.

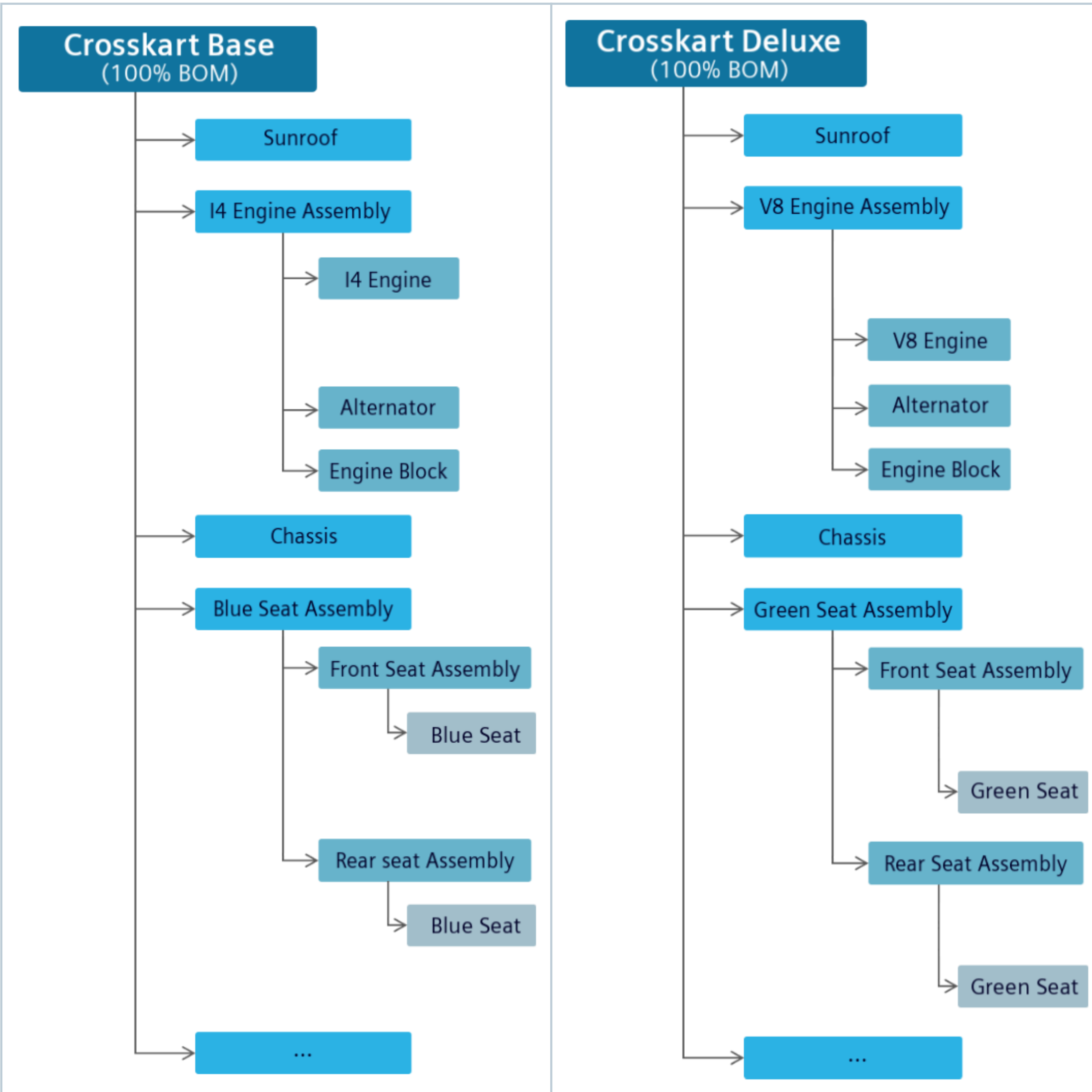

Crosskart
(150% BOM)
Sunroof
Engine Assembly
I4 Engine
V8 Engine
Alternator
Engine Block
Chassis
Seat Assembly
Front Seat Assembly
Blue Seat
Green Seat
Rear Seat Assembly
Blue Seat
Green Seat
...

To derive the base and deluxe variants, you apply a valid configuration. This is based on the variant rules already set in Product Configurator for *Crosskart*. As per these rules, *I4 Engine* and *Blue Seat* must be used in the base variant, while *V8 Engine* and *Green Seat* must be used in the deluxe variant.




In these distinct base and deluxe variants, the crosskart, engine assembly, and seat assembly are not uniquely identified even though they contain different parts. Therefore, to identify and represent them statically, you create their solution variants.

When you select *Crosskart*, *Engine Assembly*, and *Seat Assembly* to create their solution variants for the base variant, three separate solution variants are created. Each solution variant has a unique identifier. They are identified as *Crosskart Base*, *I4 Engine Assembly*, and *Blue Seat Assembly*. Similarly, three solution variants can be created for the deluxe variant, namely, *Crosskart Deluxe*, *V8 Engine Assembly*, and *Green Seat Assembly*.




The source structure, *Crosskart*, is maintained as a separate entity and is linked to each solution variant. It can continue to evolve. If you update the source structure, you must update its solution variants. When you update the source structure to correspondingly update its solution variants, solution variants that were previously created are reused. A new solution variant is created for a new part only if you choose to create a new solution variant for that part.

As solution variants are not just created at the product level but can also be created at the assembly level, this allows the reuse of variable parts. For example, consider that you want to reuse the engine assembly in another product, *Buggy*. According to the variant rules defined in the configurator, an I4 engine is used in the standard variant of Buggy. When you create a solution variant of the engine assembly for the standard variant, instead of creating a new solution variant, the one (*I4 Engine Assembly*) that was already created for *Crosskart Base* is reused. This way Teamcenter ensures that each solution variant is unique by comparing it with a list of existing solution variants before creating a new one. These solution variants, generated at the product or assembly level, can be used in downstream processes.

## Understand the types of solution variant categories

Explore the various solution variant categories in Teamcenter—Reuse, Managed, and Unmanaged—and understand their distinct synchronization and modification behaviors relative to a source product structure.

You can create the following types of solution variants for a product in Teamcenter:

**Reuse**
: Create a reuse solution variant to link it to the source product structure and synchronize its updates. This means that changes made to the source product structure are propagated to the reuse solution variant. You cannot make direct structural changes (such as adding, removing, or revising child items) to the solution variant independently.

**Managed**
: Create a managed type of solution variant when you want to manually review the updates made to the product structure and then update the solution variant. In other words, if the source product structure changes, you can compare the changes with the solution variant structure and merge the selected changes manually. You can make direct structural changes (such as adding, removing, or revising child items) to the solution variant independently.

**Unmanaged**
: Create an unmanaged solution variant to link it to the source product structure without synchronizing its updates. This means that changes made to the source product structure are not propagated to the unmanaged solution variant. You can make direct structural changes (such as adding, removing, or revising child items) to the solution variant independently.

## Create a Reuse solution variant

A Reuse type of solution variant helps you synchronize the contents of the solution variant with the contents of the source product structure, ensuring all updates are propagated.

Create a reuse solution variant to link it to the source product structure and synchronize its updates. This means that the changes made to the source product structure are propagated to the reuse solution variant.

You cannot make direct structural changes independently, such as adding, removing, or revising child items.

### Procedure

1. Set an active change notice that has the appropriate effectivity.
2. Open an engineering BOM that has the **Assembly Indicator** set as **Configurable Assembly**.
3. If not already set, select the top line of the structure and set a variability scope.
   a. In the **Details** section, click **Edit**.
   b. In **Variability Scope**, click **Add** to search for and add a configurator context.
   c. Click **Save**.

   Additionally, you can set a variability scope to any part that is set as **Configurable Assembly** or **Generic Part**.
4. To create the Reuse type of solution variant for the entire product, select the topmost line of the structure, and then set the **Solution Variant Category** as **Reuse**.

   When you set the **Solution Variant Category** as **Reuse** for the topmost line of the product structure, you must also set the **Solution Variant Category** as **Reuse** for the child assemblies.

   If the **Solution Variant Category** column is not available by default, you must first add it.
5. Enable or disable **Show Excluded by Effectivity** and **Show Suppressed** from **Configure**, as required.
6. Click the **Solution Variants** tab.

   > **Note**
   >
   > If you do not see the **Solution Variants** tab, contact your system administrator.

   The **Solution Variants** section displays a list of the existing solution variants.
7. In the **Solution Variants** section, click **Create**.

   The **Create Solution Variant** panel appears and displays a list of the variant rules that are available from the associated variability scope.The **Valid for Solution Variant** column indicates whether a solution variant can be created for a given variant rule. **Yes** indicates that all conditions are met, while **No** indicates at least one condition is invalid. Ensure that all variant rule conditions are met so that a solution variant can be created.

   By default, the following columns in the **Variants** tab act as conditions for a valid variant rule. Ensure these default conditions are met. Contact your administrator for configuring the custom validation criteria.

| Columns in the Variants tab | Required values |
|---|---|
| **Snapshot Rule Date Policy** | **Date** |
| **Snapshot Compiled Date** | **<Valid date>** |
| **Valid** | **True** |
| **Complete** | **True** |

The **Create Solution Variant** panel also displays other details for the solution variant, including its ID.

8. (Optional) Double-click the **Revision ID** column for a variant rule and select another version, if available.
   If the selected version of the variant rule is valid, the **Valid for Solution Variant** column indicates **True** and you can use it for creating the solution variant.
9. Select one or more valid variant rules for creating the solution variant.
10. If required, click the **Run in Background** check box.
11. Click **Create**.
    The Reuse type of solution variant is created and the following is carried forward: the absolute occurrence properties—**Position Designator**, **ID in Context**, and **Usage Address**—along with the absolute occurrence data properties for a part within the solution variant.
12. Release the active change. To do so, click **Changes**, select the change, and click **Release Change**.

# Create a new solution variant instead of updating an existing one for a variant

You can disable updating an existing solution variant of the type *Reuse* by specifying a status for the solution variant and create a new solution variant.

When a structure is revised multiple times and a solution variant is revised alongside, the changes to the structure might make the solution variant incompatible with the structure. In such cases, you must mark the existing solution variant as obsolete and create a new one for use instead.

### Procedure

1. In the **Solution Variants** section on the **Solution Variants** tab, select the solution variant to be marked as obsolete.
2. Specify a valid release status for the solution variant item revision. You can use a workflow, a custom flow, or a business logic configured by your administrator to set the release status on the solution variant item revision.

   After you set the release status, the solution variant is obsoleted. Teamcenter changes the solution variant category to **Unmanaged**, restricting any further updates to the solution variant.

You can create a new solution variant for the same configuration to replace the obsoleted solution variant.

3. In the **Solution Variants** section on the **Solution Variants** tab, select the variant for which you want to create a new solution variant.
4. Click **Create** ⊕.
5. In the message dialog box, you can choose to create the solution variant in the background or foreground by selecting or clearing the **Run in Background** check box, respectively. Click **Yes** to create the solution variant.
   If you choose to create the solution variant in the foreground, the solution variant is created and displayed in the **Solution Variant** column.

   If you chose to create it in the background, go to **Alerts** to open the solution variant.

### Results

For the solution variants that are obsoleted at a lower level in the hierarchy, Teamcenter automatically changes the solution variant category to **Unmanaged** and creates a replacement solution variant during the update process.

## Update a Reuse solution variant

If the source product structure is modified, you can update the solution variant that is of the reuse type so that all the changes from the source product structure are propagated to the solution variant. You cannot make structural changes, such as adding, removing, or revising child items, directly to the reuse type of solution variant.

### Procedure

1. (Optional) **Set an active change notice** that has the appropriate effectivity.
2. Open an engineering BOM that has **Assembly Indicator** set as **Configurable Assembly**.

   > **Warning**
   >
   > Do not change the state of **Show Excluded by Effectivity** and **Show Suppressed**. Their state must remain the same while creating and updating the solution variant.

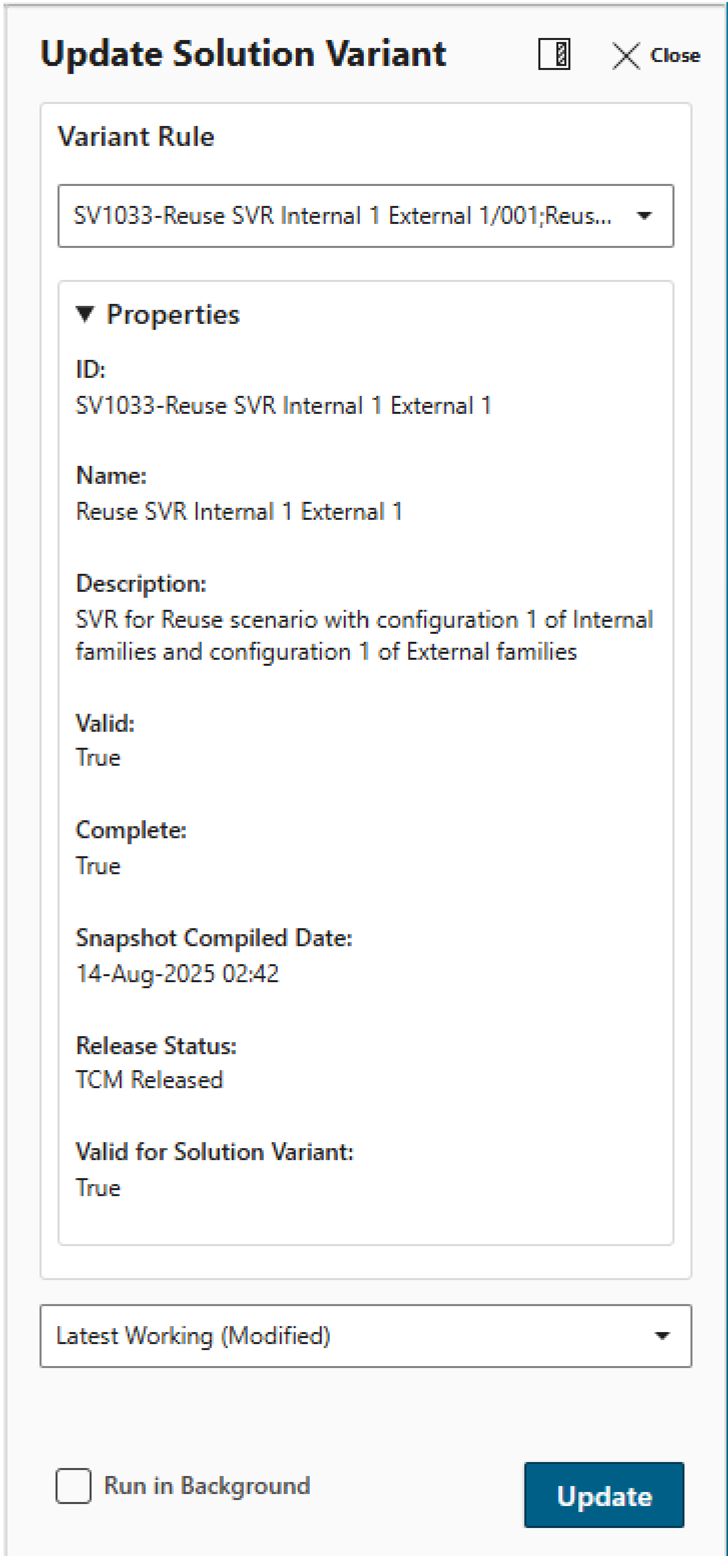
3. Open the source product structure and go to the **Solution Variants** tab.

   If the source structure is modified after the solution variant was created, the **Source Assembly Modified?** column displays **Yes** and indicates that the reuse type of the solution variant also needs to be updated.

4. Select the reuse type of the solution variant that needs to be updated and click **Update**.
5. In the Update Solution Variant panel, perform the following tasks:
   a. If required, in the **Variant Rule** list, select a different available variant rule.

b. In the **Properties** section, review the properties.

c. If required, in the **Revision Rule** list, select a different revision rule.




d. Click **Update**.

The solution variant that is impacted due to the changes in the source structure is updated. If a solution variant is already released, it is first revised and then updated.

When you update a solution variant in the context of an active change, the updates are reflected in the change summary.

After the solution variant is updated, the absolute occurrence properties for a part within the solution variant are carried forward. The absolute occurrence properties, such as **Position Designator**, **ID in Context**, and **Usage Address**, are also transferred if your administrator has configured them to be carried forward.

Consider that a solution variant with the **Solution Variant Category** set as **Reuse** is reused for more than one variant rule, and either the variant rules or the source structure is modified in such a way that source structure will not result in the same configuration for those variant rules. In such a case, when you update the solution variant, the system splits the existing solution variant to create new solution variants as needed. This ensures an efficient reuse of the existing solution variant.

**Example**

The SVR(White) and SVR(Grey) variant rules share the same solution variant. The solution variant contains beige seat covers and headrests. Later, *white thread* is added to SVR(White) for the seat cover stitches. In such a situation, a new solution variant is created for SVR(White), and it contains beige seat covers, headrests, and white thread. The SVR(Grey) will still result in the existing solution variant that contained beige seat covers and headrests.

6. To verify if a solution variant is updated correctly, open it from the **Solution Variants** column.
7. If you have set an active change notice, release the active change.
    a. Click **Changes** △, select the change, and click **Release Change**.

# Create a Managed solution variant

A Managed type of solution variant helps you manually review and merge updates from the source product structure, and allows independent modifications to be made to the solution variant.

Create a managed type of solution variant when you want to manually review the updates made to the product structure, and then update the solution variant. In other words, if the source product structure changes, you can compare the changes with the solution variant structure and merge the selected changes manually. You can make direct structural changes, such as adding, removing, or revising child items, to the solution variant independently.

### Procedure

1. Set an active change notice that has the appropriate effectivity.
2. Open an engineering BOM that has the **Assembly Indicator** set as **Configurable Assembly**.
3. If not already set, select the top line of the structure and set a variability scope:

a. In the **Details** section, click **Edit** .

b. In **Variability Scope**, click **Add** to search for and add a configurator context.

c. Click **Save** .

Additionally, you can set a variability scope to any part that is set as **Configurable Assembly** or **Generic Part**.

4. For creating the solution variant for the entire product, select the top line of the structure and then set the **Solution Variant Category** as **Managed**. For child assemblies, you can set the **Solution Variant Category** as **Reuse**, **Managed**, or **Unmanaged**.

   If the **Solution Variant Category** column is not available by default, you must first add it.

5. Enable or disable **Show Excluded by Effectivity** and **Show Suppressed** from **Configure** , as required.

6. Click the **Solution Variants** tab.

   If you do not see the **Solution Variants** tab, contact your system administrator.

7. In the **Solution Variants** section, click **Create** .

   The **Create Solution Variants** panel appears and displays the variant rules that are available from the associated variability scope.The **Valid for Solution Variant** column indicates whether a solution variant can be created for a given variant rule. **Yes** indicates that all conditions are met, while **No** indicates that at least one condition is invalid. Ensure that all variant rule conditions are met so that a solution variant can be created.

   By default, the following columns in the **Variants** tab act as conditions for a valid variant rule. Ensure these default conditions are met. Contact your administrator to configure the custom validation criteria.

| Columns in the Variants tab | Required values |
|---|---|
| Snapshot Rule Date Policy | Date |
| Snapshot Compiled Date | <Valid date> |
| Valid | True |
| Complete | True |

   The **Create Solution Variant** panel also displays other details for the solution variant, including its ID.

8. (Optional) Double-click the **Revision ID** column for a variant rule and select another version if available.
   If the selected version of the variant rule is valid, the **Valid for Solution Variant** column indicates **True** and you can use it for creating the solution variant.

9. Select one or more valid variant rules for creating the solution variant, and then click **Create**.

10. If required, click the **Run in Background** check box.

11. Click **Create**.
The Managed type of solution variant is created and the absolute occurrence properties—**Position Designator**, **ID in Context**, and **Usage Address**—along with the absolute occurrence data properties for a part within the solution variant, are carried forward.

12. Release the active change. To do so, click **Changes**, select the change, and click **Release Change**.

# Update a Managed solution variant

If the source product structure is modified, you can review the changes and merge them manually with the managed solution variant. You can also make structural changes directly to the managed type of solution variant, such as adding, removing, or revising child items.

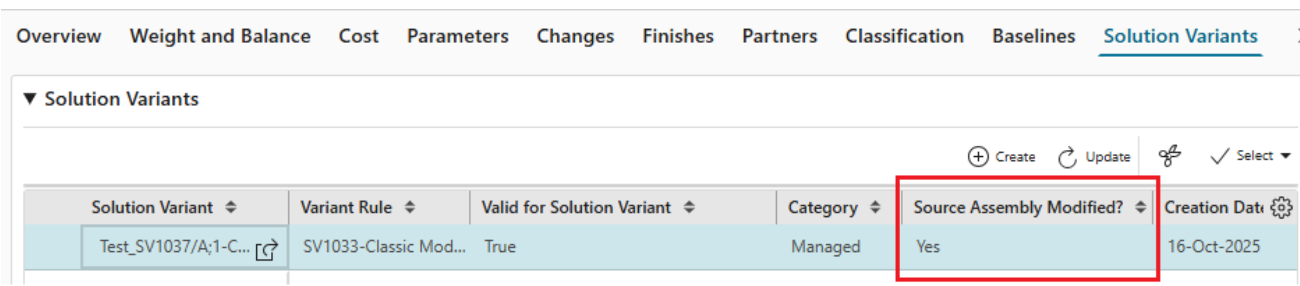
### Procedure

1. (Optional) Set an active change notice that has the appropriate effectivity.
2. Open an engineering BOM that has **Assembly Indicator** set as **Configurable Assembly**.

> **Warning**
> Do not change the state of **Show Excluded by Effectivity** and **Show Suppressed**. Their state must remain the same while creating and updating the solution variant.

3. (Optional) If you want to make structural changes directly to the managed solution variant, in the **Solution Variants** tab, click **Open** to open a managed solution variant. Then, add, remove, or revise its child items.
4. Open the source product structure and go to the **Solution Variants** tab.

   If the source structure is modified after the solution variant was created, the **Source Assembly Modified?** column displays **Yes** and indicates that the managed solution variant also needs to be updated.

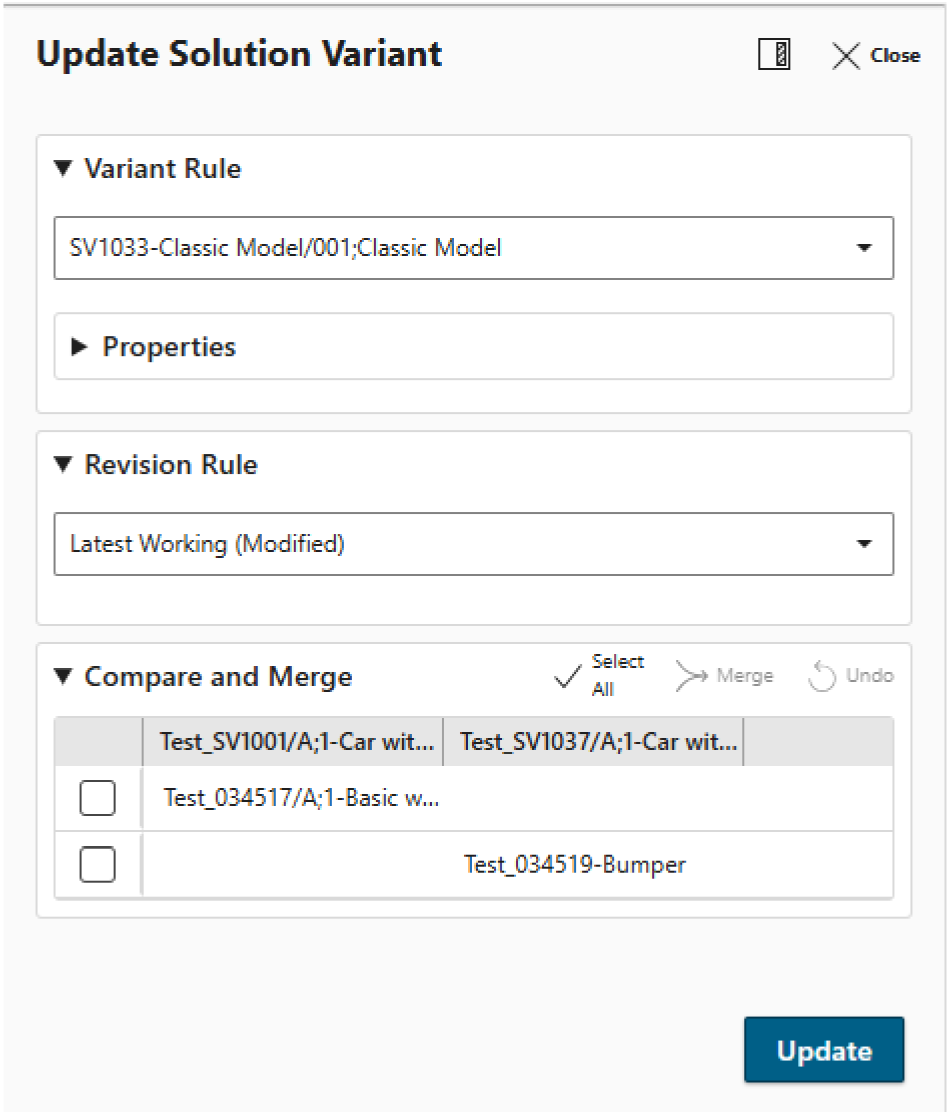



5. Select the managed type of the solution variant that needs to be updated and click **Update**.
6. In the Update Solution Variant panel, perform the following tasks.
   a. If required, in the **Variant Rule** list, select a different, available variant rule.
   b. In the **Properties** section, review the properties.

c. If required, in the **Revision Rule** list, select a different revision rule.

d. In the **Compare and Merge** section, compare the changes in the source structure and the solution variant structure.

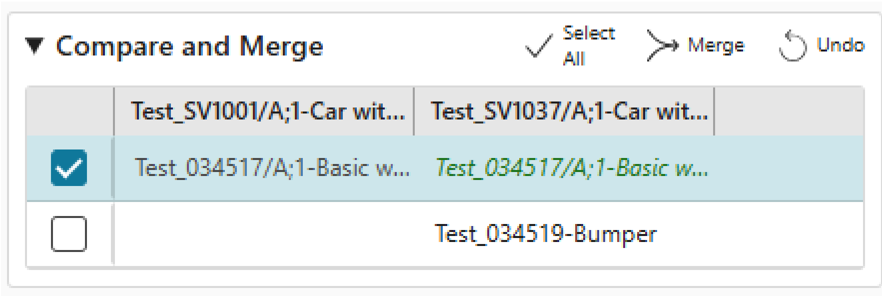
   The first column displays the items that are added, revised, or removed in the source structure. The second column displays the items that are added, revised, or removed in the selected solution variant structure. The blank cell in the column indicates that the respective structure does not contain the item.
   The following example shows that the **Test_034517/A** item is added to the source structure and the **Test_034519-Bumper** item is directly added to the managed solution variant.



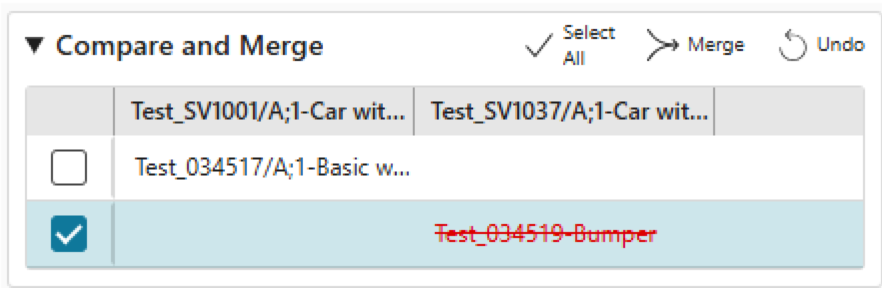

e. Select the required rows in the **Compare and Merge** section and click **Merge**.

   Click **Undo** to remove the last action.

   The following example shows that the **Test_034517/A** item is to be added to the managed solution variant.




The following example shows that the **Test_034519-Bumper** item is to be removed from the managed solution variant.




f. Click **Update**.
   The managed type of solution variant is updated.
   After the solution variant is updated, the absolute occurrence properties for a part within the solution variant are carried forward. The absolute occurrence properties, that is, **Position Designator**, **ID in Context**, and **Usage Address**, are also transferred if your administrator has configured them to be carried forward.

7. To verify if a solution variant is updated correctly, open it from the **Solution Variants** column.

8. If you have set an active change notice, release the active change.

   a. Click **Changes** △, select the change, and click **Release Change**.

# Create an Unmanaged solution variant

An Unmanaged type of solution variant helps you independently modify its structure content without synchronizing it with the source product.

Create an Unmanaged solution variant to link it to the source product structure without synchronizing its updates. This means that the changes made to the source product structure are not propagated to the Unmanaged solution variant. You can make direct structural changes, such as adding, removing, or revising child items, to the solution variant independently.

### Procedure

1. Set an active change notice that has the appropriate effectivity.

2. Open an engineering BOM that has **Assembly Indicator** set as **Configurable Assembly**.
3. If not already set, select the top line of the structure and set a variability scope:
    a. In the **Details** section, click **Edit**.
    b. In **Variability Scope**, click **Add** to search for and add a configurator context.
    c. Click **Save**.

    Additionally, you can set a variability scope to any part that is set as **Configurable Assembly** or **Generic Part**.
4. To create the solution variant for the entire product, select the top line of the structure and then set **Solution Variant Category** as **Unmanaged**. For child assemblies, you can set the **Solution Variant Category** as **Reuse**, **Managed**, or **Unmanaged**.

    If the **Solution Variant Category** column is not available by default, you must first add it.
5. Click **Configure** > **Show Excluded by Effectivity** and **Configure** > **Show Suppressed**, as required.
6. Click the **Solution Variants** tab.

    If you do not see the **Solution Variants** tab, contact your system administrator.
7. In the **Solution Variants** section, click **Create**.

    The **Create Solution Variant** panel appears and displays a list of the variant rules that are available from the associated variability scope.The **Valid for Solution Variant** column indicates whether a solution variant can be created for a given variant rule. **Yes** indicates that all conditions are met, while **No** indicates that at least one condition is invalid. Ensure that all variant rule conditions are met so that a solution variant can be created.

    By default, the following columns in the **Variants** tab act as conditions for a valid variant rule. Ensure these default conditions are met. Contact your administrator for configuring the custom validation criteria.

    | Columns in the Variants tab | Required values |
    |---|---|
    | **Snapshot Rule Date Policy** | **Date** |
    | **Snapshot Compiled Date** | **<Valid date>** |
    | **Valid** | **True** |
    | **Complete** | **True** |

    The **Create Solution Variant** panel also displays other details for the solution variant, including its ID.
8. (Optional) Double-click the **Revision ID** column for a variant rule, and select another version if available.
    If the selected version of the variant rule is valid, the **Valid for Solution Variant** column indicates **true** and you can use it for creating the solution variant.

9. Select one or more valid variant rules for creating the solution variant and click **Create**.
10. If required, click the **Run in Background** check box.
11. Click **Create**.
    The Unmanaged type of solution variant is created and the absolute occurrence properties—**Position Designator**, **ID in Context**, and **Usage Address**—along with the absolute occurrence data properties for a part within the solution variant are carried forward.
12. Release the active change. To do so, click **Changes** △, select the change, and click **Release Change**.

## Update an Unmanaged solution variant

You must make direct structural changes, such as adding, removing, or revising child items, to the Unmanaged solution variant independently. The changes made to the source product structure are not propagated to the Unmanaged solution variant.

### Procedure

1. (Optional) **Set an active change notice** that has the appropriate effectivity.
2. Open an engineering BOM that has **Assembly Indicator** set as **Configurable Assembly**.

   > **Warning**
   >
   > Do not change the state of **Show Excluded by Effectivity** and **Show Suppressed**. Their state must remain the same while creating and updating the solution variant.

3. In the **Solution Variants** tab, click to open an Unmanaged solution variant, and then add, remove, or revise its child items.
   The solution variant that is impacted due to the changes in the source structure is updated. If a solution variant is already released, it is first revised and then updated.

   When you update a solution variant in the context of an active change, the updates are reflected in the change summary.

   After the solution variant is updated, the absolute occurrence properties for a part within the solution variant are carried forward. The absolute occurrence properties, that is, the **Position Designator**, the **ID in Context**, and the **Usage Address**, are also transferred if your administrator has configured them to be carried forward.

4. If you have set an active change notice, release the active change.

   To do so, click **Changes** △, select the change, and click **Release Change**.

# Update solution variants for variability changes

When solution variants with the **Solution Variant Category** as **Reuse** and **Managed** are created for variant rules and the variant criteria is modified, you can propagate the new variability to the solution variants. This operation updates the entire hierarchy of the solution variants as applicable.

If the parent structure contains sub-assemblies that are associated with their respective configurator contexts, the variant rules that are associated with these configurator contexts must also be updated based on the modified variant rule of the parent structure.

> **Example**
>
> Consider the Front Door structure, which includes multiple solution variants for mirror assemblies. Based on business requirements, you modify the variant rule for the VX variant and the Front Door structure (150% BOM) by adding a power-folding mirror with a turn signal feature. Following this modification, you assess the impacted solution variants and update them accordingly. The VX mirror assembly variant is updated to include the power-folding mirror with the turn signal feature.

### Procedure

1. (Optional) **Set an active change notice** that has the appropriate effectivity.
2. Ensure that the variant rule for the configurator context that is associated with an assembly in the structure is already updated.

   When updating the variant rule, ensure that it is in the expanded state and validated and saved.
3. Ensure that the variant rules for the configurator contexts that are associated with the parent structure (150% BOM) and its subassemblies are also already updated.
4. Open the structure that has the impacted solution variants.

   > **Warning**
   >
   > Do not change the state of **Show Excluded by Effectivity** and **Show Suppressed**. Their state must be the same while creating and updating the solution variant.

   Because the variant rule is modified after the solution variants are created, the **Variant Rule Modified** column in the **Solution Variants** tab shows **Yes** for the solution variants. This indicates that the solution variants need to be updated to include the latest changes made to the variant rules.
5. Select the solution variant and click **Update** ↻.
   The selected solution variant and the parent structure are updated.

# Create solution variants for a product through a workflow

You can create solution variants for a design and engineering BOM structure using a workflow. You use the variant rules for creating the solution variants. You can also create Reuse, Manage, and Unmanaged types of solution variants.

### Procedure

1. Set an active change that has the appropriate effectivity.
2. Open the source structure.

   > **Warning**
   >
   > Do not change the state of the **Show Excluded by Effectivity** and **Show Suppressed** toggles. The state of these toggles must remain unchanged while creating and updating the solution variant.

3. Set the **Solution Variant Category** for the source product structure and its sub-assemblies.

   You can select **Reuse**, **Managed**, or **Unmanaged**.

4. Click **More Commands** > **Manage** > **Submit to Workflow**.
5. In the **Submit to Workflow** panel, select **Create Solution Variants** in the **Template** list.
6. In the **Targets** section, ensure that the source structure is already selected.
7. (Optional) In the **Targets** section, click **Add** to add an engineering change.
8. In the **References** section, click **Add** to add variant rules.

   If a variant rule contains multiple revisions, you can choose a revision for which the solution variant is to be created.

9. Click **Submit**.

### What to do next

To verify if the solution variants are created correctly, go to the **Solution Variants** tab and open the required solution variants to view them.

Finally, release the active change. To do so, click **Changes**, select the change, and click **Release Change**.

# Update the solution variants of a product through a workflow

When a product structure changes, you can use a workflow to update the Reuse type of solution variants for both the design and the engineering BOM. You can manage new parts and revisions, and the specific properties that must be updated in the solution variants.

When a structure is updated, you must also update the corresponding solution variants. Consider the following scenarios before starting the workflow.

| Scenarios | How to update solution variants |
|---|---|
| Source design BOM is updated. | Perform an automated or guided update to update the corresponding engineering BOM.<br>**1.** If the solution variant is not updated during the automated update, check the BOM generation report located in the **Newstuff** > **BOM Generation** folder for any errors.<br>**2.** Manually fix any error.<br>**3.** Start the workflow to update the solution variant for the engineering BOM. |
| Source engineering BOM is updated. | Send the engineering BOM through a workflow to update the corresponding solution variant. |

### Procedure

1. Set an active change that has the appropriate effectivity.
2. Open the updated source structure.

   **Warning**

   Do not change the state of the **Show Excluded by Effectivity** and **Show Suppressed** toggles. The state of these toggles must remain unchanged while creating and updating the solution variant.

3. Click **More Commands** ••• > **Manage** > **Submit to Workflow**.
4. In the **Submit to Workflow** panel, select one of the following templates in the **Template** list:

   - **Update Solution Variants**
     Select this template to update all the solution variants of the structure that are impacted due to the structure change.
   - **Update Solution Variants 2**
     Select this template, and then specify one or more solution variants in the target to update them.

     Alternatively, specify a solution variant in the target and a different revision of the variant rule in the reference. The solution variant is updated with the revised variant rule. In this case, specify only one solution variant in the target.

     If you also specify an engineering change with an effectivity in the target, the solution variants with the effectivity mentioned in the engineering change are updated.

   The update is applied to all the solution variants that are impacted due to the structure change. If a solution variant is already released, it is revised and then updated. If the effectivity of the active change notice completely overlaps with the effectivity of the change notice under which the source structure was updated, the impacted solution variant usage is revised and its **ECN In** is the same as the active change notice.

For a newly added part in the structure, Teamcenter checks if the **Assembly Indicator** of the part is marked as **Configurable Assembly** or **Generic Part**. Teamcenter then checks if a solution variant is already created for the part and reuses this solution variant in the updated solution variant structure. If a solution variant is not available, Teamcenter creates a new one for the part and reuses this new solution variant in the updated structure.

> **Note**
>
> Currently, only changes to **Quantity** in a structure are updated in the corresponding solution variants.

5. Click **Submit**.

### What to do next

To verify if the solution variants are updated correctly, go to the **Solution Variants** tab and open the required solution variants to see the updates.

After the solution variant is updated, if your administrator has set the preference with the required values, the absolute occurrence properties for a part within the solution variant are updated. These properties include **Position Designator**, **ID in Context**, and **Usage Address**

Finally, release the active change. To do so, click **Changes** △, select the change, and click **Release Change**.

## View solution variants

You can view a list of the solution variants that are available for a product structure in the **Solution Variants** tab. You can also open a solution variant and view its content.

### Procedure

1. Open the structure for which you created solution variants.
2. Go to the **Solution Variants** tab.

   The **Solution Variants** section lists the solution variants that were created for the selected structure.

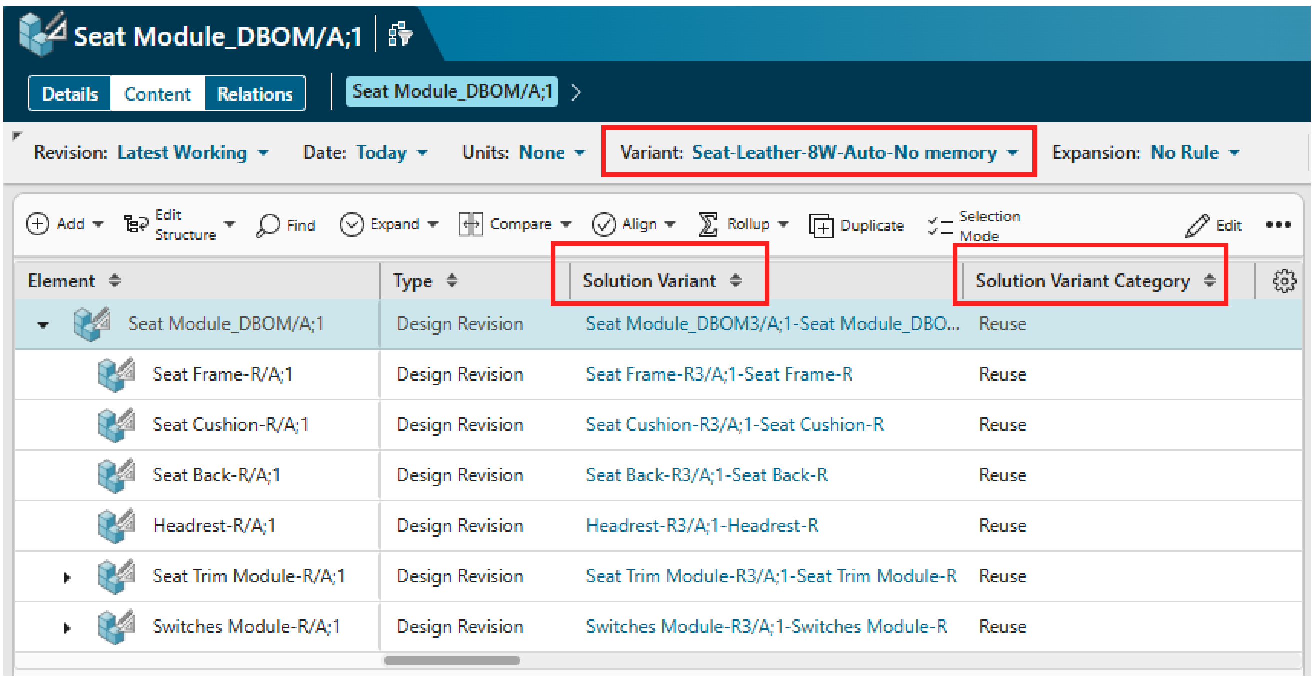
   Solution variants for a configurable product, assembly, or component are displayed in the table in the **Solution Variants** section, depending on the fully qualified match of the Saved Variant Rule (SVR) recipe of the top-level product or assembly. The solution variants that do not match the top-level product SVRs are displayed with an empty value in the **Variant Rule** column.
3. To view the content of the solution variant, in the **Solution Variants** tab, select a solution variant and click **Open**.
   The selected solution variant opens and you can view its content.

# View Reused solution variants

You can view Reused solution variants directly in the structure. This information is helpful for manufacturing engineers and process planners, ensuring accurate variant information is available for MBOM and BOP.

### Procedure

1. Open the structure for which you already created the reused type of solution variants.
2. In the **Variant** list located in the header, select a variant rule.
   The **Solution Variant** column displays the reused type of solution variant that is associated with the selected variant rule. The following screenshot displays an example of the Reused solution variants displayed in the **Solution Variant** column.




# Calculate BOM properties dynamically using the variant configuration

You can set the BOM properties based on Product Configurator feature selections so that their values are calculated dynamically by applying the variant configuration. You can then create a solution variant using these BOM properties. The values that are calculated dynamically persist while creating a solution variant.

By setting the Product Configurator feature on BOM properties, you ensure that the product structure is automatically enriched with the variant configuration-specific details that are critical for downstream processes, without manual intervention. This ensures accuracy, reduces errors, and streamlines workflows across the product lifecycle.

**Example**

When configuring a product like an industrial pump, selecting a specific motor type, such as *Explosion-Proof*, might require unique assembly instructions or special fasteners. By mapping the `Motor_Type` feature to the motor's BOM line properties, you can automatically populate these critical details. This ensures that manufacturing receives precise, configuration-specific guidance for every pump variant, reducing errors and ensuring compliance.

### Prerequisites

The administrator has configured the occurrence property that can be used for mapping the BOM property with the configurator feature.

### Procedure

1. Open a product structure that has the variability scope associated with it.
2. Understand which the occurrence property to use for mapping the BOM line property with the configurator feature.
3. Arrange the columns of the work area so that the occurrence property column is visible.
4. Create the mapping for the BOM property for which you want to set the properties:
    a. Click **Edit** to edit the occurrence property column.

       For example, **AIE Occurrence Name**.

    b. In the occurrence property column, set the value for the BOM line property.

       The BOM properties have display names or system names, and the system names are in the bl_`xxx` format. You must use the system names for the property. For example, if you choose the **AIE Occurrence Note** column to store the mapping and you want to set the value of the Motor Type family on the BOM line property, then enter the mapping in the `<bl_occurrence_note=[Teamcenter]Motor_Type>` format.

       **Tip**

       The mapping is in the `<bl_property_name=[Namespace]Feature>` format. If the mapping is not defined in this format, the mapping is not set.

    c. Click **Save Edits** to save the mapping information.
5. Click **Variant Configuration**, select the feature, and click **Apply Configuration**.

### Results

The BOM properties are set based on Product Configurator feature selections so that their values are calculated dynamically using the variant configuration. These values do not persisted on the original product structure.

### What to do next

After setting the BOM properties to calculate their values dynamically by applying the variant configuration, you can **create a solution variant**.

# 22. Review engineering BOM data

## Compare structures

### Compare the content in structures

You compare similar engineering BOM data, such as two engineering BOMs, to view the differences between them. Your administrator sets up the properties that can be compared.

**Procedure**

1. Perform the action listed in the following table depending on your task:

| Task | Action |
|---|---|
| To compare two parts | Select two parts. Click **Compare** > **Compare Structures**. |
| To compare two engineering BOMs | Select two engineering BOMs from one of the following, and then click **More Commands** ••• > **View** > **Compare Structures**:<br>• Search results<br>• **Change History** table on the **History** tab<br>• Favorites |
| To compare two revisions of a part on the **History** tab | In the **Revision History** section on the **History** tab, select two revisions of a part and click **More Commands** ••• > **View** > **Compare Structures**. |

   In the **Compare** panel, the parts that are unique to both structures are displayed with color bars under **Results**. This comparison is performed with the default comparison options.

2. (Optional) Change the comparison level:

   a. Specify the comparison level by selecting one of the following from **Options**:

| Comparison option | Description |
|---|---|
| **All Levels** | Compares the elements at every level in the two structures. If this option is selected for very large structures, performance may be adversely affected. |
| **All Levels Below Selected** | Compares all levels below the selected level in the two structures. |
| **Components Only** | Compares only the parts that are at the lowest level (components). |

| Comparison option | Description |
|---|---|
| **Current Level** | Compares only the parts that are immediate children of the selections in the two structures. This is the default level of comparison. |
| **Linked Assemblies or Components** | Compares two structures by using accountability check. This comparison level is displayed only if Multi-Structure Foundation is available in your Teamcenter environment. Accountability check is a verification mechanism to check if all the parts in one structure have equivalent parts in the other structure. You can define your own properties for equivalence. |

**b.** Specify what you want to display in the comparison results by selecting one or more of the following options from **Options** > **Display**:

| Comparison option | Description |
|---|---|
| **Matched** | Displays the parts that are a match in both structures. |
| **Different** | Displays the parts that differ across the two structures. |
| **Multiple Matches** | Displays the different numbers of occurrences of the same part in the comparison as a difference. |
| **Unique in Left View** | Displays the parts that are only listed in the structure displayed on the left. |
| **Unique in Right View** | Displays the parts that are only listed in the structure displayed on the right. |

**c.** For **Linked Assemblies or Components**, select the **Dynamic Equivalence** check box.

In some cases, the occurrences that are slightly different in the two structures might not be reported as equivalent. The dynamic equivalence check reports such occurrences.

This check box is displayed only if Multi-Structure Foundation is available in your Teamcenter environment.

**d.** (Optional) Select the **Run in Background** check box to perform the comparison in the background. This is recommended when you compare all levels or only the components in large structures. You can view the results later by selecting the relevant notification from **Alerts**.

> **Note**
>
> The **Run in Background** check box is not displayed when you select the comparison level as **Current Level**.

**e.** (Optional) Select the **Generate Report** check box to generate the comparison report.

**f.** Click **Compare**.

The comparison results are listed under **Results**. Clicking a part in the list highlights that part in the structure. If your administrator has configured it for you, the parent elements of the structures are also included in the comparison.

Comparison results are retained for a specific period. If a comparison result is older than the retention period or if a more recent result exists, the old result is automatically deleted by the system.

If you selected the **Generate Report** check box, the comparison report is generated in Excel. If you have the permissions to edit the target engineering BOM, the report is attached to it. You can click the **Attachments** tab of the target engineering BOM to download the report. If you do not have permissions to edit the target engineering BOM, the report is saved in the **Newstuff** folder. The file name for the comparison report is in the format *Compare_Report_<Source_ID>_AND_<Target_ID>_YYYYMMDD_HH_MM_SS*, where *Source_ID* is the ID of the first engineering BOM that you want to compare and *Target_ID* is the ID of the engineering BOM with which you want to compare the first engineering BOM. You can also click to understand the location of the comparison report if the BOMs are compared in the background.

> **Tip**
>
> In case you closed the **Compare** panel and want to perform any further comparison, click **Compare** on the task bar to view the **Compare** panel.

You can select some other parts either from the same structure or one each from the two structures displayed side-by-side for further comparison.

To reconcile the differences found in the structure, you can drag a part from one structure to another and compare them again.

## Compare the content in partitions

You can compare the content that is in two partitions when two different configurations are available for comparison, for example, when the content in each partition has a different revision, date, units, variants, and partition schemes. You can compare the content from a single structure where two different partition schemes are applied and where only one structure has a partition scheme applied. However, you must select the same element on both sides in a split view. You can compare the content in partitions that are present in two different structures as well. The partitions themselves are not listed in the comparison result.

### Procedure

1. Open a structure.
2. Click **Split Context**.
   The **Split View** displays the structure side by side in two views.
3. Change the structure configuration, as required. You can do this in both views.
4. Click **Compare** on the task bar.

5. In the **Compare** panel, specify the comparison level by selecting one of the following **Options**:

| Comparison option | Description |
|---|---|
| **All Levels** | Compares the elements at every level in the two structures. If this option is selected for very large structures, performance may be adversely affected. |
| **All Levels Below Selected** | Compares all levels below the selected level in the two structures. |
| **Components Only** | Compares only the elements that are at the lowest level (components). |
| **Current Level** | Compares only the elements that are immediate children of the selections in the two structures. This is the default level of comparison. |
| **Linked Assemblies or Components** | Compares two structures by using accountability check. This comparison level is displayed only if Multi-Structure Foundation is available in your Teamcenter environment. Accountability check is a verification mechanism to check if all the elements in one structure have equivalent elements in the other structure. You can define your own properties for equivalence. |

6. In the **Compare** panel, from **Options** > **Display**, select one or more of the following options to specify what you want to view in the comparison results:

| Comparison option | Description |
|---|---|
| **Matched** | Displays the elements that are a match in both views. |
| **Different** | Displays the elements that differ across the two views. |
| **Unique in Left View** | Displays the elements that are only listed in the view displayed on the left. |
| **Unique in Right View** | Displays the elements that are only listed in the view displayed on the right. |

7. For **Linked Assemblies or Components**, select the **Dynamic Equivalence** check box.

   In some cases, the occurrences that are slightly different in the two structures might not be reported as equivalent. The dynamic equivalence check reports such occurrences.

   This check box is displayed only if Multi-Structure Foundation is available in your Teamcenter environment.

8. (Optional) Select the **Run in Background** check box to perform the comparison in the background. This is recommended when you compare all levels or only the components in large structures. You can view the results later by selecting the relevant notification from **Alerts**.

   > **Note**
   >
   > The **Run in Background** check box is not displayed when you select the comparison level as **Current Level**.

9. Click **Compare**.
   The comparison results are listed under **Results**. Clicking an element in the list highlights that element in the structure. Clicking an element in the structure highlights that element in the list.

   If your system administrator has configured it for you, the top elements of the structures are also included in the comparison result.

   > **Note**
   >
   > If you close and relaunch the **Compare** panel by clicking **Compare** on the task bar, the panel is reset, and it does not display the previous results.

# Validate design and engineering BOM alignment

## About validating alignment

Learn about validating and then verifying the alignment after generating a design structure or manually aligning design and part occurrences.

After generating a design structure, you must validate if the corresponding BOM is created correctly with proper alignment. You must also validate the alignment after aligning design occurrences and part occurrences manually. To verify if the alignment is performed correctly, you can view the:

- **Aligned designs and parts**.
- **Aligned design occurrences and part occurrences**.

  You can also **view the product manufacturing information** available in an aligned engineering BOM (Usage BOM or Assembly BOM).

## View aligned parts and design

You can view aligned parts and designs to visualize a part's appearance in its floor position.

To visualize what a part looks like in the floor position, you align it with its corresponding design.

1. Open the part revision or the design revision for which you want to view the aligned designs or parts.
2. For a part, the corresponding designs are listed in the **Aligned Designs** section of the **Overview** tab.

   For a design, the corresponding parts are listed in the **Aligned Parts** section of the **Overview** tab.

   These sections also list additional details such as which design is set as the primary representation of a part or if there is some alignment mismatch.

## View the product manufacturing information

*Product manufacturing information (PMI)* includes information, such as geometric dimensions, tolerances, 3D annotation, and surface finish specifications, required for manufacturing a product. PMI is available in a design structure. You can view PMI in the context of an aligned engineering BOM (Usage BOM or Assembly BOM).

#### Procedure

1. Log on to the Teamcenter rich client.
2. Search for part revisions and open the required engineering BOM.
3. Click **Start/Open in Lifecycle Visualization**.

   Currently, you can view PMI only in the standalone viewer.
4. In the standalone Lifecycle Visualization viewer, from the **3D Load Options** dialog box, select **Open document** and click **OK**.
5. Right-click the aligned engineering BOM and select **Show PMI**.

## Visualization of electrical harness design structure and its aligned engineering BOM

You perform electrical harness design within a multi-domain environment, integrating various tools and systems to efficiently manage complex design requirements. You detail the complete specifications of the harness in an electrical CAD tool like **Capital** and perform the mechanical routing in **NX**, **CATIA**. The design structure is imported from CAD and you can directly publish the electrical harness using integrations into **Teamcenter** which generates the engineering BOM (Usage BOM or Assembly BOM). This publish action also aligns engineering BOM from electrical CAD and design structure from mechanical CAD.

The harness design structure consists of body solids (without definite geometric shape, e.g. tapes, tubes, wire bundles) and geometric components (with geometric shape).

> **Note**
>
> Do not modify the already aligned structure in the alignment view and avoid performing realignments. Instead, make all changes to the original structure in the electrical CAD and mechanical CAD tools, and then republish the aligned BOM to **Teamcenter**.

Body solids represent the physical embodiment of parts and assemblies in a 3D space, used for visualization, spatial analysis, interference checking. They help you understand the physical dimensions and interactions of each component within the assembly. They are present inside a node called the **Wiring Component (MBS)** node.

Geometric components are aligned using standard alignment process but MBS node can not be aligned because it does not have a corresponding part node in the engineering BOM.

Let's understand body solids and geometric components using an example:

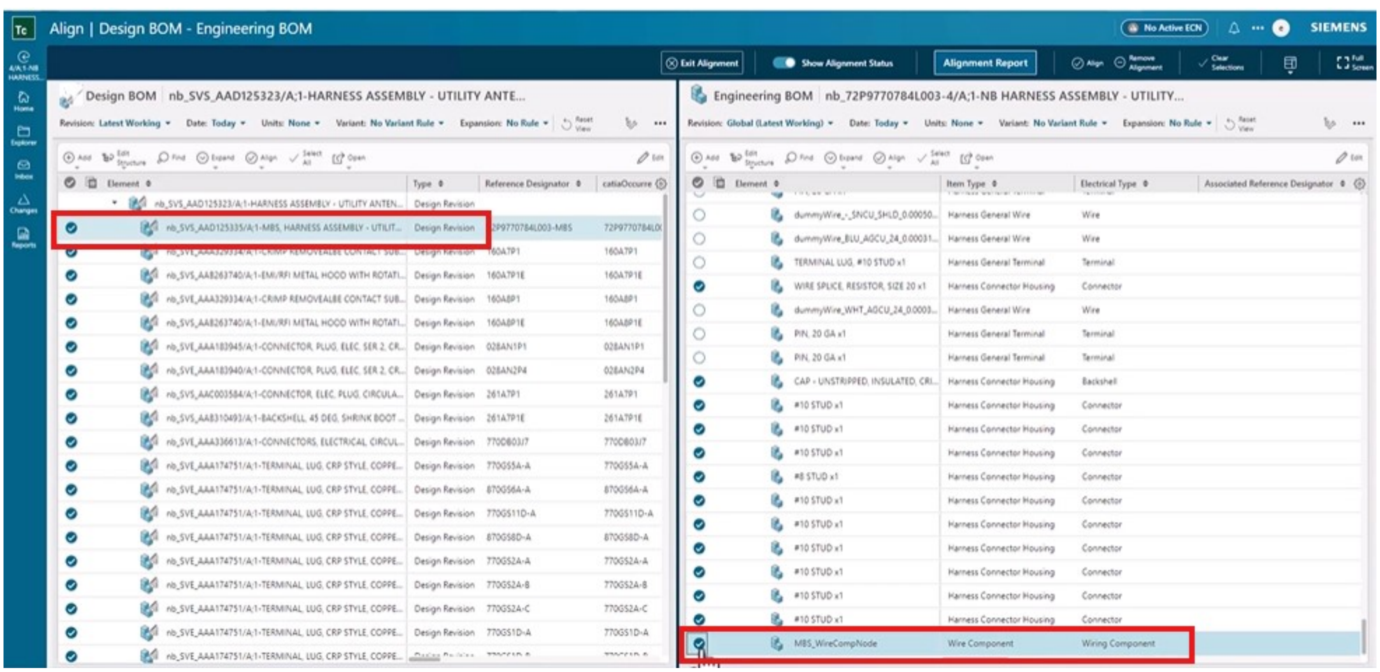
Imagine you are working on an electrical wiring harness for an aircraft. Within a CAD software like **NX**, you can model various components of the harness. These components include:

- Wire Bundles (Body Solids): It has bundle's diameter, bend radius, and its path within the aircraft structure in 3D. Body solids allow you to visualize how the bundle fits with other components, check for physical interference, and ensure optimal routing.
- Connectors (Geometric Components): These are represented as design items in **Teamcenter**. As a design item, they include information like vendor details, electrical ratings, and associated documentation without detailing the geometric shape. It serves as part of the engineering BOM.

To visualize electrical harness design structure and its aligned engineering BOM:

The **Wiring Component (MBS)** node is present in the harness design structure but does not have a corresponding part node in the engineering BOM published to **Teamcenter**. However, solid bodies present in the **Wiring Component (MBS)** node are present in the engineering BOM as parts, also known as **Geometric Proxy Components**.

To visualize these structures in **Teamcenter**, you can create an equivalent node in the engineering BOM of type **Wiring Component** and manually align it to the **Wiring Component (MBS)** node in the design structure.




## View aligned part occurrences and design occurrences

You view the aligned part occurrences and design occurrences in order to validate the alignment.

### Procedure

1. Open the required structure.

2. Click **Align** > **Open Alignment View**.

   The design structure and the engineering BOM (Usage BOM or Assembly BOM) are now displayed side by side. If you applied any variant and configuration criteria to the engineering BOM, the aligned design structure reflects the applied criteria.

   If you set a specific partition scheme on the engineering BOM, only the engineering BOM loads with that scheme in the alignment view. The partition scheme is not applied to the design structure.

   By default, **Alignment Status** shows an **Intermediate** status. This status is not set for the top BOM line and for the occurrences for which the **Design Required** or **Part Required** is set as **False**.

3. You can view the alignment between specific design and part occurrences, or for all aligned design and part occurrences.

   - To view specific aligned design and part occurrences, click the **Intermediate** indicator icon, or right-click the icon to choose **Show Alignment**.

     For a design that occurs multiple times in the design structure, the aligned part occurrence is summarized as a single instance. The quantity of the part occurrence is set as the number of times the design occurs. The part is summarized based on the grouping criteria set by the administrator.

     **Example**

     Consider a car's design structure that includes four occurrences of the wheel, each representing a different position. In the design structure, you can pack these wheel occurrences and the quantity is set to four. Similarly, you can unpack the wheel occurrences in the design structure.

     After the design structure is aligned with the engineering BOM, the part occurrences are summarized and the quantity is set to four. Thus, you cannot unpack the part occurrences as separate instances in the aligned engineering BOM.

     **Note**

     You cannot unpack the part occurrence in Active Workspace to view its separate instances. To view the separate instances, open the part in a Rich Client application such as Structure Manager or Manufacturing Process Planner.

   - To view all the aligned part and design occurrences, enable the **Show Alignment Status** option. It is recommended that you select this option if your structure is small.

     For larger structures, click **Alignment Report** to generate a report containing the alignment details. You are notified in **Alerts** when the report is generated.

   The icons in **Alignment Status** indicate the alignment status of the design occurrences and part occurrences. Additionally, the icons in **Advance Status** indicate the alignment status of the corresponding design revisions and part revisions.

| Advance Alignment Status | | Alignment Status | |
|---|---|---|---|
| Indicator | Description | Indicator | Description |
| | |  | Indicates that the design occurrence and part occurrence are not aligned with each other.<br>OR<br>If this indicator is present without the advance indicator, then it shows a missing occurrence alignment in a dual intent part. |
| | |  | Indicates that the design occurrence and part occurrence are aligned with each other.<br>OR<br>If this indicator is present without advance indicator, then it shows a complete occurrence alignment in a dual intent part. |
|  | Indicates that the corresponding design revision and part revision are not aligned with each other. |  | Indicates that the design occurrence and part occurrence are aligned with each other. |
|  | Indicates that the corresponding design and part are not aligned with the latest revisions. | | |
|  | Indicates that the corresponding design and part are aligned. | | |
| | |  | When an aligned sub-assembly is revised, this indicator is shown on its children, which indicates the alignment between the part occurrence and design occurrence under a different revision or context. Selecting a child part with this indicator displays the corresponding out-of-context design.<br>OR<br>This indicator is shown on the children when the parent is not aligned but its children are aligned. |

No alignment indicators are shown for designs present below the parent design. The parent design is marked as supplier part or skip node.

If you run the alignment report, the following additional indicators are displayed in the report:

<table>
<tr><th colspan="2">Advance Alignment Status</th><th colspan="2">Alignment Status</th></tr>
<tr><td>Indicator</td><td>Description</td><td>Indicator</td><td>Description</td></tr>
<tr><td></td><td>The corresponding design revision and part revision are not aligned. Also, it indicates that one or more of the child design occurrences are not aligned with the corresponding part occurrences.</td><td rowspan="3"></td><td rowspan="3">Indicates that the design occurrence and part occurrence are aligned with each other.</td></tr>
<tr><td></td><td>Indicates that the corresponding design and part are not aligned with the latest revisions. It also indicates that:<br>• One or more child design occurrences and part occurrences are not aligned.<br>• One or more child designs and parts are not aligned.<br>• One or more child designs and parts are not aligned with the latest revisions.</td></tr>
<tr><td></td><td>Indicates that the design and part are not aligned with the latest revisions.</td></tr>
<tr><td></td><td>Indicates that the design and part are not aligned with the latest revisions.</td><td></td><td>Indicates that the design occurrence and part occurrence are not aligned with each other.</td></tr>
</table>

### What to do next

Based on the indicators, you can take any appropriate actions, such as aligning part occurrences and design occurrences, aligning parts and designs, and **revising** a structure to fix the missing or mismatched alignments.

You can visualize the aligned design occurrences and part occurrences in **3D**. Spare parts do not participate in the alignment process. If the revision rules of the design structure and the engineering BOM are different, the designs that you visualize are based on the revision rule of the engineering BOM by default. However, your administrator can change this default behavior so that the designs are based on a different but valid revision rule.

# View and visualize structures

## View a structure in other applications

You can open and view a structure in other applications, such as NX and CATIA. If the structure exists within a partition, you can open the partition itself in some other application, so that one or more structures within that partition are opened.

| Application | Action |
|---|---|
| Teamcenter Lifecycle Visualization | Select **Open** > **Open in Visualization**. |
| Rich client | Select **Open** > **Open in Rich Client**. |
| Designcenter NX | Select **Open** > **Open in Designcenter NX** to open the selected element.<br><br>Select **Open** > **Open Product in Designcenter NX** to open the entire product structure. |
| Solid Edge | Select **Open** > **Open in Solid Edge**. |
| Illustrator | Select **Open** > **Open in Illustrator**. |
| CATIA | Select **Open** > **Open in CATIA**. |
| Supplyframe | Select **Open** > **Open in Supplyframe**. |

## View part and part usage information of an engineering BOM

Learn how to access and view information about an engineering BOM or a specific part and part usage properties. You can view these details found in tabs like **Where Used**, **Variant Configurations**, **Requirements**, **Materials**, and **Vendor Parts**.

The information related to the engineering BOM (that is, Usage BOMor Assembly BOM) and a part is available in the **PROPERTIES** section of the **Overview** tab.

The part usage information is available in the **PART USAGE PROPERTIES** section of the **Overview** tab of a part.

Other tabs such as **Where Used**, **Variant Configurations**, **Requirements**, **Materials**, and **Vendor Parts** contain additional details about an engineering BOM or a part.

## Viewing structures in the split view

You can use the split view to display the two structures or configurations side-by-side, independently modify them, compare them, and perform drag and drop operations for moving or copying occurrences. You can also filter and retain the context upon exit in the split view.

You can split the structure view into two views using the **Split Context** command. Initially, identical structures are loaded in both panels. Both panels provide a toolbar with commands relevant for the structure being displayed.

You can also open two different structures side-by-side in the split view by selecting two root nodes and then using the **Open** command.

In the split view, you can modify the structures independently. Once either structure is updated to a different configuration, then the **Compare** functionality is enabled. You can also use the split view to move or copy an occurrence from one structure to another using the drag-and-drop functionality. If the two views have the same structure with the same applied configuration, then the occurrence is moved. Otherwise the occurrence is copied.

If the structures are indexed with Smart Discovery, filter the structures by clicking **Filter** in the **Split View**. You can filter the entire structure and apply the filtering criteria to view the required content of the structure.

When you exit the **Split View**, you can continue working on the structure that you worked on the last. For example, in the **Split View**, suppose that the Headlight Assembly is available on the left view and the Front Bumper assembly is available on the right view. Then you changed the configuration of the Front Bumper assembly before exiting the **Split View**. In such a situation, the Front Bumper assembly is retained and you can continue working on it. When you click **Back** in the **Split View**, you can continue working on the structure in the left view. After clicking **Back** in the **Split View**, the structure in the left view is opened with its original configuration, that is, the configuration that was applied before splitting the view. If you applied the filter in the **Split View** and then clicked **Back**, the filter criteria is retained on the structure.

**Note**

You cannot filter Smart Discovery Indexed Usage BOM structures in the **Split View**.

## View where a part is used

A where-used analysis in Teamcenter helps you identify all parent assemblies and products in which a part is used by navigating up the structure. Starting from a selected component or assembly as the context, this analysis traces usage throughout all levels of the structure.

You can use a where-used analysis to:

- Assess the impact of engineering changes to the product structure.

- Check if changes in one assembly affect other assemblies.
- Determine which products are affected by item changes.
- View items in the context of their products through traversal paths.

### Restrictions and limitations

Your Teamcenter setup must contain Smart Discovery Indexing to access the **TOP LEVEL** section and view top-level assemblies or products.

### Procedure

1. Search for and select the part.
2. Click the **Where Used** tab.
3. Expand the tree in the **Used in Structures** section to move up in the structure.

   The **Used in Structures** section lists the immediate parents of a part.

   The table initially displays parent assemblies and any assemblies where the selected item is used as a substitute. The information displayed is based on the revision rule currently selected in the **Revision Rule** list, which defaults to **Global**. In this case, this refers to the revision that is selected in the global revision rule list.

   To view all parent items where the selected child item is configured, select **All** from the **Revision Rule** list. Click **Export to Excel** to export the information displayed in the table. The number of structure levels to be exported to Excel is set by your system administrator.

   > **Note**
   >
   > If an occurrence of an item is used as a substitute in an assembly, those assemblies are not listed in the **Used in Structures** section.

   If the selected part is a substitute, the column in the **Used in Structures** section of the **Where Used** tab displays for its parent part.
4. Expand the **Top Level** section if it is not already expanded.

   Smart Discovery Indexing is used to find the top-level assemblies or products. Therefore, if your system administrator has not installed Smart Discovery, the **Top Level** section is not displayed. You only see indexed products in the results list.

   The table initially displays the top-level products in which the selected item is used. This display is based on the revision rule currently selected in the **Revision Rule** list, which defaults to **Global** (meaning, the revision rule that is selected in the global revision rule list). This is useful for *impact analysis* when you want to see which products are affected if the item is changed. It is also useful for determining the *traversal path* when you want to see the item in the context of one of its products

   To view the top-level products where the selected child item is configured, select **All** from the **Revision Rule** list. Click **Export to Excel** to export the information displayed in the table.

5. Expand the **Contexts** section if it is not already expanded.

   The **Context** section shows various configurations for the selected component. You can select the configuration you want to open.

6. The **Part Usages** section lists all the part usages of the selected part across all the products. This is useful for *impact analysis* when you want to see which products or assemblies are affected if the part is changed.

7. Expand the **References** section if it is not already expanded.

   The **References** section lists the parts and documents that reference the selected component.

## Visualize structures

You visualize parts and product BOMs to verify if they are correctly aligned with their corresponding designs, check for clearance between parts, and verify if they are correctly place at the right proximity from other parts. However, you can perform the visualization only if **Product Master Automation** and **Visualization Data Server** are set up in your Teamcenter environment.

If the structure is indexed for massive model visualization (MMV), you can view the 3D model of it.

### Procedure

1. Open a structure and click.**Details** > **3D** .

2. Select a structure element.
   The selected element is highlighted in the **3D** tab.

3. Select an element in the **3D** tab.

   Ensure the first clause in your configured revision rule is set to **Precise** for accurate visualization of supplier parts.

   The corresponding structure element is selected. However, multi-selecting in the **3D** tab does not select the corresponding elements.

## Understand pack, unpack, and summarization of items

Understand the fundamental concepts of pack, unpack, and summarization in Teamcenter and how they simplify complex product structures.

**Pack**
: Packing refers to the process of consolidating multiple identical occurrences of an item within a structure into a single line item, along with its total quantity. This is useful for simplifying complex and large structures, especially when dealing with many instances of the same part, like four standard wheels on a car. By packing, you reduce the visual clutter and make the structure easier to read and manage.

**Unpack**

Unpack is an action of expanding a packed line item back into its individual occurrences within the structure. When you unpack an item, like the wheels, Teamcenter displays each occurrence separately in the structure. This is particularly useful when you need to inspect each instance independently, especially if they have unique positional variances or other specific attributes that differentiate them. Unpacking allows for detailed scrutiny and individual management of each occurrence.

**Summarization**

Summarization applies exclusively to the engineering BOM. An item in engineering BOM is summarized when it is aligned with more than one design occurrences (a one-to-many alignment relationship). The summarized item acts as a consolidated representation for all those alignments. Teamcenter also creates an exploded line for each individual design occurrence aligned to that summarized engineering BOM item, allowing you to inspect or manage each alignment independently. This is helpful to you when you are working with quantity-managed structures, variant formulas, and piece-part alignments, where each exploded line can carry its own variant condition derived from both the summary line and the specific design occurrence it represents.

When you open the product in the Contents tab, it shows the summarized lines; whereas, in the **3D** view, it shows the exploded lines.

The following table describes the pack, unpack, and summarization concepts.

| Concept | Applied to | Trigger | Purpose |
|---|---|---|---|
| Pack | Any structure | Multiple identical occurrences at the same level | Reduce visual clutter, shows combined quantity. |
| Unpack | Any structure | Expand a packed line. | Inspects or edits individual occurrences. |
| Summarization | Engineering BOM | One engineering BOM item is aligned to multiple design occurrences | Represents one-to-many alignments. |

**Note**

Videos are not included in PDF. To access videos, use HTML.

## Visualize structure elements located within partitions

You can visualize structure elements that are located within partitions. You can select a partition scheme to view specific partitions and their associated elements highlighted in the 3D viewer. Teamcenter dynamically updates the 3D view when you assign elements to partitions or unassign elements from partitions.

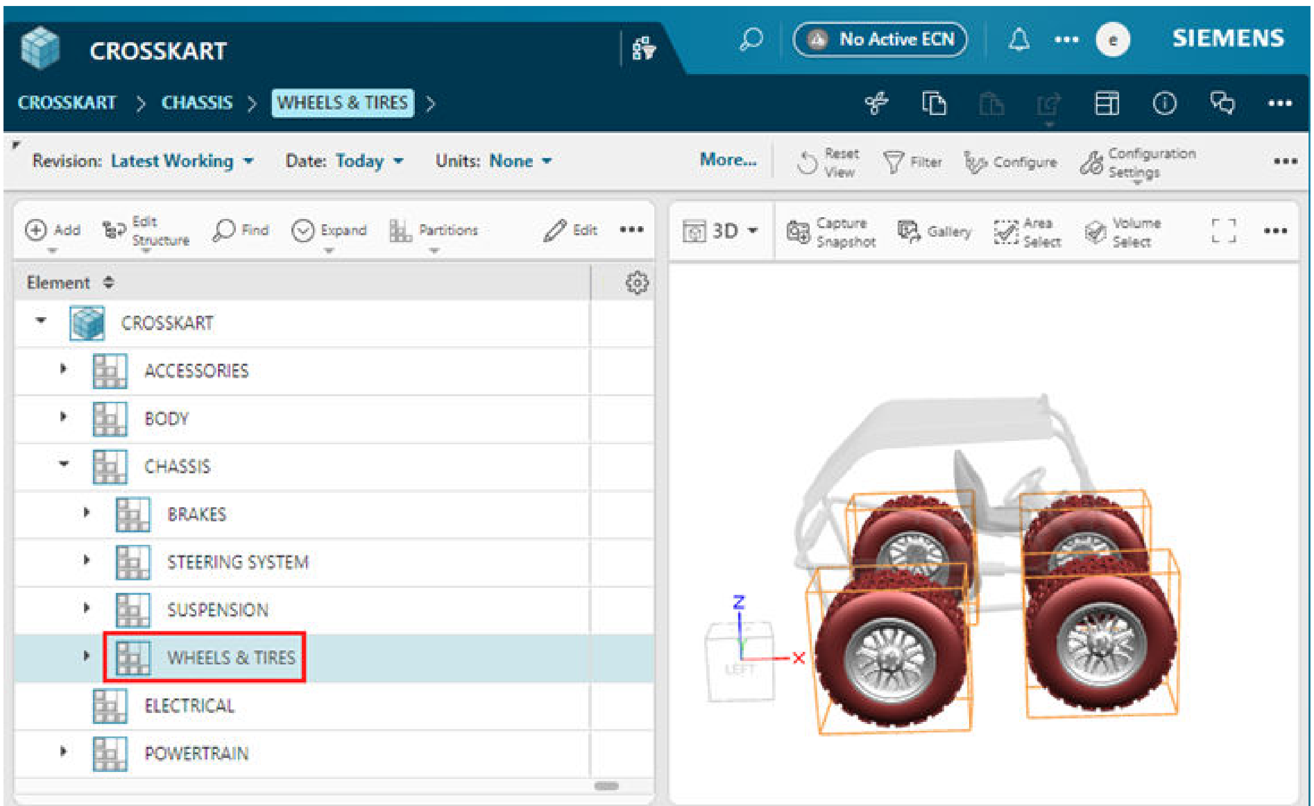
### Procedure

1. Open the structure that you want to visualize. By default, the structure opens in the partition scheme that is set as the primary scheme.

   > **Note**
   >
   > Ensure that a partition scheme exists before you open the structure that you want to visualize. For information on partition schemes and partitions, see *Structure Partitions — Usage*.

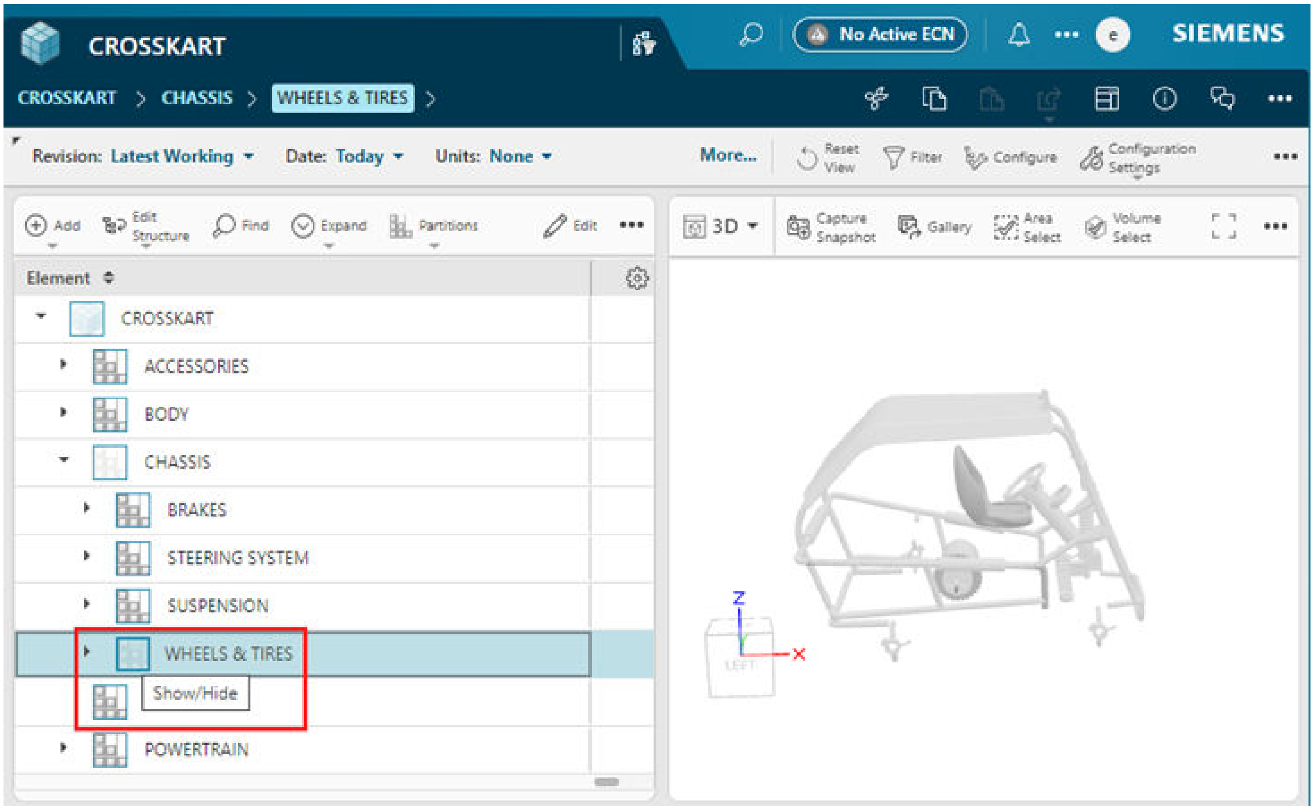
2. Select the required scheme from **Partition Scheme** that contains the required partitions.

3. To visualize the structure elements located within a partition, select the partition, and go to the 3D viewer.
   All the elements within the selected partition are highlighted with their bounding boxes.




   If the selected partition has in turn a child partition within it, all the structure elements within the child partition are also highlighted in the 3D viewer.

   You can choose to not visualize the elements of a specific partition by clicking **Show/Hide**. For example, the icon is displayed next to WHEELS & TIRES, as shown in the following image. To visualize the elements again, click **Show/Hide** again.




When you set variant criteria on partitions, the structure is configured to either show or hide the partitions and the elements within. Accordingly, the elements are displayed in the 3D viewer as well.

In several cases, the configured structure and the 3D viewer might not be synchronized.

- **Scenario 1**

  For a structure in a *workset*, a *Functional* scheme is applied and contains a variant criterion, VC1. In this case, the structure elements with a different variant criterion, VC2, are not included in the structure. However, the 3D viewer shows these elements. When you switch to a different partition scheme, the same scenario applies. In such cases, for structures in a workset, you can synchronize the partitioned structure and the 3D viewer by clicking **Replay**.

- **Scenario 2**

  A component resides in two different partitions. It is included in the structure configured for one partition, whereas it is not included in the structure configured for the other. When such a mismatch occurs, the component is displayed both in the partitioned structure and in the 3D viewer.

- **Scenario 3**

  In a partition scheme, structure elements at different hierarchical levels reside in two different partitions. When you set variant criteria on a partition to configure the structure such that the partition is not included in one case, the structure and the 3D viewer show a mismatch. Siemens recommends configuring the system to restrict the addition of structure elements in partitions in such a manner.

In the **Partition Scheme Assignment** view, the **3D** view for the **Unassigned** pane displays only the unassigned elements. When you drag one or more elements from the **Unassigned** pane to the **Assigned** pane, Teamcenter updates the elements in the **3D** view for the **Unassigned** pane accordingly. Similarly, the **3D** view for the **Assigned** pane gets updated. Thus, in both panes, the elements that you see in the tree structure match the ones you see in the **3D** view.

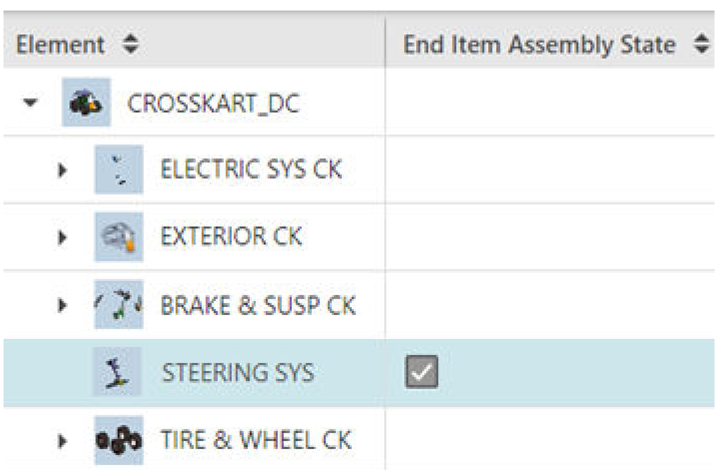
## Working with end-item assemblies

The end-item assemblies enhance the system performance and help you focus by collapsing sub-assemblies. When you search for child components within an end-item assembly, the parent end-item is highlighted.

To improve the system performance and hide information that is not relevant to the user's current task, one or more assemblies in a product structure can be set as *end-item assemblies* for that structure. When you expand a structure containing an end-item assembly, the nodes beyond the end-item assembly are not expanded.

Although you cannot set any occurrence as an end-item, you can view a structure containing end-items. When a structure containing an end-item is opened, the end-item is displayed as a single line in the assembly tree.

The **End Item Assembly State** column identifies the end-items in an assembly.




The 3D viewer displays the full assembly structure regardless of **End Item Assembly State**. To view the full structure of an end-item assembly in the tree, open the end-item as its own root node.

### What happens when you search for a child component of an end-item?

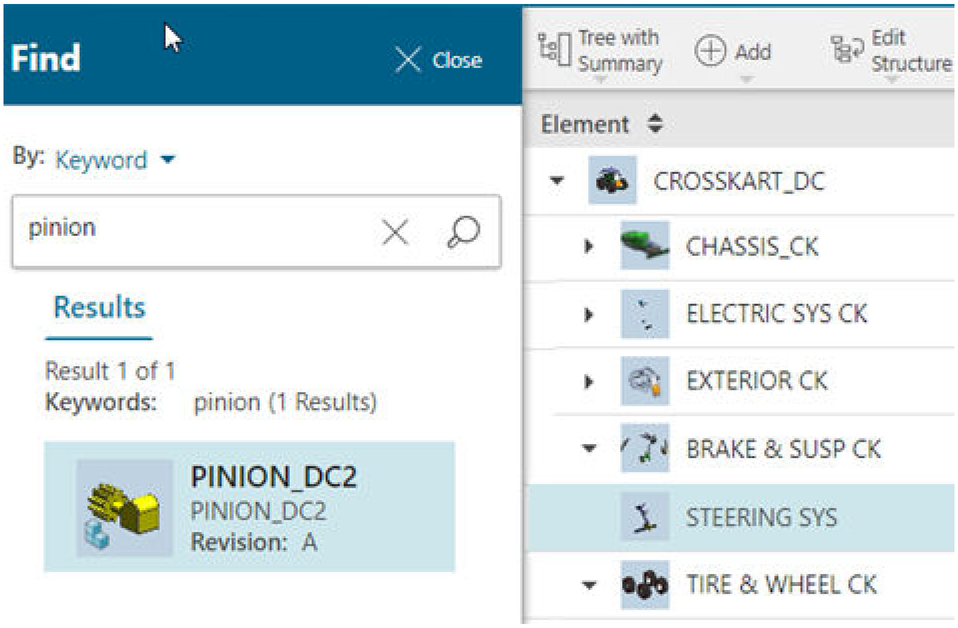
To access the components of any subassembly, you expand the relevant subassembly.


As the assemblies marked as end-items cannot be expanded within the structure, their child components are not visible when the structure is expanded.

For example, if the **steering system** assembly is marked as an end-item in a vehicle structure, its components, namely, **steering mech**, **pinion**, and **rack** are not visible in the structure.


However, when you perform an in-context search for a child component of an end-item, the search results return the child components. When you select the child component of the end-item in the search results, the parent end-item subassembly is selected in the structure.

For example, when you select the **PINION_DC2** child component from the search results, the parent **STEERING SYS** subassembly, which is an end-item, is selected in the structure.




## Control the display of configured structures

You can control the display of configured structures by showing or hiding occurrences based on effectivity, variants, filtering, and suppression. You can also manage the visibility of connections, joints, and ports within the structure.

When creating variable content or analyzing configured structures, it is important to understand which occurrences will be configured and the ones that will be excluded. When a structure is loaded, only objects that meet the applied effectivity and variant criteria are displayed. However, you can choose to hide or view the occurrences that are not configured as well.

> **Note**
>
> You can do this only when you select an individual structure within the workset or any elements within the same structure. You cannot select multiple structures within the workset or multiple elements across different structures within the workset to hide or view such occurrences.

| To perform this step | Do this |
|---|---|
| To show or hide the occurrences that are excluded by the currently applied effectivity | Click **Configuration Settings** > **Show Excluded by Effectivity**. |
| If you have **Partitions for Structure** deployed in your Teamcenter setup, and you have applied variability on partitions, to show or hide occurrences | Click **Configuration Settings** > **Show Excluded by Variants**. |

| To perform this step | Do this |
|---|---|
| and partitions that are excluded by the variant configuration | |
| To view all the structure elements including the ones that are not displayed due to the applied filters | Click **Configuration Settings** > **Show Excluded by Filtering**. |
| Occurrences in a structure can be hidden by setting the suppress property to True. When a structure is loaded, all occurrences including the suppressed occurrences are not displayed by default. To show or hide the suppressed occurrences | Click **Configuration Settings** > **Show Suppressed**. |
| Hide all secondary components of the structure. | Click **Configuration Settings** > **Show Secondary Components**.<br><br>**Note**<br>In stead of hiding all secondary component of the structure, if you want to hide the secondary components only within an assembly, right-click the assembly and click **Hide Secondary Components** . |
| To show and hide the connections and joints from the structure. For example, welds. The **3D** tab still show the 3D view of the connections and joints even if you hide them.<br><br>This button is available only when the product structure is opened without a session, workset, configuration baseline, saved working context or without a partition scheme. | Click **Show Connection Points** > **Connections and Joints** . |
| To show and hide the ports from the structure. The **3D** tab still show the 3D view of the ports even if you hide them.<br><br>This button is available only when the product structure is opened without a session, workset, configuration baseline, saved working context or without a partition scheme. | Click **Show Connection Points** > **Ports** . |

## Attach JT file for product previews

You can attach the JT file, that contains the structure's outline, to the its structure in Teamcenter so that the engineering users can preview the product outline and avoid the performance issues of loading the entire product structure in the Viewer.

#### Prerequisites

- You must have the write access to the top node of the product structure.
- You have obtained the JT file that contains the product outline. The JT file containing the product outline is typically generated from the CAD software (such as NX). Contact your CAD engineer if you need assistance generating this file.

#### Procedure

1. In Active Workspace, open the structure for which you want to attach the JT file.
2. In the **Content** tab, select the top node of the product structure, and click the **Attachments** tab.
3. In the **Files** section, click **Add** ⊕.
4. In the Add panel, click **Select File** and choose a JT file.
5. In the **Type** list, select **`Direct Model`**.
6. In the **Relation** list, select **`Product Preview`**.
7. Click **Add**.

#### Results

The JT file containing product's outline is attached to its structure. Engineering users can preview the outline by opening the product structure in the Viewer by avoiding the performance issues of loading the entire, large product structure.

## View the Supplyframe risk index for a commercial part

If your Teamcenter environment is integrated with the Supplyframe application, you can view the risk index of the vendor part.

#### Prerequisites

Ensure that the risk index information is already sent from the Supplyframe application to Teamcenter.

#### Procedure

1. Search for the vendor part in a structure.

2. In the structure table, scroll to view the **Risk Index** column.

   If the **Risk Index** column is not available in the structure table, click **Table Settings** > **Arrange** to arrange and view the column.

3. (Optional) Click the **Vendor Parts** tab to view the vendor parts for the selected part.

   If multiple vendor parts are available for the selected part, the risk index is shown for the one that is marked as **Preferred**.

# 23. Release engineering BOM data

## Release engineering BOM data not created under an active change

You release the engineering BOM data for downstream tasks, such as planning the product manufacturing with engineering BOM data.

**Procedure**

1. Open the engineering BOM data that you want to release. If the part is an assembly part, you can additionally choose its child parts to be released along with the parent part.
2. Click **More Commands** > click **Manage** > **Submit to Workflow** .
3. In the **Submit to Workflow** panel, select the appropriate release workflow (for example,**TCM Release Process**) in **Template**.
4. Click **Submit**.

## Release engineering BOM data by validating and releasing the active change

If you created or updated the engineering BOM data, by setting an active change notice, the data can be validated and released together by validating and releasing the active change notice.

**Procedure**

1. If the change is not yet submitted, **submit the change**.

   You can either use the change notice and the **EBOM Release Process** template or use a simple change and the **EBOM Simple Change Workflow** template.
2. Validate the change notice to perform validation checks on the engineering BOM before its release. This validation is based upon a set of predefined validation rules in the system. Your administrator can also create custom validation rules.
   a. From **Launcher**, click **Changes**.

      If this option is unavailable, search for it. You can pin it from the search results for future access.
   b. Select the change under which the structure is created or modified.
   c. Click **Validate Change** in the **Overview** tab.

   If the validation fails, the issues are listed in the **Validation** tab under the **Validation Summary** section. Resolve the issues, as needed.

   Once the issues are resolved and the validation is successful, you can release the change.

3. Click **Release Change** to release the change notice.

   While releasing the change, Teamcenter validates the change again to resolve any issues found during the initial validation.

   If no issues found, structure is released. Note that, the workflow (**EBOM Release Process** workflow or the **EBOM Simple Change Workflow**) also releases the BOM view revisions of the *ItemRevisions* that are tracked by the change.

# 24. Create and maintain color BOM

## About color BOM

The Color BOM complements the engineering BOM by adding color attributes to the parts included in the engineering BOM.

In several industries, the appearance of a product is an important buying criterion for customers. For example, in the automobile industry, customers expect new designs and colors. Therefore, the vehicle body, interiors, wheels, control switches, bumpers, and other prominently displayed parts must appear attractive. Manufacturers also use a combination of colors and textures based on styling trends. Such parts with color attributes are called as color parts.

Consider an example of a car with two variants, Leto BX and Leto CX. These variants have different features, and one of the differences is the color option for the bumper. The BX variant is available only with black bumpers for all cars, irrespective of the external body color, whereas the CX variant is available with body-colored bumpers.

The following image shows the red, blue, and gray color Leto BX variants, all with black bumpers.

In the CX variant, the bumper color matches the color of the car's body. Except for the color, every other technical specification of the bumper is the same in all CX variants.

## Color BOM terms

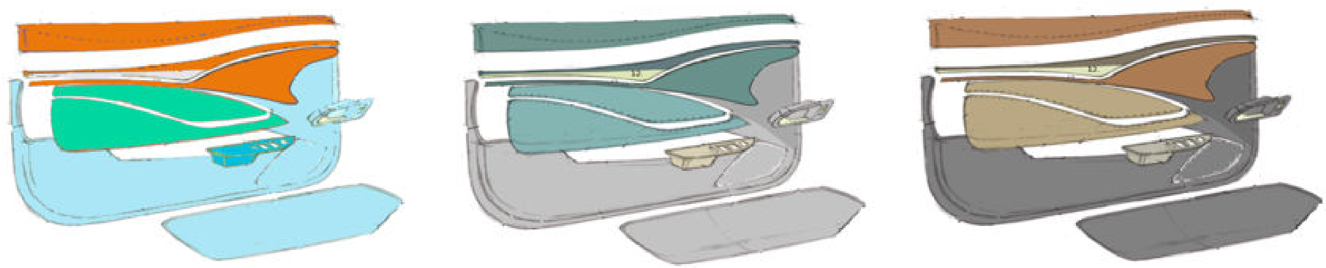
In the Color BOM, different structural elements are used to build the BOM structure.

| The term | Refers to |
| --- | --- |
| Appearance area | A color-specific region of a product that encompasses one or more part occurrences that share the same color. |
| Appearance area breakdown | A collection of appearance areas, mainly used to organize and group the relevant appearance areas together. |
| Appearance product breakdown | The root node of the appearance breakdown structure. It represents the collection of all the appearance breakdowns and areas for a product. |
| Color appearance | An object that captures all the visual appearance aspects, such as color, gloss, and texture information. |
| Color theme | A collection of color appearances for each appearance area assigned in a BOM. It has a variant condition which defines the product model configuration. |
| Less finish part | A part that can be colored. It can have one or more color parts associated with it. |
| Color part | A color part generated from a less finish part for a color specification. Each color part references at least one color specification. |

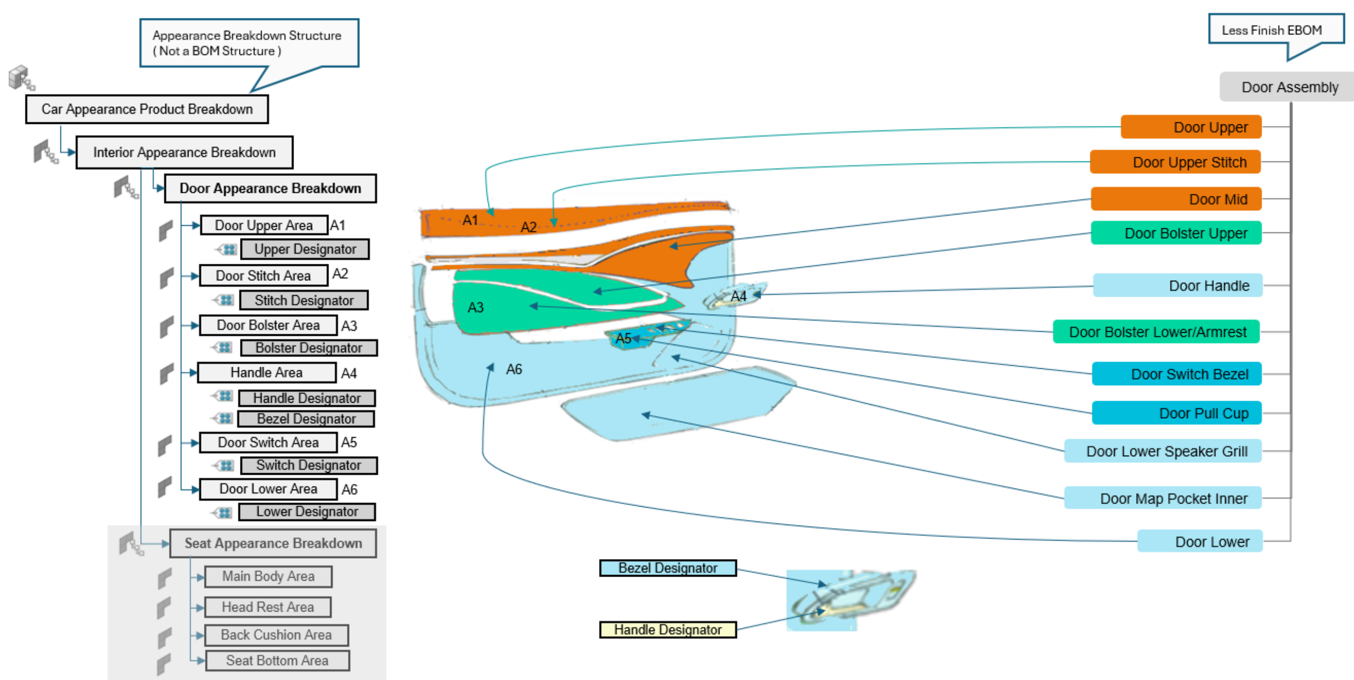
## Exploring color BOM structure through a car door example

Consider that you are building a new car. While the car’s engineering BOM tells you what parts go in the car, it does not tell you how the car will look. For example, the seat fabric type and its color, the car's exterior color and interior color, and the steering wheel texture. These visual details are defined by the color BOM, which defines the car’s appearance.

The color BOM is generated from the engineering BOM but specifically focuses on colors, textures, and finishes of parts that customers see. To understand how the color BOM works, we will use the car door assembly as an example. A car door is made of several parts, but only some parts contribute to its visual appeal, for example, the door handle, the outer door panel, the speaker grill, and the upper trim piece. These are the physical parts that will eventually get a specific color or appearance. These parts are known as *Less Finish* parts in Teamcenter.



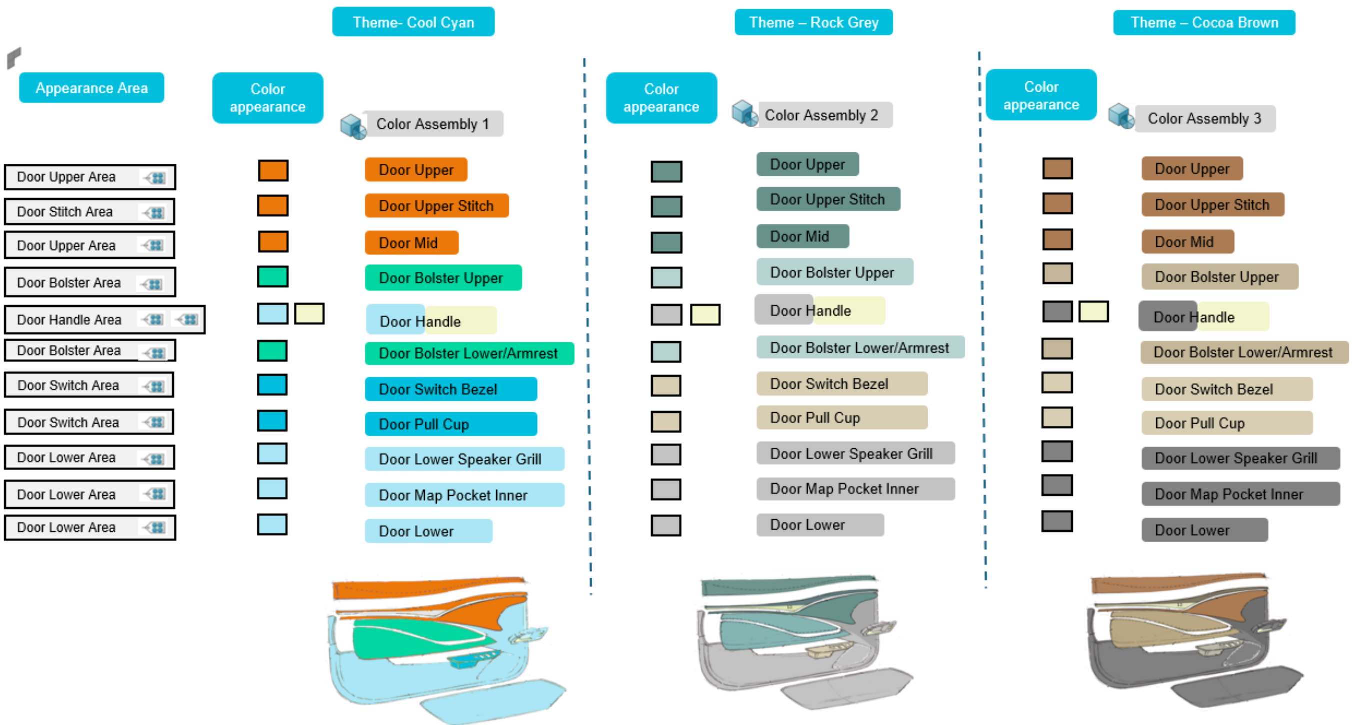
To organize all the different visual elements of the car, more than simply listing the parts is necessary. Instead, you first categorize the car's appearance in broader sections, such as the interior appearance and the exterior appearance. You can then further break down these broader sections into smaller sections, such as door appearance breakdown and seat appearance breakdown. This structured organization is known as an *appearance product breakdown* in Teamcenter.




The car door assembly will be mapped to the door appearance breakdown. In the door appearance breakdown, you define distinct zones that will share the same color or finish. For a car door, these zones can be the door upper area, the door stitch area, and the door handle area. Each of these zones, known as an *appearance area* in Teamcenter, can contain one or more Less Finish parts.

Next, you need to decide on the actual colors, textures, and gloss levels that can be applied to the Less Finish parts. These color options and styles are known as *color appearances* in Teamcenter.

Now, instead of specifying colors and styles for each part, you can define a *color theme*, which is a collection of color appearances. For example, you can define different color themes, such as Cool Cyan, Rock Grey, and Cocoa Brown. If we select the Cocoa Brown theme, it indicates that all parts of the car must use shades from the Cocoa Brown color palette, ensuring a consistent look.




Finally, once a Less Finish part is assigned a color code that belongs to a specific color theme, a *color part* is generated. It is no longer just a generic door assembly part, and it is now specifically a Cocoa Brown door assembly part. The color BOM is the final result, and it contains all the color parts, organized by their color appearance and color theme. The color BOM complements the engineering BOM by adding all the visual details that are required to manufacture the car exactly as designed.

# Identify less finish parts

To specify color-sensitive parts in the engineering BOM, you identify the parts that will have a color, and mark these as *Less Finish* parts.

### Procedure

1. Open an Assembly BOM, or a Usage BOM with a variability scope attached, and go to the **Content** tab.
2. (Optional) If the **Finish Type** column is not displayed by default, you can add it.
3. Click **Edit** .

4. Select the required part occurrence and select **Less Finish** in **Finish Type**. In this way, you can mark all the required part occurrences as **Less Finish**.
5. Click **Save Edits** .

# Filter engineering BOM based on finish type

You can filter a BOM so that you can work with a specific product definition that is smaller in size than the overall BOM.

### Restrictions and limitations

You can only filter BOM data that are indexed using Smart Discovery Indexing.

### Procedure

1. Open an Assembly BOM, or a Usage BOM with a variability scope attached, and go to the **Content** tab.
2. Click **Manage Color BOM** .
3. Click **Filter** . The BOM is filtered so that it shows all parts that have the **Finish Type** set to **Less Finish**.
4. In the **Filter** panel, apply **additional filters**, as required.

# Define appearance areas

An appearance area is a color-specific region of a product that encompasses one or more part occurrences that share the same color. You can create an appearance area in the context of an appearance product breakdown.

You can create multiple appearance areas under the appearance breakdown structure. The appearance product breakdown and appearance areas can be set by your administrator to automatically generate. If your administrator has not set these to be automatically generated, you can manually create the appearance product breakdown and appearance areas.

### Procedure

1. Open an Assembly BOM or a Usage BOM with a variability scope attached, and go to the **Content** tab.
2. Click **Manage Color BOM** .
3. Select the root node of the engineering BOM.
4. Do one of the following actions:

- To use an existing appearance product breakdown, double-click the **Appearance Area** column and select the required appearance product breakdown.
- To create a new appearance product breakdown, click ⊕. In the **Add** panel, enter the **Name** and **Product Line**. Click **Add** to create the appearance product breakdown.

**5.** For each assembly node or a part occurrence, you must set either an appearance area breakdown or an appearance area.

Do the one of the following actions:

| Scenario | Action |
| --- | --- |
| **If the selected part occurrence is an assembly node** | Do the one of the following actions:<br>• To use an existing appearance area breakdown, select the appearance area breakdown that was created for an existing appearance product breakdown for the root node.<br>• To create a new appearance area breakdown, click ⊕for the selected assembly node. In the **Add** panel, select **Type** as **Appearance Area Breakdown**, enter the **Name**, and click **Add** to create. |
| **If the selected part occurrence is a leaf node** | Do the one of the following actions:<br>• To use an existing appearance area, select the appearance area that was created for an existing appearance product breakdown for the root node.<br>• To create a new appearance area, click ⊕ for the selected part occurrence. In the **Add** panel, select **Type** as **Appearance Area** , enter the **Name**, and click **Add** to create. |

**Tip**

Appearance areas or breakdowns should always be assigned in a top-down approach to ensure an accurate alignment with the BOM. The *Less Finish* parts that require the same color should be assigned to the same appearance area. When done this way, you do not need to assign the same color multiple times within a color theme. Once the color specification is assigned to any *Less Finish* part, all other *Less Finish* parts that share the same appearance area will automatically inherit the same color specification.

# Create and manage color library

## About Color Library

Color Library serves as a single source to manage all color-related artifacts. It reduces duplication, ensures brand consistency, and streamlines updates across products. These elements are combined to create complete color appearances that can be applied to parts in your product structure.

You can create a Color Library by defining colors, gloss, texture, and appearance objects in Teamcenter. The Color Library centralizes color management to ensure consistency across all product configurations.

### Example

For example, as an appearance specialist, you want to create a *Cocoa Brown* color appearance for a car door. The *Cocoa Brown* color appearance is composed of four key specifications working together. The color definition provides the base hue and color value, in this case *Cocoa Brown*, which establishes the primary brown tone. The gloss specification controls the surface finish characteristics—in this example, a matte finish with 15% reflectivity—ensuring the appearance looks nonreflective and understated. The texture specification adds surface detail and tactile quality to the appearance. In this case, the leather grain texture mimics the feel and look of genuine leather material. Finally, the visual material specification provides the 3D rendering definition, combining RGB color values with physically based rendering (PBR) properties. In this way, the *Cocoa Brown* appearance displays correctly in digital visualizations and design tools.

### Color Themes

You can define color themes, which are collections of related color appearances and represent a complete palette of color appearances that work together visually. When you assign a color theme to parts in your product structure, all parts use the color appearances from that theme, ensuring visual consistency across your product.

### Preview color appearances

After creating color appearances, you can preview them to verify that they meet your design requirements, and then you can release them through approval workflows to make them available for use in product design and manufacturing.

## Add color to Color Library

Add a new color specification to the color library to define color values for use in product configurations. Color defines specific color values and properties that represent your brand colors or product-specific colors.

#### Prerequisites

You must be in EBOM or Author workspace to access Color Library.

#### Procedure

1. From Launcher, click **Color Library** .

   If this option is unavailable, search for it. You can pin it from the search results for future access.

2. In the **Color** tab, click **Add** to add a new color to the color library.
3. In the **Add Color** panel, specify the **Color Code** and the name of the color.
4. (Optional) Specify the RGB values, CMYK values, HSV values, and CIELab properties for the color.
5. Click **Add**.

#### Results

The new color specification is added to the color library and is available for use in color appearances.

## Add gloss to Color Library

Add a new gloss specification to the color library to define surface finish properties for product appearances. Gloss captures the surface finish or sheen level of a color, for example, matte, satin, or glossy.

#### Prerequisites

You must be in EBOM or Author workspace to access Color Library.

#### Procedure

1. From Launcher, click **Color Library** .

   If this option is unavailable, search for it. You can pin it from the search results for future access.

2. In the **Gloss** tab, click **Add** to add a new gloss to the color library.
3. In the **Add Gloss** panel, specify the **Name**, **Gloss Code**, and **Gloss Unit** properties for the gloss.
4. Click **Add**.

#### Results

The new gloss specification is added to the color library and can be combined with color and texture specifications to create color appearances.

# Add texture to Color Library

Add a new texture specification to the color library to define surface texture properties for product appearances. The specification describes the surface texture characteristics that contribute to the overall visual appearance, for example, smooth, rough, or patterned.

### Prerequisites

You must be in EBOM or Author workspace to access Color Library.

### Procedure

1. From Launcher, click **Color Library** .

   If this option is unavailable, search for it. You can pin it from the search results for future access.

2. In the **Texture** tab, click **Add** to add a new texture to the color library.
3. In the **Add Texture** panel, specify the name and **Texture Code**.
4. Click **Add**.

### Results

The new texture specification is added to the color library and can be combined with color and gloss specifications to create color appearances.

# Add color appearance to Color Library

A color appearance object captures all the visual appearance aspects of a color specification. It combines color, gloss, and texture information, along with visual material data, into one unified object. Color appearances serve as the primary objects that you assign to Bill of Materials (BOM) items. Add a new color appearance to the color library by combining color, gloss, and texture specifications to create a complete visual appearance definition.

### Prerequisites

You must be in EBOM or Author workspace to access Color Library.

### Procedure

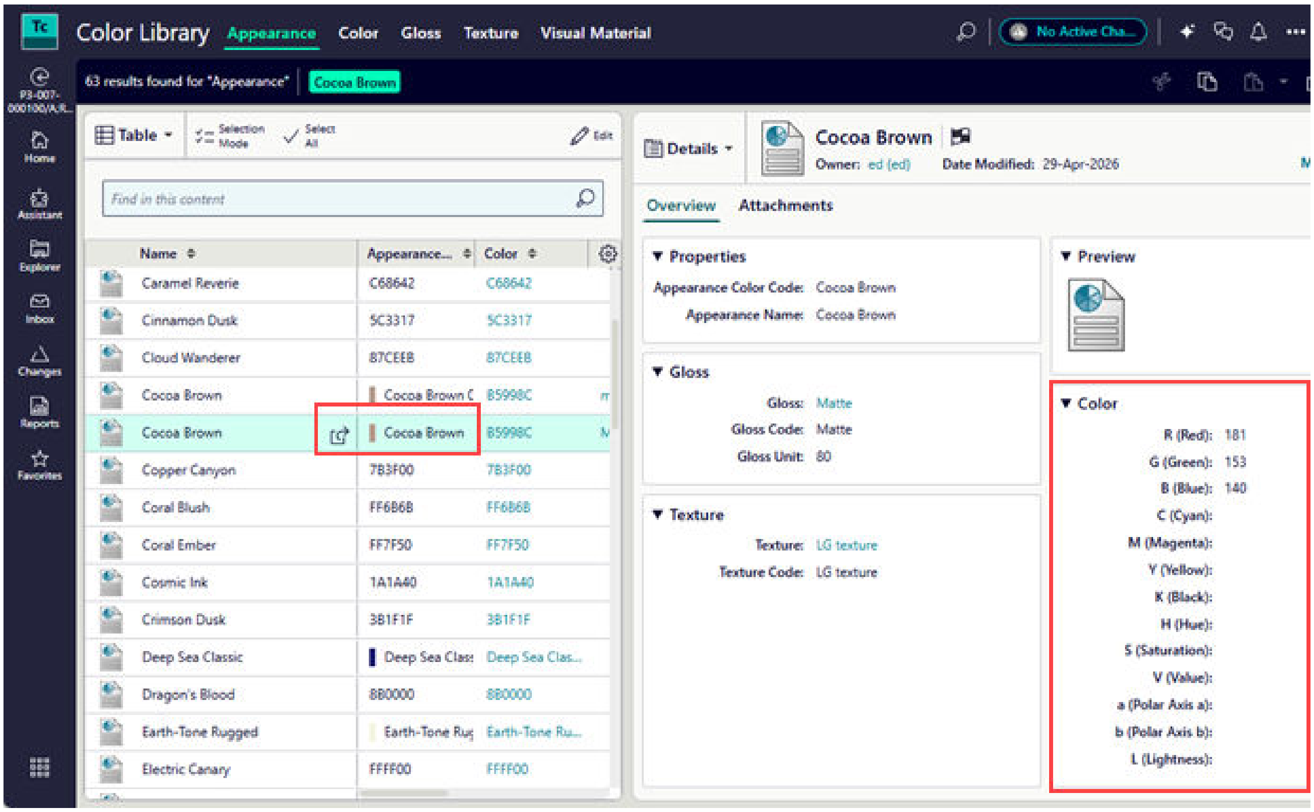
1. From Launcher, click **Color Library** .

   If this option is unavailable, search for it. You can pin it from the search results for future access.

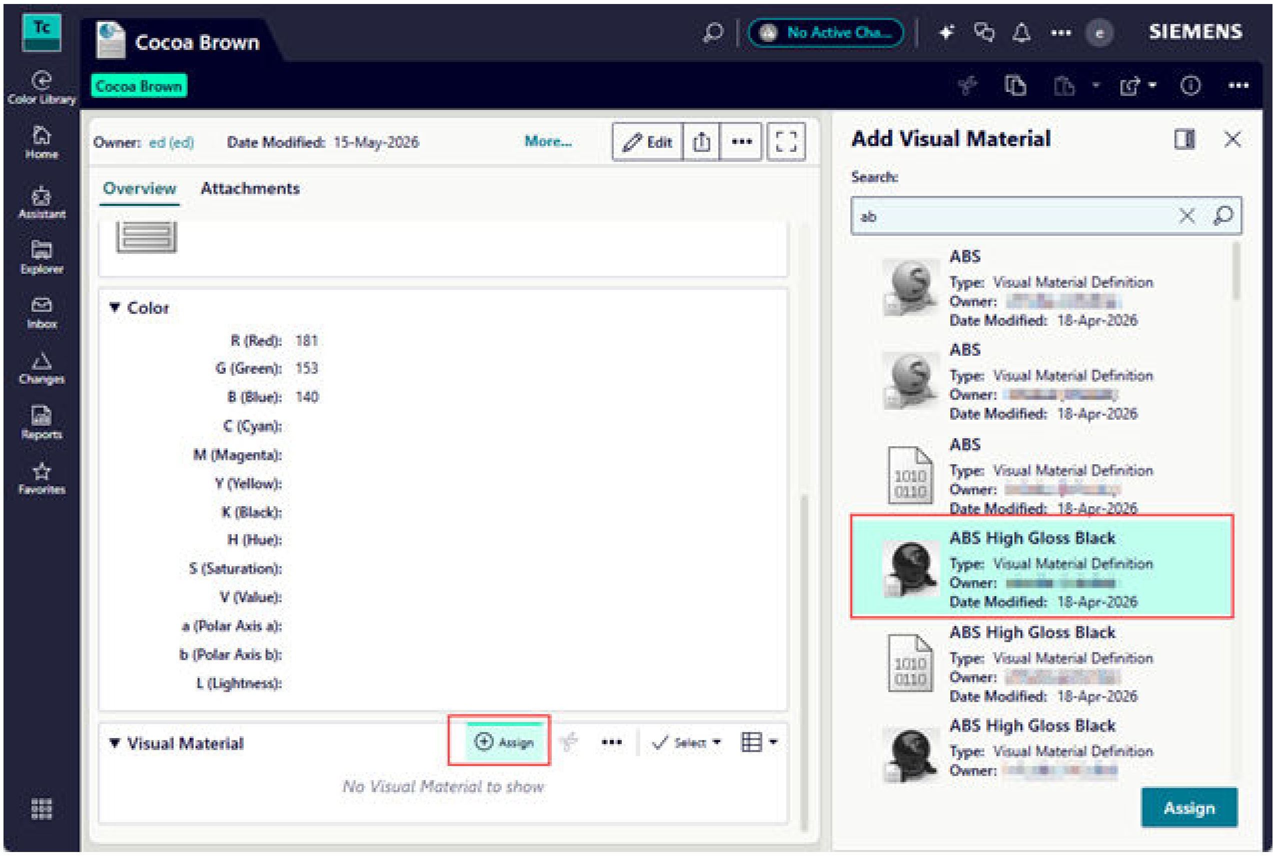
2. In the **Appearance** tab, click **Add** to add a new color appearance.

3. In the **Add Appearance Color** panel, specify the **Appearance Color Code** and the name of the appearance color.
4. (Optional) Select the **Color**, **Gloss**, and **Texture** for the appearance.
5. Click **Add**.

   In the table view, if the RGB values are specified for a color, the Appearance Color column shows a bar containing a preview of the color appearance.




6. To modify the color swatch for the appearance, open the recently created color appearance.
7. Click **Edit**.
8. (Optional) Modify the color, gloss, or texture for the appearance.
9. Under **Visual Material** for the color appearance, click **Assign** to search for and select a visual material from the **Add Visual Material** panel, and then assign it to the appearance.




This allows you to switch between RGB and Visual Material in the 3D rendering.

10. To remove the assigned visual material, select the object and click **Unassign**.

### Results

The new color appearance is added to the color library and is ready to be sent for review and approval.

## Release a color appearance

Release a color appearance through the approval workflow to make it available for scoping colors in a color theme.

### Prerequisites

Ensure you are in EBOM or Author workspace to access Color Library.

### Procedure

1. From Launcher, click **Color Library**.

   If this option is unavailable, search for it. You can pin it from the search results for future access.

2. In the **Appearance** tab, search for and open the recently created color appearance.
3. Click **More Commands** > **Manage** > **Submit to Workflow**.
4. In the **Submit to Workflow** panel, select **All** to view the available workflows.
5. From the **Template** list, select the appropriate workflow template.
6. Specify a **Name**, or accept the default.
7. (Optional) Enter the **Description**.
8. Click **Submit**.

**Results**

The color appearance is submitted to the workflow and becomes available for use in color themes on the configured approval statuses.

# Creating and managing color theme

## About color themes

A color theme is a predefined collection of color specifications assigned to appearance areas and bound to a product variant, enabling you to apply consistent color combinations across your product portfolio.

Color themes are a central component of your color management workflow. They connect your color specifications library to your product variants, allowing you to apply predefined color combinations to products systematically. Each color theme is bound to a variant rule, which means when you apply the variant to a product, the system automatically applies the associated color theme.

A color theme consists of color specifications that are assigned to appearance areas within your product structure. For example, if you have created color specifications for *Metallic Silver* and *Matte Black* in your color specifications library, you can organize these specifications into a color theme called *Modern Industrial* and assign *Metallic Silver* to the housing appearance area and *Matte Black* to the trim appearance area. When this theme is bound to a product variant, any product using that variant automatically receives the Modern Industrial color combination.

Color themes follow a lifecycle that includes creation, management, and discontinuation. You can **create new color themes** by selecting color specifications from your color specifications library and assigning them to specific appearance areas. You can **edit** existing themes to modify their properties and color code assignments as your product requirements evolve. When a color theme is no longer needed, you can **discontinue** it to prevent its use in new product variants while preserving its historical data and existing assignments in the Bill of Materials.

## Create a color theme

A color theme is a collection of color appearances that contains the visual aspects of a product, such as color, gloss, and texture. Instead of specifying a color for each part in the product, you can create a color theme, for example, *Modern Industrial*. You can optionally scope the theme to a subset of available colors, for example, *Metallic Silver* and *Matte Black*, which limits the color options that appear when you assign color appearances to parts. This streamlines the assignment process by preventing users from scrolling through large lists of irrelevant colors.

You create different themes based on product offerings.

### Procedure

1. Open an Assembly BOM or a Usage BOM with a variability scope attached.
2. Go to the **Content** tab.
3. Click **Manage Color BOM**.
4. Click **Create Theme**.
5. In the **Create Color Theme** panel:
   a. Enter a **Name** for the color theme. For example, Cocoa Brown.
   b. In **Code**, enter the theme code. For example, CCB.
   c. In **Select Models and Features**, press the space bar on your keyboard, and select the required variant criteria.

      You can concatenate additional criteria by pressing the space bar again and choosing the **AND** or **OR** operator. For example, when a seat is XL-sized and has *Metallic Silver* color, then the *Modern Industrial* theme becomes applicable. You can define this with a variant expression, such as `Color='Metallic Silver' AND Seat='Size XL'`.

      Click **Show Internal Formula** to see internal expression.

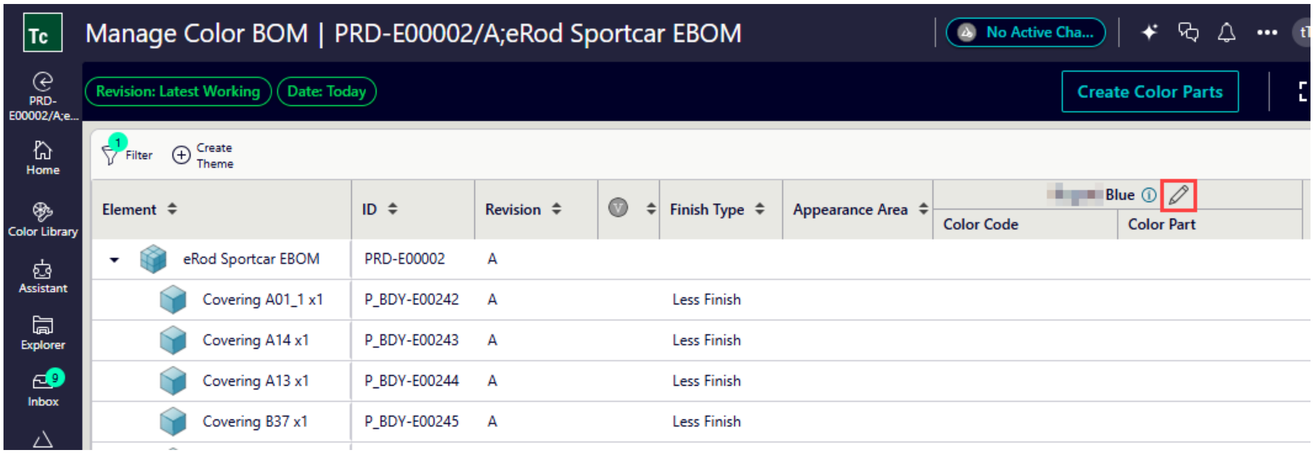
      Click **Validate Syntax** to check the correctness of the expression for the variant criteria.
   d. To define the scope of colors associated with a specific theme, select and add colors from the list of **Available Colors** to **Theme Colors**.
   e. Click **Add**.

## Edit or discontinue a color theme

You can manage your color theme portfolio by editing color theme properties and color code assignments to align with current product offerings, or discontinuing themes that are no longer relevant to prevent their use in new product variants while preserving historical data and existing Bill of Materials assignments.

### Procedure

1. Open an Assembly BOM or a Usage BOM with a variability scope attached, and go to the **Content** tab.
2. Click **Manage Color BOM** .
3. To edit a theme, click the edit button against the color theme that you want to modify.




4. In the **Edit Color Theme** panel, modify the scope of color as required and click **Save**.
5. If you want to discontinue a color theme, click **Discontinue**.

## Assign color appearance to a less finish part or an assembly

A color appearance captures all the visual appearance aspects, such as color, gloss, and texture information.

You can assign the color appearance to leaf-level *Less Finish* parts or the assembly node in accordance with the color theme.

> **Note**
>
> The color appearance at the assembly level is used only for ID generation. It does not visually represent the assembly's actual color. When you assign a color appearance to an assembly, the system extracts the color code from the assigned color appearance.

### Prerequisites

A **color theme** must exist. If the automatic generation of appearance areas is enabled, the appearance areas are created automatically. Otherwise, they must be manually created before proceeding with the mapping.

### Procedure

1. Open an Assembly BOM or a Usage BOM with a variability scope attached, and go to the **Content** tab.
2. Click **Manage Color BOM** .
3. Under the **Color Theme** for an assembly node or each part occurrence, double-click **Color Appearance** and select a color appearance that you want to assign.

   You can select the color appearances from the options available depending on the color scoping configuration:

   - Theme-level scoping only: You can select any color scoped in the color theme.
   - Both theme-level and part-level scoping: You can select only colors that are common to both the color theme and the part's color specifications. For example, if a color theme includes colors Blue, Black, and Green, but a specific part is scoped with colors Black, Green, and Red, only colors Black and Green will be available for selection for that part.
   - No scoping: You can select from all the available colors.

   > **Note**
   >
   > Part-level color scoping is configured by administrators using a command-line utility and restricts the available colors for specific parts based on supplier or sourcing constraints.

# Create and manage color parts

## Create color parts manually

A color part is generated from a *Less Finish* part for a color specification.

You can create color parts manually or **automatically**.

### Prerequisites

At least one color theme must be defined, along with the assigned **color appearances**.

### Procedure

1. Open an Assembly BOM or a Usage BOM with a variability scope attached, and go to the **Content** tab.
2. Click **Manage Color BOM** .
3. For each part occurrence, under specific color theme, click  in **Color Part**.
4. In the **Add Color Part** panel:
   a. Enter the **Name** for the color part.

b. Clear the check box for **Design Required**.

c. Select the **Finish Type** as **Color**.

d. Click **Add**.

## Create color parts automatically

A color part is generated from a *Less Finish* part for a color specification.

You can create color parts automatically or **manually**.

### Prerequisites

At least one color theme must be defined, along with the assigned **color appearance**.

### Procedure

1. Open an Assembly BOM or a Usage BOM with a variability scope attached, and go to the **Content** tab.
2. Click **Manage Color BOM** .
3. Click **Create Color Parts**.
4. Complete the details in the **Create Color Parts** panel.
    a. Select the required **Color Themes** to generate the color parts.
    b. Click **Create** to generate the color BOM.

    Once the color parts are created, you can see the color part ID in the **Color Part** column of the theme. Part occurrences with the same color specification in different themes share the same color part ID.

## Reuse color parts

You can reuse existing legacy color parts during the color part creation process. You can insert or create them in the context of the color code for *Less Finish* parts.

### Prerequisites

At least one color theme must be defined, along with the assigned **color appearances**.

### Procedure

1. Open an Assembly BOM or a Usage BOM with a variability scope attached, and go to the **Content** tab.
2. Click **Manage Color BOM** .

3. For each leaf-level part occurrence, double-click the **Color Part** column and select the required color part.

   You can only add an existing color part that satisfies the following criteria:

   a. The **Finish Type** property of the part must be either **Color** or **None**.

   b. The part must not be aligned to any design.

   c. The part must not be associated with any other *Less Finish* part.

   d. If the part is not newly created, then its color code must match that of the corresponding *Less Finish* part.

## Update color parts

Once the initial color parts are generated, you can update them whenever structural changes occur in the engineering BOM structure, such as additions, removals, or replacements within the *Less Finish* assembly.

#### Procedure

1. Open an Assembly BOM or a Usage BOM with a variability scope attached, and go to the **Content** tab.

2. Click **Manage Color BOM** .

3. For each newly added *Less Finish* part occurrence within the *Less Finish* assembly node, assign the **Appearance Area** and **Color Appearance**.

4. Select the root node of the BOM. Then, right-click and select **Submit to Workflow** .

5. In the **Submit to Workflow** panel, complete the following actions.

   a. Select **Template** for the **Update Color Parts**.

   b. Click **Submit**.

# Visualizing color parts with assigned color appearances

## Evaluating color parts with assigned color appearances

Color visualization allows you to preview products with assigned color appearances in the 3D viewer before finalizing the Bill of Materials (BOM) structure and generating color parts.

Color visualization serves as a preliminary step in the color management workflow. You can evaluate color choices and make decisions about the final appearance before committing to the BOM structure.

Color visualization supports two types of color definitions that provide different levels of visual representation.

- RGB color definitions: RGB color definitions provide a flat color representation based on Red, Green, and Blue values. RGB colors are simpler and less detailed in appearance, making them suitable for quick previews and basic color visualization.
- Siemens Visual Material (SVM) definitions: SVM definitions provide a complete appearance definition that includes color, gloss, and texture properties. SVM creates a more realistic and detailed appearance compared to RGB colors, giving you a better representation of the final product appearance.

You can switch between RGB and SVM visualizations in the 3D viewer, provided the relevant color data is available for your parts. The system uses a priority-based approach to determine which color data to display.

- If SVM data is available, the system renders using SVM properties.
- If SVM data is not available but RGB data exists, the system renders using the RGB color definition.
- If neither SVM data nor RGB data is present, the system displays the default computer-aided design (CAD)-assigned color for the part.

## Visualize color parts

Preview your products with their assigned color appearances in the 3D viewer before generating color parts in the Bill of Materials (BOM). You use RGB definitions for flat color representation and SVM definitions for realistic appearances that include color, gloss, and texture properties helping you to evaluate the final appearance of the color.

### Prerequisites

- Color appearances are created and associated with the color codes.
- Color codes are assigned to appearance areas within the color themes.
- Color themes are created.
- (Optional) Visual materials are attached to color appearance objects to allow for SVM visualization.
- Expressions are applied to the variant rules so that the associated color theme is applicable when a specific variant rule is selected.

### Procedure

1. Open an Assembly BOM or a Usage BOM with a variability scope attached, and go to the **Content** tab.
2. Navigate to your product structure in the BOM view, and click the **3D** tab to open the 3D viewer. The product is displayed in the default CAD color.
3. In the 3D viewer toolbar, click **Viewer Options**.
4. In the Viewer Options panel, select the **Use Variant Defined Color Information** check box.
5. Select one of the following color definition types:

| Option | Description |
| --- | --- |
| **Use RGB Definition** | Visualizes your product using RGB color definitions for a flat color representation. |
| **Use SVM Definition** | Visualizes your product using SVM definitions for a realistic appearance that includes color, gloss, and texture properties. |

> **Note**
>
> The available options depend on which color definitions are created for your color appearances. If SVM data is not available for a part, the system uses RGB data or the default CAD color.

6. In the BOM view, select the variant rule that specifies your color theme to apply it to the product structure.
   The 3D viewer updates to display the product with the colors defined in the selected color theme.
7. (Optional) To view only the parts with color, filter out parts with a finish type of **None** or **Less Finish** in the BOM view.

   > **Note**
   >
   > Color parts and less-finishd parts occupy the same position in the 3D model. Filtering to show only color parts provides a clearer visualization of the final appearance color.

   The 3D viewer updates to show only the parts that have color.
8. (Optional) To compare RGB and SVM visualizations, return to **Viewer Options** and change the color definition type.

   > **Note**
   >
   > If SVM data is not available for a part, the system displays the color in the RGB or default CAD color.

   The 3D viewer updates to display the product using the selected color definition type.

### Results

The 3D viewer displays your product with the assigned color appearances. You can now evaluate the colors before generating color parts in the BOM.

> **Note**
>
> If your product structure includes subassemblies with color appearances and generated color parts, the color parts and appearance information from subassemblies are automatically inherited and displayed in the complete BOM structure.